



Linux
Professional
Institute

LPIC-1

Version 5.0
English

101

Table of Contents

TOPIC 101: SYSTEM ARCHITECTURE	1
101.1 Determine and configure hardware settings	2
101.1 Lesson 1	3
Introduction	3
Device Activation	4
Device Inspection in Linux	4
Information Files and Device Files	11
Storage Devices	12
Guided Exercises	14
Explorational Exercises	15
Summary	16
Answers to Guided Exercises	17
Answers to Explorational Exercises	18
101.2 Boot the system	19
101.2 Lesson 1	21
Introduction	21
BIOS or UEFI	21
The Bootloader	23
System Initialization	25
Initialization Inspection	26
Guided Exercises	29
Explorational Exercises	30
Summary	31
Answers to Guided Exercises	32
Answers to Explorational Exercises	33
101.3 Change runlevels / boot targets and shutdown or reboot system	34
101.3 Lesson 1	36
Introduction	36
SysVinit	37
systemd	40
Upstart	43
Shutdown and Restart	44
Guided Exercises	46
Explorational Exercises	47
Summary	48
Answers to Guided Exercises	49
Answers to Explorational Exercises	50
TOPIC 102: LINUX INSTALLATION AND PACKAGE MANAGEMENT	51

102.1 Design hard disk layout	52
102.1 Lesson 1	53
Introduction	53
Mount Points	54
Keeping Things Separated	55
Swap	57
LVM	58
Guided Exercises	60
Explorational Exercises	61
Summary	62
Answers to Guided Exercises	63
Answers to Explorational Exercises	64
102.2 Install a boot manager	65
102.2 Lesson 1	66
Introduction	66
GRUB Legacy vs. GRUB 2	67
Where is the Bootloader?	67
The /boot Partition	68
GRUB 2	69
GRUB Legacy	75
Guided Exercises	80
Explorational Exercises	81
Summary	82
Answers to Guided Exercises	83
Answers to Explorational Exercises	84
102.3 Manage shared libraries	86
102.3 Lesson 1	87
Introduction	87
Concept of Shared Libraries	87
Shared Object File Naming Conventions	88
Configuration of Shared Library Paths	89
Searching for the Dependencies of a Particular Executable	92
Guided Exercises	94
Explorational Exercises	95
Summary	96
Answers to Guided Exercises	98
Answers to Explorational Exercises	99
102.4 Use Debian package management	100
102.4 Lesson 1	101
Introduction	101

The Debian Package Tool (dpkg)	102
Advanced Package Tool (apt)	106
Guided Exercises	116
Explorational Exercises	117
Summary	118
Answers to Guided Exercises	120
Answers to Explorational Exercises	121
102.5 Use RPM and YUM package management	122
102.5 Lesson 1	123
Introduction	123
The RPM Package Manager (rpm)	124
YellowDog Updater Modified (YUM)	129
DNF	134
Zypper	136
Guided Exercises	143
Explorational Exercises	144
Summary	145
Answers to Guided Exercises	146
Answers to Explorational Exercises	147
102.6 Linux as a virtualization guest	148
102.6 Lesson 1	149
Introduction	149
Virtualization Overview	149
Types of Virtual Machines	150
Working with Virtual Machine Templates	157
Deploying Virtual Machines to the Cloud	158
Containers	161
Guided Exercises	163
Explorational Exercises	164
Summary	165
Answers to Guided Exercises	166
Answers to Explorational Exercises	167
TOPIC 103: GNU AND UNIX COMMANDS	169
103.1 Work on the command line	170
103.1 Lesson 1	172
Introduction	172
Getting System Information	172
Getting Command Information	173
Using Your Command History	176
Guided Exercises	177

Explorational Exercises	178
Summary	179
Answers to Guided Exercises	180
Answers to Explorational Exercises	181
103.1 Lesson 2	182
Introduction	182
Finding Your Environment Variables	182
Creating New Environment Variables	183
Deleting Environment Variables	184
Quoting to Escape Special Characters	185
Guided Exercises	187
Explorational Exercises	188
Summary	189
Answers to Guided Exercises	190
Answers to Explorational Exercises	191
103.2 Process text streams using filters	192
103.2 Lesson 1	194
Introduction	194
A Quick Review on Redirections and Pipes	194
Processing Text Streams	196
Guided Exercises	208
Explorational Exercises	210
Summary	212
Answers to Guided Exercises	214
Answers to Explorational Exercises	219
103.3 Perform basic file management	225
103.3 Lesson 1	227
Introduction	227
Manipulating Files	228
Creating and Deleting Directories	233
Recursive Manipulation of Files and Directories	234
File Globbing and Wildcards	237
Types of Wildcards	238
Guided Exercises	242
Explorational Exercises	244
Summary	245
Answers to Guided Exercises	246
Answers to Explorational Exercises	248
103.3 Lesson 2	250
Introduction	250

How to Find Files	250
Archiving Files	254
Guided Exercises	260
Explorational Exercises	261
Summary	262
Answers to Guided Exercises	263
Answers to Explorational Exercises	264
103.4 Use streams, pipes and redirects	266
103.4 Lesson 1	267
Introduction	267
Redirects	268
Here Document and Here String	271
Guided Exercises	273
Explorational Exercises	274
Summary	275
Answers to Guided Exercises	276
Answers to Explorational Exercises	277
103.4 Lesson 2	278
Introduction	278
Pipes	278
Command Substitution	280
Guided Exercises	283
Explorational Exercises	284
Summary	285
Answers to Guided Exercises	286
Answers to Explorational Exercises	287
103.5 Create, monitor and kill processes	288
103.5 Lesson 1	290
Introduction	290
Job Control	290
Process Monitoring	295
Guided Exercises	306
Explorational Exercises	308
Summary	310
Answers to Guided Exercises	312
Answers to Explorational Exercises	315
103.5 Lesson 2	318
Introduction	318
Features of Terminal Multiplexers	318
GNU Screen	319

tmux	326
Guided Exercises	334
Explorational Exercises	338
Summary	340
Answers to Guided Exercises	341
Answers to Explorational Exercises	345
103.6 Modify process execution priorities	347
103.6 Lesson 1	348
Introduction	348
The Linux Scheduler	349
Reading Priorities	349
Process Niceness	351
Guided Exercises	353
Explorational Exercises	355
Summary	356
Answers to Guided Exercises	357
Answers to Explorational Exercises	359
103.7 Search text files using regular expressions	360
103.7 Lesson 1	361
Introduction	361
Bracket Expression	362
Quantifiers	363
Bounds	364
Branches and Back References	365
Searching with Regular Expressions	365
Guided Exercises	367
Explorational Exercises	368
Summary	369
Answers to Guided Exercises	370
Answers to Explorational Exercises	371
103.7 Lesson 2	372
Introduction	372
The Pattern Finder: grep	372
The Stream Editor: sed	376
Combining grep and sed	380
Guided Exercises	383
Explorational Exercises	384
Summary	386
Answers to Guided Exercises	387
Answers to Explorational Exercises	388

103.8 Basic file editing	390
103.8 Lesson 1	391
Introduction	391
Insert Mode	392
Normal Mode	392
Colon Commands	395
Alternative Editors	396
Guided Exercises	398
Explorational Exercises	399
Summary	400
Answers to Guided Exercises	401
Answers to Explorational Exercises	402
TOPIC 104: DEVICES, LINUX FILESYSTEMS, FILESYSTEM HIERARCHY STANDARD	403
104.1 Create partitions and filesystems	404
104.1 Lesson 1	405
Introduction	405
Understanding MBR and GPT	406
Creating File Systems	413
Managing Partitions with GNU Parted	423
Creating Swap Partitions	430
Guided Exercises	432
Explorational Exercises	433
Summary	435
Answers to Guided Exercises	436
Answers to Explorational Exercises	437
104.2 Maintain the integrity of filesystems	439
104.2 Lesson 1	440
Introduction	440
Checking Disk Usage	440
Checking for Free Space	443
Maintaining ext2, ext3 and ext4 Filesystems	447
Guided Exercises	454
Explorational Exercises	455
Summary	456
Answers to Guided Exercises	457
Answers to Explorational Exercises	459
104.3 Control mounting and unmounting of filesystems	461
104.3 Lesson 1	462
Introduction	462
Mounting and Unmounting Filesystems	462

Mounting Filesystems on Bootup	466
Using UUIDs and Labels	468
Mounting Disks with Systemd	470
Guided Exercises	474
Explorational Exercises	475
Summary	476
Answers to Guided Exercises	477
Answers to Explorational Exercises	479
104.5 Manage file permissions and ownership	481
104.5 Lesson 1	482
Introduction	482
Querying Information about Files and Directories	482
What about Directories?	483
Viewing Hidden Files	484
Understanding Filetypes	485
Understanding Permissions	485
Modifying File Permissions	487
Modifying File Ownership	491
Querying Groups	491
Default Permissions	492
Special Permissions	494
Guided Exercises	498
Explorational Exercises	500
Summary	501
Answers to Guided Exercises	502
Answers to Explorational Exercises	505
104.6 Create and change hard and symbolic links	508
104.6 Lesson 1	509
Introduction	509
Understanding Links	509
Guided Exercises	514
Explorational Exercises	515
Summary	518
Answers to Guided Exercises	519
Answers to Explorational Exercises	520
104.7 Find system files and place files in the correct location	524
104.7 Lesson 1	525
Introduction	525
The Filesystem Hierarchy Standard	525
Finding Files	528

Guided Exercises	537
Explorational Exercises	538
Summary	539
Answers to Guided Exercises	540
Answers to Explorational Exercises	542
Imprint	544



**Linux
Professional
Institute**

Topic 101: System Architecture



101.1 Determine and configure hardware settings

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 101.1

Weight

2

Key knowledge areas

- Enable and disable integrated peripherals.
- Differentiate between the various types of mass storage devices.
- Determine hardware resources for devices.
- Tools and utilities to list various hardware information (e.g. lsusb, lspci, etc.).
- Tools and utilities to manipulate USB devices.
- Conceptual understanding of sysfs, udev and dbus.

Partial list of the used files, terms and utilities

- `/sys/`
- `/proc/`
- `/dev/`
- `modprobe`
- `lsmod`
- `lspci`
- `lsusb`



101.1 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	101 System Architecture
Objective:	101.1 Determine and configure hardware settings
Lesson:	1 of 1

Introduction

Since the early years of electronic computing, business and personal computer manufacturers have integrated a variety of hardware parts in their machines, which in turn need to be supported by the operating system. That could be overwhelming from the operating system developer's perspective, unless standards for instruction sets and device communication are established by the industry. Similar to the standardized abstraction layer provided by the operating system to an application, these standards make it easier to write and maintain an operating system that is not tied to a specific hardware model. However, the complexity of the integrated underlying hardware sometimes requires adjustments on how resources should be exposed to the operating system, so it can be installed and function correctly.

Some of these adjustments can be made even without an installed operating system. Most machines offer a configuration utility that can be executed as the machine is turned on. Until the mid 2000s, the configuration utility was implemented in the BIOS (*Basic Input/Output System*), the standard for firmware containing the basic configuration routines found in x86 motherboards. From the end of the first decade of the 2000s, machines based on the x86 architecture started to replace the BIOS with a new implementation called UEFI (*Unified Extensible Firmware Interface*),

which has more sophisticated features for identification, testing, configuration and firmware upgrades. Despite the change, it is not uncommon to still call the configuration utility BIOS, as both implementations fulfill the same basic purpose.

NOTE

Further details on the similarities and differences between BIOS and UEFI will be covered in a later lesson.

Device Activation

The system configuration utility is presented after pressing a specific key when the computer is turned on. Which key to press varies from manufacturer to manufacturer, but usually it is `Del` or one of the function keys, such as `F2` or `F12`. The key combination to use is often displayed in the power on screen.

In the BIOS setup utility it is possible to enable and disable integrated peripherals, activate basic error protection and change hardware settings like IRQ (interrupt request) and DMA (direct memory access). Changing these settings is rarely needed on modern machines, but it may be necessary to make adjustments to address specific issues. There are RAM technologies, for example, that are compatible with faster data transfer rates than the default values, so it is recommended to change it to the values specified by the manufacturer. Some CPUs offer features that may not be required for the particular installation and can be deactivated. Disabled features will reduce power consumption and can increase system protection, as CPU features containing known bugs can also be disabled.

If the machine is equipped with many storage devices, it is important to define which one has the correct bootloader and should be the first entry in the device boot order. The operating system may not load if the incorrect device comes first in the BIOS boot verification list.

Device Inspection in Linux

Once devices are correctly identified, it is up to the operating system to associate the corresponding software components required by them. When a hardware feature is not working as expected, it is important to identify where exactly the problem is happening. When a piece of hardware is not detected by the operating system, it is most likely that the part — or the port to which it is connected — is defective. When the hardware part is correctly detected, but still does not properly work, there may be a problem on the operating system side. Therefore, one of the first steps when dealing with hardware-related issues is to check if the operating system is properly detecting the device. There are two basic ways to identify hardware resources on a Linux system: to use specialized commands or to read specific files inside special filesystems.

Commands for Inspection

The two essential commands to identify connected devices in a Linux system are:

lspci

Shows all devices currently connected to the PCI (*Peripheral Component Interconnect*) bus. PCI devices can be either a component attached to the motherboard, like a disk controller, or an expansion card fitted into a PCI slot, like an external graphics card.

lsusb

Lists USB (*Universal Serial Bus*) devices currently connected to the machine. Although USB devices for almost any imaginable purpose exist, the USB interface is largely used to connect input devices — keyboards, pointing devices — and removable storage media.

The output of commands `lspci` and `lsusb` consists of a list of all PCI and USB devices identified by the operating system. However, the device may not be fully operational yet, because every hardware part requires a software component to control the corresponding device. This software component is called a *kernel module* and it can be part of the official Linux kernel or added separately from other sources.

Linux kernel modules related to hardware devices are also called *drivers*, as in other operating systems. Drivers for Linux, however, are not always supplied by the device's manufacturer. Whilst some manufacturers provide their own binary drivers to be installed separately, many drivers are written by independent developers. Historically, parts that work on Windows, for example, may not have a counterpart kernel module for Linux. Nowadays, Linux-based operating systems have strong hardware support and most devices work effortlessly.

Commands directly related to hardware often require root privileges to execute or will only show limited information when executed by a normal user, so it may be necessary to login as root or to execute the command with `sudo`.

The following output of command `lspci`, for example, shows a few identified devices:

```
$ lspci
01:00.0 VGA compatible controller: NVIDIA Corporation GM107 [GeForce GTX 750 Ti] (rev a2)
04:02.0 Network controller: Ralink corp. RT2561/RT61 802.11g PCI
04:04.0 Multimedia audio controller: VIA Technologies Inc. ICE1712 [Envy24] PCI Multi-
Channel I/O Controller (rev 02)
04:0b.0 FireWire (IEEE 1394): LSI Corporation FW322/323 [TrueFire] 1394a Controller (rev 70)
```

The output of such commands can be tens of lines long, so the previous and next examples contain

only the sections of interest. The hexadecimal numbers at the beginning of each line are the unique addresses of the corresponding PCI device. The command `lspci` shows more details about a specific device if its address is given with option `-s`, accompanied by the option `-v`:

```
$ lspci -s 04:02.0 -v
04:02.0 Network controller: Ralink corp. RT2561/RT61 802.11g PCI
  Subsystem: Linksys WMP54G v4.1
  Flags: bus master, slow devsel, latency 32, IRQ 21
  Memory at e3100000 (32-bit, non-prefetchable) [size=32K]
  Capabilities: [40] Power Management version 2
  kernel driver in use: rt61pci
```

The output now shows many more details of the device on address `04:02.0`. It is a network controller, whose internal name is `Ralink corp. RT2561/RT61 802.11g PCI`. Subsystem is associated with the device's brand and model—`Linksys WMP54G v4.1`—and can be helpful for diagnostic purposes.

The kernel module can be identified in the line `kernel driver in use`, which shows the module `rt61pci`. From all the gathered information, it is correct to assume that:

1. The device was identified.
2. A matching kernel module was loaded.
3. The device should be ready for use.

Another way to verify which kernel module is in use for the specified device is provided by the option `-k`, available in more recent versions of `lspci`:

```
$ lspci -s 01:00.0 -k
01:00.0 VGA compatible controller: NVIDIA Corporation GM107 [GeForce GTX 750 Ti] (rev a2)
  kernel driver in use: nvidia
  kernel modules: nouveau, nvidia_drm, nvidia
```

For the chosen device, an NVIDIA GPU board, `lspci` tells that the module in use is named `nvidia`, at line `kernel driver in use: nvidia` and all corresponding kernel modules are listed in the line `kernel modules: nouveau, nvidia_drm, nvidia`.

Command `lsusb` is similar to `lspci`, but lists USB information exclusively:

```
$ lsusb
Bus 001 Device 029: ID 1781:0c9f Multiple Vendors USBtiny
```



```

Bus 001 Device 028: ID 093a:2521 Pixart Imaging, Inc. Optical Mouse
Bus 001 Device 020: ID 1131:1001 Integrated System Solution Corp. KY-BT100 Bluetooth Adapter
Bus 001 Device 011: ID 04f2:0402 Chicony Electronics Co., Ltd Genius LuxeMate i200 Keyboard
Bus 001 Device 007: ID 0424:7800 Standard Microsystems Corp.
Bus 001 Device 003: ID 0424:2514 Standard Microsystems Corp. USB 2.0 Hub
Bus 001 Device 002: ID 0424:2514 Standard Microsystems Corp. USB 2.0 Hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub

```

Command `lsusb` shows the available USB channels and the devices connected to them. As with `lspci`, option `-v` displays more detailed output. A specific device can be selected for inspection by providing its ID to the option `-d`:

```

$ lsusb -v -d 1781:0c9f
Bus 001 Device 029: ID 1781:0c9f Multiple Vendors USBtiny
Device Descriptor:
  bLength                18
  bDescriptorType         1
  bcdUSB                  1.01
  bDeviceClass            255 Vendor Specific Class
  bDeviceSubClass         0
  bDeviceProtocol         0
  bMaxPacketSize0         8
  idVendor                0x1781 Multiple Vendors
  idProduct               0x0c9f USBtiny
  bcdDevice               1.04
  iManufacturer          0
  iProduct               2 USBtiny
  iSerial                 0
  bNumConfigurations      1

```

With option `-t`, command `lsusb` shows the current USB device mappings as a hierarchical tree:

```

$ lsusb -t
/: Bus 01.Port 1: Dev 1, Class=root_hub, Driver=dwc_otg/lp, 480M
   |__ Port 1: Dev 2, If 0, Class=Hub, Driver=hub/4p, 480M
      |__ Port 1: Dev 3, If 0, Class=Hub, Driver=hub/3p, 480M
         |__ Port 2: Dev 11, If 1, Class=Human Interface Device, Driver=usbhid, 1.5M
         |__ Port 2: Dev 11, If 0, Class=Human Interface Device, Driver=usbhid, 1.5M
         |__ Port 3: Dev 20, If 0, Class=Wireless, Driver=btusb, 12M
         |__ Port 3: Dev 20, If 1, Class=Wireless, Driver=btusb, 12M
         |__ Port 3: Dev 20, If 2, Class=Application Specific Interface, Driver=, 12M
         |__ Port 1: Dev 7, If 0, Class=Vendor Specific Class, Driver=lan78xx, 480M

```

```
|__ Port 2: Dev 28, If 0, Class=Human Interface Device, Driver=usbhid, 1.5M
|__ Port 3: Dev 29, If 0, Class=Vendor Specific Class, Driver=, 1.5M
```

It is possible that not all devices have corresponding modules associated with them. The communication with certain devices can be handled by the application directly, without the intermediation provided by a module. Nevertheless, there is significant information in the output provided by `lsusb -t`. When a matching module exists, its name appears at the end of the line for the device, as in `Driver=btusb`. The device Class identifies the general category, like Human Interface Device, Wireless, Vendor Specific Class, Mass Storage, among others. To verify which device is using the module `btusb`, present in the previous listing, both Bus and Dev numbers should be given to the `-s` option of the `lsusb` command:

```
$ lsusb -s 01:20
```

```
Bus 001 Device 020: ID 1131:1001 Integrated System Solution Corp. KY-BT100 Bluetooth Adapter
```

It is common to have a large set of loaded kernel modules in a standard Linux system at any time. The preferable way to interact with them is to use the commands provided by the `kmod` package, which is a set of tools to handle common tasks with Linux kernel modules like insert, remove, list, check properties, resolve dependencies and aliases. The `lsmod` command, for example, shows all currently loaded modules:

```
$ lsmod
```

Module	Size	Used by
kvm_intel	138528	0
kvm	421021	1 kvm_intel
itCO_wdt	13480	0
itCO_vendor_support	13419	1 itCO_wdt
snd_usb_audio	149112	2
snd_hda_codec_realtek	51465	1
snd_ice1712	75006	3
snd_hda_intel	44075	7
arc4	12608	2
snd_cs8427	13978	1 snd_ice1712
snd_i2c	13828	2 snd_ice1712,snd_cs8427
snd_ice17xx_ak4xxx	13128	1 snd_ice1712
snd_ak4xxx_adda	18487	2 snd_ice1712,snd_ice17xx_ak4xxx
microcode	23527	0
snd_usbmidi_lib	24845	1 snd_usb_audio
gspca_pac7302	17481	0
gspca_main	36226	1 gspca_pac7302
videodev	132348	2 gspca_main,gspca_pac7302

```

rt61pci          32326  0
rt2x00pci        13083  1 rt61pci
media            20840  1 videodev
rt2x00mmio       13322  1 rt61pci
hid_dr           12776  0
snd_mpu401_uart  13992  1 snd_ice1712
rt2x00lib        67108  3 rt61pci,rt2x00pci,rt2x00mmio
snd_rawmidi      29394  2 snd_usbmidi_lib,snd_mpu401_uart

```

The output of command `lsmod` is divided into three columns:

Module

Module name.

Size

Amount of RAM occupied by the module, in bytes.

Used by

Depending modules.

Some modules require other modules to work properly, as is the case with modules for audio devices:

```

$ lsmod | fgrep -i snd_hda_intel
snd_hda_intel      42658  5
snd_hda_codec      155748  3 snd_hda_codec_hdmi,snd_hda_codec_via,snd_hda_intel
snd_pcm            81999  3 snd_hda_codec_hdmi,snd_hda_codec,snd_hda_intel
snd_page_alloc     13852  2 snd_pcm,snd_hda_intel
snd                 59132  19
snd_hwdep,snd_timer,snd_hda_codec_hdmi,snd_hda_codec_via,snd_pcm,snd_seq,snd_hda_codec,snd_h
da_intel,snd_seq_device

```

The third column, `Used by`, shows the modules that require the module in the first column to properly work. Many modules of the Linux sound architecture, prefixed by `snd`, are interdependent. When looking for problems during system diagnostics, it may be useful to unload specific modules currently loaded. Command `modprobe` can be used to both load and to unload kernel modules: to unload a module and its related modules, as long as they are not being used by a running process, command `modprobe -r` should be used. For example, to unload module `snd-hda-intel` (the module for a HDA Intel audio device) and other modules related to the sound system:

```
# modprobe -r snd-hda-intel
```

In addition to loading and unloading kernel modules while the system is running, it is possible to change module parameters when the kernel is being loaded, not so different from passing options to commands. Each module accepts specific parameters, but most times the default values are recommended and extra parameters are not needed. However, in some cases it is necessary to use parameters to change the behaviour of a module to work as expected.

Using the module name as the only argument, command `modinfo` shows a description, the file, the author, the license, the identification, the dependencies and the available parameters for the given module. Customized parameters for a module can be made persistent by including them in the file `/etc/modprobe.conf` or in individual files with the extension `.conf` in the directory `/etc/modprobe.d/`. Option `-p` will make command `modinfo` display all available parameters and ignore the other information:

```
# modinfo -p nouveau
vram_pushbuf:Create DMA push buffers in VRAM (int)
tv_norm:Default TV norm.
        Supported: PAL, PAL-M, PAL-N, PAL-Nc, NTSC-M, NTSC-J,
        hd480i, hd480p, hd576i, hd576p, hd720p, hd1080i.
        Default: PAL
        NOTE Ignored for cards with external TV encoders. (charp)
nofbaccel:Disable fbcon acceleration (int)
fbcon_bpp:fbcon bits-per-pixel (default: auto) (int)
mst:Enable DisplayPort multi-stream (default: enabled) (int)
tv_disable:Disable TV-out detection (int)
ignorelid:Ignore ACPI lid status (int)
duallink:Allow dual-link TMDS (default: enabled) (int)
hdmimhz:Force a maximum HDMI pixel clock (in MHz) (int)
config:option string to pass to driver core (charp)
debug:debug string to pass to driver core (charp)
noaccel:disable kernel/abi16 acceleration (int)
modeset:enable driver (default: auto, 0 = disabled, 1 = enabled, 2 = headless) (int)
atomic:Expose atomic ioctl (default: disabled) (int)
runpm:disable (0), force enable (1), optimus only default (-1) (int)
```

The sample output shows all the parameters available for module `nouveau`, a kernel module provided by the *Nouveau Project* as an alternative to the proprietary drivers for NVIDIA GPU cards. Option `modeset`, for example, allows to control whether display resolution and depth will be set in the kernel space rather than user space. Adding options `nouveau modeset=0` to the file `/etc/modprobe.d/nouveau.conf` will disable the `modeset` kernel feature.

If a module is causing problems, the file `/etc/modprobe.d/blacklist.conf` can be used to block the loading of the module. For example, to prevent the automatic loading of the module `nouveau`, the line `blacklist nouveau` must be added to the file `/etc/modprobe.d/blacklist.conf`. This action is required when the proprietary module `nvidia` is installed and the default module `nouveau` should be set aside.

NOTE

You can modify the `/etc/modprobe.d/blacklist.conf` file that already exists on the system by default. However, the preferred method is to create a separate configuration file, `/etc/modprobe.d/<module_name>.conf`, that will contain settings specific only to the given kernel module.

Information Files and Device Files

The commands `lspci`, `lsusb` and `lsmod` act as front-ends to read hardware information stored by the operating system. This kind of information is kept in special files in the directories `/proc` and `/sys`. These directories are mount points to filesystems not present in a device partition, but only in RAM space used by the kernel to store runtime configuration and information on running processes. Such filesystems are not intended for conventional file storage, so they are called pseudo-filefilesystems and only exist while the system is running. The `/proc` directory contains files with information regarding running processes and hardware resources. Some of the important files in `/proc` for inspecting hardware are:

`/proc/cpuinfo`

Lists detailed information about the CPU(s) found by the operating system.

`/proc/interrupts`

A list of numbers of the interrupts per IO device for each CPU.

`/proc/ioports`

Lists currently registered Input/Output port regions in use.

`/proc/dma`

Lists the registered DMA (direct memory access) channels in use.

Files inside the `/sys` directory have similar roles to those in `/proc`. However, the `/sys` directory has the specific purpose of storing device information and kernel data related to hardware, whilst `/proc` also contains information about various kernel data structures, including running processes and configuration.

Another directory directly related to devices in a standard Linux system is `/dev`. Every file inside `/dev` is associated with a system device, particularly storage devices. A legacy IDE hard drive, for

example, when connected to the motherboard's first IDE channel, is represented by the file `/dev/hda`. Every partition in this disk will be identified by `/dev/hda1`, `/dev/hda2` up to the last partition found.

Removable devices are handled by the *udev* subsystem, which creates the corresponding devices in `/dev`. The Linux kernel captures the hardware detection event and passes it to the *udev* process, which then identifies the device and dynamically creates corresponding files in `/dev`, using pre-defined rules.

In current Linux distributions, *udev* is responsible for the identification and configuration of the devices already present during machine power-up (*coldplug detection*) and the devices identified while the system is running (*hotplug detection*). *Udev* relies on *SysFS*, the pseudo filesystem for hardware related information mounted in `/sys`.

NOTE

Hotplug is the term used to refer to the detection and configuration of a device while the system is running, such as when a USB device is inserted. The Linux kernel has supported hotplug features since version 2.6, allowing most system buses (PCI, USB, etc.) to trigger hotplug events when a device is connected or disconnected.

As new devices are detected, *udev* searches for a matching rule in the pre-defined rules stored in the directory `/etc/udev/rules.d/`. The most important rules are provided by the distribution, but new ones can be added for specific cases.

Storage Devices

In Linux, storage devices are generically referred as block devices, because data is read to and from these devices in blocks of buffered data with different sizes and positions. Every block device is identified by a file in the `/dev` directory, with the name of the file depending on the device type (IDE, SATA, SCSI, etc.) and its partitions. CD/DVD and floppy devices, for example, will have their names given accordingly in `/dev`: a CD/DVD drive connected to the second IDE channel will be identified as `/dev/hdc` (`/dev/hda` and `/dev/hdb` are reserved for the master and slave devices on the first IDE channel) and an old floppy drive will be identified as `/dev/fd0`, `/dev/fd1`, etc.

From Linux kernel version 2.4 onwards, most storage devices are now identified as if they were SCSI devices, regardless of their hardware type. IDE, SSD and USB block devices will be prefixed by `sd`. For IDE disks, the `sd` prefix will be used, but the third letter will be chosen depending on whether the drive is a master or slave (in the first IDE channel, master will be `sda` and slave will be `sdb`). Partitions are listed numerically. Paths `/dev/sda1`, `/dev/sda2`, etc. are used for the first and second partitions of the block device identified first and `/dev/sdb1`, `/dev/sdb2`, etc. used to

identify the first and second partitions of the block device identified second. The exception to this pattern occurs with memory cards (SD cards) and NVMe devices (SSD connected to the PCI Express bus). For SD cards, the paths `/dev/mmcblk0p1`, `/dev/mmcblk0p2`, etc. are used for the first and second partitions of the device identified first and `/dev/mmcblk1p1`, `/dev/mmcblk1p2`, etc. used to identify the first and second partitions of the device identified second. NVMe devices receive the prefix `nvme`, as in `/dev/nvme0n1p1` and `/dev/nvme0n1p2`.

Guided Exercises

1. Suppose an operating system is unable to boot after adding a second SATA disk to the system. Knowing that all parts are not defective, what could be the possible cause for this error?

2. Suppose you want to make sure the external video card connected to the PCI bus of your newly acquired desktop computer really is the one advertised by the manufacturer, but opening the computer case will void the warranty. What command could be used to list the details of the video card, as they were detected by the operating system?

3. The following line is part of the output generated by command `lspci`:

```
03:00.0 RAID bus controller: LSI Logic / Symbios Logic MegaRAID SAS 2208 [Thunderbolt]
(rev 05)
```

What command should you execute to identify the kernel module in use for this specific device?

4. A system administrator wants to try different parameters for the `bluetooth` kernel module without rebooting the system. However, any attempt to unload the module with `modprobe -r bluetooth` results in the following error:

```
modprobe: FATAL: Module bluetooth is in use.
```

What is the possible cause for this error?

Explorational Exercises

1. It is not uncommon to find legacy machines in production environments, like when some equipment uses an outdated connection to communicate with the controlling computer, making it necessary to pay special attention to some peculiarities of these older machines. Some x86 servers with older BIOS firmware, for example, will not boot if a keyboard is not detected. How can this particular issue be avoided?

2. Operating systems built around the Linux kernel are also available for a wide variety of computer architectures other than x86, like in single board computers based on the ARM architecture. An attentive user will notice the absence of the `lspci` command on such machines, like the Raspberry Pi. What difference with x86 machines justifies that absence?

3. Many network routers have a USB port allowing for the connections of an external device, like a USB hard drive. Since most of these are using a Linux based operating system, how will an external USB hard drive be named in the `/dev/` directory, assuming no other conventional block device is present in the router?

4. In 2018, the hardware vulnerability known as *Meltdown* was discovered. It affects almost all processors of many architectures. Recent versions of the Linux kernel can inform if the current system is vulnerable. How can this information be obtained?

Summary

This lesson covers the general concepts on how the Linux kernel deals with hardware resources, mainly in the x86 architecture. The lesson goes through the following topics:

- How settings defined in BIOS or UEFI configuration utilities can affect the way the operating system interacts with hardware.
- How to use the tools provided by a standard Linux system to obtain information about the hardware.
- How to identify permanent and removable storage devices in the filesystem.

The commands and procedures addressed were:

- Commands to inspect detected hardware: `lspci` and `lsusb`.
- Commands to manage kernel modules: `lsmod` and `modprobe`.
- Hardware related special files, either the files found in the `/dev/` directory or in the pseudo-filesystems in `/proc/` and `/sys/`.

Answers to Guided Exercises

1. Suppose an operating system is unable to boot after adding a second SATA disk to the system. Knowing that all parts are not defective, what could be the possible cause for this error?

The boot device order should be configured in the BIOS setup utility, otherwise the BIOS may not be able to run the bootloader.

2. Suppose you want to make sure the external video card connected to the PCI bus of your newly acquired desktop computer really is the one advertised by the manufacturer, but opening the computer case will void the warranty. What command could be used to list the details of the video card, as they were detected by the operating system?

Command `lspci` will list detailed information about all devices currently connected to the PCI bus.

3. The following line is part of the output generated by command `lspci`:

```
03:00.0 RAID bus controller: LSI Logic / Symbios Logic MegaRAID SAS 2208 [Thunderbolt]
(rev 05)
```

What command should you execute to identify the kernel module in use for this specific device?

The command `lspci -s 03:00.0 -v` or `lspci -s 03:00.0 -k`

4. A system administrator wants to try different parameters for the `bluetooth` kernel module without rebooting the system. However, any attempt to unload the module with `modprobe -r bluetooth` results in the following error:

```
modprobe: FATAL: Module bluetooth is in use.
```

What is the possible cause for this error?

Module `bluetooth` is being used by a running process.

Answers to Explorational Exercises

1. It is not uncommon to find legacy machines in production environments, like when some equipment uses an outdated connection to communicate with the controlling computer, making it necessary to pay special attention to some peculiarities of these older machines. Some x86 servers with older BIOS firmwares, for example, will not boot if a keyboard is not detected. How this particular issue can be avoided?

The BIOS configuration utility has an option to deactivate the locking of the computer when a keyboard is not found.

2. Operating systems built around the Linux kernel are also available for a wide variety of computer architectures other than x86, like in single board computers based on the ARM architecture. An attentive user will notice the absence of the `lspci` command on such machines, like the Raspberry Pi. What difference with x86 machines justifies that absence?

Unlike most x86 machines, an ARM based computer like the Raspberry Pi lacks a PCI bus, so the command `lspci` is useless.

3. Many network routers have a USB port allowing for the connections of an external device, like a USB hard drive. Since most of these are using a Linux based operating system, how will an external USB hard drive be named in the `/dev/` directory, assuming no other conventional block device is present in the router?

Modern Linux kernels identify USB hard drives as SATA devices, so the corresponding file will be `/dev/sda` as no other conventional block device exists in the system.

4. In 2018, the hardware vulnerability known as *Meltdown* was discovered. It affects almost all processors of many architectures. Recent versions of the Linux kernel can inform if the current system is vulnerable. How can this information be obtained?

The file `/proc/cpuinfo` has a line showing the known bugs for the corresponding CPU, like bugs: `cpu_meltdown`.



101.2 Boot the system

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 101.2

Weight

3

Key knowledge areas

- Provide common commands to the boot loader and options to the kernel at boot time.
- Demonstrate knowledge of the boot sequence from BIOS/UEFI to boot completion.
- Understanding of SysVinit and systemd.
- Awareness of Upstart.
- Check boot events in the log files.

Partial list of the used files, terms and utilities

- `dmesg`
- `journalctl`
- BIOS
- UEFI
- bootloader
- kernel
- `initramfs`
- `init`
- SysVinit

- systemd



**Linux
Professional
Institute**

101.2 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	101 System Architecture
Objective:	101.2 Boot the system
Lesson:	1 of 1

Introduction

In order to control the machine, the operating system's main component — the kernel — must be loaded by a program called a *bootloader*, which itself is loaded by a pre-installed firmware such as BIOS or UEFI. The bootloader can be customized to pass parameters to the kernel, such as which partition contains the root filesystem or in which mode the operating system should execute. Once loaded the kernel continues the boot process identifying and configuring the hardware. Lastly, the kernel calls the utility responsible for starting and managing the system's services.

NOTE

On some Linux distributions, commands executed in this lesson may require root privileges.

BIOS or UEFI

The procedures executed by x86 machines to run the bootloader are different whether they use

BIOS or UEFI. The BIOS, short for *Basic Input/Output System*, is a program stored in a non-volatile memory chip attached to the motherboard, executed every time the computer is powered on. This type of program is called *firmware* and its storage location is separate from the other storage devices the system may have. The BIOS assumes that the first 440 bytes in the first storage device—following the order defined in the BIOS configuration utility—are the first stage of the bootloader (also called *bootstrap*). The first 512 bytes of a storage device are named the MBR (*Master Boot Record*) of storage devices using the standard DOS partition schema and, in addition to the first stage of the bootloader, contain the partition table. If the MBR does not contain the correct data, the system will not be able to boot, unless an alternative method is employed.

Generally speaking, the pre-operating steps to boot a system equipped with BIOS are:

1. The POST (*power-on self-test*) process is executed to identify simple hardware failures as soon as the machine is powered on.
2. The BIOS activates the basic components to load the system, like video output, keyboard and storage media.
3. The BIOS loads the first stage of the bootloader from the MBR (the first 440 bytes of the first device, as defined in the BIOS configuration utility).
4. The first stage of the bootloader calls the second stage of the bootloader, responsible for presenting boot options and loading the kernel.

The UEFI, short for *Unified Extensible Firmware Interface*, differs from BIOS in some key points. As the BIOS, the UEFI is also a firmware, but it can identify partitions and read many filesystems found in them. The UEFI does not rely on the MBR, taking into account only the settings stored in its non-volatile memory (*NVRAM*) attached to the motherboard. These definitions indicate the location of the UEFI compatible programs, called *EFI applications*, that will be executed automatically or called from a boot menu. EFI applications can be bootloaders, operating system selectors, tools for system diagnostics and repair, etc. They must be in a conventional storage device partition and in a compatible filesystem. The standard compatible filesystems are FAT12, FAT16 and FAT32 for block devices and ISO-9660 for optical media. This approach allows for the implementation of much more sophisticated tools than those possible with BIOS.

The partition containing the EFI applications is called the *EFI System Partition* or just ESP. This partition must not be shared with other system filesystems, like the root filesystem or user data filesystems. The EFI directory in the ESP partition contains the applications pointed to by the entries saved in the NVRAM.

Generally speaking, the pre-operating system boot steps on a system with UEFI are:

1. The POST (*power-on self-test*) process is executed to identify simple hardware failures as soon

as the machine is powered on.

2. The UEFI activates the basic components to load the system, like video output, keyboard and storage media.
3. UEFI's firmware reads the definitions stored in NVRAM to execute the pre-defined EFI application stored in the ESP partition's filesystem. Usually, the pre-defined EFI application is a bootloader.
4. If the pre-defined EFI application is a bootloader, it will load the kernel to start the operating system.

The UEFI standard also supports a feature called *Secure Boot*, which only allows the execution of signed EFI applications, that is, EFI applications authorized by the hardware manufacturer. This feature increases the protection against malicious software, but can make it difficult to install operating systems not covered by the manufacturer's warranty.

The Bootloader

The most popular bootloader for Linux in the x86 architecture is GRUB (*Grand Unified Bootloader*). As soon as it is called by the BIOS or by the UEFI, GRUB displays a list of operating systems available to boot. Sometimes the list does not appear automatically, but it can be invoked by pressing `Shift` while GRUB is being called by BIOS. In UEFI systems, the `Esc` key should be used instead.

From the GRUB menu it is possible to choose which one of the installed kernels should be loaded and to pass new parameters to it. Most kernel parameters follow the pattern `option=value`. Some of the most useful kernel parameters are:

acpi

Enables/disables ACPI support. `acpi=off` will disable support for ACPI.

init

Sets an alternative system initiator. For example, `init=/bin/bash` will set the Bash shell as the initiator. This means that a shell session will start just after the kernel boot process.

systemd.unit

Sets the *systemd* target to be activated. For example, `systemd.unit=graphical.target`. Systemd also accepts the numerical runlevels as defined for SysV. To activate the runlevel 1, for example, it is only necessary to include the number 1 or the letter S (short for "single") as a kernel parameter.

mem

Sets the amount of available RAM for the system. This parameter is useful for virtual machines so as to limit how much RAM will be available to each guest. Using `mem=512M` will limit to 512 megabytes the amount of available RAM to a particular guest system.

maxcpus

Limits the number of processors (or processor cores) visible to the system in symmetric multi-processor machines. It is also useful for virtual machines. A value of `0` turns off the support for multi-processor machines and has the same effect as the kernel parameter `nosmp`. The parameter `maxcpus=2` will limit the number of processors available to the operating system to two.

quiet

Hides most boot messages.

vga

Selects a video mode. The parameter `vga=ask` will show a list of the available modes to choose from.

root

Sets the root partition, distinct from the one pre-configured in the bootloader. For example, `root=/dev/sda3`.

rootflags

Mount options for the root filesystem.

ro

Makes the initial mount of the root filesystem read-only.

rw

Allows writing in the root filesystem during initial mount.

Changing the kernel parameters is not usually required, but it can be useful to detect and solve operating system related problems. Kernel parameters must be added to the file `/etc/default/grub` in the line `GRUB_CMDLINE_LINUX` to make them persistent across reboots. A new configuration file for the bootloader must be generated every time `/etc/default/grub` changes, which is accomplished by the command `grub-mkconfig -o /boot/grub/grub.cfg`. Once the operating system is running, the kernel parameters used for loading the current session are available for reading in the file `/proc/cmdline`.

NOTE | Configuring GRUB will be discussed further in a later lesson.

System Initialization

Apart from the kernel, the operating system depends on other components that provide the expected features. Many of these components are loaded during the system initialization process, varying from simple shell scripts to more complex service programs. Scripts are often used to perform short lived tasks that will run and terminate during the system initialization process. Services, also known as *daemons*, may be active all the time as they can be responsible for intrinsic aspects of the operating system.

The diversity of ways that startup scripts and daemons with the most different characteristics can be built into a Linux distribution is huge, a fact that historically hindered the development of a single solution that meets the expectations of maintainers and users of all Linux distributions. However, any tool that the distribution maintainers have chosen to perform this function will at least be able to start, stop and restart system services. These actions are often performed by the system itself after a software update, for example, but the system administrator will almost always need to manually restart the service after making modifications to its configuration file.

It is also convenient for a system administrator to be able to activate a particular set of daemons, depending on the circumstances. It should be possible, for example, to run just a minimum set of services in order to perform system maintenance tasks.

NOTE

Strictly speaking, the operating system is just the kernel and its components which control the hardware and manages all processes. It is common, however, to use the term “operating system” more loosely, to designate an entire group of distinct programs that make up the software environment where the user can perform the basic computational tasks.

The initialization of the operating system starts when the bootloader loads the kernel into RAM. Then, the kernel will take charge of the CPU and will start to detect and setup the fundamental aspects of the operating system, like basic hardware configuration and memory addressing.

The kernel will then open the *initramfs* (*initial RAM filesystem*). The *initramfs* is an archive containing a filesystem used as a temporary root filesystem during the boot process. The main purpose of an *initramfs* file is to provide the required modules so the kernel can access the “real” root filesystem of the operating system.

As soon as the root filesystem is available, the kernel will mount all filesystems configured in `/etc/fstab` and then will execute the first program, a utility named `init`. The `init` program is responsible for running all initialization scripts and system daemons. There are distinct implementations of such system initiators apart from the traditional `init`, like *systemd* and *Upstart*. Once the `init` program is loaded, the *initramfs* is removed from RAM.

SysV standard

A service manager based on the SysVinit standard controls which daemons and resources will be available by employing the concept of *runlevels*. Runlevels are numbered 0 to 6 and are designed by the distribution maintainers to fulfill specific purposes. The only runlevel definitions shared between all distributions are the runlevels 0, 1 and 6.

systemd

systemd is a modern system and services manager with a compatibility layer for the SysV commands and runlevels. systemd has a concurrent structure, employs sockets and D-Bus for service activation, on-demand daemon execution, process monitoring with *cgroups*, snapshot support, system session recovery, mount point control and a dependency-based service control. In recent years most major Linux distributions have gradually adopted systemd as their default system manager.

Upstart

Like systemd, Upstart is a substitute to init. The focus of Upstart is to speed up the boot process by parallelizing the loading process of system services. Upstart was used by Ubuntu based distributions in past releases, but today gave way to systemd.

Initialization Inspection

Errors may occur during the boot process, but they may not be so critical to completely halt the operating system. Notwithstanding, these errors may compromise the expected behaviour of the system. All errors result in messages that can be used for future investigations, as they contain valuable information about when and how the error occurred. Even when no error messages are generated, the information collected during the boot process can be useful for tuning and configuration purposes.

The memory space where the kernel stores its messages, including the boot messages, is called the *kernel ring buffer*. The messages are kept in the kernel ring buffer even when they are not displayed during the initialization process, like when an animation is displayed instead. However the kernel ring buffer loses all messages when the system is turned off or by executing the command `dmesg --clear`. Without options, command `dmesg` displays the current messages in the kernel ring buffer:

```
$ dmesg
[ 5.262389] EXT4-fs (sda1): mounted filesystem with ordered data mode. Opts: (null)
[ 5.449712] ip_tables: (C) 2000-2006 Netfilter Core Team
[ 5.460286] systemd[1]: systemd 237 running in system mode.
[ 5.480138] systemd[1]: Detected architecture x86-64.
```

```
[ 5.481767] systemd[1]: Set hostname to <torre>.
[ 5.636607] systemd[1]: Reached target User and Group Name Lookups.
[ 5.636866] systemd[1]: Created slice System Slice.
[ 5.637000] systemd[1]: Listening on Journal Audit Socket.
[ 5.637085] systemd[1]: Listening on Journal Socket.
[ 5.637827] systemd[1]: Mounting POSIX Message Queue File System...
[ 5.638639] systemd[1]: Started Read required files in advance.
[ 5.641661] systemd[1]: Starting Load Kernel Modules...
[ 5.661672] EXT4-fs (sda1): re-mounted. Opts: errors=remount-ro
[ 5.694322] lp: driver loaded but no devices found
[ 5.702609] ppdev: user-space parallel port driver
[ 5.705384] parport_pc 00:02: reported by Plug and Play ACPI
[ 5.705468] parport0: PC-style at 0x378 (0x778), irq 7, dma 3
[PCSP,TRISTATE,COMPAT,EPP,ECP,DMA]
[ 5.800146] lp0: using parport0 (interrupt-driven).
[ 5.897421] systemd-journald[352]: Received request to flush runtime journal from PID 1
```

The output of `dmesg` can be hundreds of lines long, so the previous listing contains only the excerpt showing the kernel calling the `systemd` service manager. The values in the beginning of the lines are the amount of seconds relative to when kernel load begins.

In systems based on `systemd`, command `journalctl` will show the initialization messages with options `-b`, `--boot`, `-k` or `--dmesg`. Command `journalctl --list-boots` shows a list of boot numbers relative to the current boot, their identification hash and the timestamps of the first and last corresponding messages:

```
$ journalctl --list-boots
```

```
-4 9e5b3eb4952845208b841ad4dbefa1a6 Thu 2019-10-03 13:39:23 -03-Thu 2019-10-03 13:40:30 -03
-3 9e3d79955535430aa43baa17758f40fa Thu 2019-10-03 13:41:15 -03-Thu 2019-10-03 14:56:19 -03
-2 17672d8851694e6c9bb102df7355452c Thu 2019-10-03 14:56:57 -03-Thu 2019-10-03 19:27:16 -03
-1 55c0d9439bfb4e85a20a62776d0dbb4d Thu 2019-10-03 19:27:53 -03-Fri 2019-10-04 00:28:47 -03
0 08fbbabd9f964a74b8a02bb27b200622 Fri 2019-10-04 00:31:01 -03-Fri 2019-10-04 10:17:01 -03
```

Previous initialization logs are also kept in systems based on `systemd`, so messages from prior operating system sessions can still be inspected. If options `-b 0` or `--boot=0` are provided, then messages for the current boot will be shown. Options `-b -1` or `--boot=-1` will show messages from the previous initialization. Options `-b -2` or `--boot=-2` will show the messages from the initialization before that and so on. The following excerpt shows the kernel calling the `systemd` service manager for the last initialization process:

```
$ journalctl -b 0
```

```

oct 04 00:31:01 ubuntu-host kernel: EXT4-fs (sda1): mounted filesystem with ordered data
mode. Opts: (null)
oct 04 00:31:01 ubuntu-host kernel: ip_tables: (C) 2000-2006 Netfilter Core Team
oct 04 00:31:01 ubuntu-host systemd[1]: systemd 237 running in system mode.
oct 04 00:31:01 ubuntu-host systemd[1]: Detected architecture x86-64.
oct 04 00:31:01 ubuntu-host systemd[1]: Set hostname to <torre>.
oct 04 00:31:01 ubuntu-host systemd[1]: Reached target User and Group Name Lookups.
oct 04 00:31:01 ubuntu-host systemd[1]: Created slice System Slice.
oct 04 00:31:01 ubuntu-host systemd[1]: Listening on Journal Audit Socket.
oct 04 00:31:01 ubuntu-host systemd[1]: Listening on Journal Socket.
oct 04 00:31:01 ubuntu-host systemd[1]: Mounting POSIX Message Queue File System...
oct 04 00:31:01 ubuntu-host systemd[1]: Started Read required files in advance.
oct 04 00:31:01 ubuntu-host systemd[1]: Starting Load Kernel Modules...
oct 04 00:31:01 ubuntu-host kernel: EXT4-fs (sda1): re-mounted. Opts:
commit=300,barrier=0,errors=remount-ro
oct 04 00:31:01 ubuntu-host kernel: lp: driver loaded but no devices found
oct 04 00:31:01 ubuntu-host kernel: ppdev: user-space parallel port driver
oct 04 00:31:01 ubuntu-host kernel: parport_pc 00:02: reported by Plug and Play ACPI
oct 04 00:31:01 ubuntu-host kernel: parport0: PC-style at 0x378 (0x778), irq 7, dma 3
[PCSP, TRISTATE, COMPAT, EPP, ECP, DMA]
oct 04 00:31:01 ubuntu-host kernel: lp0: using parport0 (interrupt-driven).
oct 04 00:31:01 ubuntu-host systemd-journald[352]: Journal started
oct 04 00:31:01 ubuntu-host systemd-journald[352]: Runtime journal
(/run/log/journal/abb765408f3741ae9519ab3b96063a15) is 4.9M, max 39.4M, 34.5M free.
oct 04 00:31:01 ubuntu-host systemd-modules-load[335]: Inserted module 'lp'
oct 04 00:31:01 ubuntu-host systemd-modules-load[335]: Inserted module 'ppdev'
oct 04 00:31:01 ubuntu-host systemd-modules-load[335]: Inserted module 'parport_pc'
oct 04 00:31:01 ubuntu-host systemd[1]: Starting Flush Journal to Persistent Storage...

```

Initialization and other messages issued by the operating system are stored in files inside the directory `/var/log/`. If a critical error happens and the operating system is unable to continue the initialization process after loading the kernel and the `initramfs`, an alternative boot medium could be used to start the system and to access the corresponding filesystem. Then, the files under `/var/log/` can be searched for possible reasons causing the interruption of the boot process. Options `-D` or `--directory` of command `journalctl` can be used to read log messages in directories other than `/var/log/journal/`, which is the default location for `systemd` log messages. As `systemd`'s log messages are not stored in raw text, command `journalctl` is required to read them.

Guided Exercises

1. On a machine equipped with a BIOS firmware, where is the bootstrap binary located?

2. UEFI firmware supports extended features provided by external programs, called EFI applications. These applications, however, have their own special location. Where on the system would the EFI applications be located?

3. Bootloaders allow the passing of custom kernel parameters before loading the kernel. Suppose the system is unable to boot due to a misinformed root filesystem location. How would the correct root filesystem, located at `/dev/sda3`, be given as a parameter to the kernel?

4. The boot process of a Linux machine ends up with the following message:

```
ALERT! /dev/sda3 does not exist. Dropping to a shell!
```

What is the likely cause of this problem?

Explorational Exercises

1. The bootloader will present a list of operating systems to choose from when more than one operating system is installed on the machine. However, a newly installed operating system can overwrite the MBR of the hard disk, erasing the first stage of the bootloader and making the other operating system inaccessible. Why would this not happen on a machine equipped with a UEFI firmware?

2. What is a common consequence of installing a custom kernel without providing an appropriate initramfs image?

3. The initialization log is hundreds of lines long, so the output of `dmesg` command is often piped to a pager command — like command `less` — to facilitate the reading. What `dmesg` option will automatically paginate its output, eliminating the need to use a pager command explicitly?

4. A hard drive containing the entire filesystem of an offline machine was removed and attached to a working machine as a secondary drive. Assuming its mount point is `/mnt/hd`, how would `journalctl` be used to inspect the contents of the journal files located at `/mnt/hd/var/log/journal/`?

Summary

This lesson covers the boot sequence in a standard Linux system. Proper knowledge of how the boot process of a Linux system works helps prevent errors that can make the system inaccessible. The lesson goes through the following topic areas:

- How BIOS and UEFI boot methods differ.
- Typical system initialization stages.
- Recovering boot messages.

The commands and procedures addressed were:

- Common kernel parameters.
- Commands to read boot messages: `dmesg` and `journalctl`.

Answers to Guided Exercises

1. On a machine equipped with a BIOS firmware, where is the bootstrap binary located?

In the MBR of the first storage device, as defined in the BIOS configuration utility.

2. UEFI firmware supports extended features provided by external programs, called EFI applications. These applications, however, have their own special location. Where on the system would the EFI applications be located?

EFI applications are stored in the EFI System Partition (ESP), located at any available storage block with a compatible filesystem (usually a FAT32 filesystem).

3. Bootloaders allow the passing of custom kernel parameters before loading the kernel. Suppose the system is unable to boot due to a misinformed root filesystem location. How would the correct root filesystem, located at `/dev/sda3`, be given as a parameter to the kernel?

The `root` parameter should be used, as in `root=/dev/sda3`.

4. The boot process of a Linux machine ends up with the following message:

```
ALERT! /dev/sda3 does not exist. Dropping to a shell!
```

What is the likely cause of this problem?

The kernel was unable to find the device `/dev/sda3`, informed as the root filesystem.

Answers to Explorational Exercises

1. The bootloader will present a list of operating systems to choose from when more than one operating system is installed on the machine. However, a newly installed operating system can overwrite the MBR of the hard disk, erasing the first stage of the bootloader and making the other operating system inaccessible. Why would this not happen on a machine equipped with a UEFI firmware?

UEFI machines do not use the hard disk's MBR to store the first stage of the bootloader.

2. What is a common consequence of installing a custom kernel without providing an appropriate initramfs image?

The root filesystem may be inaccessible if its type was compiled as an external kernel module.

3. The initialization log is hundreds of lines long, so the output of `dmesg` command is often piped to a pager command — like command `less` — to facilitate the reading. What `dmesg` option will automatically paginate its output, eliminating the need to use a pager command explicitly?

Commands `dmesg -H` or `dmesg --human` will enable the pager by default.

4. A hard drive containing the entire filesystem of an offline machine was removed and attached to a working machine as a secondary drive. Assuming its mount point is `/mnt/hd`, how would `journalctl` be used to inspect the contents of the journal files located at `/mnt/hd/var/log/journal/`?

With commands `journalctl -D /mnt/hd/var/log/journal` or `journalctl --directory=/mnt/hd/var/log/journal`



101.3 Change runlevels / boot targets and shutdown or reboot system

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 101.3

Weight

3

Key knowledge areas

- Set the default runlevel or boot target.
- Change between runlevels / boot targets including single user mode.
- Shutdown and reboot from the command line.
- Alert users before switching runlevels / boot targets or other major system events.
- Properly terminate processes.
- Awareness of acpid.

Partial list of the used files, terms and utilities

- `/etc/inittab`
- `shutdown`
- `init`
- `/etc/init.d/`
- `telinit`
- `systemd`
- `systemctl`
- `/etc/systemd/`

- `/usr/lib/systemd/`
- `wall`



101.3 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	101 System Architecture
Objective:	101.3 Change runlevels / boot targets and shutdown or reboot system
Lesson:	1 of 1

Introduction

A common feature among operating systems following Unix design principles is the employment of separate processes to control distinct functions of the system. These processes, called *daemons* (or, more generally, *services*), are also responsible for extended features underlying the operating system, like network application services (HTTP server, file sharing, email, etc.), databases, on-demand configuration, etc. Although Linux utilizes a monolithic kernel, many low level aspects of the operating system are affected by daemons, like load balancing and firewall configuration.

Which daemons should be active depends on the purpose of the system. The set of active daemons should also be modifiable at runtime, so services can be started and stopped without having to reboot the whole system. To tackle this issue, every major Linux distribution offers some form of service management utility to control the system.

Services can be controlled by shell scripts or by a program and its supporting configuration files. The first method is implemented by the *SysVinit* standard, also known as *System V* or just *SysV*. The second method is implemented by *systemd* and *Upstart*. Historically, SysV based service managers were the most used by Linux distributions. Today, systemd based service managers are

more often found in most Linux distributions. The service manager is the first program launched by the kernel during the boot process, so its PID (process identification number) is always 1.

SysVinit

A service manager based on the SysVinit standard will provide predefined sets of system states, called *runlevels*, and their corresponding service script files to be executed. Runlevels are numbered 0 to 6, being generally assigned to the following purposes:

Runlevel 0

System shutdown.

Runlevel 1, s or single

Single user mode, without network and other non-essential capabilities (maintenance mode).

Runlevel 2, 3 or 4

Multi-user mode. Users can log in by console or network. Runlevels 2 and 4 are not often used.

Runlevel 5

Multi-user mode. It is equivalent to 3, plus the graphical mode login.

Runlevel 6

System restart.

The program responsible for managing runlevels and associated daemons/resources is `/sbin/init`. During system initialization, the `init` program identifies the requested runlevel, defined by a kernel parameter or in the `/etc/inittab` file, and loads the associated scripts listed there for the given runlevel. Every runlevel may have many associated service files, usually scripts in the `/etc/init.d/` directory. As not all runlevels are equivalent through different Linux distributions, a short description of the runlevel's purpose can also be found in SysV based distributions.

The syntax of the `/etc/inittab` file uses this format:

```
id:runlevels:action:process
```

The `id` is a generic name up to four characters in length used to identify the entry. The `runlevels` entry is a list of runlevel numbers for which a specified action should be executed. The `action` term defines how `init` will execute the process indicated by the term `process`. The available actions are:

boot

The process will be executed during system initialization. The field `runlevels` is ignored.

bootwait

The process will be executed during system initialization and `init` will wait until it finishes to continue. The field `runlevels` is ignored.

sysinit

The process will be executed after system initialization, regardless of runlevel. The field `runlevels` is ignored.

wait

The process will be executed for the given runlevels and `init` will wait until it finishes to continue.

respawn

The process will be restarted if it is terminated.

ctrlaltdel

The process will be executed when the `init` process receives the `SIGINT` signal, triggered when the key sequence of `Ctrl` + `Alt` + `Del` is pressed.

The default runlevel — the one that will be chosen if no other is given as a kernel parameter — is also defined in `/etc/inittab`, in the entry `id:x:initdefault`. The `x` is the number of the default runlevel. This number should never be `0` or `6`, given that it would cause the system to shutdown or restart as soon as it finishes the boot process. A typical `/etc/inittab` file is shown below:

```
# Default runlevel
id:3:initdefault:

# Configuration script executed during boot
si::sysinit:/etc/init.d/rcS

# Action taken on runlevel S (single user)
~:S:wait:/sbin/sulogin

# Configuration for each execution level
l0:0:wait:/etc/init.d/rc 0
l1:1:wait:/etc/init.d/rc 1
l2:2:wait:/etc/init.d/rc 2
```



```

13:3:wait:/etc/init.d/rc 3
14:4:wait:/etc/init.d/rc 4
15:5:wait:/etc/init.d/rc 5
16:6:wait:/etc/init.d/rc 6

# Action taken upon ctrl+alt+del keystroke
ca::ctrlaltdel:/sbin/shutdown -r now

# Enable consoles for runlevels 2 and 3
1:23:respawn:/sbin/getty tty1 VC linux
2:23:respawn:/sbin/getty tty2 VC linux
3:23:respawn:/sbin/getty tty3 VC linux
4:23:respawn:/sbin/getty tty4 VC linux

# For runlevel 3, also enable serial
# terminals ttyS0 and ttyS1 (modem) consoles
S0:3:respawn:/sbin/getty -L 9600 ttyS0 vt320
S1:3:respawn:/sbin/mgetty -x0 -D ttyS1

```

The `telinit q` command should be executed every time after the `/etc/inittab` file is modified. The argument `q` (or `Q`) tells `init` to reload its configuration. Such a step is important to avoid a system halt due to an incorrect configuration in `/etc/inittab`.

The scripts used by `init` to setup each runlevel are stored in the directory `/etc/init.d/`. Every runlevel has an associated directory in `/etc/`, named `/etc/rc0.d/`, `/etc/rc1.d/`, `/etc/rc2.d/`, etc., with the scripts that should be executed when the corresponding runlevel starts. As the same script can be used by different runlevels, the files in those directories are just symbolic links to the actual scripts in `/etc/init.d/`. Furthermore, the first letter of the link filename in the runlevel's directory indicates if the service should be started or terminated for the corresponding runlevel. A link's filename starting with letter `K` determines that the service will be killed when entering the runlevel (kill). Starting with letter `S`, the service will be started when entering the runlevel (start). The directory `/etc/rc1.d/`, for example, will have many links to network scripts beginning with letter `K`, considering that the runlevel 1 is the single user runlevel, without network connectivity.

The command `runlevel` shows the current runlevel for the system. The `runlevel` command shows two values, the first is the previous runlevel and the second is the current runlevel:

```

$ runlevel
N 3

```

The letter `N` in the output shows that the runlevel has not changed since last boot. In the example,

the `runlevel 3` is the current runlevel of the system.

The same `init` program can be used to alternate between runlevels in a running system, without the need to reboot. The command `telinit` can also be used to alternate between runlevels. For example, commands `telinit 1`, `telinit s` or `telinit S` will change the system to runlevel 1.

systemd

Currently, `systemd` is the most widely used set of tools to manage system resources and services, which are referred to as *units* by `systemd`. A unit consists of a name, a type and a corresponding configuration file. For example, the unit for a `httpd` server process (like the Apache web server) will be `httpd.service` on Red Hat based distributions and its configuration file will also be called `httpd.service` (on Debian based distributions this unit is named `apache2.service`).

There are seven distinct types of `systemd` units:

service

The most common unit type, for active system resources that can be initiated, interrupted and reloaded.

socket

The socket unit type can be a filesystem socket or a network socket. All socket units have a corresponding service unit, loaded when the socket receives a request.

device

A device unit is associated with a hardware device identified by the kernel. A device will only be taken as a `systemd` unit if a `udev` rule for this purpose exists. A device unit can be used to resolve configuration dependencies when certain hardware is detected, given that properties from the `udev` rule can be used as parameters for the device unit.

mount

A mount unit is a mount point definition in the filesystem, similar to an entry in `/etc/fstab`.

automount

An automount unit is also a mount point definition in the filesystem, but mounted automatically. Every automount unit has a corresponding mount unit, which is initiated when the automount mount point is accessed.

target

A target unit is a grouping of other units, managed as a single unit.

snapshot

A snapshot unit is a saved state of the systemd manager (not available on every Linux distribution).

The main command for controlling systemd units is `systemctl`. Command `systemctl` is used to execute all tasks regarding unit activation, deactivation, execution, interruption, monitoring, etc. For a fictitious unit called `unit.service`, for example, the most common `systemctl` actions will be:

systemctl start unit.service

Starts unit.

systemctl stop unit.service

Stops unit.

systemctl restart unit.service

Restarts unit.

systemctl status unit.service

Shows the state of unit, including if it is running or not.

systemctl is-active unit.service

Shows *active* if unit is running or inactive otherwise.

systemctl enable unit.service

Enables unit, that is, unit will load during system initialization.

systemctl disable unit.service

unit will not start with the system.

systemctl is-enabled unit.service

Verifies if unit starts with the system. The answer is stored in the variable `$?`. The value `0` indicates that unit starts with the system and the value `1` indicates that unit does not start with the system.

NOTE

Newer installations of systemd will actually list a unit's configuration for boot time. For example:

```
$ systemctl is-enabled apparmor.service
enabled
```

If no other units with the same name exist in the system, then the suffix after the dot can be dropped. If, for example, there is only one `httpd` unit of type `service`, then only `httpd` is enough as the unit parameter for `systemctl`.

The `systemctl` command can also control *system targets*. The `multi-user.target` unit, for example, combines all units required by the multi-user system environment. It is similar to the runlevel number 3 in a system utilizing SysV.

Command `systemctl isolate` alternates between different targets. So, to manually alternate to target `multi-user`:

```
# systemctl isolate multi-user.target
```

There are corresponding targets to SysV runlevels, starting with `runlevel0.target` up to `runlevel6.target`. However, `systemd` does not use the `/etc/inittab` file. To change the default system target, the option `systemd.unit` can be added to the kernel parameters list. For example, to use `multi-user.target` as the standard target, the kernel parameter should be `systemd.unit=multi-user.target`. All kernel parameters can be made persistent by changing the bootloader configuration.

Another way to change the default target is to modify the symbolic link `/etc/systemd/system/default.target` so it points to the desired target. The redefinition of the link can be done with the `systemctl` command by itself:

```
# systemctl set-default multi-user.target
```

Likewise, you can determine what your system's default boot target is with the following command:

```
$ systemctl get-default  
graphical.target
```

Similar to systems adopting SysV, the default target should never point to `shutdown.target`, as it corresponds to the runlevel 0 (shutdown).

The configuration files associated with every unit can be found in the `/lib/systemd/system/` directory. The command `systemctl list-unit-files` lists all available units and shows if they are enabled to start when the system boots. The option `--type` will select only the units for a given type, as in `systemctl list-unit-files --type=service` and `systemctl list-unit-`

`files --type=target`.

Active units or units that have been active during the current system session can be listed with command `systemctl list-units`. Like the `list-unit-files` option, the `systemctl list-units --type=service` command will select only units of type `service` and command `systemctl list-units --type=target` will select only units of type `target`.

`systemd` is also responsible for triggering and responding to power related events. The `systemctl suspend` command will put the system in low power mode, keeping current data in memory. Command `systemctl hibernate` will copy all memory data to disk, so the current state of the system can be recovered after powering it off. The actions associated with such events are defined in the file `/etc/systemd/logind.conf` or in separate files inside the directory `/etc/systemd/logind.conf.d/`. However, this `systemd` feature can only be used when there is no other power manager running in the system, like the `acpid` daemon. The `acpid` daemon is the main power manager for Linux and allows finer adjustments to the actions following power related events, like closing the laptop lid, low battery or battery charging levels.

Upstart

The initialization scripts used by Upstart are located in the directory `/etc/init/`. System services can be listed with command `initctl list`, which also shows the current state of the services and, if available, their PID number.

```
# initctl list
avahi-cups-reload stop/waiting
avahi-daemon start/running, process 1123
mountall-net stop/waiting
mountnfs-bootclean.sh start/running
nmbd start/running, process 3085
passwd stop/waiting
rc stop/waiting
rsyslog start/running, process 1095
tty4 start/running, process 1761
udev start/running, process 1073
upstart-udev-bridge start/running, process 1066
console-setup stop/waiting
irqbalance start/running, process 1842
plymouth-log stop/waiting
smbd start/running, process 1457
tty5 start/running, process 1764
failsafe stop/waiting
```

Every Upstart action has its own independent command. For example, command `start` can be used to initiate a sixth virtual terminal:

```
# start tty6
```

The current state of a resource can be verified with command `status`:

```
# status tty6
tty6 start/running, process 3282
```

And the interruption of a service is done with the command `stop`:

```
# stop tty6
```

Upstart does not use the `/etc/inittab` file to define runlevels, but the legacy commands `runlevel` and `telinit` can still be used to verify and alternate between runlevels.

NOTE

Upstart was developed for the Ubuntu Linux distribution to help facilitate parallel startup of processes. Ubuntu has stopped using Upstart since 2015 when it switched from Upstart to systemd.

Shutdown and Restart

A very traditional command used to shutdown or restart the system is unsurprisingly called `shutdown`. The `shutdown` command adds extra functions to the power off process: it automatically notifies all logged-in users with a warning message in their shell sessions and new logins are prevented. Command `shutdown` acts as an intermediary to SysV or systemd procedures, that is, it executes the requested action by calling the corresponding action in the services manager adopted by the system.

After `shutdown` is executed, all processes receive the `SIGTERM` signal, followed by the `SIGKILL` signal, then the system shuts down or changes its runlevel. By default, when neither options `-h` or `-r` are used, the system alternates to runlevel 1, that is, the single user mode. To change the default options for `shutdown`, the command should be executed with the following syntax:

```
$ shutdown [option] time [message]
```

Only the parameter `time` is required. The `time` parameter defines when the requested action will

be executed, accepting the following formats:

hh:mm

This format specifies the execution time as hour and minutes.

+m

This format specifies how many minutes to wait before execution.

now or +0

This format determines immediate execution.

The `message` parameter is the warning text sent to all terminal sessions of logged-in users.

The SysV implementation allows for the limiting of users that will be able to restart the machine by pressing `Ctrl + Alt + Del`. This is possible by placing option `-a` for the `shutdown` command present at the line regarding `ctrlaltdel` in the `/etc/inittab` file. By doing this, only users whose usernames are in the `/etc/shutdown.allow` file will be able to restart the system with the `Ctrl + Alt + Del` keystroke combination.

The `systemctl` command can also be used to turn off or to restart the machine in systems employing `systemd`. To restart the system, the command `systemctl reboot` should be used. To turn off the system, the command `systemctl poweroff` should be used. Both commands require root privileges to run, as ordinary users can not perform such procedures.

Some Linux distributions will link `poweroff` and `reboot` to `systemctl` as individual commands. For example:

NOTE

```
$ sudo which poweroff
/usr/sbin/poweroff
$ sudo ls -l /usr/sbin/poweroff
lrwxrwxrwx 1 root root 14 Aug 20 07:50 /usr/sbin/poweroff -> /bin/systemctl
```

Not all maintenance activities require the system to be turned off or restarted. However, when it is necessary to change the system's state to single-user mode, it is important to warn logged-in users so that they are not harmed by an abrupt termination of their activities.

Similar to what the `shutdown` command does when powering off or restarting the system, the `wall` command is able to send a message to terminal sessions of all logged-in users. To do so, the system administrator only needs to provide a file or directly write the message as a parameter to command `wall`.

Guided Exercises

1. How could the `telinit` command be used to reboot the system?

2. What will happen to the services related to the file `/etc/rc1.d/K90network` when the system enters runlevel 1?

3. Using command `systemctl`, how could a user verify if the unit `sshd.service` is running?

4. In a `systemd` based system, what command must be executed to enable activation of the unit `sshd.service` during system initialization?

Explorational Exercises

1. In a SysV based system, suppose the default runlevel defined in `/etc/inittab` is 3, but the system always starts in runlevel 1. What is the probable cause for that?

2. Although the file `/sbin/init` can be found in systemd based systems, it is only a symbolic link to another executable file. In such systems, what is the file pointed by `/sbin/init`?

3. How can the default system target be verified in a systemd based system?

4. How can a system reboot scheduled with the `shutdown` command be canceled?

Summary

This lesson covers the main utilities used as service managers by Linux distributions. The SysVinit, systemd and Upstart utilities each have their own approach to control system services and system states. The lesson goes through the following topics:

- What system services are and their role in the operating system.
- Concepts and basic usage of SysVinit, systemd and Upstart commands.
- How to properly start, stop and restart system services and the system itself.

The commands and procedures addressed were:

- Commands and files related to SysVinit, like `init`, `/etc/inittab` and `telinit`.
- The main systemd command: `systemctl`.
- Upstart commands: `initctl`, `status`, `start`, `stop`.
- Traditional power management commands, like `shutdown`, and command `wall`.

Answers to Guided Exercises

1. How could the `telinit` command be used to reboot the system?

The command `telinit 6` will alternate to runlevel 6, that is, reboot the system.

2. What will happen to the services related to the file `/etc/rc1.d/K90network` when the system enters runlevel 1?

Due to the letter `K` in the beginning of the file name, the related services will be stopped.

3. Using command `systemctl`, how could a user verify if the unit `sshd.service` is running?

With the command `systemctl status sshd.service` or `systemctl is-active sshd.service`.

4. In a `systemd` based system, what command must be executed to enable activation of the unit `sshd.service` during system initialization?

Command `systemctl enable sshd.service`, executed by root.

Answers to Explorational Exercises

1. In a SysV based system, suppose the default runlevel defined in `/etc/inittab` is 3, but the system always starts in runlevel 1. What is the probable cause for that?

The parameters `1` or `S` may be present in the kernel's parameter list.

2. Although file `/sbin/init` can be found in systemd based systems, it is only a symbolic link to another executable file. In such systems, what is the file pointed by `/sbin/init`?

The main systemd binary: `/lib/systemd/systemd`.

3. How can the default system target be verified in a systemd based system?

The symbolic link `/etc/systemd/system/default.target` will point to the unit file defined as the default target. Command `systemctl get-default` can also be used.

4. How can a system reboot scheduled with the `shutdown` command be canceled?

The command `shutdown -c` should be used.



**Linux
Professional
Institute**

Topic 102: Linux Installation and Package Management



102.1 Design hard disk layout

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 102.1

Weight

2

Key knowledge areas

- Allocate filesystems and swap space to separate partitions or disks.
- Tailor the design to the intended use of the system.
- Ensure the /boot partition conforms to the hardware architecture requirements for booting.
- Knowledge of basic features of LVM.

Partial list of the used files, terms and utilities

- / (root) filesystem
- /var filesystem
- /home filesystem
- /boot filesystem
- EFI System Partition (ESP)
- swap space
- mount points
- partitions



Linux
Professional
Institute

102.1 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	102 Linux Installation and Package Management
Objective:	102.1 Design hard disk layout
Lesson:	1 of 1

Introduction

To succeed in this objective, you need to understand the relationship between *disks*, *partitions*, *filesystems* and *volumes*.

Think of a disk (or *storage device*, since modern devices do not contain any “disks” at all) as a “physical container” for your data.

Before a disk can be used by a computer it needs to be partitioned. A partition is a logical subset of the physical disk, like a logical “fence”. Partitioning is a way to “compartmentalize” information stored on the disk, separating, for example, operating system data from user data.

Every disk needs at least one partition, but can have multiple partitions if needed, and information about them is stored in a partition table. This table includes information about the first and last sectors of the partition and its type, as well as further details on each partition.

Inside each partition there is a filesystem. The filesystem describes the way the information is actually stored on the disk. This information includes how the directories are organized, what is the relationship between them, where is the data for each file, etc.

Partitions cannot span multiple disks. But using the *Logical Volume Manager* (LVM) multiple partitions can be combined, even across disks, to form a single logical volume.

Logical volumes abstract the limitations of the physical devices and let you work with “pools” of disk space that can be combined or distributed in a much more flexible way than traditional partitions. LVM is useful in situations where you would need to add more space to a partition without having to migrate the data to a larger device.

In this objective you will learn how to design a disk partitioning scheme for a Linux system, allocating filesystems and swap space to separate partitions or disks when needed.

How to *create* and *manage* partitions and filesystems will be discussed in other lessons. We will discuss an overview of LVM in this objective, but a detailed explanation is out of the scope.

Mount Points

Before a filesystem can be accessed on Linux it needs to be *mounted*. This means attaching the filesystem to a specific point in your system’s directory tree, called a *mount point*.

When mounted, the contents of the filesystem will be available under the mount point. For example, imagine you have a partition with your users’ personal data (their home directories), containing the directories `/john`, `/jack` and `/carol`. When mounted under `/home`, the contents of those directories will be available under `/home/john`, `/home/jack` and `/home/carol`.

The mount point must exist before mounting the filesystem. You cannot mount a partition under `/mnt/userdata` if this directory does not exist. However if the directory does exist and contains files, those files will be unavailable until you unmount the filesystem. If you list the contents of the directory, you will see the files stored on the mounted filesystem, not the original contents of the directory.

Filesystems can be mounted anywhere you want. However, there are some good practices that should be followed to make system administration easier.

Traditionally, `/mnt` was the directory under which all external devices would be mounted and a number of pre-configured *anchor points* for common devices, like CD-ROM drives (`/mnt/cdrom`) and floppy disks (`/mnt/floppy`) existed under it.

This has been superseded by `/media`, which is now the default mount point for any user-removable media (e.g. external disks, USB flash drives, memory card readers, optical disks, etc.) connected to the system.

On most modern Linux distributions and desktop environments, removable devices are

automatically mounted under `/media/USER/LABEL` when connected to the system, where `USER` is the username and `LABEL` is the device label. For example, a USB flash drive with the label `FlashDrive` connected by the user `john` would be mounted under `/media/john/FlashDrive/`. The way this is handled is different depending on the desktop environment.

That being said, whenever you need to *manually* mount a filesystem, it is good practice to mount it under `/mnt`. The specific commands to control the mounting and unmounting of filesystems under Linux will be discussed in another lesson.

Keeping Things Separated

On Linux, there are some directories that you should consider keeping on separate partitions. There are many reasons for this: for example, by keeping bootloader-related files (stored on `/boot`) on a *boot partition*, you ensure your system will still be able to boot in case of a crash on the root filesystem.

Keeping user's personal directories (under `/home`) on a separate partition makes it easier to reinstall the system without the risk of accidentally touching user data. Keeping data related to a web or database server (usually under `/var`) on a separate partition (or even a separate disk) makes system administration easier should you need to add more disk space for those use cases.

There may even be performance reasons to keep certain directories on separate partitions. You may want to keep the root filesystem (`/`) on a speedy SSD unit, and bigger directories like `/home` and `/var` on slower hard disks which offer much more space for a fraction of the cost.

The Boot Partition (`/boot`)

The boot partition contains files used by the bootloader to load the operating system. On Linux systems the bootloader is usually GRUB2 or, on older systems, GRUB Legacy. The partition is usually mounted under `/boot` and its files are stored in `/boot/grub`.

Technically a boot partition is not needed, since in most cases GRUB can mount the root partition (`/`) and load the files from a separate `/boot` directory.

However, a separate boot partition may be desired for safety (ensuring the system will boot even in case of a root filesystem crash), or if you wish to use a filesystem which the bootloader cannot understand in the root partition, or if it uses an unsupported encryption or compression method.

The boot partition is usually the first partition on the disk. This is because the original IBM PC BIOS addressed disks using *cylinders*, *heads* and *sectors* (CHS), with a maximum of 1024 cylinders, 256 heads and 63 sectors, resulting in a maximum disk size of 528 MB (504 MB under MS-DOS).

This means that anything past this mark would not be accessible on legacy systems, unless a different disk addressing scheme (like *Logical Block Addressing*, LBA) was used.

So for maximum compatibility, the boot partition is usually located at the start of the disk and ends before cylinder 1024 (528 MB), ensuring that no matter what, the machine will be always able to load the kernel.

Since the boot partition only stores the files needed by the bootloader, the initial RAM disk and kernel images, it can be quite small by today's standards. A good size is around 300 MB.

The EFI System Partition (ESP)

The *EFI System Partition* (ESP) is used by machines based on the *Unified Extensible Firmware Interface* (UEFI) to store boot loaders and kernel images for the operating systems installed.

This partition is formatted in a FAT-based filesystem. On a disk partitioned with a GUID Partition Table it has a globally unique identifier of C12A7328-F81F-11D2-BA4B-00A0C93EC93B. If the disk was formatted under the MBR partitioning scheme the partition ID is 0xEF.

On machines running Microsoft Windows this partition is usually the first one on the disk, although this is not required. The ESP is created (or populated) by the operating system upon installation, and on a Linux system is mounted under `/boot/efi`.

The `/home` Partition

Each user in the system has a home directory to store personal files and preferences, and most of them are located under `/home`. Usually the home directory is the same as the username, so the user John would have his directory under `/home/john`.

However there are exceptions. For example the home directory for the root user is `/root` and some system services may have associated users with home directories elsewhere.

There is no rule to determine the size of a partition for the `/home` directory (the home partition). You should take into account the number of users in the system and how it will be used. A user which only does web browsing and word processing will require less space than one who works with video editing, for example.

Variable Data (`/var`)

This directory contains “variable data”, or files and directories the system must be able to write to during operation. This includes system logs (in `/var/log`), temporary files (`/var/tmp`) and cached application data (in `/var/cache`).

`/var/www/html` is also the default directory for the data files for the Apache Web Server and `/var/lib/mysql` is the default location for database files for the MySQL server. However, both of these can be changed.

One good reason for putting `/var` in a separate partition is stability. Many applications and processes write to `/var` and subdirectories, like `/var/log` or `/var/tmp`. A misbehaved process may write data until there is no free space left on the filesystem.

If `/var` is under `/` this may trigger a kernel panic and filesystem corruption, causing a situation that is difficult to recover from. But if `/var` is kept under a separate partition, the root filesystem will be unaffected.

Like in `/home` there is no universal rule to determine the size of a partition for `/var`, as it will vary with how the system is used. On a home system, it may take only a few gigabytes. But on a database or web server much more space may be needed. In such scenarios, it may be wise to put `/var` on a partition on a different disk than the root partition adding an extra layer of protection against physical disk failure.

Swap

The swap partition is used to swap memory pages from RAM to disk as needed. This partition needs to be of a specific type, and set-up with a proper utility called `mkswap` before it can be used.

The swap partition cannot be mounted like the others, meaning that you cannot access it like a normal directory and peek at its contents.

A system can have multiple swap partitions (though this is uncommon) and Linux also supports the use of swap *files* instead of partitions, which can be useful to quickly increase swap space when needed.

The size of the swap partition is a contentious issue. The old rule from the early days of Linux (“twice the amount of RAM”) may not apply anymore depending on how the system is being used and the amount of physical RAM installed.

On the documentation for Red Hat Enterprise Linux 7, Red Hat recommends the following:

Amount of RAM	Recommended Swap Size	Recommended Swap Size with Hibernation
< 2 GB of RAM	2x the amount of RAM	3x the amount of RAM
2-8 GB of RAM	Equal to the amount of RAM	2x the amount of RAM

Amount of RAM	Recommended Swap Size	Recommended Swap Size with Hibernation
8-64 GB of RAM	At least 4 GB	1.5x the amount of RAM
> 64 GB of RAM	At least 4 GB	Not recommended

Of course the amount of swap can be workload dependent. If the machine is running a critical service, such as a database, web or SAP server, it is wise to check the documentation for these services (or your software vendor) for a recommendation.

NOTE

For more on creating and enabling swap partitions and swap files, see Objective 104.1 of LPIC-1.

LVM

We have already discussed how disks are organized into one or more partitions, with each partition containing a filesystem which describes how files and associated metadata are stored. One of the downsides of partitioning is that the system administrator has to decide beforehand how the available disk space on a device will be distributed. This can present some challenges later, if a partition requires more space than originally planned. Of course partitions can be resized, but this may not be possible if, for example, there is no free space on the disk.

Logical Volume Management (LVM) is a form of storage virtualization that offers system administrators a more flexible approach to managing disk space than traditional partitioning. The goal of LVM is to facilitate managing the storage needs of your end users. The basic unit is the *Physical Volume* (PV), which is a block device on your system like a disk partition or a RAID array.

PVs are grouped into *Volume Groups* (VG) which abstract the underlying devices and are seen as a single logical device, with the combined storage capacity of the component PVs.

Each volume in a Volume Group is subdivided into fixed-sized pieces called *extents*. Extents on a PV are called *Physical Extents* (PE), while those on a Logical Volume are *Logical Extents* (LE). Generally, each Logical Extent is mapped to a Physical Extent, but this can change if features like disk mirroring are used.

Volume Groups can be subdivided into Logical Volumes (LVs), which functionally work in a similar way to partitions but with more flexibility.

The size of a Logical Volume, as specified during its creation, is in fact defined by the size of the physical extents (4 MB by default) multiplied by the number of extents on the volume. From this it is easy to understand that to grow a Logical Volume, for example, all that the system

administrator has to do is add more extents from the pool available in the Volume Group. Likewise, extents can be removed to shrink the LV.

After a Logical Volume is created it is seen by the operating system as a normal block device. A device will be created in `/dev`, named as `/dev/VGNAME/LVNAME`, where `VGNAME` is the name of the Volume Group, and `LVNAME` is the name of the Logical Volume.

These devices can be formatted with a desired filesystem using standard utilities (like `mkfs.ext4`, for example) and mounted using the usual methods, either manually with the `mount` command or automatically by adding them to the `/etc/fstab` file.

Guided Exercises

1. On Linux systems, where are the files for the GRUB bootloader stored?

2. Where should the boot partition end to ensure that a PC will always be able to load the kernel?

3. Where is the EFI partition usually mounted?

4. When manually mounting a filesystem, under which directory should it usually be mounted?

Explorational Exercises

1. What is the smallest unit inside of a Volume Group?

2. How is the size of a Logical Volume defined?

3. On a disk formatted with the MBR partitioning scheme, which is the ID of the EFI System Partition?

4. Besides swap partitions, how can you quickly increase swap space on a Linux system?

Summary

In this lesson you learned about partitioning, which directories are usually kept in separate partitions and why this is done. Also, we discussed an overview of LVM (Logical Volume Management) and how it can offer a more flexible way to allocate your data and disk space compared to traditional partitioning.

The following files, terms and utilities have been discussed:

/

The Linux root filesystem.

/var

The standard location for “variable data”, data that can shrink and grow over time.

/home

The standard parent directory for home directories of regular users on a system.

/boot

The standard location for the boot loader files, Linux kernel and initial RAM disk.

EFI System Partition (ESP)

Used by systems that have UEFI implemented for the storage of the system’s boot files.

Swap space

Used to swap out kernel memory pages when RAM is heavily used.

Mount points

Directory locations where a device (such as a hard disk) will be mounted to.

Partitions

Divisions on a hard disk.

Answers to Guided Exercises

1. On Linux systems, where are the files for the GRUB bootloader stored?

Under `/boot/grub`.

2. Where should the boot partition end to ensure that a PC will always be able to load the kernel?

Before cylinder 1024.

3. Where is the EFI partition usually mounted?

Under `/boot/efi`.

4. When manually mounting a filesystem, under which directory should it usually be mounted?

Under `/mnt`. However, this is not mandatory. You can mount a partition under any directory you want.

Answers to Explorational Exercises

1. What is the smallest unit inside of a Volume Group?

Volume Groups are subdivided into extents.

2. How is the size of a Logical Volume defined?

By the size of the physical extents multiplied by the number of extents on the volume.

3. On a disk formatted with the MBR partitioning scheme, which is the ID of the EFI System Partition?

The ID is 0xEF.

4. Besides swap partitions, how can you quickly increase swap space on a Linux system?

Swap files can be used.



**Linux
Professional
Institute**

102.2 Install a boot manager

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 102.2

Weight

2

Key knowledge areas

- Providing alternative boot locations and backup boot options.
- Install and configure a boot loader such as GRUB Legacy.
- Perform basic configuration changes for GRUB 2.
- Interact with the boot loader.

Partial list of the used files, terms and utilities

- `menu.lst`, `grub.cfg` and `grub.conf`
- `grub-install`
- `grub-mkconfig`
- MBR



102.2 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	102 Linux Installation and Package Management
Objective:	102.2 Install a boot manager
Lesson:	1 of 1

Introduction

When a computer is powered on the first software to run is the boot loader. This is a piece of code whose sole purpose is to load an operating system kernel and hand over control to it. The kernel will load the necessary drivers, initialize the hardware and then load the rest of the operating system.

GRUB is the boot loader used on most Linux distributions. It can load the Linux kernel or other operating systems, such as Windows, and can handle multiple kernel images and parameters as separate menu entries. Kernel selection at boot is done via a keyboard-driven interface, and there is a command-line interface for editing boot options and parameters.

Most Linux distributions install and configure GRUB (actually, GRUB 2) automatically, so a regular user does not need to think about that. However, as a system administrator, it is vital to know how to control the boot process so you can recover the system from a boot failure after a failed kernel upgrade, for example.

In this lesson you will learn about how to install, configure and interact with GRUB.

GRUB Legacy vs. GRUB 2

The original version of GRUB (*Grand Unified Bootloader*), now known as *GRUB Legacy* was developed in 1995 as part of the GNU Hurd project, and later was adopted as the default boot loader of many Linux distributions, replacing earlier alternatives such as LILO.

GRUB 2 is a complete rewrite of GRUB aiming to be cleaner, safer, more robust, and more powerful. Among the many advantages over GRUB Legacy are a much more flexible configuration file (with many more commands and conditional statements, similar to a scripting language), a more modular design and better localization/internationalization.

There is also support for themes and graphical boot menus with splash screens, the ability to boot LiveCD ISOs directly from the hard drive, better support for non-x86 architectures, universal support for UUIDs (making it easier to identify disks and partitions) and much more.

GRUB Legacy is no longer under active development (the last release was 0.97, in 2005), and today most major Linux distributions install GRUB 2 as the default boot loader. However, you may still find systems using GRUB Legacy, so it is important to know how to use it and where it is different from GRUB 2.

Where is the Bootloader?

Historically, hard disks on IBM PC compatible systems were partitioned using the MBR partitioning scheme, created in 1982 for IBM PC-DOS (MS-DOS) 2.0.

In this scheme, the first 512-byte sector of the disk is called the *Master Boot Record* and contains a table describing the partitions on the disk (the partition table) and also bootstrap code, called a bootloader.

When the computer is turned on, this very minimal (due to size restrictions) bootloader code is loaded, executed and passes control to a secondary boot loader on disk, usually located in a 32 KB space between the MBR and the first partition, which in turn will load the operating system(s).

On an MBR-partitioned disk, the boot code for GRUB is installed to the MBR. This loads and passes control to a “core” image installed between the MBR and the first partition. From this point, GRUB is capable of loading the rest of the needed resources (menu definitions, configuration files and extra modules) from disk.

However, MBR has limitations on the number of partitions (originally a maximum of 4 primary partitions, later a maximum of 3 primary partitions with 1 extended partition subdivided into a number of logical partitions) and maximum disk sizes of 2 TB. To overcome these limitations a new partitioning scheme called GPT (*GUID Partition Table*), part of the UEFI (*Unified Extensible*

Firmware Interface) standard, was created.

GPT-partitioned disks can be used either with computers with the traditional PC BIOS or ones with UEFI firmware. On machines with a BIOS, the second part of GRUB is stored in a special BIOS boot partition.

On systems with UEFI firmware, GRUB is loaded by the firmware from the files `grubia32.efi` (for 32-Bit systems) or `grubx64.efi` (for 64-Bit systems) from a partition called the ESP (*EFI System Partition*).

The `/boot` Partition

On Linux the files necessary for the boot process are usually stored on a boot partition, mounted under the root file system and colloquially referred to as `/boot`.

A boot partition is not needed on current systems, as boot loaders such as GRUB can usually mount the root file system and look for the needed files inside a `/boot` directory, but it is good practice as it separates the files needed for the boot process from the rest of the filesystem.

This partition is usually the first one on the disk. This is because the original IBM PC BIOS addressed disks using *Cylinders, Heads* and *Sectors* (CHS), with a maximum of 1024 cylinders, 256 heads and 63 sectors, resulting in a maximum disk size of 528 MB (504 MB under MS-DOS). This means that anything past this mark would not be accessible, unless a different disk addressing scheme (like LBA, *Logical Block Addressing*) was used.

So for maximum compatibility, the `/boot` partition is usually located at the start of the disk and ends before cylinder 1024 (528 MB), ensuring that the machine will always be able to load the kernel. The recommended size for this partition on a current machine is 300 MB.

Other reasons for a separate `/boot` partition are encryption and compression since some methods may not be supported by GRUB 2 yet, or if you need to have the system root partition (`/`) formatted using an unsupported file system.

Contents of the Boot Partition

The contents of the `/boot` partition may vary with system architecture or the boot loader in use, but on a x86-based system you will usually find the files below. Most of these are named with a `-VERSION` suffix, where `-VERSION` is the version of the corresponding Linux kernel. So, for example, a configuration file for the Linux kernel version `4.15.0-65-generic` would be called `config-4.15.0-65-generic`.

Config file

This file, usually called `config-VERSION` (see example above), stores configuration parameters for the Linux kernel. This file is generated automatically when a new kernel is compiled or installed and *should not be* directly modified by the user.

System map

This file is a look-up table matching symbol names (like variables or functions) to their corresponding position in memory. This is useful when debugging a kind of system failure known as a *kernel panic*, as it allows the user to know which variable or function was being called when the failure occurred. Like the config file, the name is usually `System.map-VERSION` (e.g. `System.map-4.15.0-65-generic`).

Linux kernel

This is the operating system kernel proper. The name is usually `vmlinux-VERSION` (e.g. `vmlinux-4.15.0-65-generic`). You may also find the name `vmlinuz` instead of `vmlinux`, the `z` at the end meaning that the file has been compressed.

Initial RAM disk

This is usually called `initrd.img-VERSION` and contains a minimal root file system loaded into a RAM disk, containing utilities and kernel modules needed so the kernel can mount the real root filesystem.

Boot loader related files

On systems with GRUB installed, these are usually located on `/boot/grub` and include the GRUB configuration file (`/boot/grub/grub.cfg` for GRUB 2 or `/boot/grub/menu.lst` in case of GRUB Legacy), modules (in `/boot/grub/i386-pc`), translation files (in `/boot/grub/locale`) and fonts (in `/boot/grub/fonts`).

GRUB 2

Installing GRUB 2

GRUB 2 can be installed using the `grub-install` utility. If you have a non-booting system you will need to boot using a Live CD or rescue disc, find out which is the boot partition for your system, mount it and then run the utility.

NOTE

The commands below assume that you are logged in as root. If not, first run `sudo su -` to “become” root. When finished, type `exit` to log out and return to a regular user.

The first disk on a system is usually the *boot device* and you may need to know whether there is a *boot partition* on the disk. This can be done with the `fdisk` utility. To list all the partitions on the first disk of your machine, use:

```
# fdisk -l /dev/sda
Disk /dev/sda: 111,8 GiB, 120034123776 bytes, 234441648 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x97f8fef5

Device      Boot    Start        End    Sectors    Size Id Type
/dev/sda1   *         2048    2000895    1998848    976M 83 Linux
/dev/sda2             2002942 234440703 232437762 110,9G  5 Extended
/dev/sda5             2002944 18008063  16005120    7,6G 82 Linux swap / Solaris
/dev/sda6             18010112 234440703 216430592 103,2G 83 Linux
```

The boot partition is identified with the `*` under the boot column. In the example above, it is `/dev/sda1`.

Now, create a temporary directory under `/mnt` and mount the partition under it:

```
# mkdir /mnt/tmp
# mount /dev/sda1 /mnt/tmp
```

Then run `grub-install`, pointing it to the boot *device* (*not* the partition) and the directory where the boot partition is mounted. If your system does have a dedicated boot partition, the command is:

```
# grub-install --boot-directory=/mnt/tmp /dev/sda
```

If you are installing to a system which does not have a boot partition, but just a `/boot` directory on the root filesystem, point `grub-install` to it. So, the command is:

```
# grub-install --boot-directory=/boot /dev/sda
```


Configuring GRUB 2

The default configuration file for GRUB 2 is `/boot/grub/grub.cfg`. This file is automatically generated and manual editing is not recommended. To make changes to the GRUB configuration, you need to edit the file `/etc/default/grub` and then run the `update-grub` utility to generate a compliant file.

NOTE

`update-grub` is usually a shortcut to `grub-mkconfig -o /boot/grub/grub.cfg`, so they produce the same results.

There are some options in the file `/etc/default/grub` that control the behaviour of GRUB 2, like the default kernel to boot, timeout, extra command line parameters, etc. The most important ones are:

GRUB_DEFAULT=

The default menu entry to boot. This can be a numeric value (like `0`, `1`, etc.), the name of a menu entry (like `debian`) or `saved`, which is used in conjunction with `GRUB_SAVEDEFAULT=`, explained below. Keep in mind that menu entries start at zero, so the first menu entry is `0`, the second is `1`, etc.

GRUB_SAVEDEFAULT=

If this option is set to `true` and `GRUB_DEFAULT=` is set to `saved`, then the default boot option will always be the last one selected in the boot menu.

GRUB_TIMEOUT=

The timeout, in seconds, before the default menu entry is selected. If set to `0` the system will boot the default entry without showing a menu. If set to `-1` the system will wait until the user selects an option, no matter how long it takes.

GRUB_CMDLINE_LINUX=

This lists command line options that will be added to entries for the Linux kernel.

GRUB_CMDLINE_LINUX_DEFAULT=

By default, two menu entries are generated for each Linux kernel, one with the default options and one entry for recovery. With this option you can add extra parameters that will be added only to the default entry.

GRUB_ENABLE_CRYPTODISK=

If set to `y`, commands like `grub-mkconfig`, `update-grub` and `grub-install` will look for encrypted disks and add the commands needed to access them during boot. This disables automatic booting (`GRUB_TIMEOUT=` with any value other than `-1`) because a passphrase is

needed to decrypt the disks before they can be accessed.

Managing Menu Entries

When `update-grub` is run, GRUB 2 will scan for kernels and operating systems on the machine and generate the corresponding menu entries on the `/boot/grub/grub.cfg` file. New entries can be manually added to the script files within the `/etc/grub.d` directory.

These files should be executable, and are processed in numerical order by `update-grub`. Therefore `05_debian_theme` is processed before `10_linux` and so on. Custom menu entries are usually added to the `40_custom` file.

The basic syntax for a menu entry is shown below:

```
menuentry "Default OS" {  
    set root=(hd0,1)  
    linux /vmlinuz root=/dev/sda1 ro quiet splash  
    initrd /initrd.img  
}
```

The first line always starts with `menuentry` and ends with `}`. The text between quotes will be shown as the entry label on the GRUB 2 boot menu.

The `set root` parameter defines the disk and partition where the root file system for the operating system is located. Please note that on GRUB 2 disks are numbered from zero, so `hd0` is the first disk (`sda` under Linux), `hd1` the second, and so on. Partitions, however, are numbered starting from one. In the example above the root file system is located at the first disk (`hd0`), first partition (`, 1`), or `sda1`.

Instead of directly specifying the device and partition you can also have GRUB 2 search for a file system with a specific label or UUID (*Universally Unique Identifier*). For that, use the parameter `search --set=root` followed by the `--label` parameter and file system label to search for, or `--fs-uuid` followed by the UUID of the file system.

You can find the UUID of a filesystem with the command below:

```
$ ls -l /dev/disk/by-uuid/  
total 0  
lrwxrwxrwx 1 root root 10 nov  4 08:40 3e0b34e2-949c-43f2-90b0-25454ac1595d -> ../../sda5  
lrwxrwxrwx 1 root root 10 nov  4 08:40 428e35ee-5ad5-4dcB-adca-539aba6c2d84 -> ../../sda6  
lrwxrwxrwx 1 root root 10 nov  5 19:10 56C11DCC5D2E1334 -> ../../sdb1
```

```
lrwxrwxrwx 1 root root 10 nov  4 08:40 ae71b214-0aec-48e8-80b2-090b6986b625 -> ../../sda1
```

In the example above, the UUID for `/dev/sda1` is `ae71b214-0aec-48e8-80b2-090b6986b625`. Should you wish to set it as the root device for GRUB 2, the command would be `search --set=root --fs-uuid ae71b214-0aec-48e8-80b2-090b6986b625`.

When using the `search` command it is common to add the `--no-floppy` parameter so GRUB will not waste time searching on floppy disks.

The `linux` line indicates where the kernel for the operating system is located (in this case, the `vmlinuz` file at the root of the file system). After that, you can pass command line parameters to the kernel.

In the example above we specified the root partition (`root=/dev/sda1`) and passed three kernel parameters: the root partition should be mounted read-only (`ro`), most of the log messages should be disabled (`quiet`) and a splash screen should be shown (`splash`).

The `initrd` line indicates where the initial RAM disk is located. In the example above the file is `initrd.img`, located at the root of the file system.

NOTE

Most Linux distributions do not actually put the kernel and `initrd` at the root directory of the root file system. Instead, these are links to the actual files inside the `/boot` directory or partition.

The last line of a menu entry should contain only the character `}`.

Interacting with GRUB 2

When booting a system with GRUB 2 you will see a menu of options. Use the arrow keys to select an option and `Enter` to confirm and boot the selected entry.

TIP

If you see just a countdown, but not a menu, press `Shift` to bring up the menu.

To edit an option, select it with the arrow keys and press `E`. This will show an editor window with the contents of the `menuentry` associated with that option, as defined in `/boot/grub/grub.cfg`.

After editing an option, type `Ctrl + X` or `F10` to boot, or `Esc` to return to the menu.

To enter the GRUB 2 shell, press `C` on the menu screen (or `Ctrl + C` on the editing window). You will see a command prompt like this: `grub >`

Type `help` to see a list of all available commands, or press `Esc` to exit the shell and return to the menu screen.

NOTE

Remember this menu will not appear if `GRUB_TIMEOUT` is set to `0` in `/etc/default/grub`.

Booting from the GRUB 2 Shell

You can use the GRUB 2 shell to boot the system in case a misconfiguration in a menu entry causes it to fail.

The first thing you should do is find out where the boot partition is. You can do that with the command `ls`, which will show you a list of the partitions and disks GRUB 2 has found.

```
grub> ls
(proc) (hd0) (hd0,msdos1)
```

In the example above, things are easy. There is only one disk (`hd0`) with only one partition on it: (`hd0,msdos1`).

The disks and partitions listed will be different in your system. In our example the first partition of `hd0` is called `msdos1` because the disk was partitioned using the MBR partitioning scheme. If it were partitioned using GPT, the name would be `gpt1`.

To boot Linux, we need a kernel and initial RAM disk (`initrd`). Let us check the contents of (`hd0,msdos1`):

```
grub> ls (hd0,msdos1)/
lost+found/ swapfile etc/ media/ bin/ boot/ dev/ home/ lib/ lib64/ mnt/ opt/ proc/ root/
run/ sbin/ srv/ sys/ tmp/ usr/ var/ initrd.img initrd.img.old vmlinuz cdrom/
```

You can add the `-l` parameter to `ls` to get a long listing, similar to what you would get on a Linux terminal. Use `Tab` to autocomplete disk, partition and file names.

Note that we have a kernel (`vmlinuz`) and `initrd` (`initrd.img`) images right on the root directory. If not, we could check the contents of `/boot` with `list (hd0,msdos1)/boot/`.

Now, set the boot partition:

```
grub> set root=(hd0,msdos1)
```

Load the Linux kernel with the `linux` command, followed by the path to the kernel and the `root=` option to tell the kernel where the root filesystem for the operating system is located.

```
grub> linux /vmlinuz root=/dev/sda1
```

Load the initial RAM disk with `initrd`, followed by the full path to the `initrd.img` file:

```
grub> initrd /initrd.img
```

Now, boot the system with `boot`.

Booting from the Rescue Shell

In case of a boot failure GRUB 2 may load a rescue shell, a simplified version of the shell we mentioned before. You will recognize it by the command prompt, which displays as `grub rescue>`.

The process to boot a system from this shell is almost the same as shown before. However, you will need to load a few GRUB 2 modules to make things work.

After finding out which partition is the boot partition (with `ls`, as shown before), use the `set prefix=` command, followed by the full path to the directory containing the GRUB 2 files. Usually `/boot/grub`. In our example:

```
grub rescue> set prefix=(hd0,msdos1)/boot/grub
```

Now, load the modules `normal` and `linux` with the `insmod` command:

```
grub rescue> insmod normal
grub rescue> insmod linux
```

Next, set the boot partition with `set root=` as directed before, load the linux kernel (with `linux`), the initial RAM disk (`initrd`) and try to boot with `boot`.

GRUB Legacy

Installing GRUB Legacy from a Running System

To install GRUB Legacy on a disk from a running system we will use the `grub-install` utility. The basic command is `grub-install DEVICE` where `DEVICE` is the disk where you wish to install GRUB Legacy. An example would be `/dev/sda`.

```
# grub-install /dev/sda
```

Note that you need to specify the *device* where GRUB Legacy will be installed, like `/dev/sda/`, *not the partition* as in `/dev/sda1`.

By default GRUB will copy the needed files to the `/boot` directory on the specified device. If you wish to copy them to another directory, use the `--boot-directory=` parameter, followed by the full path to where the files should be copied to.

Installing GRUB Legacy from a GRUB Shell

If you cannot boot the system for some reason and need to reinstall GRUB Legacy, you can do so from the GRUB shell on a GRUB Legacy boot disk.

From the GRUB shell (type `c` at the boot menu to get to the `grub>` prompt), the first step is to set the boot device, which contains the `/boot` directory. For example, if this directory is in the first partition of the first disk, the command would be:

```
grub> root (hd0,0)
```

If you do not know which device contains the `/boot` directory, you can ask GRUB to search for it with the `find` command, like below:

```
grub> find /boot/grub/stage1
(hd0,0)
```

Then, set the boot partition as instructed above and use the `setup` command to install GRUB Legacy to the MBR and copy the needed files to the disk:

```
grub> setup (hd0)
```

When finished reboot the system and it should boot normally.

Configuring GRUB Legacy Menu Entries and Settings

GRUB Legacy menu entries and settings are stored in the file `/boot/grub/menu.lst`. This is a simple text file with a list of commands and parameters, which can be edited directly with your favourite text editor.

Lines starting with `#` are considered comments, and blank lines are ignored.

A menu entry has at least three commands. The first one, `title`, sets the title of the operating system on the menu screen. The second one, `root`, tells GRUB Legacy which device or partition to boot from.

The third entry, `kernel`, specifies the full path to the kernel image that should be loaded when the corresponding entry is selected. Note that this path is relative to the device specified on the `root` parameter.

A simple example follows:

```
# This line is a comment
title My Linux Distribution
root (hd0,0)
kernel /vmlinuz root=/dev/hda1
```

Unlike GRUB 2, in GRUB Legacy both disks *and* partitions are numbered from zero. So, the command `root (hd0,0)` will set the boot partition as the first partition (0) of the first disk (hd0).

You can omit the `root` statement if you specify the boot device before the path on the `kernel` command. The syntax is the same, so:

```
kernel (hd0,0)/vmlinuz root=dev/hda1
```

is equivalent to:

```
root (hd0,0)
kernel /vmlinuz root=/dev/hda1
```

Both will load the file `vmlinuz` from the root directory (`/`) of the first partition of the first disk (`hd0,0`).

The parameter `root=/dev/hda1` after the `kernel` command tells the Linux kernel which partition should be used as the root filesystem. This is a Linux kernel parameter, not a GRUB Legacy command.

NOTE

To learn more about kernel parameters, see <https://www.kernel.org/doc/html/v4.14/admin-guide/kernel-parameters.html>.

You may need to specify the location of the initial RAM Disk image for the operating system with the `initrd` parameter. The full path to the file can be specified like in the `kernel` parameter, and you can also specify a device or partition before the path, e.g.:

```
# This line is a comment
title My Linux Distribution
root (hd0,0)
kernel /vmlinuz root=/dev/hda1
initrd /initrd.img
```

GRUB Legacy has a modular design, where modules (usually stored as `.mod` files in `/boot/grub/i386-pc`) can be loaded to add extra features, like support for unusual hardware, filesystems or new compression algorithms.

Modules are loaded using the `module` command, followed by the full path to the corresponding `.mod` file. Keep in mind that, like kernels and `initrd` images, this path is relative to the device specified in the `root` command.

The example below will load the module `915resolution`, needed to correctly set the framebuffer resolution on systems with Intel 800 or 900 series video chipsets.

```
module /boot/grub/i386-pc/915resolution.mod
```

Chainloading other Operating Systems

GRUB Legacy can be used to load unsupported operating systems, like Windows, using a process called *chainloading*. GRUB Legacy is loaded first, and when the corresponding option is selected the bootloader for the desired system is loaded.

A typical entry for chainloading Windows would look like the one below:

```
# Load Windows
title Windows XP
root (hd0,1)
makeactive
chainload +1
boot
```

Let us walk through each parameter. As before, `root (hd0,1)` specifies the device and partition where the boot loader for the operating system we wish to load is located. In this example, the

second partition of the first disk.

makeactive

will set a flag indicating that this is an active partition. This only works on DOS primary partitions.

chainload +1

tells GRUB to load the first sector of the boot partition. This is where bootloaders are usually located.

boot

will run the bootloader and load the corresponding operating system.

Guided Exercises

1. What is the default location for the GRUB 2 configuration file?

2. What are the steps needed to change the settings for GRUB 2?

3. Into which file should custom GRUB 2 menu entries be added?

4. Where are the menu entries for GRUB Legacy stored?

5. From a GRUB 2 or GRUB Legacy menu, how can you enter the GRUB Shell?

Explorational Exercises

1. Imagine a user configuring GRUB Legacy to boot from the second partition of the first disk. He writes the following custom menu entry:

```
title My Linux Distro
root (hd0,2)
kernel /vmlinuz root=/dev/hda1
initrd /initrd.img
```

However, the system will not boot. What is wrong?

2. Imagine you have a disk identified as `/dev/sda` with multiple partitions. Which command can be used to find out which is the boot partition on a system?

3. Which command can be used to find out the UUID of a partition?

4. Consider the following entry for GRUB 2

```
menuentry "Default OS" {
    set root=(hd0,1)
    linux /vmlinuz root=/dev/sda1 ro quiet splash
    initrd /initrd.img
}
```

Change it so the system will boot from a disk with the UUID `5dda0af3-c995-481a-a6f3-46dcd3b6998d`

5. How can you set GRUB 2 to wait for 10 seconds before booting the default menu entry?

6. From a GRUB Legacy shell, what are the commands to install GRUB to the first partition of the second disk?

Summary

In this lesson we learned

- What is a boot loader.
- The differences between GRUB Legacy and GRUB 2.
- What is a boot partition, and what are its contents.
- How to install GRUB Legacy and GRUB 2.
- How to configure GRUB Legacy and GRUB 2.
- How to add custom menu entries to GRUB Legacy and GRUB 2.
- How to interact with the menu screen and console of GRUB Legacy and GRUB 2.
- How to boot a system from a GRUB Legacy or GRUB 2 shell or rescue shell.

The following commands were discussed in this lesson:

- `grub-install`
- `update-grub`
- `grub-mkconfig`

Answers to Guided Exercises

1. What is the default location for the GRUB 2 configuration file?

`/boot/grub/grub.cfg`

2. What are the steps needed to change the settings for GRUB 2?

Make your changes to the file `/etc/default/grub`, then update the configuration with `update-grub`.

3. Into which file should custom GRUB 2 menu entries be added?

`/etc/grub.d/40_custom`

4. Where are the menu entries for GRUB Legacy stored?

`/boot/grub/menu.lst`

5. From a GRUB 2 or GRUB Legacy menu, how can you enter the GRUB Shell?

Press `c` in the menu screen.

Answers to Explorational Exercises

1. Imagine a user configuring GRUB Legacy to boot from the second partition of the first disk. He writes the following custom menu entry:

```
title My Linux Distro
root (hd0,2)
kernel /vmlinuz root=/dev/hda1
initrd /initrd.img
```

However, the system will not boot. What is wrong?

The boot partition is wrong. Remember that, unlike GRUB 2, GRUB Legacy counts the partitions from *zero*. So, the correct command for the second partition of the first disk should be `root (hd0,1)`.

2. Imagine you have a disk identified as `/dev/sda` with multiple partitions. Which command can be used to find out which is the boot partition on a system?

Use `fdisk -l /dev/sda`. The boot partition will be marked with an asterisk (*) in the listing.

3. Which command can be used to find out the UUID of a partition?

Use `ls -la /dev/disk/by-uuid/` and look for the UUID that points to the partition.

4. Consider the following entry for GRUB 2

```
menuentry "Default OS" {
    set root=(hd0,1)
    linux /vmlinuz root=/dev/sda1 ro quiet splash
    initrd /initrd.img
}
```

Change it so the system will boot from a disk with the UUID `5dda0af3-c995-481a-a6f3-46dcd3b6998d`

You will need to change the `set root` statement. Instead of specifying a disk and partition, tell grub to search for the partition with the desired UUID.

```
menuentry "Default OS" {
    search --set=root --fs-uuid 5dda0af3-c995-481a-a6f3-46dcd3b6998d
```

```
linux /vmlinuz root=/dev/sda1 ro quiet splash
initrd /initrd.img
}
```

5. How can you set GRUB 2 to wait for 10 seconds before booting the default menu entry?

Add the parameter `GRUB_TIMEOUT=10` to `/etc/default/grub`.

6. From a GRUB Legacy shell, what are the commands to install GRUB to the first partition of the second disk?

```
grub> root (hd1,0)
grub> setup (hd1)
```



102.3 Manage shared libraries

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 102.3

Weight

1

Key knowledge areas

- Identify shared libraries.
- Identify the typical locations of system libraries.
- Load shared libraries.

Partial list of the used files, terms and utilities

- `ldd`
- `ldconfig`
- `/etc/ld.so.conf`
- `LD_LIBRARY_PATH`



102.3 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	102 Linux Installation and Package Management
Objective:	102.3 Manage shared libraries
Lesson:	1 of 1

Introduction

In this lesson we will be discussing *shared libraries*, also known as *shared objects*: pieces of compiled, reusable code like functions or classes, that are recurrently used by various programs.

To start with, we will explain what shared libraries are, how to identify them and where they are found. Next, we will go into how to configure their storage locations. Finally, we will show how to search for the shared libraries on which a particular program depends.

Concept of Shared Libraries

Similar to their physical counterparts, software libraries are collections of code that are intended to be used by many different programs; just as physical libraries keep books and other resources to be used by many different people.

To build an executable file from a program's source code, two important steps are necessary. First, the *compiler* turns the source code into machine code that is stored in so-called *object files*. Secondly, the *linker* combines the object files and *links* them to libraries in order to generate the

final executable file. This linking can be done *statically* or *dynamically*. Depending on which method we go for, we will be talking about static libraries or, in case of dynamic linking, about shared libraries. Let us explain their differences.

Static libraries

A static library is merged with the program at link time. A copy of the library code is embedded into the program and becomes part of it. Thus, the program has no dependencies on the library at run time because the program already contains the libraries code. Having no dependencies can be seen as an advantage since you do not have to worry about making sure the used libraries are always available. On the downside, statically linked programs are heavier.

Shared (or dynamic) libraries

In the case of shared libraries, the linker simply takes care that the program references libraries correctly. The linker does, however, not copy any library code into the program file. At run time, though, the shared library must be available to satisfy the program's dependencies. This is an economical approach to managing system resources as it helps reduce the size of program files and only one copy of the library is loaded in memory, even when it is used by multiple programs.

Shared Object File Naming Conventions

The name of a shared library, also known as *soname*, follows a pattern which is made up of three elements:

- Library name (normally prefixed by `lib`)
- `so` (which stands for “shared object”)
- Version number of the library

Here an example: `libpthread.so.0`

By contrast, static library names end in `.a`, e.g. `libpthread.a`.

NOTE

Because the files containing shared libraries must be available when the program is executed, most Linux systems contain shared libraries. Since static libraries are only required in a dedicated file when a program is linked, they might not be present on an end user system.

`glibc` (GNU C library) is a good example of a shared library. On a Debian GNU/Linux 9.9 system, its file is named `libc.so.6`. Such rather general file names are normally symbolic links that point to the actual file containing a library, whose name contains the exact version number. In case of

`glibc`, this symbolic link looks like this:

```
$ ls -l /lib/x86_64-linux-gnu/libc.so.6
lrwxrwxrwx 1 root root 12 feb  6 22:17 /lib/x86_64-linux-gnu/libc.so.6 -> libc-2.24.so
```

This pattern of referencing shared library files named by a specific version by more general file names is common practice.

Other examples of shared libraries include `libreadline` (which allows users to edit command lines as they are typed in and includes support for both Emacs and vi editing modes), `libcrypt` (which contains functions related to encryption, hashing, and encoding), or `libcurl` (which is a multiprotocol file transfer library).

Common locations for shared libraries in a Linux system are:

- `/lib`
- `/lib32`
- `/lib64`
- `/usr/lib`
- `/usr/local/lib`

NOTE

The concept of shared libraries is not exclusive to Linux. In Windows, for example, they are called DLL which stands for *dynamic linked libraries*.

Configuration of Shared Library Paths

The references contained in dynamically linked programs are resolved by the dynamic linker (`ld.so` or `ld-linux.so`) when the program is run. The dynamic linker searches for libraries in a number of directories. These directories are specified by the *library path*. The library path is configured in the `/etc` directory, namely in the file `/etc/ld.so.conf` and, more common nowadays, in files residing in the `/etc/ld.so.conf.d` directory. Normally, the former includes just a single `include` line for `*.conf` files in the latter:

```
$ cat /etc/ld.so.conf
include /etc/ld.so.conf.d/*.conf
```

The `/etc/ld.so.conf.d` directory contains `*.conf` files:

```
$ ls /etc/ld.so.conf.d/
libc.conf  x86_64-linux-gnu.conf
```

These *.conf files must include the absolute paths to the shared library directories:

```
$ cat /etc/ld.so.conf.d/x86_64-linux-gnu.conf
# Multiarch support
/lib/x86_64-linux-gnu
/usr/lib/x86_64-linux-gnu
```

The `ldconfig` command takes care of reading these config files, creating the aforementioned set of symbolic links that help to locate the individual libraries and finally of updating the cache file `/etc/ld.so.cache`. Thus, `ldconfig` must be run every time configuration files are added or updated.

Useful options for `ldconfig` are:

-v, --verbose

Display the library version numbers, the name of each directory, and the links that are created:

```
$ sudo ldconfig -v
/usr/local/lib:
/lib/x86_64-linux-gnu:
    libnss_myhostname.so.2 -> libnss_myhostname.so.2
    libfuse.so.2 -> libfuse.so.2.9.7
    libidn.so.11 -> libidn.so.11.6.16
    libnss_mdns4.so.2 -> libnss_mdns4.so.2
    libparted.so.2 -> libparted.so.2.0.1
    (...)
```

So we can see, for example, how `libfuse.so.2` is linked to the actual shared object file `libfuse.so.2.9.7`.

-p, --print-cache

Print the lists of directories and candidate libraries stored in the current cache:

```
$ sudo ldconfig -p
1094 libs found in the cache `/etc/ld.so.cache'
    libzvbi.so.0 (libc6,x86-64) => /usr/lib/x86_64-linux-gnu/libzvbi.so.0
    libzvbi-chains.so.0 (libc6,x86-64) => /usr/lib/x86_64-linux-gnu/libzvbi-chains.so.0
```

```
libzmq.so.5 (libc6,x86-64) => /usr/lib/x86_64-linux-gnu/libzmq.so.5
libzeitgeist-2.0.so.0 (libc6,x86-64) => /usr/lib/x86_64-linux-gnu/libzeitgeist-
2.0.so.0
(...)
```

Note how the cache uses the fully qualified soname of the links:

```
$ sudo ldconfig -p |grep libfuse
libfuse.so.2 (libc6,x86-64) => /lib/x86_64-linux-gnu/libfuse.so.2
```

If we long list `/lib/x86_64-linux-gnu/libfuse.so.2`, we will find the reference to the actual shared object file `libfuse.so.2.9.7` which is stored in the same directory:

```
$ ls -l /lib/x86_64-linux-gnu/libfuse.so.2
lrwxrwxrwx 1 root root 16 Aug 21 2018 /lib/x86_64-linux-gnu/libfuse.so.2 ->
libfuse.so.2.9.7
```

NOTE

Since it requires write access to `/etc/ld.so.cache` (owned by root), you must either be root or use `sudo` to invoke `ldconfig`. For more information about `ldconfig` switches, refer to its manual page.

In addition to the configuration files described above, the `LD_LIBRARY_PATH` environment variable can be used to add new paths for shared libraries temporarily. It is made up of a colon-separated (:) set of directories where libraries are looked up. To add, for example, `/usr/local/mylib` to the library path in the current shell session, you could type:

```
$ LD_LIBRARY_PATH=/usr/local/mylib
```

Now you can check its value:

```
$ echo $LD_LIBRARY_PATH
/usr/local/mylib
```

To add `/usr/local/mylib` to the library path in the current shell session and have it exported to all child processes spawned from that shell, you would type:

```
$ export LD_LIBRARY_PATH=/usr/local/mylib
```

To remove the `LD_LIBRARY_PATH` environment variable, just type:

```
$ unset LD_LIBRARY_PATH
```

To make the changes permanent, you can write the line

```
export LD_LIBRARY_PATH=/usr/local/mylib
```

in one of Bash's initialization scripts such as `/etc/bash.bashrc` or `~/.bashrc`.

NOTE

`LD_LIBRARY_PATH` is to shared libraries what `PATH` is to executables. For more information about environment variables and shell configuration, refer to the respective lessons.

Searching for the Dependencies of a Particular Executable

To look up the shared libraries required by a specific program, use the `ldd` command followed by the absolute path to the program. The output shows the path of the shared library file as well as the hexadecimal memory address at which it is loaded:

```
$ ldd /usr/bin/git
linux-vdso.so.1 => (0x00007ffcb310000)
libpcrc.so.3 => /lib/x86_64-linux-gnu/libpcrc.so.3 (0x00007f18241eb000)
libz.so.1 => /lib/x86_64-linux-gnu/libz.so.1 (0x00007f1823fd1000)
libresolv.so.2 => /lib/x86_64-linux-gnu/libresolv.so.2 (0x00007f1823db6000)
libpthread.so.0 => /lib/x86_64-linux-gnu/libpthread.so.0 (0x00007f1823b99000)
librt.so.1 => /lib/x86_64-linux-gnu/librt.so.1 (0x00007f1823991000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007f18235c7000)
/lib64/ld-linux-x86-64.so.2 (0x00007f182445b000)
```

Likewise, we use `ldd` to search for the dependencies of a shared object:

```
$ ldd /lib/x86_64-linux-gnu/libc.so.6
/lib64/ld-linux-x86-64.so.2 (0x00007fbfed578000)
linux-vdso.so.1 (0x00007fffb7bf5000)
```

With the `-u` (or `--unused`) option `ldd` prints the unused direct dependencies (if they exist):

```
$ ldd -u /usr/bin/git
```

Unused direct dependencies:

```
/lib/x86_64-linux-gnu/libz.so.1  
/lib/x86_64-linux-gnu/libpthread.so.0  
/lib/x86_64-linux-gnu/librt.so.1
```

The reason for unused dependencies is related to the options used by the linker when building the binary. Although the program does not need an unused library, it was still linked and labelled as **NEEDED** in the information about the object file. You can investigate this using commands such as **readelf** or **objdump**, which you will soon use in the explorational exercise.

Guided Exercises

1. Divide the following shared library names into their parts:

Complete file name	Library name	so suffix	Version number
linux-vdso.so.1			
libprocps.so.6			
libdl.so.2			
libc.so.6			
libsystemd.so.0			
ld-linux-x86-64.so.2			

2. You have developed a piece of software and want to add a new shared library directory to your system (`/opt/lib/mylib`). You write its absolute path in a file called `mylib.conf`.

- In what directory should you put this file?

- What command should you run to make the changes fully effective?

3. What command would you use to list the shared libraries required by `kill`?

Explorational Exercises

1. `objdump` is a command line utility that displays information from object files. Check if it is installed in your system with `which objdump`. If it is not, please, install it.

- Use `objdump` with the `-p` (or `--private-headers`) option and `grep` to print the dependencies of `glibc`:

- Use `objdump` with the `-p` (or `--private-headers`) option and `grep` to print the soname of `glibc`:

- Use `objdump` with the `-p` (or `--private-headers`) option and `grep` to print the dependencies of `Bash`:

Summary

In this lesson you have learned:

- What a shared (or dynamic) library is.
- The differences between shared and static libraries.
- The names of shared libraries (*sonames*).
- The preferred locations for shared libraries in a Linux system such as `/lib` or `/usr/lib`.
- The purpose of the dynamic linker `ld.so` (or `ld-linux.so`).
- How to configure shared library paths by means of files in `/etc/` such as `ld.so.conf` or the ones in the `ld.so.conf.d` directory.
- How to configure shared library paths by means of the `LD_LIBRARY_PATH` environment variable.
- How to look up for executable and shared library dependencies.

Commands used in this lesson:

ls

List directory contents.

cat

Concatenate files and print on the standard output.

sudo

Have the superuser executing the command with administrative privileges.

ldconfig

Configure dynamic linker run-time bindings.

echo

Display value of environment variable.

export

Export value of environment variable to child shells.

unset

Remove environment variable.

ldd

Print shared objects dependencies of a program.

readelf

Display information about ELF files (ELF stands for *executable and linkable format*).

objdump

Print information from object files.

Answers to Guided Exercises

1. Divide the following shared library names into their parts:

Complete file name	Library name	so suffix	Version number
linux-vdso.so.1	linux-vdso	so	1
libprocps.so.6	libprocps	so	6
libdl.so.2	libdl	so	2
libc.so.6	libc	so	6
libsystemd.so.0	libsystemd	so	0
ld-linux-x86-64.so.2	ld-linux-x86-64	so	2

2. You have developed a piece of software and want to add a new shared library directory to your system (`/opt/lib/mylib`). You write its absolute path in a file called `mylib.conf`.

- In what directory should you put this file?

`/etc/ld.so.conf.d`

- What command should you run to make the changes fully effective?

`ldconfig`

3. What command would you use to enumerate the shared libraries required by `kill`?

`ldd /bin/kill`

Answers to Explorational Exercises

1. `objdump` is a command line utility that displays information from object files. Check if it is installed in your system with `which objdump`. If it is not, please, install it.

- Use `objdump` with the `-p` (or `--private-headers`) option and `grep` to print the dependencies of `glibc`:

```
objdump -p /lib/x86_64-linux-gnu/libc.so.6 | grep NEEDED
```

- Use `objdump` with the `-p` (or `--private-headers`) option and `grep` to print the soname of `glibc`:

```
objdump -p /lib/x86_64-linux-gnu/libc.so.6 | grep SONAME
```

- Use `objdump` with the `-p` (or `--private-headers`) option and `grep` to print the dependencies of `Bash`:

```
objdump -p /bin/bash | grep NEEDED
```



102.4 Use Debian package management

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 102.4

Weight

3

Key knowledge areas

- Install, upgrade and uninstall Debian binary packages.
- Find packages containing specific files or libraries which may or may not be installed.
- Obtain package information like version, content, dependencies, package integrity and installation status (whether or not the package is installed).
- Awareness of apt.

Partial list of the used files, terms and utilities

- `/etc/apt/sources.list`
- `dpkg`
- `dpkg-reconfigure`
- `apt-get`
- `apt-cache`



102.4 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	102 Linux Installation and Package Management
Objective:	102.4 Use Debian package management
Lesson:	1 of 1

Introduction

A long time ago, when Linux was still in its infancy, the most common way to distribute software was a compressed file (usually a `.tar.gz` archive) with source code, which you would unpack and compile yourself.

However, as the amount and complexity of software grew, the need for a way to distribute pre-compiled software became clear. After all, not everyone had the resources, both in time and computing power, to compile large projects like the Linux kernel or an X Server.

Soon, efforts to standardize a way to distribute these software “packages” grew, and the first package managers were born. These tools made it much easier to install, configure or remove software from a system.

One of those was the Debian package format (`.deb`) and its package tool (`dpkg`). Today, they are widely used not only on Debian itself, but also on its derivatives, like Ubuntu and those derived from it.

Another package management tool that is popular on Debian-based systems is the *Advanced*

Package Tool (apt), which can streamline many of the aspects of the installation, maintenance and removal of packages, making it much easier.

In this lesson, we will learn how to use both `dpkg` and `apt` to obtain, install, maintain and remove software on a Debian-based Linux system.

The Debian Package Tool (dpkg)

The *Debian Package* (dpkg) tool is the essential utility to install, configure, maintain and remove software packages on Debian-based systems. The most basic operation is to install a `.deb` package, which can be done with:

```
# dpkg -i PACKAGENAME
```

Where `PACKAGENAME` is the name of the `.deb` file you want to install.

Package upgrades are handled the same way. Before installing a package, `dpkg` will check if a previous version already exists in the system. If so, the package will be upgraded to the new version. If not, a fresh copy will be installed.

Dealing with Dependencies

More often than not, a package may depend on others to work as intended. For example, an image editor may need libraries to open JPEG files, or another utility may need a widget toolkit like Qt or GTK for its user interface.

`dpkg` will check if those dependencies are installed on your system, and will fail to install the package if they are not. In this case, `dpkg` will list which packages are missing. However it *cannot* solve dependencies by itself. It is up to the user to find the `.deb` packages with the corresponding dependencies and install them.

In the example below, the user tries to install the OpenShot video editor package, but some dependencies were missing:

```
# dpkg -i openshot-qt_2.4.3+dfsg1-1_all.deb
(Reading database ... 269630 files and directories currently installed.)
Preparing to unpack openshot-qt_2.4.3+dfsg1-1_all.deb ...
Unpacking openshot-qt (2.4.3+dfsg1-1) over (2.4.3+dfsg1-1) ...
dpkg: dependency problems prevent configuration of openshot-qt:
 openshot-qt depends on fonts-cantarell; however:
  Package fonts-cantarell is not installed.
```



```

openshot-qt depends on python3-openshot; however:
  Package python3-openshot is not installed.
openshot-qt depends on python3-pyqt5; however:
  Package python3-pyqt5 is not installed.
openshot-qt depends on python3-pyqt5.qtsvg; however:
  Package python3-pyqt5.qtsvg is not installed.
openshot-qt depends on python3-pyqt5.qtwebkit; however:
  Package python3-pyqt5.qtwebkit is not installed.
openshot-qt depends on python3-zmq; however:
  Package python3-zmq is not installed.

```

```

dpkg: error processing package openshot-qt (--install):
  dependency problems - leaving unconfigured
Processing triggers for mime-support (3.60ubuntu1) ...
Processing triggers for gnome-menus (3.32.0-1ubuntu1) ...
Processing triggers for desktop-file-utils (0.23-4ubuntu1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for man-db (2.8.5-2) ...
Errors were encountered while processing:
 openshot-qt

```

As shown above, OpenShot depends on the `fonts-cantarell`, `python3-openshot`, `python3-pyqt5`, `python3-pyqt5.qtsvg`, `python3-pyqt5.qtwebkit` and `python3-zmq` packages. All of those need to be installed before the installation of OpenShot can succeed.

Removing Packages

To remove a package, pass the `-r` parameter to `dpkg`, followed by the package name. For example, the following command will remove the `unrar` package from the system:

```

# dpkg -r unrar
(Reading database ... 269630 files and directories currently installed.)
Removing unrar (1:5.6.6-2) ...
Processing triggers for man-db (2.8.5-2) ...

```

The removal operation also runs a dependency check, and a package cannot be removed unless every other package that depends on it is also removed. If you try to do so, you will get an error message like the one below:

```

# dpkg -r p7zip
dpkg: dependency problems prevent removal of p7zip:

```

```

winetricks depends on p7zip; however:
Package p7zip is to be removed.
p7zip-full depends on p7zip (= 16.02+dfsg-6).

dpkg: error processing package p7zip (--remove):
dependency problems - not removing
Errors were encountered while processing:
p7zip

```

You can pass multiple package names to `dpkg -r`, so they will all be removed at once.

When a package is removed, the corresponding configuration files are left on the system. If you want to remove *everything* associated with the package, use the `-P` (purge) option instead of `-r`.

NOTE

You can force `dpkg` to install or remove a package, even if dependencies are not met, by adding the `--force` parameter like in `dpkg -i --force PACKAGENAME`. However, doing so will most likely leave the installed package, or even your system, in a broken state. *Do not* use `--force` unless you are absolutely sure of what you are doing.

Getting Package Information

To get information about a `.deb` package, such as its version, architecture, maintainer, dependencies and more, use the `dpkg` command with the `-I` parameter, followed by the filename of the package you want to inspect:

```

# dpkg -I google-chrome-stable_current_amd64.deb
new Debian package, version 2.0.
size 59477810 bytes: control archive=10394 bytes.
    1222 bytes,   13 lines   control
    16906 bytes,  457 lines *  postinst          #!/bin/sh
    12983 bytes,  344 lines *  postrm           #!/bin/sh
    1385 bytes,   42 lines *  prerm            #!/bin/sh
Package: google-chrome-stable
Version: 76.0.3809.100-1
Architecture: amd64
Maintainer: Chrome Linux Team <chromium-dev@chromium.org>
Installed-Size: 205436
Pre-Depends: dpkg (>= 1.14.0)
Depends: ca-certificates, fonts-liberation, libappindicator3-1, libasound2 (>= 1.0.16),
libatk-bridge2.0-0 (>= 2.5.3), libatk1.0-0 (>= 2.2.0), libatspi2.0-0 (>= 2.9.90), libc6 (>=
2.16), libcairo2 (>= 1.6.0), libcups2 (>= 1.4.0), libdbus-1-3 (>= 1.5.12), libexpat1 (>=

```

```

2.0.1), libgcc1 (>= 1:3.0), libgdk-pixbuf2.0-0 (>= 2.22.0), libglib2.0-0 (>= 2.31.8),
libgtk-3-0 (>= 3.9.10), libnspr4 (>= 2:4.9-2~), libnss3 (>= 2:3.22), libpango-1.0-0 (>=
1.14.0), libpangocairo-1.0-0 (>= 1.14.0), libuuid1 (>= 2.16), libx11-6 (>= 2:1.4.99.1),
libx11-xcb1, libxcb1 (>= 1.6), libxcomposite1 (>= 1:0.3-1), libxcursor1 (>= 1.1.2),
libxdamage1 (>= 1:1.1), libxext6, libxfixes3, libxi6 (>= 2:1.2.99.4), libxrandr2 (>=
2:1.2.99.3), libxrender1, libxss1, libxtst6, lsb-release, wget, xdg-utils (>= 1.0.2)
Recommends: libu2f-udev
Provides: www-browser
Section: web
Priority: optional
Description: The web browser from Google
  Google Chrome is a browser that combines a minimal design with sophisticated technology to
  make the web faster, safer, and easier.

```

Listing Installed Packages and Package Contents

To get a list of every package installed on your system, use the `--get-selections` option, as in `dpkg --get-selections`. You can also get a list of every file installed by a specific package by passing the `-L PACKAGENAME` parameter to `dpkg`, like below:

```

# dpkg -L unrar
/.
/usr
/usr/bin
/usr/bin/unrar-nonfree
/usr/share
/usr/share/doc
/usr/share/doc/unrar
/usr/share/doc/unrar/changelog.Debian.gz
/usr/share/doc/unrar/copyright
/usr/share/man
/usr/share/man/man1
/usr/share/man/man1/unrar-nonfree.1.gz

```

Finding Out Which Package Owns a Specific File

Sometimes you may need to find out which package owns a specific file in your system. You can do so by using the `dpkg-query` utility, followed by the `-S` parameter and the path to the file in question:

```

# dpkg-query -S /usr/bin/unrar-nonfree

```

```
unrar: /usr/bin/unrar-nonfree
```

Reconfiguring Installed Packages

When a package is installed there is a configuration step called *post-install* where a script runs to set-up everything needed for the software to run such as permissions, placement of configuration files, etc. This may also ask some questions of the user to set preferences on how the software will run.

Sometimes, due to a corrupt or malformed configuration file, you may wish to restore a package's settings to its “fresh” state. Or you may wish to change the answers you gave to the initial configuration questions. To do this run the `dpkg-reconfigure` utility, followed by the package name.

This program will back-up the old configuration files, unpack the new ones in the correct directories and run the *post-install* script provided by the package, as if the package had been installed for the first time. Try to reconfigure the `tzdata` package with the following example:

```
# dpkg-reconfigure tzdata
```

Advanced Package Tool (apt)

The *Advanced Package Tool* (APT) is a package management system, including a set of tools, that greatly simplifies package installation, upgrade, removal and management. APT provides features like advanced search capabilities and automatic dependency resolution.

APT is not a “substitute” for `dpkg`. You may think of it as a “front end”, streamlining operations and filling gaps in `dpkg` functionality, like dependency resolution.

APT works in concert with software repositories which contain the packages available to install. Such repositories may be a local or remote server, or (less common) even a CD-ROM disc.

Linux distributions, such as Debian and Ubuntu, maintain their own repositories, and other repositories may be maintained by developers or user groups to provide software not available from the main distribution repositories.

There are many utilities that interact with APT, the main ones being:

apt-get

used to download, install, upgrade or remove packages from the system.

apt-cache

used to perform operations, like searches, in the package index.

apt-file

used for searching for files inside packages.

There is also a “friendlier” utility named simply `apt`, combining the most used options of `apt-get` and `apt-cache` in one utility. Many of the commands for `apt` are the same as the ones for `apt-get`, so they are in many cases interchangeable. However, since `apt` may not be installed on a system, it is recommended to learn how to use `apt-get` and `apt-cache`.

NOTE

`apt` and `apt-get` may require a network connection, because packages and package indexes may need to be downloaded from a remote server.

Updating the Package Index

Before installing or upgrading software with APT, it is recommended to update the package index first in order to retrieve information about new and updated packages. This is done with the `apt-get` command, followed by the `update` parameter:

```
# apt-get update
Ign:1 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 https://repo.skype.com/deb stable InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu disco InRelease
Hit:4 http://repository.spotify.com stable InRelease
Hit:5 http://dl.google.com/linux/chrome/deb stable Release
Hit:6 http://apt.pop-os.org/proprietary disco InRelease
Hit:7 http://ppa.launchpad.net/system76/pop/ubuntu disco InRelease
Hit:8 http://us.archive.ubuntu.com/ubuntu disco-security InRelease
Hit:9 http://us.archive.ubuntu.com/ubuntu disco-updates InRelease
Hit:10 http://us.archive.ubuntu.com/ubuntu disco-backports InRelease
Reading package lists... Done
```

TIP

Instead of `apt-get update`, you can also use `apt update`.

Installing and Removing Packages

With the package index updated you may now install a package. This is done with `apt-get install`, followed by the name of the package you wish to install:

```
# apt-get install xournal
```

```
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  xournal
0 upgraded, 1 newly installed, 0 to remove and 75 not upgraded.
Need to get 285 kB of archives.
After this operation, 1041 kB of additional disk space will be used.
```

Similarly, to remove a package use `apt-get remove`, followed by the package name:

```
# apt-get remove xournal
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages will be REMOVED:
  xournal
0 upgraded, 0 newly installed, 1 to remove and 75 not upgraded.
After this operation, 1041 kB disk space will be freed.
Do you want to continue? [Y/n]
```

Be aware that when installing or removing packages, APT will do automatic dependency resolution. This means that any additional packages needed by the package you are installing *will also be installed*, and that packages that depend on the package you are removing *will also be removed*. APT will always show what will be installed or removed before asking if you want to continue:

```
# apt-get remove p7zip
Reading package lists... Done
Building dependency tree
The following packages will be REMOVED:
  android-libbacktrace android-libunwind android-libutils
  android-libziparchive android-sdk-platform-tools fastboot p7zip p7zip-full
0 upgraded, 0 newly installed, 8 to remove and 75 not upgraded.
After this operation, 6545 kB disk space will be freed.
Do you want to continue? [Y/n]
```

Note that when a package is removed the corresponding configuration files are left on the system. To remove the package *and* any configuration files, use the `purge` parameter instead of `remove` or the `remove` parameter with the `--purge` option:

```
# apt-get purge p7zip
```

or

```
# apt-get remove --purge p7zip
```

TIP | You can also use `apt install` and `apt remove`.

Fixing Broken Dependencies

It is possible to have “broken dependencies” on a system. This means that one or more of the installed packages depend on other packages that have not been installed, or are not present anymore. This may happen due to an APT error, or because of a manually installed package.

To solve this, use the `apt-get install -f` command. This will try to “fix” the broken packages by installing the missing dependencies, ensuring that all packages are consistent again.

TIP | You can also use `apt install -f`.

Upgrading Packages

APT can be used to automatically upgrade any installed packages to the latest versions available from the repositories. This is done with the `apt-get upgrade` command. Before running it, first update the package index with `apt-get update`:

```
# apt-get update
Hit:1 http://us.archive.ubuntu.com/ubuntu disco InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu disco-security InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu disco-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu disco-backports InRelease
Reading package lists... Done

# apt-get upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages have been kept back:
  gnome-control-center
The following packages will be upgraded:
  cups cups-bsd cups-client cups-common cups-core-drivers cups-daemon
```

```
cups-ipp-utils cups-ppdc cups-server-common firefox-locale-ar (...)
```

```
74 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.  
Need to get 243 MB of archives.  
After this operation, 30.7 kB of additional disk space will be used.  
Do you want to continue? [Y/n]
```

The summary at the bottom of the output shows how many packages will be upgraded, how many will be installed, removed or kept back, the total download size and how much extra disk space will be needed to complete the operation. To complete the upgrade, just answer `Y` and wait for `apt-get` to finish the task.

To upgrade a single package, just run `apt-get upgrade` followed by the package name. As in `dpkg`, `apt-get` will first check if a previous version of a package is installed. If so, the package will be upgraded to the newest version available in the repository. If not, a fresh copy will be installed.

TIP | You can also use `apt upgrade` and `apt update`.

The Local Cache

When you install or update a package, the corresponding `.deb` file is downloaded to a local cache directory before the package is installed. By default, this directory is `/var/cache/apt/archives`. Partially downloaded files are copied to `/var/cache/apt/archives/partial/`.

As you install and upgrade packages, the cache directory can get quite large. To reclaim space, you can empty the cache by using the `apt-get clean` command. This will remove the contents of the `/var/cache/apt/archives` and `/var/cache/apt/archives/partial/` directories.

TIP | You can also use `apt clean`.

Searching for Packages

The `apt-cache` utility can be used to perform operations on the package index, such as searching for a specific package or listing which packages contain a specific file.

To conduct a search, use `apt-cache search` followed by a search pattern. The output will be a list of every package that contains the pattern, either in its package name, description or files provided.

```
# apt-cache search p7zip  
liblzma-dev - XZ-format compression library - development files  
liblzma5 - XZ-format compression library
```



```
forensics-extra - Forensics Environment - extra console components (metapackage)
p7zip - 7zr file archiver with high compression ratio
p7zip-full - 7z and 7za file archivers with high compression ratio
p7zip-rar - non-free rar module for p7zip
```

In the example above, the entry `liblzma5 - XZ-format compression library` does not seem to match the pattern. However, if we show the full information, including description, for the package using the `show` parameter, we will find the pattern there:

```
# apt-cache show liblzma5
Package: liblzma5
Architecture: amd64
Version: 5.2.4-1
Multi-Arch: same
Priority: required
Section: libs
Source: xz-utils
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Jonathan Nieder <jrnieder@gmail.com>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 259
Depends: libc6 (>= 2.17)
Breaks: liblzma2 (<< 5.1.1alpha+20110809-3~)
Filename: pool/main/x/xz-utils/liblzma5_5.2.4-1_amd64.deb
Size: 92352
MD5sum: 223533a347dc76a8cc9445cfc6146ec3
SHA1: 8ed14092fb1caecfebc556fda0745e1e74ba5a67
SHA256: 01020b5a0515dbc9a7c00b464a65450f788b0258c3fbb733ecad0438f5124800
Homepage: https://tukaani.org/xz/
Description-en: XZ-format compression library
XZ is the successor to the Lempel-Ziv/Markov-chain Algorithm
compression format, which provides memory-hungry but powerful
compression (often better than bzip2) and fast, easy decompression.
.
The native format of liblzma is XZ; it also supports raw (headerless)
streams and the older LZMA format used by lzma. (For 7-Zip's related
format, use the p7zip package instead.)
```

You can use *regular expressions* with the search pattern, allowing for very complex (and precise) searches. However, this topic is out of scope for this lesson.

TIP

You can also use `apt search` instead of `apt-cache search` and `apt show` instead of `apt-cache show`.

The Sources List

APT uses a list of sources to know where to get packages from. This list is stored in the file `sources.list`, located inside the `/etc/apt` directory. This file can be edited directly with a text editor, like `vi`, `pico` or `nano`, or with graphical utilities like `aptitude` or `synaptic`.

A typical line inside `sources.list` looks like this:

```
deb http://us.archive.ubuntu.com/ubuntu/ disco main restricted universe multiverse
```

The syntax is archive type, URL, distribution and one or more components, where:

Archive type

A repository may contain packages with ready-to-run software (binary packages, type `deb`) or with the source code to this software (source packages, type `deb-src`). The example above provides binary packages.

URL

The URL for the repository.

Distribution

The name (or codename) for the distribution for which packages are provided. One repository may host packages for multiple distributions. In the example above, `disco` is the codename for Ubuntu 19.04 *Disco Dingo*.

Components

Each component represents a set of packages. These components may be different on different Linux distributions. For example, on Ubuntu and derivatives, they are:

main

contains officially supported, open-source packages.

restricted

contains officially supported, closed-source software, like device drivers for graphic cards, for example.

universe

contains community maintained open-source software.

multiverse

contains unsupported, closed-source or patent-encumbered software.

On Debian, the main components are:

main

consists of packages compliant with the *Debian Free Software Guidelines* (DFSG), which do not rely on software outside this area to operate. Packages included here are considered to be part of the Debian distribution.

contrib

contains DFSG-compliant packages, but which depend on other packages that are not in **main**.

non-free

contains packages that are not compliant with the DFSG.

security

contains security updates.

backports

contains more recent versions of packages that are in **main**. The development cycle of the stable versions of Debian is quite long (around two years), and this ensures that users can get the most up-to-date packages without having to modify the **main** core repository.

NOTE

You can learn more about the *Debian Free Software Guidelines* at: https://www.debian.org/social_contract#guidelines

To add a new repository to get packages from, you can simply add the corresponding line (usually provided by the repository maintainer) to the end of `sources.list`, save the file and reload the package index with `apt-get update`. After that, the packages in the new repository will be available for installation using `apt-get install`.

Keep in mind that lines starting with the `#` character are considered comments, and are ignored.

The `/etc/apt/sources.list.d` Directory

Inside the `/etc/apt/sources.list.d` directory you can add files with additional repositories to be used by APT, without the need to modify the main `/etc/apt/sources.list` file. These are

simple text files, with the same syntax described above and the `.list` file extension.

Below you see the contents of a file called `/etc/apt/sources.list.d/buster-backports.list`:

```
deb http://deb.debian.org/debian buster-backports main contrib non-free
deb-src http://deb.debian.org/debian buster-backports main contrib non-free
```

Listing Package Contents and Finding Files

A utility called `apt-file` can be used to perform more operations in the package index, like listing the contents of a package or finding a package that contains a specific file. This utility may not be installed by default in your system. In this case, you can usually install it using `apt-get`:

```
# apt-get install apt-file
```

After installation, you will need to update the package cache used for `apt-file`:

```
# apt-file update
```

This usually takes only a few seconds. After that, you are ready to use `apt-file`.

To list the contents of a package, use the `list` parameter followed by the package name:

```
# apt-file list unrar
unrar: /usr/bin/unrar-nonfree
unrar: /usr/share/doc/unrar/changelog.Debian.gz
unrar: /usr/share/doc/unrar/copyright
unrar: /usr/share/man/man1/unrar-nonfree.1.gz
```

TIP You can also use `apt list` instead of `apt-file list`.

You can search all packages for a file using the `search` parameter, followed by the file name. For example, if you wish to know which package provides a file called `libSDL2.so`, you can use:

```
# apt-file search libSDL2.so
libsdl2-dev: /usr/lib/x86_64-linux-gnu/libSDL2.so
```

The answer is the package `libsdl2-dev`, which provides the file `/usr/lib/x86_64-linux-`

`gnu/libSDL2.so`.

The difference between `apt-file search` and `dpkg-query` is that `apt-file search` will also search uninstalled packages, while `dpkg-query` can only show files that belong to an installed package.

Guided Exercises

1. What is the command to install a package named `package.deb` using `dpkg`?

2. Using `dpkg-query`, find which package contains a file named `7zr.1.gz`.

3. Can you remove a package called `unzip` from the system using `dpkg -r unzip` if the package `file-roller` depends on it? If not, what would be the correct way to do it?

4. Using `apt-file`, how can you find out which package contains the file `unrar`?

5. Using `apt-cache`, what is the command to show information for the package `gimp`?

Explorational Exercises

1. Consider a repository with Debian source packages for the `xenial` distribution, hosted at `http://us.archive.ubuntu.com/ubuntu/` and with packages for the `universe` component. What would be the corresponding line to be added to `/etc/apt/sources.list`?
2. While compiling a program, you come across an error message complaining that the header file `zzip-io.h` is not present in your system. How can you find out which package provides that file?
3. How can you ignore a dependency warning and remove a package using `dpkg`, even if there are packages that depend on it in the system?
4. How can you get more information about a package called `midori` using `apt`?
5. Before installing or updating packages with `apt`, which command should be used to ensure that the package index is up-to-date?

Summary

In this lesson, you learned:

- How to use `dpkg` to install and remove packages.
- How to list installed packages and package contents.
- How to reconfigure an installed package.
- What is `apt`, and how to install, upgrade and remove packages using it.
- How to use `apt-cache` to search for packages.
- How the `/etc/apt/sources.list` file works.
- How to use `apt-file` to show the contents of a package, or how to find which package contains a specific file.

The following commands were discussed:

`dpkg -i`

Installs a single package, or a space-separated list of packages.

`dpkg -r`

Removes a package, or a space-separated list of packages.

`dpkg -I`

Inspects a package, providing details on the software it installs and any needed dependencies.

`dpkg --get-selections`

Lists out every package that `dpkg` has installed on the system.

`dpkg -L`

Prints out a list of every file that a particular package installs.

`dpkg-query`

With a specified file name, this command will print out the package that installed the file.

`dpkg-reconfigure`

This command will re-run a package's *post-install* script so that an administrator can make configuration adjustments to the package's installation.

apt-get update

This command will update the local package index to match what is available within the configured repositories under the `/etc/apt/` directory.

apt-get install

This command will download a package from a remote repository and install it along with its dependencies, can also be used to install a Debian package that has already been downloaded.

apt-get remove

This command will uninstall the specified package(s) from the system.

apt-cache show

Just like the `dpkg -I` command, this command can be used to show details on a specific package.

apt-cache search

This command will search your local APT cached database for a particular package.

apt-file update

This command will update the package cache so that the `apt-file` command can query its contents.

apt-file search

This command will search in which package a file is included. A list of all packages containing the pattern is returned.

apt-file list

This command is used to list the contents of a package, just like the `dpkg -L` command.

Answers to Guided Exercises

1. What is the command to install a package named `package.deb` using `dpkg`?

Pass the `-i` parameter to `dpkg`:

```
# dpkg -i package.deb
```

2. Using `dpkg-query`, find which package contains a file named `7zr.1.gz`.

Add the `-S` parameter to `dpkg-query`:

```
# dpkg-query -S 7zr.1.gz
```

3. Can you remove a package called `unzip` from the system using `dpkg -r unzip` if the package `file-roller` depends on it? If not, what would be the correct way to do it?

No. `dpkg` will not resolve dependencies, and will not let you remove a package if another installed package depends on it. In this example, you could first remove `file-roller` (assuming nothing depends on it) and then remove `unzip`, or remove both at the same time with:

```
# dpkg -r unzip file-roller
```

4. How can you find out which package contains the file `/usr/bin/unrar` using the `apt-file` utility?

Use the `search` parameter followed by the path (or filename):

```
# apt-file search /usr/bin/unrar
```

5. Using `apt-cache`, what is the command to show information for the package `gimp`?

Use the `show` parameter followed by the package name:

```
# apt-cache show gimp
```

Answers to Explorational Exercises

1. Consider a repository with Debian source packages for the `xenial` distribution, hosted at `http://us.archive.ubuntu.com/ubuntu/` and with packages for the `universe` component. What would be the corresponding line to be added to `/etc/apt/sources.list`?

Source packages are of the `deb-src` type, so the line should be:

```
deb-src http://us.archive.ubuntu.com/ubuntu/ xenial universe
```

This line could also be added inside a `.list` file in `/etc/apt/sources.list.d/`. The name is up to you but should be descriptive, something like `xenial_sources.list`.

2. While compiling a program, you come across an error message complaining that the header file `zzip-io.h` is not present on your system. How can you find out which package provides that file?

Use `apt-file search` to find which package contains a file not present in the system:

```
# apt-file search zzip-io.h
```

3. How can you ignore a dependency warning and remove a package using `dpkg`, even if there are packages that depend on it in the system?

The parameter `--force` can be used, but this should *never* be done unless you know exactly what you are doing, since there is a great risk that your system will be left in an inconsistent or “broken” state.

4. How can you get more information about a package called `midori` using `apt-cache`?

Use `apt-cache show` followed by the package name:

```
# apt-cache show midori
```

5. Before installing or updating packages with `apt-get`, which command should be used to ensure that the package index is up-to-date?

`apt-get update` should be used. This will download the latest package indexes from the repositories described in the `/etc/apt/sources.list` file or in the `/etc/apt/sources.list.d/` directory.



102.5 Use RPM and YUM package management

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 102.5

Weight

3

Key knowledge areas

- Install, re-install, upgrade and remove packages using RPM, YUM and Zypper.
- Obtain information on RPM packages such as version, status, dependencies, integrity and signatures.
- Determine what files a package provides, as well as find which package a specific file comes from.
- Awareness of dnf.

Partial list of the used files, terms and utilities

- rpm
- rpm2cpio
- /etc/yum.conf
- /etc/yum.repos.d/
- yum
- zypper



102.5 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	102 Linux Installation and Package Management
Objective:	102.5 Use RPM and YUM package management
Lesson:	1 of 1

Introduction

A long time ago, when Linux was still in its infancy, the most common way to distribute software was a compressed file (usually as a `.tar.gz` archive) with source code, which you would unpack and compile yourself.

However, as the amount and complexity of software grew, the need for a way to distribute pre-compiled software became clear. After all, not everyone had the resources, both in time and computing power, to compile large projects like the Linux kernel or an X Server.

Soon, efforts to standardize a way to distribute these software “packages” grew, and the first package managers were born. These tools made it much easier to install, configure or remove software from a system.

One of those was the *RPM Package Manager* and its corresponding tool (`rpm`), developed by Red Hat. Today, they are widely used not only on Red Hat Enterprise Linux (RHEL) itself, but also on its descendants, like Fedora, CentOS and Oracle Linux, other distributions like openSUSE and even other operating systems, like IBM’S AIX.

Other package management tools popular on Red Hat compatible distros are `yum` (YellowDog Updater Modified), `dnf` (Dandified YUM) and `zypper`, which can streamline many of the aspects of the installation, maintenance and removal of packages, making package management much easier.

In this lesson, we will learn how to use `rpm`, `yum`, `dnf` and `zypper` to obtain, install, manage and remove software on a Linux system.

NOTE

Despite using the same package format, there are internal differences between distributions so a package made specifically for openSUSE might not work on a RHEL system, and vice-versa. When searching for packages, always check for compatibility and try to find one tailored for your specific distribution, if possible.

The RPM Package Manager (rpm)

The RPM Package Manager (`rpm`) is the essential tool for managing software packages on Red Hat-based (or derived) systems.

Installing, Upgrading and Removing Packages

The most basic operation is to install a package, which can be done with:

```
# rpm -i PACKAGENAME
```

Where `PACKAGENAME` is the name of the `.rpm` package you wish to install.

If there is a previous version of a package on the system, you can upgrade to a newer version using the `-U` parameter:

```
# rpm -U PACKAGENAME
```

If there is no previous version of `PACKAGENAME` installed, then a fresh copy will be installed. To avoid this and *only* upgrade an *installed* package, use the `-F` option.

In both operations you can add the `-v` parameter to get a verbose output (more information is shown during the installation) and `-h` to get hash signs (#) printed as a visual aid to track installation progress. Multiple parameters can be combined into one, so `rpm -i -v -h` is the same as `rpm -ivh`.

To remove an installed package, pass the `-e` parameter (as in “erase”) to `rpm`, followed by the

name of the package you wish to remove:

```
# rpm -e wget
```

If an installed package depends on the package being removed, you will get an error message:

```
# rpm -e unzip
error: Failed dependencies:
    /usr/bin/unzip is needed by (installed) file-roller-3.28.1-2.el7.x86_64
```

To complete the operation, first you will need to remove the packages that depend on the one you wish to remove (in the example above, `file-roller`). You can pass multiple package names to `rpm -e` to remove multiple packages at once.

Dealing with Dependencies

More often than not, a package may depend on others to work as intended. For example, an image editor may need libraries to open JPG files, or a utility may need a widget toolkit like Qt or GTK for its user interface.

`rpm` will check if those dependencies are installed on your system, and will fail to install the package if they are not. In this case, `rpm` will list what is missing. However it *cannot* solve dependencies by itself.

In the example below, the user tried to install a package for the GIMP image editor, but some dependencies were missing:

```
# rpm -i gimp-2.8.22-1.el7.x86_64.rpm
error: Failed dependencies:
    babl(x86-64) >= 0.1.10 is needed by gimp-2:2.8.22-1.el7.x86_64
    gegl(x86-64) >= 0.2.0 is needed by gimp-2:2.8.22-1.el7.x86_64
    gimp-libs(x86-64) = 2:2.8.22-1.el7 is needed by gimp-2:2.8.22-1.el7.x86_64
    libbabl-0.1.so.0()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
    libgegl-0.2.so.0()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
    libgimp-2.0.so.0()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
    libgimpbase-2.0.so.0()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
    libgimpcolor-2.0.so.0()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
    libgimpconfig-2.0.so.0()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
    libgimpmath-2.0.so.0()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
    libgimpmodule-2.0.so.0()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
    libgimpthumb-2.0.so.0()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
```

```
libgimpui-2.0.so.0()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
libgimpwidgets-2.0.so.0()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
libmng.so.1()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
libwmf-0.2.so.7()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
libwmflite-0.2.so.7()(64bit) is needed by gimp-2:2.8.22-1.el7.x86_64
```

It is up to the user to find the `.rpm` packages with the corresponding dependencies and install them. Package managers such as `yum`, `zypper` and `dnf` have tools that can tell which package provides a specific file. Those will be discussed later in this lesson.

Listing Installed Packages

To get a list of all installed packages on your system, use the `rpm -qa` (think of “query all”).

```
# rpm -qa
selinux-policy-3.13.1-229.el7.noarch
pciutils-libs-3.5.1-3.el7.x86_64
redhat-menus-12.0.2-8.el7.noarch
grubby-8.28-25.el7.x86_64
hunspell-en-0.20121024-6.el7.noarch
dejavu-fonts-common-2.33-6.el7.noarch
xorg-x11-drv-dummy-0.3.7-1.el7.1.x86_64
libevdev-1.5.6-1.el7.x86_64
[...]
```

Getting Package Information

To get information about an *installed* package, such as its version number, architecture, install date, packager, summary, etc., use `rpm` with the `-qi` (think of “query info”) parameters, followed by the package name. For example:

```
# rpm -qi unzip
Name       : unzip
Version    : 6.0
Release    : 19.el7
Architecture: x86_64
Install Date: Sun 25 Aug 2019 05:14:39 PM EDT
Group      : Applications/Archiving
Size       : 373986
License    : BSD
Signature   : RSA/SHA256, Wed 25 Apr 2018 07:50:02 AM EDT, Key ID 24c6a8a7f4a80eb5
```



```
Source RPM : unzip-6.0-19.el7.src.rpm
Build Date : Wed 11 Apr 2018 01:24:53 AM EDT
Build Host : x86-01.bsys.centos.org
Relocations : (not relocatable)
Packager   : CentOS BuildSystem <http://bugs.centos.org>
Vendor     : CentOS
URL        : http://www.info-zip.org/UnZip.html
Summary    : A utility for unpacking zip files
Description:
The unzip utility is used to list, test, or extract files from a zip
archive. Zip archives are commonly found on MS-DOS systems. The zip
utility, included in the zip package, creates zip archives. Zip and
unzip are both compatible with archives created by PKWARE(R)'s PKZIP
for MS-DOS, but the programs' options and default behaviors do differ
in some respects.

Install the unzip package if you need to list, test or extract files from
a zip archive.
```

To get a list of what files are inside an *installed* package use the `-ql` parameters (think of “query list”) followed by the package name:

```
# rpm -ql unzip
/usr/bin/funzip
/usr/bin/unzip
/usr/bin/unzipsfx
/usr/bin/zipgrep
/usr/bin/zipinfo
/usr/share/doc/unzip-6.0
/usr/share/doc/unzip-6.0/BUGS
/usr/share/doc/unzip-6.0/LICENSE
/usr/share/doc/unzip-6.0/README
/usr/share/man/man1/funzip.1.gz
/usr/share/man/man1/unzip.1.gz
/usr/share/man/man1/unzipsfx.1.gz
/usr/share/man/man1/zipgrep.1.gz
/usr/share/man/man1/zipinfo.1.gz
```

If you wish to get information or a file listing from a package that has *not* been installed yet, just add the `-p` parameter to the commands above, followed by the name of the RPM file (FILENAME). So `rpm -qi PACKAGENAME` becomes `rpm -qip FILENAME`, and `rpm -ql PACKAGENAME` becomes `rpm -qlp FILENAME`, as shown below.

```
# rpm -qip atom.x86_64.rpm
Name       : atom
Version    : 1.40.0
Release    : 0.1
Architecture: x86_64
Install Date: (not installed)
Group      : Unspecified
Size       : 570783704
License    : MIT
Signature  : (none)
Source RPM : atom-1.40.0-0.1.src.rpm
Build Date : sex 09 ago 2019 12:36:31 -03
Build Host : b01bbeaf3a88
Relocations : /usr
URL        : https://atom.io/
Summary    : A hackable text editor for the 21st Century.
Description :
A hackable text editor for the 21st Century.
```

```
# rpm -qlp atom.x86_64.rpm
/usr/bin/apm
/usr/bin/atom
/usr/share/applications/atom.desktop
/usr/share/atom
/usr/share/atom/LICENSE
/usr/share/atom/LICENSES.chromium.html
/usr/share/atom/atom
/usr/share/atom/atom.png
/usr/share/atom/blink_image_resources_200_percent.pak
/usr/share/atom/content_resources_200_percent.pak
/usr/share/atom/content_shell.pak

(listing goes on)
```

Finding Out Which Package Owns a Specific File

To find out which installed package owns a file, use the `-qf` (think “query file”) followed by the full path to the file:

```
# rpm -qf /usr/bin/unzip
unzip-6.0-19.el7.x86_64
```

In the example above, the file `/usr/bin/unzip` belongs to the `unzip-6.0-19.el7.x86_64` package.

YellowDog Updater Modified (YUM)

`yum` was originally developed as the *Yellow Dog Updater* (YUP), a tool for package management on the Yellow Dog Linux distribution. Over time, it evolved to manage packages on other RPM based systems, such as Fedora, CentOS, Red Hat Enterprise Linux and Oracle Linux.

Functionally, it is similar to the `apt` utility on Debian-based systems, being able to search for, install, update and remove packages and automatically handle dependencies. `yum` can be used to install a single package, or to upgrade a whole system at once.

Searching for Packages

In order to install a package, you need to know its name. For this you can perform a search with `yum search PATTERN`, where `PATTERN` is the name of the package you are searching for. The result is a list of packages whose name or summary contain the search pattern specified. For example, if you need a utility to handle 7Zip compressed files (with the `.7z` extension) you can use:

```
# yum search 7zip
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
 * base: mirror.ufscar.br
 * epel: mirror.globo.com
 * extras: mirror.ufscar.br
 * updates: mirror.ufscar.br
===== N/S matchyutr54ed: 7zip =====
p7zip-plugins.x86_64 : Additional plugins for p7zip
p7zip.x86_64 : Very high compression ratio file archiver
p7zip-doc.noarch : Manual documentation and contrib directory
p7zip-gui.x86_64 : 7zG - 7-Zip GUI version
```

Name and summary matches only, use "search all" for everything.

Installing, Upgrading and Removing Packages

To install a package using `yum`, use the command `yum install PACKAGENAME`, where `PACKAGENAME` is the name of the package. `yum` will fetch the package and corresponding dependencies from an online repository, and install everything in your system.

```
# yum install p7zip
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
 * base: mirror.ufscar.br
 * epel: mirror.globo.com
 * extras: mirror.ufscar.br
 * updates: mirror.ufscar.br
Resolving Dependencies
--> Running transaction check
---> Package p7zip.x86_64 0:16.02-10.el7 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package            Arch             Version           Repository        Size
=====
Installing:
p7zip              x86_64           16.02-10.el7      epel               604 k

Transaction Summary
=====
Install 1 Package

Total download size: 604 k
Installed size: 1.7 M
Is this ok [y/d/N]:
```

To upgrade an installed package, use `yum update PACKAGENAME`, where `PACKAGENAME` is the name of the package you want to upgrade. For example:

```
# yum update wget
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
 * base: mirror.ufscar.br
 * epel: mirror.globo.com
 * extras: mirror.ufscar.br
 * updates: mirror.ufscar.br
Resolving Dependencies
--> Running transaction check
---> Package wget.x86_64 0:1.14-18.el7 will be updated
---> Package wget.x86_64 0:1.14-18.el7_6.1 will be an update
--> Finished Dependency Resolution
```

Dependencies Resolved

```

=====
Package      Arch          Version           Repository        Size
=====
Updating:
wget         x86_64        1.14-18.el7_6.1   updates          547 k

```

Transaction Summary

```

=====
Upgrade 1 Package

```

```

Total download size: 547 k

```

```

Is this ok [y/d/N]:

```

If you omit the name of a package, you can update every package on the system for which an update is available.

To check if an update is available for a specific package, use `yum check-update PACKAGENAME`. As before, if you omit the package name, `yum` will check for updates for every installed package on the system.

To remove an installed package, use `yum remove PACKAGENAME`, where `PACKAGENAME` is the name of the package you wish to remove.

Finding Which Package Provides a Specific File

In a previous example we showed an attempt to install the `gimp` image editor, which failed because of unmet dependencies. However, `rpm` shows which files are missing, but does not list the name of the packages that provide them.

For example, one of the dependencies missing was `libgimpui-2.0.so.0`. To see what package provides it, you can use `yum whatprovides`, followed by the name of the file you are searching for:

```

# yum whatprovides libgimpui-2.0.so.0
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
* base: mirror.ufscar.br
* epel: mirror.globo.com
* extras: mirror.ufscar.br

```

```
* updates: mirror.ufscar.br
2:gimp-libs-2.8.22-1.el7.i686 : GIMP libraries
Repo      : base
Matched from:
Provides   : libgimpui-2.0.so.0
```

The answer is `gimp-libs-2.8.22-1.el7.i686`. You can then install the package with the command `yum install gimp-libs`.

This also works for files already in your system. For example, if you wish to know where the file `/etc/hosts` came from, you can use:

```
# yum whatprovides /etc/hosts
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
* base: mirror.ufscar.br
* epel: mirror.globo.com
* extras: mirror.ufscar.br
* updates: mirror.ufscar.br
setup-2.8.71-10.el7.noarch : A set of system configuration and setup files
Repo      : base
Matched from:
Filename   : /etc/hosts
```

The answer is `setup-2.8.71-10.el7.noarch`.

Getting Information About a Package

To get information about a package, such as its version, architecture, description, size and more, use `yum info PACKAGENAME` where `PACKAGENAME` is the name of the package you want information for:

```
# yum info firefox
Last metadata expiration check: 0:24:16 ago on Sat 21 Sep 2019 02:39:43 PM -03.
Installed Packages
Name      : firefox
Version   : 69.0.1
Release   : 3.fc30
Architecture : x86_64
Size      : 268 M
Source    : firefox-69.0.1-3.fc30.src.rpm
Repository : @System
```

```

From repo      : updates
Summary       : Mozilla Firefox Web browser
URL           : https://www.mozilla.org/firefox/
License       : MPLv1.1 or GPLv2+ or LGPLv2+
Description   : Mozilla Firefox is an open-source web browser, designed
                  : for standards compliance, performance and portability.

```

Managing Software Repositories

For yum the “repos” are listed in the directory `/etc/yum.repos.d/`. Each repository is represented by a `.repo` file, like `CentOS-Base.repo`.

Additional, extra repositories can be added by the user by adding a `.repo` file in the directory mentioned above, or at the end of `/etc/yum.conf`. However, the recommended way to add or manage repositories is with the `yum-config-manager` tool.

To add a repository, use the `--add-repo` parameter, followed by the URL to a `.repo` file.

```

# yum-config-manager --add-repo https://rpms.remirepo.net/enterprise/remi.repo
Loaded plugins: fastestmirror, langpacks
adding repo from: https://rpms.remirepo.net/enterprise/remi.repo
grabbing file https://rpms.remirepo.net/enterprise/remi.repo to /etc/yum.repos.d/remi.repo
repo saved to /etc/yum.repos.d/remi.repo

```

To get a list of all available repositories use `yum repolist all`. You will get an output similar to this:

```

# yum repolist all
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
* base: mirror.ufscar.br
* epel: mirror.globo.com
* extras: mirror.ufscar.br
* updates: mirror.ufscar.br

```

repo id	repo name	status
updates/7/x86_64	CentOS-7 - Updates	enabled: 2,500
updates-source/7	CentOS-7 - Updates Sources	disabled

disabled repositories will be ignored when installing or upgrading software. To enable or disable a repository, use the `yum-config-manager` utility, followed by the repository id.

In the output above, the repository id is shown on the first column (repo id) of each line. Use only the part before the first /, so the id for the CentOS-7 - Updates repo is updates, and not updates/7/x86_64.

```
# yum-config-manager --disable updates
```

The command above will disable the updates repo. To re-enable it use:

```
# yum-config-manager --enable updates
```

NOTE

Yum stores downloaded packages and associated metadata in a cache directory (usually /var/cache/yum). As the system gets upgraded and new packages are installed, this cache can get quite large. To clean the cache and reclaim disk space you can use the `yum clean` command, followed by what to clean. The most useful parameters are `packages` (`yum clean packages`) to delete downloaded packages and `metadata` (`yum clean metadata`) to delete associated metadata. See the manual page for yum (type `man yum`) for more information.

DNF

dnf is the package management tool used on Fedora, and is a fork of yum. As such, many of the commands and parameters are similar. This section will give you just a quick overview of dnf.

Searching for packages

`dnf search PATTERN`, where PATTERN is what you are searching for. For example, `dnf search unzip` will show all packages that contain the word unzip in the name or description.

Getting information about a package

```
dnf info PACKAGENAME
```

Installing packages

`dnf install PACKAGENAME`, where PACKAGENAME is the name of the package you wish to install. You can find the name by performing a search.

Removing packages

```
dnf remove PACKAGENAME
```

Upgrading packages

`dnf upgrade PACKAGENAME` to update only one package. Omit the package name to upgrade

all the packages in the system.

Finding out which package provides a specific file

```
dnf provides FILENAME
```

Getting a list of all the packages installed in the system

```
dnf list --installed
```

Listing the contents of a package

```
dnf repoquery -l PACKAGENAME
```

NOTE

dnf has a built-in help system, which shows more information (such as extra parameters) for each command. To use it, type `dnf help` followed by the command, like `dnf help install`.

Managing Software Repositories

Just as with `yum` and `zypper`, `dnf` works with software repositories (repos). Each distribution has a list of default repositories, and administrators can add or remove repos as needed.

To get a list of all available repositories, use `dnf repolist`. To list only enabled repositories, add the `--enabled` option, and to list only disabled repositories, add the `--disabled` option.

dnf repolist

Last metadata expiration check: 0:20:09 ago on Sat 21 Sep 2019 02:39:43 PM -03.

repo id	repo name	status
*fedora	Fedora 30 - x86_64	56,582
*fedora-modular	Fedora Modular 30 - x86_64	135
*updates	Fedora 30 - x86_64 - Updates	12,774
*updates-modular	Fedora Modular 30 - x86_64 - Updates	145

To add a repository, use `dnf config-manager --add-repo URL`, where `URL` is the full URL to the repository. To enable a repository, use `dnf config-manager --set-enabled REPO_ID`.

Likewise, to disable a repository use `dnf config-manager --set-disabled REPO_ID`. In both cases `REPO_ID` is the unique ID for the repository, which you can get using `dnf repolist`. Added repositories are enabled by default.

Repositories are stored in `.repo` files in the directory `/etc/yum.repos.d/`, with exactly the same syntax used for `yum`.

Zypper

zypper is the package management tool used on SUSE Linux and OpenSUSE. Feature-wise it is similar to apt and yum, being able to install, update and remove packages from a system, with automated dependency resolution.

Updating the Package Index

Like other package management tools, zypper works with repositories containing packages and metadata. This metadata needs to be refreshed from time to time, so that the utility will know about the latest packages available. To do a refresh, simply type:

```
# zypper refresh
Repository 'Non-OSS Repository' is up to date.
Repository 'Main Repository' is up to date.
Repository 'Main Update Repository' is up to date.
Repository 'Update Repository (Non-Oss)' is up to date.
All repositories have been refreshed.
```

zypper has an auto-refresh feature that can be enabled on a per-repository basis, meaning that some repos may be refreshed automatically before a query or package installation, and others may need to be refreshed manually. You will learn how to control this feature shortly.

Searching for Packages

To search for a package, use the search (or se) operator, followed by the package name:

```
# zypper se gnumeric
Loading repository data...
Reading installed packages...

S | Name                | Summary                               | Type
--+-----+-----+-----+-----+
  | gnumeric            | Spreadsheet Application                | package
  | gnumeric-devel      | Spreadsheet Application                | package
  | gnumeric-doc         | Documentation files for Gnumeric      | package
  | gnumeric-lang        | Translations for package gnumeric     | package
```

The search operator can also be used to obtain a list of all the installed packages in the system. To do so, use the `-i` parameter without a package name, as in `zypper se -i`.

To see if a specific package is installed, add the package name to the command above. For example, the following command will search among the installed packages for any containing “firefox” in the name:

```
# zypper se -i firefox
Loading repository data...
Reading installed packages...

S | Name | Summary | Type
--+-----+-----+-----
i | MozillaFirefox | Mozilla Firefox Web B-> | package
i | MozillaFirefox-branding-openSUSE | openSUSE branding of -> | package
i | MozillaFirefox-translations-common | Common translations f-> | package
```

To search only among *non-installed* packages, add the `-u` parameter to the `se` operator.

Installing, Upgrading and Removing Packages

To install a software package, use the `install` (or `in`) operator, followed by the package name. Like so:

```
# zypper in unrar
zypper in unrar
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following NEW package is going to be installed:
  unrar

1 new package to install.
Overall download size: 141.2 KiB. Already cached: 0 B. After the operation, additional 301.6 KiB will be used.
Continue? [y/n/v/...? shows all options] (y): y
Retrieving package unrar-5.7.5-lp151.1.1.x86_64
(1/1), 141.2 KiB (301.6 KiB unpacked)
Retrieving: unrar-5.7.5-lp151.1.1.x86_64.rpm .....[done]
Checking for file conflicts: .....[done]
(1/1) Installing: unrar-5.7.5-lp151.1.1.x86_64 .....[done]
```

`zypper` can also be used to install an RPM package on disk, while trying to satisfy its dependencies

using packages from the repositories. To do so, just provide the full path to the package instead of a package name, like `zypper in /home/john/newpackage.rpm`.

To update packages installed on the system, use `zypper update`. As in the installation process, this will show a list of packages to be installed/upgraded before asking if you want to proceed.

If you wish to only list the available updates, without installing anything, you can use `zypper list-updates`.

To remove a package, use the `remove` (or `rm`) operator, followed by the package name:

```
# zypper rm unrar
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following package is going to be REMOVED:
  unrar

1 package to remove.
After the operation, 301.6 KiB will be freed.
Continue? [y/n/v/...? shows all options] (y): y
(1/1) Removing unrar-5.7.5-lp151.1.1.x86_64 .....[done]
```

Keep in mind that removing a package also removes any other packages that depend on it. For example:

```
# zypper rm libgimp-2_0-0
Loading repository data...
Warning: No repositories defined. Operating only with the installed resolvables. Nothing can
be installed.
Reading installed packages...
Resolving package dependencies...

The following 6 packages are going to be REMOVED:
  gimp gimp-help gimp-lang gimp-plugins-python libgimp-2_0-0
  libgimpui-2_0-0

6 packages to remove.
After the operation, 98.0 MiB will be freed.
Continue? [y/n/v/...? shows all options] (y):
```

Finding Which Packages Contain a Specific File

To see which packages contain a specific file, use the search operator followed by the `--provides` parameter and the name of the file (or full path to it). For example, if you want to know which packages contain the file `libgimpmodule-2.0.so.0` in `/usr/lib64/` you would use:

```
# zypper se --provides /usr/lib64/libgimpmodule-2.0.so.0
Loading repository data...
Reading installed packages...

S | Name | Summary | Type
--+-+-----+-----+-----
i | libgimp-2_0-0 | The GNU Image Manipulation Program - Libra-> | package
```

Getting Package Information

To see the metadata associated with a package, use the `info` operator followed by the package name. This will provide you with the origin repository, package name, version, architecture, vendor, installed size, if it is installed or not, the status (if it is up-to-date), the source package and a description.

```
# zypper info gimp
Loading repository data...
Reading installed packages...

Information for package gimp:
-----
Repository      : Main Repository
Name            : gimp
Version        : 2.8.22-lp151.4.6
Arch           : x86_64
Vendor         : openSUSE
Installed Size  : 29.1 MiB
Installed      : Yes (automatically)
Status         : up-to-date
Source package : gimp-2.8.22-lp151.4.6.src
Summary        : The GNU Image Manipulation Program
Description    :
    The GIMP is an image composition and editing program, which can be
    used for creating logos and other graphics for Web pages. The GIMP
    offers many tools and filters, and provides a large image
    manipulation toolbox, including channel operations and layers,
```

effects, subpixel imaging and antialiasing, and conversions, together with multilevel undo. The GIMP offers a scripting facility, but many of the included scripts rely on fonts that we cannot distribute.

Managing Software Repositories

zypper can also be used to manage software repositories. To see a list of all the repositories currently registered in your system, use `zypper repos`:

zypper repos

Repository priorities are without effect. All enabled repositories share the same priority.

#	Alias	Name	Enabled	GPG Check
Refresh				
-----+-----+-----+-----+-----				
1	openSUSE-Leap-15.1-1	openSUSE-Leap-15.1-1	No	----
2	repo-debug	Debug Repository	No	----
3	repo-debug-non-oss	Debug Repository (Non-OSS)	No	----
4	repo-debug-update	Update Repository (Debug)	No	----
5	repo-debug-update-non-oss	Update Repository (Debug, Non-OSS)	No	----
6	repo-non-oss	Non-OSS Repository	Yes	(r) Yes
7	repo-oss	Main Repository	Yes	(r) Yes
8	repo-source	Source Repository	No	----
9	repo-source-non-oss	Source Repository (Non-OSS)	No	----
10	repo-update	Main Update Repository	Yes	(r) Yes
11	repo-update-non-oss	Update Repository (Non-Oss)	Yes	(r) Yes

See in the `Enabled` column that some repositories are enabled, while others are not. You can change this with the `modifyrepo` operator, followed by the `-e` (enable) or `-d` (disable) parameter and the repository alias (the second column in the output above).

```
# zypper modifyrepo -d repo-non-oss
Repository 'repo-non-oss' has been successfully disabled.

# zypper modifyrepo -e repo-non-oss
Repository 'repo-non-oss' has been successfully enabled.
```

Previously we mentioned that `zypper` has an *auto refresh* capability that can be enabled on a per-repository basis. When enabled, this flag will make `zypper` run a refresh operation (the same as running `zypper refresh`) before working with the specified repo. This can be controlled with the `-f` and `-F` parameters of the `modifyrepo` operator:

```
# zypper modifyrepo -F repo-non-oss
Autorefresh has been disabled for repository 'repo-non-oss'.

# zypper modifyrepo -f repo-non-oss
Autorefresh has been enabled for repository 'repo-non-oss'.
```

Adding and Removing Repositories

To add a new software repository for `zypper`, use the `addrepo` operator followed by the repository URL and repository name, like below:

```
# zypper addrepo http://packman.inode.at/suse/openSUSE_Leap_15.1/ packman
Adding repository 'packman' .....[done]
Repository 'packman' successfully added

URI          : http://packman.inode.at/suse/openSUSE_Leap_15.1/
Enabled      : Yes
GPG Check    : Yes
Autorefresh  : No
Priority     : 99 (default priority)

Repository priorities are without effect. All enabled repositories share the same priority.
```

While adding a repository, you can enable auto-updates with the `-f` parameter. Added repositories are enabled by default, but you can add and disable a repository at the same time by using the `-d` parameter.

To remove a repository, use the `removerepo` operator, followed by the repository name (Alias). To remove the repository added in the example above, the command would be:

```
# zypper removerepo packman
Removing repository 'packman' .....[done]
Repository 'packman' has been removed.
```


Guided Exercises

1. Using `rpm` on a Red Hat Enterprise Linux system, how would you install the package `file-roller-3.28.1-2.el7.x86_64.rpm` showing a progress bar during the installation?
2. Using `rpm`, find out which package contains the file `/etc/redhat-release`.
3. How would you use `yum` to check for updates for all packages in the system?
4. Using `zypper`, how would you disable a repository called `repo-extras`?
5. If you have a `.repo` file describing a new repository, where this file should be put so that it is recognized by DNF?

Explorational Exercises

1. How would you use `zypper` to find out which package owns the file `/usr/sbin/swapon`?

2. How can you get a list of all installed packages in the system using `dnf`?

3. Using `dnf`, what is the command to add a repository located at `https://www.example.url/home:reponame.repo` to the system?

4. How can you use `zypper` to check if the package `unzip` is installed?

5. Using `yum`, find out which package provides the file `/bin/wget`.

Summary

In this lesson, you learned:

- How to use `rpm` to install, upgrade and remove packages.
- How to use `yum`, `zypper` and `dnf`.
- How to get information about a package.
- How to get a list of the package contents.
- How to find out which package a file came from.
- How to list, add, remove, enable or disable software repositories.

The following commands were discussed:

- `rpm`
- `yum`
- `dnf`
- `zypper`

Answers to Guided Exercises

1. Using `rpm` on a Red Hat Enterprise Linux system, how would you install the package `file-roller-3.28.1-2.el7.x86_64.rpm` showing a progress bar during the installation?

Use the `-i` parameter to install a package, and the `-h` option to enable “hash marks” showing installation progress. So, the answer is: `rpm -ih file-roller-3.28.1-2.el7.x86_64.rpm`.

2. Using `rpm`, find out which package contains the file `/etc/redhat-release`.

You are querying information about a file, so use the `-qf` parameter: `rpm -qf /etc/redhat-release`.

3. How would you use `yum` to check for updates for all packages in the system?

Use the `check-update` operation *without* a package name: `yum check-update`.

4. Using `zypper`, how would you disable a repository called `repo-extras`?

Use the `modifyrepo` operation to change the parameters of a repo, and the `-d` parameter to disable it: `zypper modifyrepo -d repo-extras`.

5. If you have a `.repo` file describing a new repository, where this file should be put so that it is recognized by DNF?

`.repo` files for DNF should be put on the same place used by YUM, inside `/etc/yum.repos.d/`.

Answers to Explorational Exercises

1. How would you use `zypper` to find out which package owns the file `/usr/sbin/swapon`?

Use the `se` (search) operator and the `--provides` parameter: `zypper se --provides /usr/sbin/swapon`.

2. How can you get a list of all installed packages in the system using `dnf`?

Use the `list` operator, followed by the `--installed` parameter: `dnf list --installed`.

3. Using `dnf`, what is the command to add a repository located at `https://www.example.url/home:reponame.repo` to the system?

Working with repositories is a “configuration change”, so use the `config-manager` and the `--add_repo` parameter: `dnf config-manager --add_repo https://www.example.url/home:reponame.repo`.

4. How can you use `zypper` to check if the package `unzip` is installed?

You need to do a search (`se`) on the installed (`-i`) packages: `zypper se -i unzip`.

5. Using `yum`, find out which package provides the file `/bin/wget`.

To find out what provides a file, use `whatprovides` and the filename: `yum whatprovides /bin/wget`.



102.6 Linux as a virtualization guest

Reference to LPI objectives

LPIC-1, Exam 101, Objective 102.6

Weight

1

Key knowledge areas

- Understand the general concept of virtual machines and containers
- Understand common elements virtual machines in an IaaS cloud, such as computing instances, block storage and networking
- Understand unique properties of a Linux system which have to be changed when a system is cloned or used as a template
- Understand how system images are used to deploy virtual machines, cloud instances and containers
- Understand Linux extensions which integrate Linux with a virtualization product
- Awareness of cloud-init

Partial list of the used files, terms and utilities

- Virtual machine
- Linux container
- Application container
- Guest drivers
- SSH host keys
- D-Bus machine id



102.6 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	102 Linux Installation and Package Management
Objective:	102.6 Linux as a virtualization guest
Lesson:	1 of 1

Introduction

One of the great strengths of Linux is its versatility. One aspect of this versatility is the ability to use Linux as a means of hosting other operating systems, or individual applications, in a completely isolated and secure environment. This lesson will focus on the concepts of virtualization and container technologies, along with some technical details that should be taken into consideration when deploying a virtual machine on a cloud platform.

Virtualization Overview

Virtualization is a technology that allows for a software platform, called a *hypervisor*, to run processes that contain a fully emulated computer system. The hypervisor is responsible for managing the physical hardware's resources that can be used by individual virtual machines. These virtual machines are called *guests* of the hypervisor. A virtual machine has many aspects of a physical computer emulated in software, such as a system's BIOS and hard drive disk controllers. A virtual machine will often use hard disk images that are stored as individual files, and will have access to the host machine's RAM and CPU through the hypervisor software. The hypervisor separates the access to the host system's hardware resources among the guests, thus

allowing for multiple operating systems to run on a single host system.

Commonly used hypervisors for Linux include:

Xen

Xen is an open source Type-1 hypervisor, meaning that it does not rely on an underlying operating system to function. A hypervisor of this sort is known as a *bare-metal hypervisor* since the computer can boot directly into the hypervisor.

KVM

The Kernel Virtual Machine is a Linux kernel module for virtualization. KVM is a hypervisor of both Type-1 and Type-2 hypervisor because, although it needs a generic Linux operating system to work, it is able to perform as a hypervisor perfectly well by integrating with a running Linux installation. Virtual machines deployed with KVM use the `libvirt` daemon and associated software utilities to be created and managed.

VirtualBox

A popular desktop application that makes it easy to create and manage virtual machines. Oracle VM VirtualBox is cross-platform, and will work on Linux, macOS, and Microsoft Windows. Since VirtualBox requires an underlying operating system to run, it is a Type-2 hypervisor.

Some hypervisors allow for the dynamic relocation of a virtual machine. The process of moving a virtual machine from one hypervisor installation to another is called a *migration*, and the techniques involved differ between hypervisor implementations. Some migrations can only be performed when the guest system has been completely shut down, and other can be performed while the guest is running (called a *live migration*). Such techniques can be of aid during maintenance windows for hypervisors, or for system resiliency when a hypervisor becomes non-functional and the guest can be moved to a hypervisor that is working.

Types of Virtual Machines

There are three main types of virtual machines, the *fully virtualized* guest, the *paravirtualized* guest and the *hybrid* guest.

Fully Virtualized

All instructions that a guest operating system is expected to execute must be able to run within a fully virtualized operating system installation. The reason for this is that no additional software drivers are installed within the guest to translate the instructions to either simulated or real hardware. A fully virtualized guest is one where the guest (or HardwareVM) is unaware that it is a running virtual machine instance. In order for this type of virtualization to take

place on x86 based hardware the Intel VT-x or AMD-V CPU extensions must be enabled on the system that has the hypervisor installed. This can be done from a BIOS or UEFI firmware configuration menu.

Paravirtualized

A paravirtualized guest (or PVM) is one where the guest operating system is aware that it is a running virtual machine instance. These types of guests will make use of a modified kernel and special drivers (known as *guest drivers*) that will help the guest operating system utilize software and hardware resources of the hypervisor. The performance of a paravirtualized guest is often better than that of the fully virtualized guest due to the advantage that these software drivers provide.

Hybrid

Paravirtualization and full virtualization can be combined to allow unmodified operating systems to receive near native I/O performance by using paravirtualized drivers on fully virtualized operating systems. The paravirtualized drivers contain storage and network device drivers with enhanced disk and network I/O performance.

Virtualization platforms often provide packaged guest drivers for virtualized operating systems. The KVM utilizes drivers from the *Virtio* project, whereas Oracle VM VirtualBox uses *Guest Extensions* available from a downloadable ISO CD-ROM image file.

Example libvirt Virtual Machine

We will look at an example virtual machine that is managed by `libvirt` and uses the KVM hypervisor. A virtual machine often consists of a group of files, primarily an XML file that *defines* the virtual machine (such as its hardware configuration, network connectivity, display capabilities, and more) and an associated hard disk image file that contains the installation of the operating system and its software.

First, let us start to examine an example XML configuration file for a virtual machine and its network environment:

```
$ ls /etc/libvirt/qemu
total 24
drwxr-xr-x 3 root root 4096 Oct 29 17:48 networks
-rw----- 1 root root 5667 Jun 29 17:17 rhel8.0.xml
```

NOTE

The `qemu` portion of the directory path refers to the underlying software that KVM-based virtual machines rely on. The QEMU project provides software for the

hypervisor to emulate hardware devices that the virtual machine will use, such as disk controllers, access to the host CPU, network card emulation, and more.

Note that there is a directory named `networks`. This directory contains definition files (also using XML) that create network configurations that the virtual machines can use. This hypervisor is only using one network, and so there is only one definition file that contains a configuration for a virtual network segment that these systems will utilize.

```
$ ls -l /etc/libvirt/qemu/networks/
total 8
drwxr-xr-x 2 root root 4096 Jun 29 17:15 autostart
-rw----- 1 root root 576 Jun 28 16:39 default.xml
$ sudo cat /etc/libvirt/qemu/networks/default.xml
<!--
WARNING: THIS IS AN AUTO-GENERATED FILE. CHANGES TO IT ARE LIKELY TO BE
OVERWRITTEN AND LOST. Changes to this xml configuration should be made using:
    virsh net-edit default
or other application using the libvirt API.
-->

<network>
  <name>default</name>
  <uuid>55ab064f-62f8-49d3-8d25-8ef36a524344</uuid>
  <forward mode='nat' />
  <bridge name='virbr0' stp='on' delay='0' />
  <mac address='52:54:00:b8:e0:15' />
  <ip address='192.168.122.1' netmask='255.255.255.0'>
    <dhcp>
      <range start='192.168.122.2' end='192.168.122.254' />
    </dhcp>
  </ip>
</network>
```

This definition includes a Class C private network and an emulated hardware device to act as a router for this network. There is also a range of IP addresses for the hypervisor to use with a DHCP server implementation that can be assigned to the virtual machines that use this network. This network configuration also utilizes *network address translation* (NAT) to forward packets to other networks, such as the hypervisor's LAN.

Now we turn our attention to a Red Hat Enterprise Linux 8 virtual machine definition file. (sections of special note are in bold text):

```
$ sudo cat /etc/libvirt/qemu/rhel8.0.xml
<!--
WARNING: THIS IS AN AUTO-GENERATED FILE. CHANGES TO IT ARE LIKELY TO BE
OVERWRITTEN AND LOST. Changes to this xml configuration should be made using:
    virsh edit rhel8.0
or other application using the libvirt API.
-->

<domain type='kvm'>
  <name>rhel8.0</name>
  <uuid>fadd8c5d-c5e1-410e-b425-30da7598d0f6</uuid>
  <metadata>
    <libosinfo:libosinfo xmlns:libosinfo="http://libosinfo.org/xmlns/libvirt/domain/1.0">
      <libosinfo:os id="http://redhat.com/rhel/8.0"/>
    </libosinfo:libosinfo>
  </metadata>
  <memory unit='KiB'>4194304</memory>
  <currentMemory unit='KiB'>4194304</currentMemory>
  <vcpu placement='static'>2</vcpu>
  <os>
    <type arch='x86_64' machine='pc-q35-3.1'>hvm</type>
    <boot dev='hd' />
  </os>
  <features>
    <acpi/>
    <apic/>
    <vmport state='off' />
  </features>
  <cpu mode='host-model' check='partial'>
    <model fallback='allow' />
  </cpu>
  <clock offset='utc'>
    <timer name='rtc' tickpolicy='catchup' />
    <timer name='pit' tickpolicy='delay' />
    <timer name='hpet' present='no' />
  </clock>
  <on_poweroff>destroy</on_poweroff>
  <on_reboot>restart</on_reboot>
  <on_crash>destroy</on_crash>
  <pm>
    <suspend-to-mem enabled='no' />
    <suspend-to-disk enabled='no' />
  </pm>
  <devices>
```

```

<emulator>/usr/bin/qemu-system-x86_64</emulator>
<disk type='file' device='disk'>
  <driver name='qemu' type='qcow2'/>
  <source file='/var/lib/libvirt/images/rhel8'/>
  <target dev='vda' bus='virtio'/>
  <address type='pci' domain='0x0000' bus='0x04' slot='0x00' function='0x0'/>
</disk>
<controller type='usb' index='0' model='qemu-xhci' ports='15'>
  <address type='pci' domain='0x0000' bus='0x02' slot='0x00' function='0x0'/>
</controller>
<controller type='sata' index='0'>
  <address type='pci' domain='0x0000' bus='0x00' slot='0x1f' function='0x2'/>
</controller>
<controller type='pci' index='0' model='pcie-root'/>
<controller type='pci' index='1' model='pcie-root-port'>
  <model name='pcie-root-port'/>
  <target chassis='1' port='0x10'/>
  <address type='pci' domain='0x0000' bus='0x00' slot='0x02' function='0x0'
multifunction='on'/>
</controller>
<controller type='pci' index='2' model='pcie-root-port'>
  <model name='pcie-root-port'/>
  <target chassis='2' port='0x11'/>
  <address type='pci' domain='0x0000' bus='0x00' slot='0x02' function='0x1'/>
</controller>
<controller type='pci' index='3' model='pcie-root-port'>
  <model name='pcie-root-port'/>
  <target chassis='3' port='0x12'/>
  <address type='pci' domain='0x0000' bus='0x00' slot='0x02' function='0x2'/>
</controller>
<controller type='pci' index='4' model='pcie-root-port'>
  <model name='pcie-root-port'/>
  <target chassis='4' port='0x13'/>
  <address type='pci' domain='0x0000' bus='0x00' slot='0x02' function='0x3'/>
</controller>
<controller type='pci' index='5' model='pcie-root-port'>
  <model name='pcie-root-port'/>
  <target chassis='5' port='0x14'/>
  <address type='pci' domain='0x0000' bus='0x00' slot='0x02' function='0x4'/>
</controller>
<controller type='pci' index='6' model='pcie-root-port'>
  <model name='pcie-root-port'/>
  <target chassis='6' port='0x15'/>
  <address type='pci' domain='0x0000' bus='0x00' slot='0x02' function='0x5'/>

```

```

</controller>
<controller type='pci' index='7' model='pcie-root-port'>
  <model name='pcie-root-port' />
  <target chassis='7' port='0x16' />
  <address type='pci' domain='0x0000' bus='0x00' slot='0x02' function='0x6' />
</controller>
<controller type='virtio-serial' index='0'>
  <address type='pci' domain='0x0000' bus='0x03' slot='0x00' function='0x0' />
</controller>
<interface type='network'>
  <mac address='52:54:00:50:a7:18' />
  <source network='default' />
  <model type='virtio' />
  <address type='pci' domain='0x0000' bus='0x01' slot='0x00' function='0x0' />
</interface>
<serial type='pty'>
  <target type='isa-serial' port='0'>
    <model name='isa-serial' />
  </target>
</serial>
<console type='pty'>
  <target type='serial' port='0' />
</console>
<channel type='unix'>
  <target type='virtio' name='org.qemu.guest_agent.0' />
  <address type='virtio-serial' controller='0' bus='0' port='1' />
</channel>
<channel type='spicevmc'>
  <target type='virtio' name='com.redhat.spice.0' />
  <address type='virtio-serial' controller='0' bus='0' port='2' />
</channel>
<input type='tablet' bus='usb'>
  <address type='usb' bus='0' port='1' />
</input>
<input type='mouse' bus='ps2' />
<input type='keyboard' bus='ps2' />
<graphics type='spice' autoport='yes'>
  <listen type='address' />
  <image compression='off' />
</graphics>
<sound model='ich9'>
  <address type='pci' domain='0x0000' bus='0x00' slot='0x1b' function='0x0' />
</sound>
<video>

```

```

    <model type='virtio' heads='1' primary='yes' />
    <address type='pci' domain='0x0000' bus='0x00' slot='0x01' function='0x0' />
</video>
<redirdev bus='usb' type='spicevmc'>
    <address type='usb' bus='0' port='2' />
</redirdev>
<redirdev bus='usb' type='spicevmc'>
    <address type='usb' bus='0' port='3' />
</redirdev>
<memballoon model='virtio'>
    <address type='pci' domain='0x0000' bus='0x05' slot='0x00' function='0x0' />
</memballoon>
<rng model='virtio'>
    <backend model='random'>/dev/urandom</backend>
    <address type='pci' domain='0x0000' bus='0x06' slot='0x00' function='0x0' />
</rng>
</devices>
</domain>

```

This file defines a number of hardware settings that are used by this guest, such as the amount of RAM that it will have assigned to it, the number of CPU cores from the hypervisor that the guest will have access to, the hard disk image file that is associated with this guest (under the disk stanza), its display capabilities (via the SPICE protocol), and the guest's access to USB devices as well as emulated keyboard and mice input.

Example Virtual Machine Disk Storage

This virtual machine's hard disk image resides at `/var/lib/libvirt/images/rhel8`. Here is the disk image itself on this hypervisor:

```

$ sudo ls -lh /var/lib/libvirt/images/rhel8
-rw----- 1 root root 5.5G Oct 25 15:57 /var/lib/libvirt/images/rhel8

```

The current size of this disk image consumes only 5.5 GB of space on the hypervisor. However, the operating system within this guest sees a 23.3 GB sized disk, as evidenced by the output of the following command from within the running virtual machine:

```

$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINT
vda         252:0    0 23.3G  0 disk
└─vda1      252:1    0   1G   0 part /boot

```

```

└─vda2      252:2    0 22.3G  0 part
  └─rhel-root 253:0    0  20G   0 lvm  /
    └─rhel-swap 253:1    0  2.3G   0 lvm  [SWAP]

```

This is due to the type of disk provisioning used for this guest. There are multiple types of disk images that a virtual machine can use but the two primary types are:

COW

Copy-on-write (also referred to as *thin-provisioning* or *sparse images*) is a method where a disk file is created with a pre-defined upper size limit. The disk image size only increases as new data is written to the disk. Just like the previous example, the guest operating system sees the predefined disk limit of 23.3 GB, but has only written 5.5 GB of data to the disk file. The disk image format used for the example virtual machine is `qcow2` which is a QEMU COW file format.

RAW

A *raw* or *full* disk type is a file that has all of its space pre-allocated. For example, a 10 GB raw disk image file consumes 10 GB of actual disk space on the hypervisor. There is a performance benefit to this style of disk as all of the needed disk space already exists, so the underlying hypervisor can just write data to the disk without the performance hit of monitoring the disk image to ensure that it has not yet reached its limit and extending the size of the file as new data is written to it.

There are other virtualization management platforms such as *Red Hat Enterprise Virtualization* and *oVirt* that can make use of physical disks to act as backing storage locations for a virtual machine's operating system. These systems can utilize storage area network (SAN) or network attached storage (NAS) devices to write their data to, and the hypervisor keeps track of what storage locations belong to which virtual machines. These storage systems can use technologies such as logical volume management (LVM) to grow or shrink the size of a virtual machine's disk storage as needed, and to aid in the creation and management of storage snapshots.

Working with Virtual Machine Templates

Since virtual machines are typically just files running on a hypervisor, it is easy to create *templates* that can be customized for particular deployment scenarios. Often a virtual machine will have a basic operating system installation and some pre-configured authentication configuration settings set up to ease future system launches. This cuts down on the amount of time it takes to build a new system by reducing the amount of work that is often repeated, such as base package installation and locale settings.

This virtual machine template could then later get copied to a new guest system. In this case, the

new guest would get renamed, a new MAC address generated for its network interface, and other modifications can be made depending on its intended use.

The D-Bus Machine ID

Many Linux installations will utilize a machine identification number generated at install time, called the *D-Bus machine ID*. However, if a virtual machine is *cloned* to be used as a template for other virtual machine installations, a new D-Bus machine ID would need to be created to ensure that system resources from the hypervisor get directed to the appropriate guest system.

The following command can be used to validate that a D-Bus machine ID exists for the running system:

```
$ dbus-uuidgen --ensure
```

If no error messages are displayed, then an ID exists for the system. To view the current D-Bus machine ID, run the following:

```
$ dbus-uuidgen --get  
17f2e0698e844e31b12ccd3f9aa4d94a
```

The string of text that is displayed is the current ID number. No two Linux systems running on a hypervisor should have the same D-Bus machine ID.

The D-Bus machine ID is located at `/var/lib/dbus/machine-id` and is symbolically linked to `/etc/machine-id`. Changing this ID number on a running system is discouraged as system instability and crashes are likely to occur. If two virtual machines do have the same D-Bus machine ID, follow the procedure below to generate a new one:

```
$ sudo rm -f /etc/machine-id  
$ sudo dbus-uuidgen --ensure=/etc/machine-id
```

In the event that `/var/lib/dbus/machine-id` is not a symbolic link back to `/etc/machine-id`, then `/var/lib/dbus/machine-id` will need to be removed.

Deploying Virtual Machines to the Cloud

There are a multitude of IaaS (*infrastructure as a service*) providers available that run hypervisor systems and that can deploy virtual guest images for an organization. Practically all of these

providers have tools in place that allows an administrator to build, deploy and configure custom virtual machines based on a variety of Linux distributions. Many of these companies also have systems in place that allow for the deployment and migrations of virtual machines built from within a customer's organization.

When assessing a deployment of a Linux system in an IaaS environment, there are some key elements that an administrator should be cognizant of:

Computing Instances

Many cloud providers will charge usage rates based on “computing instances”, or how much CPU time your cloud-based infrastructure will use. Careful planning of how much processing time applications will actually require will aid in keeping the costs of a cloud solution manageable.

Computing instances will also often refer to the number of virtual machines that are provisioned in a cloud environment. Again, the more instances of systems that are running at one time will also factor into how much overall CPU time an organization will be charged for.

Block Storage

Cloud providers also have various levels of block storage available for an organization to use. Some offerings are simply meant to be web-based network storage for files, and other offerings relate to external storage for a cloud provisioned virtual machine to use for hosting files.

The cost for such offerings will vary based on the amount of storage used, and the speed of the storage within the provider's data centers. Faster storage access typically will cost more, and conversely data “at rest” (as in archival storage) is often very inexpensive.

Networking

One of the main components of working with a cloud solutions provider is how the virtual network will be configured. Many IaaS providers will have some form of web-based utilities that can be utilized for the design and implementation of different network routes, subnetting, and firewall configurations. Some will even provide DNS solutions so that publicly accessible FQDN (*fully qualified domain names*) can be assigned to your internet facing systems. There are even “hybrid” solutions available that can connect an existing, on-premise network infrastructure to a cloud-based infrastructure through the means of a VPN (*virtual private network*), thus tying the two infrastructures together.

Securely Accessing Guests in the Cloud

The most prevalent method in use for accessing a remote virtual guest on a cloud platform is through the use of OpenSSH software. A Linux system that resides in the cloud would have the

OpenSSH server running, while an administrator would use an OpenSSH client with pre-shared keys for remote access.

An administrator would run the following command

```
$ ssh-keygen
```

and follow the prompts to create a public and private SSH key pair. The private key remains on the administrator's local system (stored in `~/.ssh/`) and the public key gets copied to the remote cloud system, the exact same method one would use when working with networked machines on a corporate LAN.

The administrator would then run the following command:

```
$ ssh-copy-id -i <public_key> user@cloud_server
```

This will copy the public SSH key from the key pair just generated to the remote cloud server. The public key will be recorded in the `~/.ssh/authorized_keys` file of the cloud server, and set the appropriate permissions on the file.

NOTE

If there is only one public key file in the `~/.ssh/` directory, then the `-i` switch can be omitted, as the `ssh-copy-id` command will default to the public key file in the directory (typically the file ending with the `.pub` extension).

Some cloud providers will automatically generate a key pair when a new Linux system is provisioned. The administrator will then need to download the private key for the new system from the cloud provider and store it on their local system. Note that the permissions for SSH keys must be `0600` for a private key, and `0644` for a public key.

Preconfiguring Cloud Systems

A useful tool that simplifies the deployments of cloud-based virtual machine is the `cloud-init` utility. This command, along with the associated configuration files and pre-defined virtual machine image, is a vendor-neutral method for deploying a Linux guest across a plethora of IaaS providers. Utilizing YAML (*YAML Ain't Markup Language*) plain-text files an administrator can pre-configure network settings, software package selections, SSH key configuration, user account creations, locale settings, along with a myriad of other options to quickly build out new systems.

During a new system's initial boot up, `cloud-init` will read in the settings from YAML configurations files and apply them. This process only needs to apply to a system's initial setup,

and makes deploying a fleet of new systems on a cloud provider's platform easy.

The YAML file syntax used with `cloud-init` is called *cloud-config*. Here is a sample `cloud-config` file:

```
#cloud-config
timezone: Africa/Dar_es_Salaam
hostname: test-system

# Update the system when it first boots up
apt_update: true
apt_upgrade: true

# Install the Nginx web server
packages:
  - nginx
```

Note that on the top line there is no space between the hash symbol (#) and the term `cloud-config`.

NOTE

`cloud-init` is not just for virtual machines. The `cloud-init` tool suite can also be used to pre-configure containers (such as LXD Linux containers) prior to deployment.

Containers

Container technology is similar in some aspects to a virtual machine, where you get an isolated environment to easily deploy an application. Whereas with a virtual machine an entire computer is emulated, a container uses just enough software to run an application. In this way, there is far less overhead.

Containers allow for greater flexibility over that of a virtual machine. An application container can be migrated from one host to another, just as a virtual machine can be migrated from one hypervisor to another. However, sometimes a virtual machine will need to be powered off before it could be migrated, whereas with a container the application is always running while it is getting migrated. Containers also make it easy to deploy new versions of applications in tandem with an existing version. As users close their sessions with running containers, these containers can get automatically removed from the system by the container orchestration software and replaced with the new version thus reducing downtime.

NOTE

There are numerous container technologies available for Linux, such as *Docker*,

Kubernetes, LXD/LXC, systemd-nspawn, OpenShift and more. The exact implementation of a container software package is beyond the scope of the LPIC-1 exam.

Containers make use of the *control groups* (better known as *cgroups*) mechanism within the Linux kernel. The cgroup is a way to partition system resources such as memory, processor time as well as disk and network bandwidth for an individual application. An administrator can use cgroups directly to set system resource limits on an application, or a group of applications that could exist within a single cgroup. In essence this is what container software does for the administrator, along with providing tools that ease the management and deployment of cgroups.

NOTE

Currently, knowledge of cgroups are not necessary for passing the LPIC-1 exam. The concept of the cgroup is mentioned here so that the candidate would at least have some background knowledge of how an application is segregated for the sake of system resource utilization.

Guided Exercises

1. What CPU extensions are necessary on an x86 based hardware platform that will run fully virtualized guests?

2. A mission-critical server installation that will require the fastest performance will likely use what type of virtualization?

3. Two virtual machines that have been cloned from the same template and that utilize D-Bus are performing erratically. They both have separate hostnames and network configuration settings. What command would be used to determine if each of the virtual machines have different D-Bus Machine IDs?

Explorational Exercises

1. Run the following command to see if your system already has CPU extensions enabled to run a virtual machine (your results may vary depending on your CPU):

```
grep --color -E "vmx|svm" /proc/cpuinfo
```

Depending on the output, you may have `vmx` highlighted (for Intel VT-x enabled CPU's) or `svm` highlighted (for AMD SVM enabled CPU's). Should you get no results, consult your BIOS or UEFI firmware instructions on how to enable virtualization for your processor.

2. If your processor supports virtualizations, seek out your distribution's documentation for running a KVM hypervisor.

- Install the necessary packages to run a KVM hypervisor.
-

- If you are using a graphical desktop environment, it is recommended to also install the `virt-manager` application which is a graphical front-end that can be used on a KVM installation. This will aid in virtual machine installations and management.
-

- Download a Linux distribution ISO image of your choice, and following your distribution's documentation create a new virtual machine using this ISO.
-

Summary

In this lesson we covered the conceptual basics of virtual machines and containers, and how these technologies can be used with Linux.

We briefly described the following commands:

dbus-uuidgen

Used to verify and view a system's DBus ID.

ssh-keygen

Used to generate a public and private SSH key pair for use when accessing remote, cloud-based systems.

ssh-copy-id

Used to copy a system's public SSH key to a remote system to facilitate remote authentication.

cloud-init

Used to aid in the configuration and deployment of virtual machines and containers to a cloud environment.

Answers to Guided Exercises

1. What CPU extensions are necessary on an x86 based hardware platform that will run fully virtualized guests?

VT-x for Intel CPUs or AMD-V for AMD CPUs

2. A mission-critical server installation that will require the fastest performance will likely use what type of virtualization?

An operating system that makes use of paravirtualization, such as Xen, as the guest operating system can make better use of hardware resources available to it through the use of software drivers designed to work with the hypervisor.

3. Two virtual machines that have been cloned from the same template and that utilize D-Bus are performing erratically. They both have separate hostnames and network configuration settings. What command would be used to determine if each of the virtual machines have different D-Bus Machine IDs?

```
dbus-uuidgen --get
```


Answers to Explorational Exercises

1. Run the following command to see if your system already has CPU extensions enabled to run a virtual machine (your results may vary depending on your CPU): `grep --color -E "vmx|svm" /proc/cpuinfo`. Depending on the output, you may have "vmx" highlighted (for Intel VT-x enabled CPU's) or "svm" highlighted (for AMD SVM enabled CPU's). Should you get no results, consult your BIOS or UEFI firmware instructions on how to enable virtualization for your processor.

The results will vary depending on your CPU that you have. Here is example output from a computer with an Intel CPU with virtualization extensions enabled in the UEFI firmware:

```
$ grep --color -E "vmx|svm" /proc/cpuinfo
flags          : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36
clflush dts acpi mmx fxsr sse sse2 ss ht tm pbe syscall nx pdpe1gb rdtscp lm constant_tsc
art arch_perfmon pebs bts rep_good nopl xtopology nonstop_tsc cpuid aperfmperf pni
pclmulqdq dtes64 monitor ds_cpl vmx smx est tm2 ssse3 sdbg fma cx16 xtpr pdcm pcid sse4_1
sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand lahf_lm abm
3dnowprefetch cpuid_fault epb invpcid_single pti ssbd ibrs ibpb stibp tpr_shadow vnmi
flexpriority ept vpid ept_ad fsgsbase tsc_adjust bmi1 hle avx2 smep bmi2 erms invpcid rtm
mpx rdseed adx smap clflushopt intel_pt xsaveopt xsavec xgetbv1 xsaves dtherm ida arat
pln pts hwp hwp_notify hwp_act_window hwp_epp md_clear flush_l1d
```

2. If your processor supports virtualizations, seek out your distribution's documentation for running a KVM hypervisor.
 - Install the necessary packages to run a KVM hypervisor.

This will vary depending on your distribution, but here are some starting points:

Ubuntu — <https://help.ubuntu.com/lts/serverguide/libvirt.html>

Fedora — <https://docs.fedoraproject.org/en-US/quick-docs/getting-started-with-virtualization/>

Arch Linux — <https://wiki.archlinux.org/index.php/KVM>

- If you are using a graphical desktop environment, it is recommended to also install the `virt-manager` application which is a graphical front-end that can be used on a KVM installation. This will aid in virtual machine installations and management.

Again, this will vary by distribution. An example using Ubuntu looks like this:

```
$ sudo apt install virt-manager
```

- Download a Linux distribution ISO image of your choice, and following your distribution's documentation create a new virtual machine using this ISO.

This task is easily handled by the `virt-manager` package. However a virtual machine can be created from the command-line using the `virt-install` command. Try both methods to get an understanding of how virtual machines are deployed.



**Linux
Professional
Institute**

Topic 103: GNU and Unix Commands



103.1 Work on the command line

Reference to LPI objectives

LPIC-1 version 5.0, Exam 101, Objective 103.1

Weight

4

Key knowledge areas

- Use single shell commands and one line command sequences to perform basic tasks on the command line.
- Use and modify the shell environment including defining, referencing and exporting environment variables.
- Use and edit command history.
- Invoke commands inside and outside the defined path.

Partial list of the used files, terms and utilities

- `bash`
- `echo`
- `env`
- `export`
- `pwd`
- `set`
- `unset`
- `type`
- `which`

- `man`
- `uname`
- `history`
- `.bash_history`
- Quoting



103.1 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.1 Work on the command line
Lesson:	1 of 2

Introduction

Newcomers to the world of Linux administration and the Bash shell often feel a bit lost without the reassuring comforts of a GUI interface. They are used to having right-click access to the visual cues and contextual information that graphic file manager utilities make available. So it is important to quickly learn and master the relatively small set of command line tools through which you can instantly tap into all the data offered by your old GUI — and more.

Getting System Information

While staring at the flashing rectangle of a command line prompt, your first question will probably be “Where am I?” Or, more precisely, “Where in the Linux filesystem am I right now and if, say, I created a new file, where would it live?” What you are after here is your *present work directory*, and the `pwd` command will tell you what you want to know:

```
$ pwd
/home/frank
```

Assuming that Frank is currently logged in to the system and he is now in his home directory: `/home/frank/`. Should Frank create an empty file using the `touch` command without specifying any other location in the filesystem, the file will be created within `/home/frank/`. Listing the directory contents using `ls` will show us that new file:

```
$ touch newfile
$ ls
newfile
```

Besides your location in the filesystem, you will often want information about the Linux system you are running. This might include the exact release number of your distribution or the Linux kernel version that is currently loaded. The `uname` tool is what you are after here. And, in particular, `uname` using the `-a` (“all”) option.

```
$ uname -a
Linux base 4.18.0-18-generic #19~18.04.1-Ubuntu SMP Fri Apr 5 10:22:13 UTC 2019 x86_64
x86_64 x86_64 GNU/Linux
```

Here, `uname` shows that Frank’s machine has the Linux kernel version 4.18.0 installed and is running Ubuntu 18.04 on a 64-bit (x86_64) CPU.

Getting Command Information

You will often come across documentation talking about Linux commands with which you are not yet familiar. The command line itself offers all kinds of helpful information on what commands do and how to effectively use them. Perhaps the most useful information is found within the many files of the `man` system.

As a rule, Linux developers write `man` files and distribute them along with the utilities they create. `man` files are highly structured documents whose contents are intuitively divided by standard section headings. Typing `man` followed by the name of a command will give you information that includes the command name, a brief usage synopsis, a more detailed description, and some important historical and licensing background. Here is an example:

```
$ man uname
NAME(1)                User Commands                UNAME(1)
NAME
    uname - print system information
SYNOPSIS
    uname [OPTION]...
```

DESCRIPTION

Print certain system information. With no OPTION, same as -s.

-a, --all

print all information, in the following order, except omit -p and -i if unknown:

-s, --kernel-name

print the kernel name

-n, --nodename

print the network node hostname

-r, --kernel-release

print the kernel release

-v, --kernel-version

print the kernel version

-m, --machine

print the machine hardware name

-p, --processor

print the processor type (non-portable)

-i, --hardware-platform

print the hardware platform (non-portable)

-o, --operating-system

print the operating system

--help display this help and exit

--version

output version information and exit

AUTHOR

Written by David MacKenzie.

REPORTING BUGS

GNU coreutils online help: <<http://www.gnu.org/software/coreutils/>>

Report uname translation bugs to

<<http://translationproject.org/team/>>

COPYRIGHT

Copyright©2017 Free Software Foundation, Inc. License GPLv3+: GNU

GPL version 3 or later <<http://gnu.org/licenses/gpl.html>>.

This is free software: you are free to change and redistribute it.

There is NO WARRANTY, to the extent permitted by law.

SEE ALSO

arch(1), uname(2)

Full documentation at: <<http://www.gnu.org/software/coreutils/uname>>

or available locally via: info '(coreutils) uname invocation'

GNU coreutils 8.28

January 2018

UNAME(1)

man only works when you give it an exact command name. If, however, you are not sure about the name of the command you are after, you can use the `apropos` command to search through the

man page names and descriptions. Assuming, for instance, that you cannot remember that it is `uname` that will give you your current Linux kernel version, you can pass the word `kernel` to `apropos`. You will probably get many lines of output, but they should include these:

```
$ apropos kernel
systemd-udev-kernel.socket (8) - Device event managing daemon
uname (2)                - get name and information about current kernel
urandom (4)              - kernel random number source devices
```

If you do not need a command's full documentation, you can quickly get basic data about a command using `type`. This example uses `type` to query four separate commands at once. The results show us that `cp` ("copy") is a program that lives in `/bin/cp` and that `kill` (change the state of a running process) is a shell builtin — meaning that it is actually a part of the Bash shell itself:

```
$ type uname cp kill which
uname is hashed (/bin/uname)
cp is /bin/cp
kill is a shell builtin
which is /usr/bin/which
```

Notice that, besides being a regular binary command like `cp`, `uname` is also "hashed." That is because Frank recently used `uname` and, to increase system efficiency, it was added to a hash table to make it more accessible the next time you run it. If he would run `type uname` after a system boot, Frank would find that `type` once again describes `uname` as a regular binary.

NOTE A quicker way to clean up the hash table is to run the command `hash -d`.

Sometimes — particularly when working with automated scripts — you will need a simpler source of information about a command. The `which` command that our previous `type` command traced for us, will return nothing but the absolute location of a command. This example locates both the `uname` and `which` commands.

```
$ which uname which
/bin/uname
/usr/bin/which
```

NOTE If you want to display information about "builtin" commands, you can use the `help` command.

Using Your Command History

You will often carefully research the proper usage for a command and successfully run it along with a complicated trail of options and arguments. But what happens a few weeks later when you need to run the same command with the same options and arguments but cannot remember the details? Rather than having to start your research over again from scratch, you will often be able to recover the original command using `history`.

Typing `history` will return the most recent commands you have executed with the most recent of those appearing last. You can easily search through those commands by piping a specific string to the `grep` command. This example will search for any command that included the text `bash_history`:

```
$ history | grep bash_history
1605 sudo find /home -name ".bash_history" | xargs grep sudo
```

Here a single command is returned along with its sequence number, 1605.

And speaking of `bash_history`, that is actually the name of a hidden file you should find within your user's home directory. Since it is a hidden file (designated as such by the dot that precedes its filename), it will only be visible by listing the directory contents using `ls` with the `-a` argument:

```
$ ls /home/frank
newfile
$ ls -a /home/frank
.  ..  .bash_history  .bash_logout  .bashrc  .profile  .ssh  newfile
```

What is in the `.bash_history` file? Take a look for yourself: you will see hundreds and hundreds of your most recent commands. You might, however, be surprised to find that some of your *most* recent commands are missing. That is because, while they are instantly added to the dynamic history database, the latest additions to your command history are not written to the `.bash_history` file until you exit your session.

You can leverage the contents of `history` to make your command line experience much faster and more efficient using the up and down arrow keys on your keyboard. Hitting the up key multiple times will populate the command line with recent commands. When you get to the one you would like to execute a second time, you can run it by pressing Enter. This makes it easy to recall and, if desired, modify commands multiple times during a shell session.

Guided Exercises

1. Use the `man` system to determine how to tell `apropos` to output a brief command so that it outputs only a brief usage message and then exits.

2. Use the `man` system to determine which copyright license is assigned to the `grep` command.

Explorational Exercises

1. Identify the hardware architecture and Linux kernel version being used on your computer in an easy-to-read output format.

2. Print out the last twenty lines of the dynamic `history` database and the `.bash_history` file to compare them.

3. Use the `apropos` tool to identify the `man` page where you will find the command you will need to display the size of an attached physical block device in bytes rather than megabytes or gigabytes.

Summary

In this lesson, you learned:

- How to get information about your filesystem location and OS software stack.
- How to find help for command usage.
- How to identify the filesystem location and type of command binaries.
- How to find and reuse previously-executed commands.

The following commands were discussed in this lesson:

pwd

Print the path to the current working directory.

uname

Print your system's hardware architecture, Linux kernel version, distribution, and distribution release.

man

Access help files documenting command usage.

type

Print the filesystem location and type for one or more commands.

which

Print the filesystem location for a command.

history

Print or reuse commands that you have previously executed.

Answers to Guided Exercises

1. Use the `man` system to determine how to tell `apropos` to output a brief command so that it outputs only a brief usage message and then exits.

Run `man apropos` and scroll down through the “Options” section until you get to the `--usage` paragraph.

2. Use the `man` system to determine which copyright license is assigned to the `grep` command.

Run `man grep` and scroll down to the “Copyright” section of the document. Note that the program uses a copyright from the Free Software Foundation.

Answers to Explorational Exercises

1. Identify the hardware architecture and Linux kernel version being used on your computer in an easy-to-read output format.

Run `man uname`, read through the “Description” section, and identify the command arguments that will display only the exact results you want. Note how `-v` will give you the kernel version and `-i` will provide hardware platform.

```
$ man uname
$ uname -v
$ uname -i
```

2. Print out the last twenty lines of the dynamic `history` database and the `.bash_history` file to compare them.

```
$ history 20
$ tail -n 20 .bash_history
```

3. Use the `apropos` tool to identify the `man` page where you will find the command you will need to display the size of an attached physical block device in bytes rather than megabytes or gigabytes.

One way, would be to run `apropos` with the string `block`, read through the results, note that `lsblk` lists block devices (and would, therefore, be the most likely tool for our needs), run `man lsblk`, scroll through the “Description” section and note that `-b` will display a device size in bytes. Finally, run `lsblk -b` to see what comes out.

```
$ apropos block
$ man lsblk
$ lsblk -b
```



103.1 Lesson 2

Certificate:	LPIC-1
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.1 Work on the command line
Lesson:	2 of 2

Introduction

An operating system environment includes the basic tools—like command line shells and sometimes a GUI—that you will need in order to get stuff done. But your environment will also come with a catalog of shortcuts and preset values. Here is where we will learn how to list, invoke, and manage those values.

Finding Your Environment Variables

So just how do we identify the current values for each of our environment variables? One way is through the `env` command:

```
$ env
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
XDG_RUNTIME_DIR=/run/user/1000
XAUTHORITY=/run/user/1000/gdm/Xauthority
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
```



```
GJS_DEBUG_TOPICS=JS ERROR;JS LOG
[...]
```

You will get a lot of output — much more than what is included in the above excerpt. But for now note the `PATH` entry, which contains the directories where your shell (and other programs) will look for other programs without having to specify a complete path. With that set, you could run a binary program that lives, say, in `/usr/local/bin` from within your home directory and it would run just as though the file was local.

Let us change the subject for a moment. The `echo` command will print to the screen whatever you tell it to. Believe it or not, there will be many times when getting `echo` to literally repeat something will be very useful.

```
$ echo "Hi. How are you?"
Hi. How are you?
```

But there is something else you can do with `echo`. When you feed it the name of an environment variable — and tell it that this is a variable by prefixing the variable name with a `$` — then, instead of just printing the variable's name, the shell will expand it giving you the value. Not sure whether your favorite directory is currently in the path? You can quickly check by running it through `echo`:

```
$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
```

Creating New Environment Variables

You can add your own custom variables to your environment. The simplest way is to use the `=` character. The string to the left will be the name of your new variable, and the string to the right will be its value. You can now feed the variable name to `echo` to confirm it worked:

```
$ myvar=hello
$ echo $myvar
hello
```

NOTE

Notice that there is no space on either side of the equal sign during variable assignment.

But did it really work? Type `bash` into the terminal to open a new shell. This new shell looks exactly like the one you were just in, but it is actually a *child* of the original one (which we call the *parent*). Now, inside this new child shell, try to get `echo` to do its magic the way it did before. Nothing. What's going on?

```
$ bash
$ echo $myvar

$
```

A variable you create the way we just did is only going to be available locally—within the immediate shell session. If you start up a new shell—or close down the session using `exit`—the variable will not go along with you. Typing `exit` here will take you back to your original parent shell which, right now, is where we want to be. You can run `echo $myvar` once again if you like just to confirm that the variable is still valid. Now type `export myvar` to pass the variable to any child shells that you may subsequently open. Try it out: type `bash` for a new shell and then `echo`:

```
$ exit
$ export myvar
$ bash
$ echo $myvar
hello
```

All this may feel a bit silly when we are creating shells for no real purpose. But understanding how shell variables are propagated through your system will become very important once you start writing serious scripts.

Deleting Environment Variables

Want to know how to clean up all those ephemeral variables you have created? One way is to simply close your parent shell—or reboot your computer. But there are simpler ways. Like, for instance, `unset`. Typing `unset` (without the `$`) will kill the variable. `echo` will now prove that.

```
$ unset myvar
$ echo $myvar

$
```

If there is an `unset` command, then you can bet there must be a `set` command to go with it.

Running `set` by itself will display lots of output, but it is really not all that different from what `env` gave you. Look at the first line of output you will get when you filter for `PATH`:

```
$ set | grep PATH
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/game
s:/snap/bin
[...]
```

What is the difference between `set` and `env`? For our purposes, the main thing is that `set` will output all variables and functions. Let us illustrate that. We will create a new variable called `mynewvar` and then confirm it is there:

```
$ mynewvar=goodbye
$ echo $mynewvar
goodbye
```

Now, running `env` while using `grep` to filter for the string `mynewvar` will not display any output. But running `set` the same way will show us our local variable.

```
$ env | grep mynewvar

$ set | grep mynewvar
mynewvar=goodbye
```

Quoting to Escape Special Characters

Now is as good a time as any other to introduce you to the problem of special characters. Alphanumeric characters (a-z and 0-9) will normally be read literally by Bash. If you try to create a new file called `myfile` you would just type `touch` followed by `myfile` and Bash will know what to do with it. But if you want to include a special character in your filename, you will need to do a bit more work.

To illustrate this, we will type `touch` and follow it by the title: `my big file`. The problem is that there are two spaces there between words which Bash will interpret. While, technically, you would not call a space a “character,” it is like one in the sense that Bash will not read it literally. If you list the contents of your current directory, rather than one file called `my big file`, you will see three files named, respectively, `my`, `big`, and `file`. That is because Bash thought you wanted to create multiple files whose names you were passing in a list:

```
$ touch my big file
$ ls
my big file
```

The spaces will be interpreted the same way if you delete (`rm`) the three files all in one command:

```
$ rm my big file
```

Now let us try it the right way. Type `touch` and the three parts of your filename but this time enclose the name in quotation marks. This time it worked. Listing the directory contents will show you a single file with the proper name.

```
$ touch "my big file"
$ ls
'my big file'
```

There are other ways to get the same effect. Single quotes, for instance, work just as well as double quotes. (Note that single quotes will preserve the literal value of all characters, while double quotes will preserve all characters *except* for `$`, ```, `\` and, on certain cases, `!`.)

```
$ rm 'my big file'
```

Prepending each special character with the backslash will “escape” the specialness of the character and cause Bash to read it literally.

```
$ touch my\ big\ file
```

Guided Exercises

1. Use the `export` command to add a new directory to your path (this will not survive a reboot).

2. Use the `unset` command to delete the `PATH` variable. Try running a command (like `sudo cat /etc/shadow`) using `sudo`. What happened? Why? (Exiting your shell will return you to your original state.)

Explorational Exercises

1. Search the internet to find and explore the complete list of special characters.
2. Try running commands using strings made up of special characters and using various methods to escape them. Are there differences between the way those methods behave?

Summary

In this lesson, you learned:

- How to identify your system's environment variables.
- How to create your own environment variables and export them to other shells.
- How to remove environment variables and how to use both the `env` and `set` commands.
- How to escape special characters so Bash will read them literally.

The following commands were discussed in this lesson:

echo

Print input strings and variables.

env

Understand and modify your environment variables.

export

Pass an environment variable to child shells.

unset

Unset values and attributes of shell variables and functions.

Answers to Guided Exercises

1. Use the `export` command to add a new directory to your path (this will not survive a reboot).

You can temporarily add a new directory (perhaps one called `myfiles` that lives in your home directory) to your path using `export PATH="/home/yourname/myfiles:$PATH"`. Create a simple script in the `myfiles/` directory, make it executable, and try to run it from a different directory. These commands assume you're in your home directory which contains a directory called `myfiles`.

```
$ touch myfiles/myscript.sh
$ echo '#!/bin/bash' >> myfiles/myscript.sh
$ echo 'echo Hello' >> myfiles/myscript.sh
$ chmod +x myfiles/myscript.sh
$ myscript.sh
Hello
```

2. Use the `unset` command to delete the `PATH` variable. Try running a command (like `sudo cat /etc/shadow`) using `sudo`. What happened? Why? (Exiting your shell will return you to your original state.)

Typing `unset PATH` will erase the current path settings. Trying to invoke a binary without its absolute address will fail. For that reason, trying to run a command using `sudo` (which itself is a binary program located in `/usr/bin/sudo`) will fail—unless you specify the absolute location, as in: `/usr/bin/sudo /bin/cat /etc/shadow`. You can reset your `PATH` using `export` or by simply exiting from the shell.

Answers to Explorational Exercises

1. Search the internet to find and explore the complete list of special characters.

Here is a list: `& ; | * ? " ' [ ] ( ) $ < > { } # / \ ! ~.`

2. Try running commands using strings made up of special characters and using various methods to escape them. Are there differences between the way those methods behave?

Escaping using `"` characters will preserve the special values of the dollar sign, a backtick, and the backslash. Escaping using a `'` character, on the other hand, will render *all* characters as literal.

```
$ echo "$mynewvar"  
goodbye  
$ echo '$mynewvar'  
$mynewvar
```



103.2 Process text streams using filters

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 103.2

Weight

2

Key knowledge areas

- Send text files and output streams through text utility filters to modify the output using standard UNIX commands found in the GNU textutils package.

Partial list of the used files, terms and utilities

- `bzcat`
- `cat`
- `cut`
- `head`
- `less`
- `md5sum`
- `nl`
- `od`
- `paste`
- `sed`
- `sha256sum`
- `sha512sum`
- `sort`

- `split`
- `tail`
- `tr`
- `uniq`
- `wc`
- `xzcat`
- `zcat`



103.2 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.2 Process text streams using filters
Lesson:	1 of 1

Introduction

Dealing with text is a major part of every systems administrator's job. Doug McIlroy, a member of the original Unix development team, summarized the Unix philosophy and said (among other important things): "Write programs to handle text streams, because that is a universal interface." Linux is inspired by the Unix operating system and it firmly adopts its philosophy, so an administrator must expect lots of text manipulation tools within a Linux distribution.

A Quick Review on Redirections and Pipes

Also from the Unix philosophy:

- Write programs that do one thing and do it well.
- Write programs to work together.

One major way of making programs work together is through *piping* and *redirections*. Pretty much all of your text manipulation programs will get text from a standard input (*stdin*), output it to a standard output (*stdout*) and send eventual errors to a standard error output (*stderr*). Unless you specify otherwise, the standard input will be what you type on your keyboard (the program will

read it after you press the Enter key). Similarly, the standard output and errors will be displayed in your terminal screen. Let us see how this works.

In your terminal, type `cat` and then hit the Enter key. Then type some random text.

```
$ cat
This is a test
This is a test
Hey!
Hey!
It is repeating everything I type!
It is repeating everything I type!
(I will hit ctrl+c so I will stop this nonsense)
(I will hit ctrl+c so I will stop this nonsense)
^C
```

For more information about the `cat` command (the term comes from “concatenate”) please refer to the man pages.

NOTE

If you are working on a really plain installation of a Linux server, some commands such as `info` and `less` might not be available. If this is the case, install these tools using the proper procedure in your system as described in the corresponding lessons.

As demonstrated above if you do not specify where `cat` should read from, it will read from the standard input (whatever you type) and output whatever it reads to your terminal window (its standard output).

Now try the following:

```
$ cat > mytextfile
This is a test
I hope cat is storing this to mytextfile as I redirected the output
I will hit ctrl+c now and check this
^C

$ cat mytextfile
This is a test
I hope cat is storing this to mytextfile as I redirected the output
I will hit ctrl+c now and check this
```

The `>` (greater than) tells `cat` to direct its output to the `mytextfile` file, not the standard output.

Now try this:

```
$ cat mytextfile > mynewtextfile
$ cat mynewtextfile
This is a test
I hope cat is storing this to mytextfile as I redirected the output
I will hit ctrl+c now and check this
```

This has the effect of copying `mytextfile` to `mynewtextfile`. You can actually verify that these two files have the same content by performing a `diff`:

```
$ diff mynewtextfile mytextfile
```

As there is no output, the files are equal. Now try the append redirection operator (`>>`):

```
$ echo 'This is my new line' >> mynewtextfile
$ diff mynewtextfile mytextfile
4d3
< This is my new line
```

So far we have used redirections to create and manipulate files. We can also use pipes (represented by the symbol `|`) to redirect the output of one program to another program. Let us find the lines where the word “this” is found:

```
$ cat mytextfile | grep this
I hope cat is storing this to mytextfile as I redirected the output
I will hit ctrl+c now and check this

$ cat mytextfile | grep -i this
This is a test
I hope cat is storing this to mytextfile as I redirected the output
I will hit ctrl+c now and check this
```

Now we have piped the output of `cat` to another command: `grep`. Notice when we ignore the case (using the `-i` option) we get an extra line as a result.

Processing Text Streams

Reading a Compressed File

We will create a file called `ftu.txt` containing a list of the following commands:

```
bzcat
cat
cut
head
less
md5sum
nl
od
paste
sed
sha256sum
sha512sum
sort
split
tail
tr
uniq
wc
xzcat
zcat
```

Now we will use the `grep` command to print all of the lines containing the string `cat`:

```
$ cat ftu.txt | grep cat
bzcat
cat
xzcat
zcat
```

Another way to get this information is to just use the `grep` command to filter the text directly, without the need to use another application to send the text stream to `stdout`.

```
$ grep cat ftu.txt
bzcat
cat
xzcat
zcat
```

NOTE Remember there are many ways to perform the same task using Linux.

There are other commands that handle compressed files (`bzcat` for `bzip` compressed files, `xzcat` for `xz` compressed files and `zcat` for `gzip` compressed files) and each one is used to view the contents of a compressed file based on the compression algorithm used.

Verify that the newly created file `ftu.txt` is the only one in the directory, then create a `gzip` compressed version of the file:

```
$ ls ftu*
ftu.txt

$ gzip ftu.txt
$ ls ftu*
ftu.txt.gz
```

Next, use the `zcat` command to view the contents of the gzipped compressed file:

```
$ zcat ftu.txt.gz
bzcat
cat
cut
head
less
md5sum
nl
od
paste
sed
sha256sum
sha512sum
sort
split
tail
tr
uniq
wc
xzcat
zcat
```

Note that `gzip` will compress `ftu.txt` into `ftu.txt.gz` and it will remove the original file. By default, no output from the `gzip` command will be displayed. However, if you do want `gzip` to tell

you what it is doing, use the `-v` option for the “verbose” output.

Viewing a File in a Pager

You know `cat` concatenates a file to the standard output (once a file is provided after the command). The file `/var/log/syslog` is where your Linux system stores everything important going on in your system. Using the `sudo` command to elevate privileges so as to be able to read the `/var/log/syslog` file:

```
$ sudo cat /var/log/syslog
```

...you will see messages scrolling very fast within your terminal window. You can pipe the output to the program `less` so the results will be paginated. By using `less` you can use the arrow keys to navigate through the output and also use `vi` like commands to navigate and search throughout the text.

However, rather than pipe the `cat` command into a pagination program it is more pragmatic to just use the pagination program directly:

```
$ sudo less /var/log/syslog
... (output omitted for clarity)
```

Getting a Portion of a Text File

If only the start or end of a file needs to be reviewed, there are other methods available. The command `head` is used to read the first ten lines of a file by default, and the command `tail` is used to read the last ten lines of a file by default. Now try:

```
$ sudo head /var/log/syslog
Nov 12 08:04:30 hypatia rsyslogd: [origin software="rsyslogd" swVersion="8.1910.0" x-
pid="811" x-info="https://www.rsyslog.com"] rsyslogd was HUPed
Nov 12 08:04:30 hypatia systemd[1]: logrotate.service: Succeeded.
Nov 12 08:04:30 hypatia systemd[1]: Started Rotate log files.
Nov 12 08:04:30 hypatia vdr: [928] video directory scanner thread started (pid=882, tid=928,
prio=low)
Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'A - ATSC'
Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'C - DVB-C'
Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'S - DVB-S'
Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'T - DVB-T'
Nov 12 08:04:30 hypatia vdr[882]: vdr: no primary device found - using first device!
```

```
Nov 12 08:04:30 hypatia vdr: [929] epg data reader thread started (pid=882, tid=929,
prio=high)
$ sudo tail /var/log/syslog
Nov 13 10:24:45 hypatia kernel: [ 8001.679238] mce: CPU7: Core temperature/speed normal
Nov 13 10:24:46 hypatia dbus-daemon[2023]: [session uid=1000 pid=2023] Activating via
systemd: service name='org.freedesktop.Tracker1.Miner.Extract' unit='tracker-
extract.service' requested by ':1.73' (uid=1000 pid=2425 comm="/usr/lib/tracker/tracker-
miner-fs ")
Nov 13 10:24:46 hypatia systemd[2004]: Starting Tracker metadata extractor...
Nov 13 10:24:47 hypatia dbus-daemon[2023]: [session uid=1000 pid=2023] Successfully
activated service 'org.freedesktop.Tracker1.Miner.Extract'
Nov 13 10:24:47 hypatia systemd[2004]: Started Tracker metadata extractor.
Nov 13 10:24:54 hypatia kernel: [ 8010.462227] mce: CPU0: Core temperature above threshold,
cpu clock throttled (total events = 502907)
Nov 13 10:24:54 hypatia kernel: [ 8010.462228] mce: CPU4: Core temperature above threshold,
cpu clock throttled (total events = 502911)
Nov 13 10:24:54 hypatia kernel: [ 8010.469221] mce: CPU0: Core temperature/speed normal
Nov 13 10:24:54 hypatia kernel: [ 8010.469222] mce: CPU4: Core temperature/speed normal
Nov 13 10:25:03 hypatia systemd[2004]: tracker-extract.service: Succeeded.
```

To help illustrate the number of lines displayed, we can pipe the output of the `head` command to the `nl` command, which will display the number of lines of text streamed into the command:

```
$ sudo head /var/log/syslog | nl
1 Nov 12 08:04:30 hypatia rsyslogd: [origin software="rsyslogd" swVersion="8.1910.0" x-
pid="811" x-info="https://www.rsyslog.com"] rsyslogd was HUPed
2 Nov 12 08:04:30 hypatia systemd[1]: logrotate.service: Succeeded.
3 Nov 12 08:04:30 hypatia systemd[1]: Started Rotate log files.
4 Nov 12 08:04:30 hypatia vdr: [928] video directory scanner thread started (pid=882,
tid=928, prio=low)
5 Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'A - ATSC'
6 Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'C - DVB-C'
7 Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'S - DVB-S'
8 Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'T - DVB-T'
9 Nov 12 08:04:30 hypatia vdr[882]: vdr: no primary device found - using first device!
10 Nov 12 08:04:30 hypatia vdr: [929] epg data reader thread started (pid=882, tid=929,
prio=high)
```

And we can do the same by piping the output of the `tail` command to the `wc` command, which by default will count the number of words within a document, and using the `-l` switch to print out the number of lines of text that the command has read:

```
$ sudo tail /var/log/syslog | wc -l
10
```

Should an administrator need to review more (or less) of the beginning or end of a file, the `-n` option can be used to limit the commands' output:

```
$ sudo tail -n 5 /var/log/syslog
Nov 13 10:37:24 hypatia systemd[2004]: tracker-extract.service: Succeeded.
Nov 13 10:37:42 hypatia dbus-daemon[2023]: [session uid=1000 pid=2023] Activating via
systemd: service name='org.freedesktop.Tracker1.Miner.Extract' unit='tracker-
extract.service' requested by ':1.73' (uid=1000 pid=2425 comm="/usr/lib/tracker/tracker-
miner-fs ")
Nov 13 10:37:42 hypatia systemd[2004]: Starting Tracker metadata extractor...
Nov 13 10:37:43 hypatia dbus-daemon[2023]: [session uid=1000 pid=2023] Successfully
activated service 'org.freedesktop.Tracker1.Miner.Extract'
Nov 13 10:37:43 hypatia systemd[2004]: Started Tracker metadata extractor.
$ sudo head -n 12 /var/log/syslog
Nov 12 08:04:30 hypatia rsyslogd: [origin software="rsyslogd" swVersion="8.1910.0" x-
pid="811" x-info="https://www.rsyslog.com"] rsyslogd was HUPed
Nov 12 08:04:30 hypatia systemd[1]: logrotate.service: Succeeded.
Nov 12 08:04:30 hypatia systemd[1]: Started Rotate log files.
Nov 12 08:04:30 hypatia vdr: [928] video directory scanner thread started (pid=882, tid=928,
prio=low)
Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'A - ATSC'
Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'C - DVB-C'
Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'S - DVB-S'
Nov 12 08:04:30 hypatia vdr: [882] registered source parameters for 'T - DVB-T'
Nov 12 08:04:30 hypatia vdr[882]: vdr: no primary device found - using first device!
Nov 12 08:04:30 hypatia vdr: [929] epg data reader thread started (pid=882, tid=929,
prio=high)
Nov 12 08:04:30 hypatia vdr: [882] no DVB device found
Nov 12 08:04:30 hypatia vdr: [882] initializing plugin: vnsiserver (1.8.0): VDR-Network-
Streaming-Interface (VNSI) Server
```

The Basics of sed, the Stream Editor

Let us take a look at the other files, terms and utilities that do not have `cat` in their names. We can do this by passing the `-v` option to `grep`, which instructs the command to output only the lines not containing `cat`:

```
$ zcat ftu.txt.gz | grep -v cat
```

```
cut
head
less
md5sum
nl
od
paste
sed
sha256sum
sha512sum
sort
split
tail
tr
uniq
wc
```

Most of what we can do with `grep` we can also do with `sed`—the stream editor for filtering and transforming text (as stated in the `sed` manual page). First we will recover our `ftu.txt` file by decompressing our `gzip` archive of the file:

```
$ gunzip ftu.txt.gz
$ ls ftu*
ftu.txt
```

Now, we can use `sed` to list only the lines containing the string `cat`:

```
$ sed -n /cat/p < ftu.txt
bzcat
cat
xzcat
zcat
```

We have used the less-than sign `<` to direct the contents of the file `ftu.txt` into our `sed` command. The word enclosed between slashes (i.e. `/cat/`) is the term we are searching for. The `-n` option instructs `sed` to produce no output (unless the ones later instructed by the `p` command). Try running this same command without the `-n` option to see what happens. Then try this:

```
$ sed /cat/d < ftu.txt
cut
head
```

```
less
md5sum
nl
od
paste
sed
sha256sum
sha512sum
sort
split
tail
tr
uniq
wc
```

If we do not use the `-n` option, `sed` will print everything from the file except for what the `d` instructs `sed` to delete from its output.

A common use of `sed` is to find and replace text within a file. Suppose you want to change every occurrence of `cat` to `dog`. You can use `sed` to do this by supplying the `s` option to swap out each instance of the first term, `cat`, for the second term, `dog`:

```
$ sed s/cat/dog/ < ftu.txt
bzdog
dog
cut
head
less
md5sum
nl
od
paste
sed
sha256sum
sha512sum
sort
split
tail
tr
uniq
wc
xzdog
```

```
zdog
```

Rather than using a redirection operator (<) to pass the `ftu.txt` file into our `sed` command, we can just have the `sed` command operate on the file directly. We will try that next, while simultaneously creating a backup of the original file:

```
$ sed -i.backup s/cat/dog/ ftu.txt
$ ls ftu*
ftu.txt  ftu.txt.backup
```

The `-i` option will perform an in-place `sed` operation on your *original* file. If you do not use the `.backup` after the `-i` parameter, you would just have rewritten your *original* file. Whatever you use as text after the `-i` parameter will be the name the original file will be saved to prior to the modifications you asked `sed` to perform.

Ensuring Data Integrity

We have demonstrated how easy it is to manipulate files in Linux. There are times where you may wish to distribute a file to someone else, and you want to be sure that the recipient ends up with a true copy of the original file. A very common use of this technique is practiced when Linux distribution servers host downloadable CD or DVD images of their software along with files that contain the calculated checksum values of those disc images. Here is an example listing from a Debian download mirror:

[PARENTDIR]	Parent Directory	-
[SUM]	MD5SUMS	2019-09-08 17:46 274
[CRT]	MD5SUMS.sign	2019-09-08 17:52 833
[SUM]	SHA1SUMS	2019-09-08 17:46 306
[CRT]	SHA1SUMS.sign	2019-09-08 17:52 833
[SUM]	SHA256SUMS	2019-09-08 17:46 402
[CRT]	SHA256SUMS.sign	2019-09-08 17:52 833
[SUM]	SHA512SUMS	2019-09-08 17:46 658
[CRT]	SHA512SUMS.sign	2019-09-08 17:52 833
[ISO]	debian-10.1.0-amd64-netinst.iso	2019-09-08 04:37 335M
[ISO]	debian-10.1.0-amd64-xfce-CD-1.iso	2019-09-08 04:38 641M
[ISO]	debian-edu-10.1.0-amd64-netinst.iso	2019-09-08 04:38 405M
[ISO]	debian-mac-10.1.0-amd64-netinst.iso	2019-09-08 04:38 334M

In the listing above, the Debian installer image files are accompanied by text files that contain checksums of the files from the various algorithms (MD5, SHA1, SHA256 and SHA512).

NOTE

A checksum is a value derived from a mathematical computation, based on a cryptographic hash function, against a file. There are different types of cryptographic hash functions that vary in strength. The exam will expect you to be familiar with using `md5sum`, `sha256sum` and `sha512sum`.

Once you download a file (for example, the `debian-10.1.0-amd64-netinst.iso` image) you would then compare the checksum of the file that was downloaded against a checksum value that was provided for you.

Here is an example to illustrate the point. We will calculate the SHA256 value of the `ftu.txt` file using the `sha256sum` command:

```
$ sha256sum ftu.txt
345452304fc26999a715652543c352e5fc7ee0c1b9deac6f57542ec91daf261c  ftu.txt
```

The long string of characters preceding the file name is the SHA256 checksum value of this text file. Let us create a file that contains that value, so that we can use it to verify the integrity of our original text file. We can do this with the same `sha256sum` command and redirect the output to a file:

```
$ sha256sum ftu.txt > sha256.txt
```

Now, to verify the `ftu.txt` file, we just use the same command and supply the filename that contains our checksum value along with the `-c` switch:

```
$ sha256sum -c sha256.txt
ftu.txt: OK
```

The value contained within the file matches the calculated SHA256 checksum for our `ftu.txt` file, just as we would expect. However, if the original file were modified (such as a few bytes lost during a file download, or someone had deliberately tampered with it) the value check would fail. In such cases we know that our file is bad or corrupted, and we can not trust the integrity of its contents. To prove the point, we will add some text at the end of the file:

```
$ echo "new entry" >> ftu.txt
```

Now we will make an attempt to verify the file's integrity:

```
$ sha256sum -c sha256.txt
ftu.txt: FAILED
sha256sum: WARNING: 1 computed checksum did NOT match
```

And we see that the checksum does not match what was expected for the file. Therefore, we could not trust the integrity of this file. We could attempt to download a new copy of a file, report the failure of the checksum to the sender of the file, or report it to a data center security team depending on the importance of the file.

Looking Deeper into Files

The octal dump (`od`) command is often used for debugging applications and various files. By itself, the `od` command will just list out a file's contents in octal format. We can use our `ftu.txt` file from earlier to practice with this command:

```
$ od ftu.txt
0000000 075142 060543 005164 060543 005164 072543 005164 062550
0000020 062141 066012 071545 005163 062155 071465 066565 067012
0000040 005154 062157 070012 071541 062564 071412 062145 071412
0000060 060550 032462 071466 066565 071412 060550 030465 071462
0000100 066565 071412 071157 005164 070163 064554 005164 060564
0000120 066151 072012 005162 067165 070551 073412 005143 075170
0000140 060543 005164 061572 072141 000012
0000151
```

The first column of output is the *byte offset* for each line of output. Since `od` prints out information in octal format by default, each line begins with the byte offset of eight bits, followed by eight columns, each containing the octal value of the data within that column.

TIP | Recall that a *byte* is 8 bits in length.

Should you need to view a file's contents in hexadecimal format, use the `-x` option:

```
$ od -x ftu.txt
0000000 7a62 6163 0a74 6163 0a74 7563 0a74 6568
0000020 6461 6c0a 7365 0a73 646d 7335 6d75 6e0a
0000040 0a6c 646f 700a 7361 6574 730a 6465 730a
0000060 6168 3532 7336 6d75 730a 6168 3135 7332
0000100 6d75 730a 726f 0a74 7073 696c 0a74 6174
```



```
0000120 6c69 740a 0a72 6e75 7169 770a 0a63 7a78
0000140 6163 0a74 637a 7461 000a
0000151
```

Now each of the eight columns after the byte offset are represented by their hexadecimal equivalents.

One handy use of the `od` command is for debugging scripts. For example, the `od` command can show us characters that are not normally seen that exist within a file, such as *newline* entries. We can do this with the `-c` option, so that instead of displaying the numerical notation for each byte, these column entries will instead be shown as their character equivalents:

```
$ od -c ftu.txt
0000000 b z c a t \n c a t \n c u t \n h e
0000020 a d \n l e s s \n m d 5 s u m \n n
0000040 l \n o d \n p a s t e \n s e d \n s
0000060 h a 2 5 6 s u m \n s h a 5 1 2 s
0000100 u m \n s o r t \n s p l i t \n t a
0000120 i l \n t r \n u n i q \n w c \n x z
0000140 c a t \n z c a t \n
0000151
```

All of the newline entries within the file are represented by the hidden `\n` characters. If you just want to view all of the characters within a file, and do not need to see the byte offset information, the byte offset column can be removed from the output like so:

```
$ od -An -c ftu.txt
b z c a t \n c a t \n c u t \n h e
a d \n l e s s \n m d 5 s u m \n n
l \n o d \n p a s t e \n s e d \n s
h a 2 5 6 s u m \n s h a 5 1 2 s
u m \n s o r t \n s p l i t \n t a
i l \n t r \n u n i q \n w c \n x z
c a t \n z c a t \n
```

Guided Exercises

1. Someone just donated a laptop to your school and now you wish to install Linux on it. There is no manual and you were forced to boot it from a USB thumb drive with no graphics whatsoever. You do get a shell terminal and you know, for every processor you have there will be a line for it in the `/proc/cpuinfo` file:

```
processor      : 0
vendor_id     : GenuineIntel
cpu family    : 6
model         : 158
```

(lines skipped)

```
processor      : 1
vendor_id     : GenuineIntel
cpu family    : 6
model         : 158
```

(more lines skipped)

- Using the commands `grep` and `wc` display how many processors you have.

- Do the same thing with `sed` instead of `grep`.

2. Explore your local `/etc/passwd` file with the `grep`, `sed`, `head` and `tail` commands per the tasks below:

- Which users have access to a Bash shell?

- Your system has various users that exist to handle specific programs or for administrative purposes. They do not have access to a shell. How many of those exist in your system?

- How many users and groups exist in your system (remember: use only the `/etc/passwd` file)?

- List only the first line, the last line and the tenth line of your `/etc/passwd` file.

3. Consider this `/etc/passwd` file example. Copy the lines below to a local file named `mypasswd` for this exercise.

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
nvidia-persistenced:x:121:128:NVIDIA Persistence Daemon,,,:/nonexistent:/sbin/nologin
libvirt-qemu:x:64055:130:Libvirt Qemu,,,:/var/lib/libvirt:/usr/sbin/nologin
libvirt-dnsmasq:x:122:133:Libvirt Dnsmasq,,,:/var/lib/libvirt/dnsmasq:/usr/sbin/nologin
carol:x:1000:2000:Carol Smith,Finance,,Main Office:/home/carol:/bin/bash
dave:x:1001:1000:Dave Edwards,Finance,,Main Office:/home/dave:/bin/ksh
emma:x:1002:1000:Emma Jones,Finance,,Main Office:/home/emma:/bin/bash
frank:x:1003:1000:Frank Cassidy,Finance,,Main Office:/home/frank:/bin/bash
grace:x:1004:1000:Grace Kearns,Engineering,,Main Office:/home/grace:/bin/ksh
henry:x:1005:1000:Henry Adams,Sales,,Main Office:/home/henry:/bin/bash
john:x:1006:1000:John Chapel,Sales,,Main Office:/home/john:/bin/bash
```

- List all users in group `1000` (use `sed` to select only the appropriate field) from your `mypasswd` file.

- List only the full names of all the users for this group (use `sed` and `cut`).

Explorational Exercises

1. Once more using the `mypasswd` file from the previous exercises, devise a Bash command that will select one individual from the Main Office to win a raffle contest. Use the `sed` command to only print out the lines for the Main Office, and then a `cut` command sequence to retrieve the first name of each user from these lines. Next you will want to randomly sort these names and only print out the top name from the list.

2. How many people work in Finance, Engineering and Sales? (Consider exploring the `uniq` command.)

3. Now you want to prepare a CSV (comma separated values) file so you can easily import, from the `mypasswd` file in the previous example, the file `names.csv` into LibreOffice. The file contents will have the following format:

```
First Name,Last Name,Position
Carol,Smith,Finance
...
John,Chapel,Sales
```

Tip: Use the `sed`, `cut`, and `paste` commands to achieve the desired results. Note that the comma (,) will be the delimiter for this file.

4. Suppose that the `names.csv` spreadsheet created in the previous exercise is an important file and we want to make sure nobody will tamper with it from the moment we send it to someone and the moment our recipient receives it. How can we insure the integrity of this file using `md5sum`?

5. You promised yourself that you would read a classic book 100 lines per day and you decided to start with *Mariner and Mystic* by Herman Melville. Devise a command using `split` that will separate this book into sections of 100 lines each. In order to get the book in plain text format, search for it at <https://www.gutenberg.org>.

6. Using `ls -l` on the `/etc` directory, what kind of listing do you get? Using the `cut` command on the output of the given `ls` command how would you display only the file names? What about the filename and the owner of the files? Along with the `ls -l` and `cut` commands, utilize the

`tr` command to *squeeze* multiple occurrences of a space into a single space to aid in formatting the output with a `cut` command.

7. This exercise assumes you are on a real machine (not a virtual machine). You must also have a USB stick with you. Review the manual pages for the `tail` command and find out how to follow a file as text is appended to it. While monitoring the output of a `tail` command on the `/var/log/syslog` file, insert a USB stick. Write out the full command that you would use to get the Product, Manufacturer and the total amount of memory of your USB stick.

Summary

Dealing with text streams is of great importance when administering any Linux system. Text streams can be processed using scripts to automate daily tasks or finding relevant debugging information in log files. Here is a short summary of the commands covered in this lesson:

cat

Used to combine or read plain text files.

bzcat

Allows for the processing or reading of files compressed using the `bzip2` method.

xzcat

Allows for the processing or reading of files compressed using the `xz` method.

zcat

Allows for the processing or reading of files compressed using the `gzip` method.

less

This command paginates the contents of a file, and allows for navigation and search functionality.

head

This command will display the first 10 lines of a file by default. With the use of the `-n` switch fewer or more lines can be displayed.

tail

This command will display the last 10 lines of a file by default. With the use of the `-n` switch fewer or more lines can be displayed. The `-f` option is used to follow the output of a text file has new data is being written to it.

wc

Short for “word count” but depending on the parameters you use it will count characters, words and lines.

sort

Used for the organizing the output of a listing alphabetically, reverse alphabetically, or in a random order.

uniq

Used to list (and count) matching strings.

od

The “octal dump” command is used to display a binary file in either octal, decimal, or hexadecimal notation.

nl

The “number line” command will display the number of lines in a file as well as recreate a file with each line prepended by its line number.

sed

The stream editor can be used to find matching occurrences of strings using Regular Expressions as well as editing files using pre-defined patterns.

tr

The translate command can replace characters and also removes and compresses repeating characters.

cut

This command can print columns of text files as fields based on a file’s character delimiter.

paste

Join files in columns based on the usage of field separators.

split

This command can split larger files into smaller ones depending on the criteria set by the command’s options.

md5sum

Used for calculating the MD5 hash value of a file. Also used to verify a file against an existing hash value to ensure a file’s integrity.

sha256sum

Used for calculating the SHA256 hash value of a file. Also used to verify a file against an existing hash value to ensure a file’s integrity.

sha512sum

Used for calculating the SHA512 hash value of a file. Also used to verify a file against an existing hash value to ensure a file’s integrity.

Answers to Guided Exercises

1. Someone just donated a laptop to your school and now you wish to install Linux on it. There is no manual and you were forced to boot it from a USB thumb drive with no graphics whatsoever. You do get a shell terminal and you know, for every processor you have there will be a line for it in the `/proc/cpuinfo` file:

```
processor      : 0
vendor_id     : GenuineIntel
cpu family    : 6
model         : 158
```

(lines skipped)

```
processor      : 1
vendor_id     : GenuineIntel
cpu family    : 6
model         : 158
```

(more lines skipped)

- Using the commands `grep` and `wc` display how many processors you have.

Here are two options:

```
$ cat /proc/cpuinfo | grep processor | wc -l
$ grep processor /proc/cpuinfo | wc -l
```

Now that you know there are several ways you can do the same thing, when should you be using one or the other? It really depends on several factors, the two most important ones being performance and readability. Most of the time you will use shell commands inside shell scripts to automate your tasks and the larger and more complex your scripts become the more you need to worry about keeping them fast.

- Do the same thing with `sed` instead of `grep`

Now, instead of `grep` we will try this with `sed`:

```
$ sed -n /processor/p /proc/cpuinfo | wc -l
```


Here we used `sed` with the `-n` parameter so `sed` will not print anything except for what matches with the expression processor, as instructed by the `p` command. As we did in the `grep` solutions, `wc -l` will count the number of lines, thus the number of processors we have.

Study this next example:

```
$ sed -n /processor/p /proc/cpuinfo | sed -n '$='
```

This command sequence provides identical results to the previous example where the output of `sed` was piped into a `wc` command. The difference here is that instead of using `wc -l` to count the number of lines, `sed` is again invoked to provide equivalent functionality. Once more, we are suppressing the output of `sed` with the `-n` option, except for the expression that we are explicitly calling, which is `'$='`. This expression tells `sed` to match the last line (\$) and then to print that line number (=).

2. Explore your local `/etc/passwd` file with the `grep`, `sed`, `head` and `tail` commands per the tasks below:

- Which users have access to a Bash shell?

```
$ grep ":/bin/bash$" /etc/passwd
```

We will improve this answer by only displaying the name of the user that utilizes the Bash shell.

```
$ grep ":/bin/bash$" /etc/passwd | cut -d: -f1
```

The user name is the first field (`-f1` parameter of the `cut` command) and the `/etc/passwd` file uses `:` as separators (`-d:` parameter of the `cut` command) we just pipe the output of the `grep` command to the appropriate `cut` command.

- Your system has various users that exists to handle specific programs or for administrative purposes. They do not have access to a shell. How many of those exist in your system?

The easiest way to find this is by printing out the lines for accounts that do not use the Bash shell:

```
$ grep -v ":/bin/bash$" /etc/passwd | wc -l
```

- How many users and groups exist in your system (remember: use only the `/etc/passwd` file)

The first field of any given line in your `/etc/passwd` file is the user name, the second is typically an `x` indicating the user password is not stored here (it is encrypted in the `/etc/shadow` file). The third is the user id (UID) and the fourth is the group id (GID). So this should give us the number of users:

```
$ cut -d: -f3 /etc/passwd | wc -l
```

Well, most of the time it will. However, there are situations where you will set different super users or other special kinds of users sharing the same UID (user id). So, to be on the safe side we will pipe the result of our `cut` command to the `sort` command and then count the number of lines.

```
$ cut -d: -f3 /etc/passwd | sort -u | wc -l
```

Now, for the number of groups:

```
$ cut -d: -f4 /etc/passwd | sort -u | wc -l
```

- List only the first line, the last line and the tenth line of your `/etc/passwd` file

This will do:

```
$ sed -n -e '1'p -e '10'p -e '$'p /etc/passwd
```

Remember that the parameter `-n` tells `sed` not to print anything other than what is specified by the `p` command. The dollar sign (\$) used here is a regular expression meaning the last line of the file.

3. Consider this `/etc/passwd` file example. Copy the lines below to a local file named `mypasswd` for this exercise.

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
```

```
nvidia-persistenced:x:121:128:NVIDIA Persistence Daemon,,,:/nonexistent:/sbin/nologin
libvirt-qemu:x:64055:130:Libvirt Qemu,,,:/var/lib/libvirt:/usr/sbin/nologin
libvirt-dnsmasq:x:122:133:Libvirt Dnsmasq,,,:/var/lib/libvirt/dnsmasq:/usr/sbin/nologin
carol:x:1000:2000:Carol Smith,Finance,,Main Office:/home/carol:/bin/bash
dave:x:1001:1000:Dave Edwards,Finance,,Main Office:/home/dave:/bin/ksh
emma:x:1002:1000:Emma Jones,Finance,,Main Office:/home/emma:/bin/bash
frank:x:1003:1000:Frank Cassidy,Finance,,Main Office:/home/frank:/bin/bash
grace:x:1004:1000:Grace Kearns,Engineering,,Main Office:/home/grace:/bin/ksh
henry:x:1005:1000:Henry Adams,Sales,,Main Office:/home/henry:/bin/bash
john:x:1006:1000:John Chapel,Sales,,Main Office:/home/john:/bin/bash
```

- List all users in group 1000 (use `sed` to select only the appropriate field) from your `mypasswd` file:

The GID is the fourth field in the `/etc/passwd` file. You might be tempted to try this:

```
$ sed -n /1000/p mypasswd
```

In this case you will also get this line:

```
carol:x:1000:2000:Carol Smith,Finance,,Main Office:/home/carol:/bin/bash
```

You know this is not correct since Carol Smith is a member of GID 2000 and the the match occurred because of the UID. However, you may have noticed that after the GID the next field starts with an upper case character. We can use a regular expression to solve this problem.

```
$ sed -n /:1000:[A-Z]/p mypasswd
```

The expression `[A-Z]` will match any single upper case character. You will learn more about this in the respective lesson.

- List only the full names of all the users for this group (use `sed` and `cut`):

Use the same technique you used to solve the first part of this exercise and pipe it to a `cut` command.

```
$ sed -n /:1000:[A-Z]/p mypasswd | cut -d: -f5
Dave Edwards,Finance,,Main Office
Emma Jones,Finance,,Main Office
```

```
Frank Cassidy,Finance,,,Main Office
Grace Kearns,Engineering,,,Main Office
Henry Adams,Sales,,,Main Office
John Chapel,Sales,,,Main Office
```

Not quite there! Do note how the fields inside your results can be separated by `,`. So we will pipe the output to another `cut` command, using the `,` as a delimiter.

```
$ sed -n /:1000:[A-Z]/p mypasswd | cut -d: -f5 | cut -d, -f1
Dave Edwards
Emma Jones
Frank Cassidy
Grace Kearns
Henry Adams
John Chapel
```

Answers to Explorational Exercises

1. Once more using the `mypasswd` file from the previous exercises, devise a Bash command that will select one individual from the Main Office to win a raffle contest. Use the `sed` command to only print out the lines for the Main Office, and then a `cut` command sequence to retrieve the first name of each user from these lines. Next you will want to randomly sort these names and only print out the top name from the list.

First explore how the parameter `-R` manipulates the output of the `sort` command. Repeat this command a couple times on your machine (note you will need to enclose 'Main Office' within single quotes, so `sed` will handle it as a single string):

```
$ sed -n /'Main Office'/p mypasswd | cut -d: -f5 | cut -d, -f1 | sort -R
```

Here is a solution to the problem:

```
$ sed -n /'Main Office'/p mypasswd | cut -d: -f5 | cut -d, -f1 | sort -R | head -1
```

2. How many people work in Finance, Engineering and Sales? (Consider exploring the `uniq` command.)

Keep building on top of what you learned from the previous exercises. Try the following:

```
$ sed -n /'Main Office'/p mypasswd
$ sed -n /'Main Office'/p mypasswd | cut -d, -f2
```

Notice now we do not care about the `:` as a delimiter. We just want the second field when we split the lines by the `,` characters.

```
$ sed -n /'Main Office'/p mypasswd | cut -d, -f2 | uniq -c
  4 Finance
  1 Engineering
  2 Sales
```

The `uniq` command will only output the unique lines (not the repeating lines) and the parameter `-c` tells `uniq` to count the occurrences of the equal lines. There is a caveat here: `uniq` will only consider adjacent lines. When this is not the case you will have to use the `sort` command.

3. Now you want to prepare a CSV (comma separated values) file so you can easily import, from the `mypasswd` file in the previous example, the file `names.csv` into LibreOffice. The file contents will have the following format:

```
First Name,Last Name,Position
Carol,Smith,Finance
...
John,Chapel,Sales
```

Tip: Use the `sed`, `cut`, and `paste` commands to achieve the desired results. Note that the comma (,) will be the delimiter for this file.

Start with the `sed` and `cut` commands, building on top of what we learned from the previous exercises:

```
$ sed -n '/Main Office'/p mypasswd | cut -d: -f5 | cut -d" " -f1 > firstname
```

Now we have the file `firstname` with the first names of our employees.

```
$ sed -n '/Main Office'/p mypasswd | cut -d: -f5 | cut -d" " -f2 | cut -d, -f1 > lastname
```

Now we have the file `lastname` containing the surnames of each employee.

Next we determine which department each employee works in:

```
$ sed -n '/Main Office'/p mypasswd | cut -d: -f5 | cut -d, -f2 > department
```

Before we work on the final solution, try the following commands to see what type of output they generate:

```
$ cat firstname lastname department
$ paste firstname lastname department
```

And now for the final solution:

```
$ paste firstname lastname department | tr '\t' ,
$ paste firstname lastname department | tr '\t' , > names.csv
```

Here we use the command `tr` to *translate* `\t`, the tab separator, by a `,`. `tr` is quite useful when we need to exchange one character for another. Be sure to review the man pages for both `tr` and `paste`. For example, we can use the `-d` option for the delimiter to make the previous command less complex:

```
$ paste -d, firstname lastname department
```

We used the `paste` command here once we needed to get you familiar with it. However we could have easily performed all of the tasks in a single command chain:

```
$ sed -n /'Main Office'/p mypasswd | cut -d: -f5 | cut -d, -f1,2 | tr ' ' , > names.csv
```

- Suppose that the `names.csv` spreadsheet created in the previous exercise is an important file and we want to make sure nobody will tamper with it from the moment we send it to someone and the moment our recipient receives it. How can we insure the integrity of this file using `md5sum`?

If you look into the man pages for `md5sum`, `sha256sum` and `sha512sum` you will see they all start with the following text:

“compute and check XXX message digest”

Where “XXX” is the algorithm that will be used to create this *message digest*.

We will use `md5sum` as an example and later you can try with the other commands.

```
$ md5sum names.csv
61f0251fcab61d9575b1d0cbf0195e25  names.csv
```

Now, for instance, you can make the file available through a secure ftp service and send the generated *message digest* using another secure means of communication. If the file has been slightly modified the *message digest* will be completely different. Just to prove it, edit `names.csv` and change Jones to James as demonstrated here:

```
$ sed -i.backup s/Jones/James/ names.csv
$ md5sum names.csv
f44a0d68cb480466099021bf6d6d2e65  names.csv
```

Whenever you make files available for download, it is always a good practice to also distribute

a *message digest* correspondent so people who download your file can produce a new *message digest* and check against the original. If you browse through <https://kernel.org> you will find the page <https://mirrors.edge.kernel.org/pub/linux/kernel/v5.x/sha256sums.asc> where you can obtain the sha256sum for all files available for download.

5. You promised yourself that you would read a classic book 100 lines per day and you decided to start with *Mariner and Mystic* by Herman Melville. Devise a command using `split` that will separate this book into sections of 100 lines each. In order to get the book in plain text format, search for it at <https://www.gutenberg.org>.

First we will get the whole book from the Project Gutenberg site, where you can get this and other books that are available in the public domain.

```
$ wget https://www.gutenberg.org/files/50461/50461-0.txt
```

You might need to install `wget` if it is not already installed in your system. Alternatively, you can also use `curl`. Use `less` to verify the book:

```
$ less 50461-0.txt
```

Now we will split the book into chunks of 100 lines each:

```
$ split -l 100 -d 50461-0.txt melville
```

`50461-0.txt` is the file we will be splitting. `melville` will be the prefix for the split files. The `-l 100` specifies the number of lines and the `-d` option tells `split` to number the files (using the provided suffix). You can use `nl` on any of the splitted files (probably not on the last one) and confirm each one of them have 100 lines.

6. Using `ls -l` on the `/etc` directory, what kind of listing do you get? Using the `cut` command on the output of the given `ls` command how would you display only the file names? What about the filename and the owner of the files? Along with the `ls -l` and `cut` commands, utilize the `tr` command to *squeeze* multiple occurrences of a space into a single space to aid in formatting the output with a `cut` command.

The `ls` command by itself will give you just the names of the files. We can, however, prepare the output of the `ls -l` (the long listing) to extract more specific information.

```
$ ls -l /etc | tr -s ' ' ,
```



```
drwxr-xr-x,3,root,root,4096,out,24,16:58,acpi
-rw-r--r--,1,root,root,3028,dez,17,2018,adduser.conf
-rw-r--r--,1,root,root,10,out,2,17:38,adjtime
drwxr-xr-x,2,root,root,12288,out,31,09:40,alternatives
-rw-r--r--,1,root,root,401,mai,29,2017,anacrontab
-rw-r--r--,1,root,root,433,out,1,2017,apg.conf
drwxr-xr-x,6,root,root,4096,dez,17,2018,apm
drwxr-xr-x,3,root,root,4096,out,24,16:58,apparmor
drwxr-xr-x,9,root,root,4096,nov,6,20:20,apparmor.d
```

The `-s` parameter instructs `tr` to shrink the repeated spaces into a single instance of a space. The `tr` command works for any kind of repeating character you specify. Then we replace the spaces with a comma `,`. We actually do not need to replace the spaces in our example so we will just omit the `,`.

```
$ ls -l /etc | tr -s ' '
drwxr-xr-x 3 root root 4096 out 24 16:58 acpi
-rw-r--r-- 1 root root 3028 dez 17 2018 adduser.conf
-rw-r--r-- 1 root root 10 out 2 17:38 adjtime
drwxr-xr-x 2 root root 12288 out 31 09:40 alternatives
-rw-r--r-- 1 root root 401 mai 29 2017 anacrontab
-rw-r--r-- 1 root root 433 out 1 2017 apg.conf
drwxr-xr-x 6 root root 4096 dez 17 2018 apm
drwxr-xr-x 3 root root 4096 out 24 16:58 apparmor
```

If I want just the filenames then all that we need displayed is the ninth field:

```
$ ls -l /etc | tr -s ' ' | cut -d" " -f9
```

For the filename and the owner of a file we will need the ninth and the third fields:

```
$ ls -l /etc | tr -s ' ' | cut -d" " -f9,3
```

What if we just need the folder names and its owner?

```
$ ls -l /etc | grep ^d | tr -s ' ' | cut -d" " -f9,3
```

7. This exercise assumes you are on a real machine (not a virtual machine). You must also have a USB stick with you. Review the manual pages for the `tail` command and find out how to

follow a file as text is appended to it. While monitoring the output of a `tail` command on the `/var/log/syslog` file, insert a USB stick. Write out the full command that you would use to get the Product, Manufacturer and the total amount of memory of your USB stick.

```
$ tail -f /var/log/syslog | grep -i 'product\:\|\blocks\|manufacturer'
Nov  8 06:01:35 brod-avell kernel: [124954.369361] usb 1-4.3: Product: Cruzer Blade
Nov  8 06:01:35 brod-avell kernel: [124954.369364] usb 1-4.3: Manufacturer: SanDisk
Nov  8 06:01:37 brod-avell kernel: [124955.419267] sd 2:0:0:0: [sdc] 61056064 512-byte
logical blocks: (31.3 GB/29.1 GiB)
```

Of course this is an example and the results may vary depending on your USB memory stick manufacturer. Notice now we use the `-i` parameter with the `grep` command as we are not sure if the strings we are searching for were in upper or lower case. We also used the `|` as a logical OR so we search for lines containing product OR blocks OR manufacturer.



**Linux
Professional
Institute**

103.3 Perform basic file management

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 103.3

Weight

4

Key knowledge areas

- Copy, move and remove files and directories individually.
- Copy multiple files and directories recursively.
- Remove files and directories recursively.
- Use simple and advanced wildcard specifications in commands.
- Using find to locate and act on files based on type, size or time.
- Usage of tar, cpio and dd.

Partial list of the used files, terms and utilities

- `cp`
- `find`
- `mkdir`
- `mv`
- `ls`
- `rm`
- `rmdir`
- `touch`

- tar
- cpio
- dd
- file
- gzip
- gunzip
- bzip2
- bunzip2
- file globbing



Linux
Professional
Institute

103.3 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.3 Perform basic file management
Lesson:	1 of 2

Introduction

Everything in Linux is a file, so knowing how to manipulate them is very important. In this lesson, we shall cover basic operations on files.

In general, as a Linux user, you will be called upon to navigate through the file system, copy files from one location to another and delete files. We shall also cover the commands associated with file management.

A file is an entity that stores data and programs. It consists of content and meta data (file size, owner, creation date, permissions). Files are organized in directories. A directory is a file that stores other files.

The different types of files include:

Regular files

which store data and programs.

Directories

which contain other files.

Special files

which are used for input and output.

Of course, other kinds of files exist but are beyond the scope of this lesson. Later, we shall discuss how to identify these different file types.

Manipulating Files

Using `ls` to List Files

The `ls` command is one of the most important command line tools you should learn in order to navigate the file system.

In its basic form, `ls` will list file and directory names *only*:

```
$ ls
Desktop Downloads  emp_salary file1  Music  Public  Videos
Documents  emp_name  examples.desktop  file2  Pictures  Templates
```

When used with `-l`, referred to as “long listing” format, it shows file or directory permissions, owner, size, modified date, time and name:

```
$ ls -l
total 60
drwxr-xr-x 2 frank frank 4096 Apr 1 2018 Desktop
drwxr-xr-x 2 frank frank 4096 Apr 1 2018 Documents
drwxr-xr-x 2 frank frank 4096 Apr 1 2018 Downloads
-rw-r--r-- 1 frank frank 21 Sep 7 12:59 emp_name
-rw-r--r-- 1 frank frank 20 Sep 7 13:03 emp_salary
-rw-r--r-- 1 frank frank 8980 Apr 1 2018 examples.desktop
-rw-r--r-- 1 frank frank 10 Sep 1 2018 file1
-rw-r--r-- 1 frank frank 10 Sep 1 2018 file2
drwxr-xr-x 2 frank frank 4096 Apr 1 2018 Music
drwxr-xr-x 2 frank frank 4096 Apr 1 2018 Pictures
drwxr-xr-x 2 frank frank 4096 Apr 1 2018 Public
drwxr-xr-x 2 frank frank 4096 Apr 1 2018 Templates
drwxr-xr-x 2 frank frank 4096 Apr 1 2018 Videos
```

The first character in the output indicates the file type:

-
for a regular file.

d
for a directory.

c
for a special file.

To show the file sizes in a human readable format add the option **-h**:

```
$ ls -lh
total 60K
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Desktop
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Documents
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Downloads
-rw-r--r-- 1 frank frank 21 Sep 7 12:59 emp_name
-rw-r--r-- 1 frank frank 20 Sep 7 13:03 emp_salary
-rw-r--r-- 1 frank frank 8.8K Apr 1 2018 examples.desktop
-rw-r--r-- 1 frank frank 10 Sep 1 2018 file1
-rw-r--r-- 1 frank frank 10 Sep 1 2018 file2
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Music
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Pictures
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Public
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Templates
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Videos
```

To list all files including hidden files (those starting with **.**) use the option **-a**:

```
$ ls -a
.          .dbus  file1  .profile
..         Desktop file2  Public
.bash_history .dmrc  .gconf .sudo_as_admin_successful
```

Configuration files such as **.bash_history** which are by default hidden are now visible.

In general, the **ls** command syntax is given by:

```
ls OPTIONS FILE
```

Where `OPTIONS` are any of the options shown previously (to view all the possible options run `man ls`), and `FILE` is the name of the file or directory's details you wish to list.

NOTE When `FILE` is not specified, the current directory is implied.

Creating, Copying, Moving and Deleting Files

Creating files with `touch`

The `touch` command is the easiest way to create new, empty files. You can also use it to change the timestamps (i.e., modification time) of existing files and directories. The syntax for using `touch` is:

```
touch OPTIONS FILE_NAME(S)
```

Without any options, `touch` would create new files for any file names that are supplied as arguments, provided that files with such names do not already exist. `touch` can create any number of files simultaneously:

```
$ touch file1 file2 file3
```

This would create three new empty files named `file1`, `file2` and `file3`.

Several `touch` options are specifically designed to allow the user to change the timestamps for files. For example, the `-a` option changes only the access time, while the `-m` option changes only the modification time. The use of both options together changes the access and also the modification times to the current time:

```
$ touch -am file3
```

Copying Files with `cp`

As a Linux user, you will often copy files from one location to another. Whether it is moving a music file from one directory to another or a system file, use `cp` for all copy tasks:

```
$ cp file1 dir2
```


This command can be literally interpreted as copy `file1` into directory `dir2`. The result is the presence of `file1` inside `dir2`. For this command to be executed successfully `file1` should be existent in the user's current directory. Otherwise, the system reports an error with the message `No such file or directory`.

```
$ cp dir1/file1 dir2
```

In this case, observe that the path to `file1` is more explicit. The source path can be expressed either as a *relative* or *absolute path*. Relative paths are given in reference to a specific directory, while absolute paths are not given with a reference. Below we shall further clarify this notion.

For the moment, just observe that this command copies `file1` into the directory `dir2`. The path to `file1` is given with more detail since the user is currently not located in `dir1`.

```
$ cp /home/frank/Documents/file2 /home/frank/Documents/Backup
```

In this third case, `file2` located at `/home/frank/Documents` is copied into the directory `/home/frank/Documents/Backup`. The source path provided here is *absolute*. In the two examples above, the source paths are *relative*. When a path starts with the character `/` it is an absolute path, otherwise it is a relative path.

The general syntax for `cp` is:

```
cp OPTIONS SOURCE DESTINATION
```

`SOURCE` is the file to copy and `DESTINATION` the directory into which the file would be copied. `SOURCE` and `DESTINATION` can be specified either as absolute or relative paths.

Moving Files with `mv`

Just like `cp` for copying, Linux provides a command for moving and renaming files. It is called `mv`.

The move operation is analogue to the cut and paste operation you generally perform through a Graphical User Interface (GUI).

If you wish to move a file into a new location, use `mv` in the following way:

```
mv FILENAME DESTINATION_DIRECTORY
```

Here is an example:

```
$ mv myfile.txt /home/frank/Documents
```

The result is that `myfile.txt` is moved into destination `/home/frank/Documents`.

To rename a file, `mv` is used in the following way:

```
$ mv old_file_name new_file_name
```

This changes the name of the file from `old_file_name` to `new_file_name`.

By default, `mv` would not seek your confirmation (technically said “would not prompt”) if you wish to overwrite (rename) an existing file. However, you can allow the system to prompt, by using the option `-i`:

```
$ mv -i old_file_name new_file_name
mv: overwrite 'new_file_name'?
```

This command would ask the user’s permission before overwriting `old_file_name` to `new_file_name`.

Conversely, using the `-f`:

```
$ mv -f old_file_name new_file_name
```

would forcefully overwrite the file, without asking any permission.

Deleting Files with `rm`

`rm` is used to delete files. Think of it as an abbreviated form of the word “remove”. Note that the action of removing a file is usually irreversible thus this command should be used with caution.

```
$ rm file1
```

This command would delete `file1`.

```
$ rm -i file1
```

```
rm: remove regular file 'file1'?
```

This command would request the user for confirmation before deleting `file1`. Remember, we saw the `-i` option when using `mv` above.

```
$ rm -f file1
```

This command forcefully deletes `file1` without seeking your confirmation.

Multiple files can be deleted at the same time:

```
$ rm file1 file2 file3
```

In this example `file1`, `file2` and `file3` are deleted simultaneously.

The syntax for `rm` is generally given by:

```
rm OPTIONS FILE
```

Creating and Deleting Directories

Creating Directories with `mkdir`

Creating directories is critical to organizing your files and folders. Files may be grouped together in a logical way by keeping them inside a directory. To create a directory, use `mkdir`:

```
mkdir OPTIONS DIRECTORY_NAME
```

where `DIRECTORY_NAME` is the name of the directory to be created. Any number of directories can be created simultaneously:

```
$ mkdir dir1
```

would create the directory `dir1` in the user's current directory.

```
$ mkdir dir1 dir2 dir3
```

The preceding command would create three directories `dir1`, `dir2` and `dir3` at the same time.

To create a directory together with its subdirectories use the option `-p` (“parents”):

```
$ mkdir -p parents/children
```

This command would create the directory structure `parents/children`, i.e. it would create the directories `parents` and `children`. `children` would be located inside `parents`.

Removing Directories with `rmdir`

`rmdir` deletes a directory *if it is empty*. Its syntax is given by:

```
rmdir OPTIONS DIRECTORY
```

where `DIRECTORY` could be a single argument or a list of arguments.

```
$ rmdir dir1
```

This command would delete `dir1`.

```
$ rmdir dir1 dir2
```

This command would simultaneously delete `dir1` and `dir2`.

You may remove a directory with its subdirectory:

```
$ rmdir -p parents/children
```

This would remove the directory structure `parents/children`. Note that if any of the directories are not empty, they will not be deleted.

Recursive Manipulation of Files and Directories

To manipulate a directory and its contents, you need to apply *recursion*. Recursion means, do an action and repeat that action all down the directory tree. In Linux, the options `-r` or `-R` or `--recursive` are generally associated with recursion.

The following scenario would help you better understand recursion:

You list the contents of a directory `students`, which contains two subdirectories `level 1` and `level 2` and the file named `frank`. By applying recursion, the `ls` command would list the content of `students` i.e. `level 1`, `level 2` and `frank`, but would not end there. It would equally enter subdirectories `level 1` and `level 2` and list their contents and so on down the directory tree.

Recursive Listing with `ls -R`

`ls -R` is used to list the contents of a directory together with its subdirectories and files.

```
$ ls -R mydirectory
mydirectory/:
file1  newdirectory

mydirectory/newdirectory:
```

In the listing above, `mydirectory` including all its content are listed. You can observe `mydirectory` contains the subdirectory `newdirectory` and the file `file1`. `newdirectory` is empty that is why no content is shown.

In general, to list the contents of a directory including its subdirectories, use:

```
ls -R DIRECTORY_NAME
```

Adding a trailing slash to `DIRECTORY_NAME` has no effect:

```
$ ls -R animal
```

is similar to

```
$ ls -R animal/
```

Recursive Copy with `cp -r`

`cp -r` (or `-R` or `--recursive`) allows you to copy a directory together with its all subdirectories and files.

```
$ tree mydir
```

```
mydir
|_file1
|_newdir
    |_file2
    |_insidenew
        |_lastdir

3 directories, 2 files
$ mkdir newcopy
$ cp mydir newcopy
cp: omitting directory 'mydir'
$ cp -r mydir newcopy
* tree newcopy
newcopy
|_mydir
    |_file1
    |_newdir
        |_file2
        |_insidenew
            |_lastdir

4 directories, 2 files
```

In the listing above, we observe that trying to copy `mydir` into `newcopy`, using `cp` without `-r`, the system displays the message `cp: omitting directory 'mydir'`. However, by adding the option `-r` all the contents of `mydir` including itself are copied into `newcopy`.

To copy directories and subdirectories use:

```
cp -r SOURCE DESTINATION
```

Recursive Deletion with `rm -r`

`rm -r` will remove a directory and all its contents (subdirectories and files).

WARNING

Be very careful with either the `-r` or the option combination of `-rf` when used with the `rm` command. A recursive remove command on an important system directory could render the system unusable. Employ the recursive remove command only when absolutely certain that the contents of a directory are safe to remove from a computer.

In trying to delete a directory without using `-r` the system would report an error:

```
$ rm newcopy/  
rm: cannot remove 'newcopy/': Is a directory  
$ rm -r newcopy/
```

You have to add `-r` as in the second command for the deletion to take effect.

NOTE

You may be wondering why we do not use `rmdir` in this case. There is a subtle difference between the two commands. `rmdir` would succeed in deleting only if the given directory is empty whereas `rm -r` can be used irrespective of whether this directory is empty or not.

Add the option `-i` to seek confirmation before the file is deleted:

```
$ rm -ri mydir/  
rm: remove directory 'mydir/'?
```

The system prompts before trying to delete `mydir`.

File Globbing and Wildcards

File *globbing* is a feature provided by the Unix/Linux shell to represent multiple filenames by using special characters called *wildcards*. Wildcards are essentially symbols which may be used to substitute for one or more characters. They allow, for example, to show all files that start with the letter `A` or all files that end with the letters `.conf`.

Wildcards are very useful as they can be used with commands such as `cp`, `ls` or `rm`.

The following are some examples of file globbing:

`rm *`

Delete all files in current working directory.

`ls l?st`

List all files with names beginning with `l` followed by any single character and ending with `st`.

`rmdir [a-z]*`

Remove all directories whose name starts with a letter.

Types of Wildcards

There are three characters that can be used as wildcards in Linux:

*** (asterisk)**

which represents zero, one or more occurrences of any character.

? (question mark)

which represents a single occurrence of any character.

[] (bracketed characters)

which represents any occurrence of the character(s) enclosed in the square brackets. It is possible to use different types of characters whether numbers, letters, other special characters. For example, the expression `[0-9]` matches all digits.

The Asterisk

An asterisk (*) matches zero, one or more occurrences of any character.

For example:

```
$ find /home -name *.png
```

This would find all files that end with `.png` such as `photo.png`, `cat.png`, `frank.png`. The `find` command will be explored further in a following lesson.

Similarly:

```
$ ls lpic-*.txt
```

would list all text files that start with the characters `lpic-` followed by any number of characters and end with `.txt`, such as `lpic-1.txt` and `lpic-2.txt`.

The asterisk wildcard can be used to manipulate (copy, delete or move) all the contents of a directory:

```
$ cp -r animal/* forest
```

In this example, all the contents of `animal` is copied into `forest`.

In general to copy all the contents of a directory we use:

```
cp -r SOURCE_PATH/* DEST_PATH
```

where `SOURCE_PATH` can be omitted if we are already in the required directory.

The asterisk, just as any other wildcard, could be used repeatedly in the same command and at any position:

```
$ rm *ate*
```

Filenames prefixed with zero, one or more occurrence of any character, followed by the letters `ate` and ending with zero, one or more occurrence of any character will be removed.

The Question Mark

The question mark (?) matches a *single* occurrence of a character.

Consider the listing:

```
$ ls
last.txt    lest.txt    list.txt    third.txt   past.txt
```

To return only the files that start with `l` followed by any single character and the characters `st.txt`, we use the question mark (?) wildcard:

```
$ ls l?st.txt
last.txt    lest.txt    list.txt
```

Only the files `last.txt`, `lest.txt` and `list.txt` are displayed as they match the given criteria.

Similarly,

```
$ ls ??st.txt
last.txt    lest.txt    list.txt    past.txt
```

output files that are prefixed with any two characters followed by the text `st.txt`.

Bracketed Characters

The bracketed wildcards matches any occurrence of the character(s) enclosed in the square brackets:

```
$ ls l[ae]st.txt
last.txt  lest.txt
```

This command would list all files starting with `l` followed by *any* of the characters in the set `ae` and ending with `st.txt`.

The square brackets could also take ranges:

```
$ ls l[a-z]st.txt
last.txt  lest.txt  list.txt
```

This outputs all files with names starting with `l` followed by any lower case letter in the range `a` to `z` and ending with `st.txt`.

Multiple ranges could also be applied in the square brackets:

```
$ ls
student-1A.txt student-2A.txt student-3.txt
$ ls student-[0-9][A-Z].txt
student-1A.txt student-2A.txt
```

The listing shows a school directory with a list of registered students. To list only those students whose registration numbers meet the following criteria:

- begin with `student-`
- followed by a number, and an uppercase character
- and end with `.txt`

Combining Wildcards

Wildcards can be combined as in:

```
$ ls
last.txt  lest.txt  list.txt  third.txt  past.txt
```

```
$ ls [plf]?st*  
last.txt    lest.txt    list.txt    past.txt
```

The first wildcard component (`[plf]`) matches any of the characters `p`, `l` or `f`. The second wildcard component (`?`) matches any single character. The third wildcard component (`*`) matches zero, one or many occurrences of any character.

```
$ ls  
file1.txt file.txt file23.txt fom23.txt  
$ ls f*[0-9].txt  
file1.txt file23.txt fom23.txt
```

The previous command displays all files that begin with the letter `f`, followed by any set of letters, at least one occurrence of a digit and ends with `.txt`. Note that `file.txt` is not displayed as it does not match this criteria.

Guided Exercises

1. Consider the listing below:

```
$ ls -lh
total 60K
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Desktop
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Documents
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Downloads
-rw-r--r-- 1 frank frank 21 Sep 7 12:59 emp_name
-rw-r--r-- 1 frank frank 20 Sep 7 13:03 emp_salary
-rw-r--r-- 1 frank frank 8.8K Apr 1 2018 examples.desktop
-rw-r--r-- 1 frank frank 10 Sep 1 2018 file1
-rw-r--r-- 1 frank frank 10 Sep 1 2018 file2
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Music
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Pictures
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Public
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Templates
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Videos
```

- What does the character `d` represent in the output?
- Why are the sizes given in human readable format?
- What would be the difference in the output if `ls` was used with no argument?

2. Consider the command below:

```
$ cp /home/frank/emp_name /home/frank/backup
```

- What would happen to the file `emp_name` if this command is executed successfully?
- If `emp_name` was a directory what option should be added to `cp` to execute the command?
- If `cp` is now changed to `mv` what results do you expect?

3. Consider the listing :

```
$ ls
file1.txt file2.txt file3.txt file4.txt
```

Which wildcard would help to delete all the contents of this directory?

4. Based on the previous listing, what files would be displayed by the following command?

```
$ ls file*.txt
```

5. Complete the command by adding the appropriate digits and characters in the square brackets that would list all the content above:

```
$ ls file[ ].txt
```

Explorational Exercises

1. In your home directory, create the files called `dog` and `cat`.
 2. Still in your home directory, create the directory called `animal`. Move `dog` and `cat` into `animal`.
 3. Go to the `Documents` folder found in your home directory and inside, create the directory `backup`.
 4. Copy `animal` and its contents into `backup`.
 5. Rename `animal` in `backup` to `animal.bkup`.
 6. The `/home/lpi/databases` directory contains many files which includes: `db-1.tar.gz`, `db-2.tar.gz` and `db-3.tar.gz`. Which single command can you use to list only the files mentioned above?
-

7. Consider the listing:

```
$ ls
cne1222223.pdf cne12349.txt cne1234.pdf
```

With the use of a single globbing character, what command would remove only the pdf files?

Summary

In this lesson, we explored how to view what is within a directory with the `ls` command, how to copy (`cp`) files and folders and how to move (`mv`) them as well. We also looked at how new directories can be created with the `mkdir` command. The commands for removing files (`rm`) and folders (`rmdir`) was also discussed.

In this lesson, you also learned about file globbing and wildcards. File globbing is used to represent multiple file names by using special characters called wildcards. The basic wildcards and their meanings:

? (question mark)

represents a single occurrence of a character.

[] (square brackets)

represents any occurrence of the character(s) enclosed in the square brackets.

*** (asterisk)**

represents zero, one or more occurrences of any character.

You may combine any these wildcards in the same statement.

Answers to Guided Exercises

1. Consider the listing below:

```
$ ls -lh
total 60K
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Desktop
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Documents
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Downloads
-rw-r--r-- 1 frank frank 21 Sep 7 12:59 emp_name
-rw-r--r-- 1 frank frank 20 Sep 7 13:03 emp_salary
-rw-r--r-- 1 frank frank 8.8K Apr 1 2018 examples.desktop
-rw-r--r-- 1 frank frank 10 Sep 1 2018 file1
-rw-r--r-- 1 frank frank 10 Sep 1 2018 file2
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Music
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Pictures
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Public
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Templates
drwxr-xr-x 2 frank frank 4.0K Apr 1 2018 Videos
```

- What does the character `d` represent in the output?

`d` is the character that identifies a directory.

- Why are the sizes given in human readable format?

Due the option `-h`.

- What would be the difference in the output if `ls` was used with no argument?

Directory and file names would be provided only.

2. Consider the command below:

```
$ cp /home/frank/emp_name /home/frank/backup
```

- What would happen to the file `emp_name` if this command is executed successfully?

`emp_name` would be copied into `backup`.

- If `emp_name` was a directory what option should be added to `cp` to execute the command?

-r

- If `cp` is now changed to `mv` what results do you expect?

`emp_name` would be moved into `backup`. It would no longer be present inside the home directory of user `frank`.

3. Consider the listing :

```
$ ls
file1.txt file2.txt file3.txt file4.txt
```

Which wildcard would help to delete all the contents of this directory?

The asterisk `*`.

4. Based on the previous listing, what files would be displayed by the following command?

```
$ ls file*.txt
```

All of them, since the asterisk character represents any number of characters.

5. Complete the command by adding the appropriate digits and characters in the square brackets that would list all the content above:

```
$ ls file[ ].txt
```

`file[0-9].txt`

Answers to Explorational Exercises

1. In your home directory, create the files called `dog` and `cat`.

```
$ touch dog cat
```

2. Still in your home directory, create the directory called `animal`. Move `dog` and `cat` into `animal`.

```
$ mkdir animal  
$ mv dog cat -t animal/
```

3. Go to the `Documents` folder found in your home directory and inside, create the directory `backup`.

```
$ cd ~/Documents  
$ mkdir backup
```

4. Copy `animal` and its contents into `backup`.

```
$ cp -r animal ~/Documents/backup
```

5. Rename `animal` in `backup` to `animal.bkup`.

```
$ mv animal/ animal.bkup
```

6. The `/home/lpi/databases` directory contains many files which includes: `db-1.tar.gz`, `db-2.tar.gz` and `db-3.tar.gz`. Which single command can you use to list only the files mention above?

```
$ ls db-[1-3].tar.gz
```

7. Consider the listing:

```
$ ls  
cne1222223.pdf cne12349.txt cne1234.pdf
```

With the use of a single globbing character, what command would remove only the pdf files?

```
$ rm *.pdf
```



103.3 Lesson 2

Certificate:	LPIC-1
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.3 Perform basic file management
Lesson:	2 of 2

Introduction

How to Find Files

As you use your machine, files progressively grow in number and size. Sometimes it becomes difficult to locate a particular file. Fortunately, Linux provides `find` to quickly search and locate files. `find` uses the following syntax:

```
find STARTING_PATH OPTIONS EXPRESSION
```

STARTING_PATH

defines the directory where the search begins.

OPTIONS

controls the behavior and adds specific criteria to optimize the search process.

EXPRESSION

defines the search query.

```
$ find . -name "myfile.txt"
./myfile.txt
```

The starting path in this case is the current directory. The option `-name` specifies that the search is based on the name of the file. `myfile.txt` is the name of the file to search. When using file globbing, be sure to include the expression in quotation marks:

```
$ find /home/frank -name "*.png"
/home/frank/Pictures/logo.png
/home/frank/screenshot.png
```

This command finds all files ending with `.png` starting from `/home/frank/` directory and beneath. If you do not understand the usage of the asterisk (`*`), it is covered in the previous lesson.

Using Criteria to Speed Search

Use `find` to locate files based on *type*, *size* or *time*. By specifying one or more options, the desired results are obtained in less time.

Switches to finding files based on type include:

-type f

file search.

-type d

directory search.

-type l

symbolic link search.

```
$ find . -type d -name "example"
```

This command finds all directories in the current directory and below, that have the name `example`.

Other criteria which could be used with `find` include:

-name

performs a search based on the given name.

-iname

searches based on the name, however, the case is not important (i.e. the test case `myFile` is similar to `MYFILE`).

-not

returns those results that do *not* match the test case.

-maxdepth N

searches the current directory as well as subdirectories `N` levels deep.

Locating Files by Modification Time

`find` also allows to filter a directory hierarchy based on when the file was modified:

```
$ sudo find / -name "*.conf" -mtime 7
/etc/logrotate.conf
```

This command would search for all files in the entire file system (the starting path is the root directory, i.e. `/`) that end with the characters `.conf` and have been modified in the last seven days. This command would require elevated privileges to access directories starting at the base of the system's directory structure, hence the use of `sudo` here. The argument passed to `mtime` represents the *number of days* since the file was last modified.

Locating Files by Size

`find` can also locate files by size. For example, searching for files larger than 2G in `/var`:

```
$ sudo find /var -size +2G
/var/lib/libvirt/images/debian10.qcow2
/var/lib/libvirt/images/rhel8.qcow2
```

The `-size` option displays files of sizes corresponding to the argument passed. Some example arguments include:

-size 100c

files which are exactly 100 bytes.

-size +100k

files taller than 100 kilobytes.

-size -20M

files smaller than 20 megabytes.

-size +2G

files larger than 2 gigabytes.

NOTE To find empty files we can use: `find . -size 0c` or `find . -empty`.

Acting on the Result Set

Once a search is done, it is possible to perform an action on the resulting set by using `-exec`:

```
$ find . -name "*.conf" -exec chmod 644 '{}' \;
```

This filters every object in the current directory (.) and below for file names ending with `.conf` and then executes the `chmod 644` command to modify file permissions on the results.

For now, do not bother with the meaning of `'{}' \;` as it will be discussed later.

Using grep to Filter for Files Based on Content

`grep` is used to search for the occurrence of a keyword.

Consider a situation where we are to find files based on content:

```
$ find . -type f -exec grep "lpi" '{}' \; -print
./bash_history
Alpine/M
helping/M
```

This would search every object in the current directory hierarchy (.) that is a file (`-type f`) and then executes the command `grep "lpi"` for every file that satisfies the conditions. The files that match these conditions are printed on the screen (`-print`). The curly braces ({}) are a placeholder for the `find` match results. The {} are enclosed in single quotes (') to avoid passing `grep` files with names containing special characters. The `-exec` command is terminated with a semicolon (;), which should be escaped (\;) to avoid interpretation by the shell.

Adding the option `-delete` to the end of an expression would delete all files that match. This option should be used when you are certain that the results only match the files that you wish to delete.

In the example below, `find` locates all files in the hierarchy starting at the current directory then deletes all files that end with the characters `.bak`:

```
$ find . -name "*.bak" -delete
```

Archiving Files

The `tar` Command (Archiving and Compression)

The `tar` command, short for “tape archive(r)”, is used to create tar archives by converting a group of files into an archive. Archives are created so as to easily move or backup a group of files. Think of `tar` as a tool that creates a glue onto which files can be attached, grouped and easily moved.

`tar` also has the ability to extract tar archives, display a list of the files included in the archive as well as add additional files to an existing archive.

The `tar` command syntax is as follows:

```
tar [OPERATION_AND_OPTIONS] [ARCHIVE_NAME] [FILE_NAME(S)]
```

OPERATION

Only one operation argument is allowed and required. The most frequently used operations are:

--create (-c)

Create a new tar archive.

--extract (-x)

Extract the entire archive or one or more files from an archive.

--list (-t)

Display a list of the files included in the archive.

OPTIONS

The most frequently used options are:

--verbose (-v)

Show the files being processed by the `tar` command.

--file=archive-name (-f archive-name)

Specifies the archive file name.

ARCHIVE_NAME

The name of the archive.

FILE_NAME(S)

A space-separated list of file names to be extracted. If not provided the entire archive is extracted.

Creating an Archive

Let's say we have a directory named `stuff` in the current directory and we want to save it to a file named `archive.tar`. We would run the following command:

```
$ tar -cvf archive.tar stuff
stuff/
stuff/service.conf
```

Here's what those switches actually mean:

-c

Create an archive.

-v

Display progress in the terminal while creating the archive, also known as “verbose” mode. The `-v` is always optional in these commands, but it is helpful.

-f

Allows to specify the filename of the archive.

In general to archive a single directory or a single file on Linux, we use:

```
tar -cvf NAME-OF-ARCHIVE.tar /PATH/TO/DIRECTORY-OR-FILE
```

NOTE

`tar` works recursively. It will perform the required action on every subsequent directory inside the directory specified.

To archive multiple directories at once, we list all the directories delimiting them by a space in the section `/PATH/TO/DIRECTORY-OR-FILE`:

```
$ tar -cvf archive.tar stuff1 stuff2
```

This would produce an archive of `stuff1` and `stuff2` in `archive.tar`

Extracting an Archive

We can extract an archive using `tar`:

```
$ tar -xvf archive.tar
stuff/
stuff/service.conf
```

This will extract the contents of `archive.tar` to the current directory.

This command is the same as the archive creation command used above, except the `-x` switch that replaces the `-c` switch.

To extract the contents of the archive to a specific directory we use `-C`:

```
$ tar -xvf archive.tar -C /tmp
```

This will extract the contents of `archive.tar` to the `/tmp` directory.

```
$ ls /tmp
stuff
```

Compressing with tar

The GNU `tar` command included with Linux distributions can create a `.tar` archive and then compress it with `gzip` or `bzip2` compression in a single command:

```
$ tar -czvf name-of-archive.tar.gz stuff
```

This command would create a compressed file using the `gzip` algorithm (`-z`).

While `gzip` compression is most frequently used to create `.tar.gz` or `.tgz` files, `tar` also supports `bzip2` compression. This allows the creation of `bzip2` compressed files, often named `.tar.bz2`, `.tar.bz` or `.tbz` files.

To do so, we replace `-z` for `gzip` with `-j` for `bzip2`:

```
$ tar -cjvf name-of-archive.tar.bz stuff
```

To decompress the file, we replace `-c` with `-x`, where `x` stands for “extract”:

```
$ tar -xzvf archive.tar.gz
```

`gzip` is faster, but it generally compresses a bit less, so you get a somewhat larger file. `bzip2` is slower, but it compresses a bit more, so you get a somewhat smaller file. In general, though, `gzip` and `bzip2` are practically the same thing and both will work similarly.

Alternatively we may apply `gzip` or `bzip2` compression using `gzip` command for `gzip` compressions and the `bzip` command for `bzip` compressions. For example, to apply `gzip` compression, use:

```
gzip FILE-TO-COMPRESS
```

`gzip`

creates the compressed file with the same name but with a `.gz` ending.

`gzip`

removes the original files after creating the compressed file.

The `bzip2` command works in a similar fashion.

To uncompress the files we use either `gunzip` or `bunzip2` depending on the algorithm used to compress a file.

The `cpio` Command

`cpio` stands for “copy in, copy out”. It is used to process archive files such as `*.cpio` or `*.tar` files.

`cpio` performs the following operations:

- Copying files to an archive.
- Extracting files from an archive.

It takes the list of files from the standard input (mostly output from `ls`).

To create a `cpio` archive, we use:

```
$ ls | cpio -o > archive.cpio
```

The `-o` option instructs `cpio` to create an output. In this case, the output file created is `archive.cpio`. The `ls` command lists the contents of the current directory which are to be archived.

To extract the archive we use :

```
$ cpio -id < archive.cpio
```

The `-i` option is used to perform the extract. The `-d` option would create the destination folder. The character `<` represents standard input. The input file to be extracted is `archive.cpio`.

The `dd` Command

`dd` copies data from one location to another. The command line syntax of `dd` differs from many other Unix programs, it uses the syntax `option=value` for its command line options rather than the GNU standard `-option value` or `--option=value` formats:

```
$ dd if=oldfile of=newfile
```

This command would copy the content of `oldfile` into `newfile`, where `if=` is the input file and `of=` refers to the output file.

NOTE

The `dd` command typically will not output anything to the screen until the command has finished. By providing the `status=progress` option, the console will display the amount of work getting done by the command. For example: `dd status=progress if=oldfile of=newfile`.

`dd` is also used in changing data to upper/lower case or writing directly to block devices such as `/dev/sdb`:

```
$ dd if=oldfile of=newfile conv=ucase
```

This would copy all the contents of `oldfile` into `newfile` and capitalise all of the text.

The following command will backup the whole hard disk located at `/dev/sda` to a file named

backup.dd:

```
$ dd if=/dev/sda of=backup.dd bs=4096
```

Guided Exercises

1. Consider the following listing:

```
$ find /home/frank/Documents/ -type d
/home/frank/Documents/
/home/frank/Documents/animal
/home/frank/Documents/animal/domestic
/home/frank/Documents/animal/wild
```

- What kind of files would this command output?

- In which directory does the search begin?

2. A user wishes to compress his backup folder. He uses the following command:

```
$ tar cvf /home/frank/backup.tar.gz /home/frank/dir1
```

Which option is lacking to compress the backup using the `gzip` algorithm?

Explorational Exercises

1. As system administrator, it is required to perform regular checks in order to remove voluminous files. These voluminous files are located in `/var` and end with a `.backup` extension.

- Write down the command, using `find`, to locate these files:

- An analysis of the sizes of these files reveals that they range from `100M` to `1000M`. Complete the previous command with this new information, so that you may locate those backup files ranging from `100M` to `1000M`:

- Finally, complete this command, with the delete action so that these files will be removed:

2. In the `/var` directory, there exist four backup files:

```
db-jan-2018.backup  
db-feb-2018.backup  
db-march-2018.backup  
db-apr-2018.backup
```

- Using `tar`, specify the command that would create an archive file with the name `db-first-quarter-2018.backup.tar`:

- Using `tar`, specify the command that would create the archive and compress it using `gzip`. Take note that the resulting file name should end with `.gz`:

Summary

In this section, you learned:

- How to find files with `find`.
- How to add search criteria based on time, file type or size by supplying argument to `find`.
- How to act on a returned set.
- How to archive, compress and decompress files using `tar`.
- Processing archives with `cpio`.
- Copying files with `dd`.

Answers to Guided Exercises

1. Consider the following listing:

```
$ find /home/frank/Documents/ -type d
/home/frank/Documents/
/home/frank/Documents/animal
/home/frank/Documents/animal/domestic
/home/frank/Documents/animal/wild
```

- What kind of files would this command output?

Directories.

- In which directory does the search begins?

/home/frank/Documents

2. A user wishes to compress his backup folder. He uses the following command:

```
$ tar cvf /home/frank/backup.tar.gz /home/frank/dir1
```

Which option is lacking to compress the backup using the `gzip` algorithm?

Option `-z`.

Answers to Explorational Exercises

1. As system administrator, it is required of you to perform regular checks in order to remove voluminous files. These voluminous files are located in `/var` and end with a `.backup` extension.

- Write down the command, using `find`, to locate these files:

```
$ find /var -name *.backup
```

- An analysis of the sizes of these files reveals that they range from `100M` to `1000M`. Complete the previous command with this new information, so that you may locate those backup files ranging from `100M` to `1000M`:

```
$ find /var -name *.backup -size +100M -size -1000M
```

- Finally, complete this command, with the delete action so that these files will be removed:

```
$ find /var -name *.backup -size +100M -size -1000M -delete
```

2. In the `/var` directory, there exist four backup files:

```
db-jan-2018.backup  
db-feb-2018.backup  
db-march-2018.backup  
db-apr-2018.backup
```

- Using `tar`, specify the command that would create an archive file with the name `db-first-quarter-2018.backup.tar`:

```
$ tar -cvf db-first-quarter-2018.backup.tar db-jan-2018.backup db-feb-2018.backup db-march-2018.backup db-apr-2018.backup
```

- Using `tar`, specify the command that would create the archive and compress it using `gzip`. Take note that the resulting file name should end with `.gz`:

```
$ tar -zcvf db-first-quarter-2018.backup.tar.gz db-jan-2018.backup db-feb-2018.backup
```

```
db-march-2018.backup db-apr-2018.backup
```



103.4 Use streams, pipes and redirects

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 103.4

Weight

4

Key knowledge areas

- Redirecting standard input, standard output and standard error.
- Pipe the output of one command to the input of another command.
- Use the output of one command as arguments to another command.
- Send output to both stdout and a file.

Partial list of the used files, terms and utilities

- tee
- xargs



103.4 Lesson 1

Certificate:	LPIC-1 (101)
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.4 Use streams, pipes and redirects
Lesson:	1 of 2

Introduction

All computer programs follow the same general principle: data received from some source is transformed to generate an intelligible outcome. In Linux shell context, the data source can be a local file, a remote file, a device (like a keyboard), etc. The program's output is usually rendered on a screen, but is also common to store the output data in a local filesystem, send it to a remote device, play it through audio speakers, etc.

Operating systems inspired by Unix, like Linux, offer a great variety of input/output methods. In particular, the method of *file descriptors* allows to dynamically associate integer numbers with data channels, so that a process can reference them as its input/output data streams.

Standard Linux processes have three communication channels opened by default: the *standard input* channel (most times simply called *stdin*), the *standard output* channel (*stdout*) and the *standard error* channel (*stderr*). The numerical file descriptors assigned to these channels are `0` to *stdin*, `1` to *stdout* and `2` to *stderr*. Communication channels are also accessible through the special devices `/dev/stdin`, `/dev/stdout` and `/dev/stderr`.

These three standard communication channels allow programmers to write code that reads and

writes data without worrying about the kind of media it's coming from or going to. For example, if a program needs a set of data as its input, it can just ask for data from the standard input and whatever is being used as the standard input will provide that data. Likewise, the simplest method a program can use to display its output is to write it in the standard output. In a standard shell session, the keyboard is defined as the `stdin` and the monitor screen is defined as the `stdout` and `stderr`.

The Bash shell has the ability to reassign the communication channels when loading a program. It allows, for example, to override the screen as the standard output and use a file in the local filesystem as `stdout`.

Redirects

The reassignment of a channel's file descriptor in the shell environment is called a *redirect*. A redirect is defined by a special character within the command line. For example, to redirect the standard output of a process to a file, the *greater than* symbol `>` is positioned at the end of the command and followed by the path to the file that will receive the redirected output:

```
$ cat /proc/cpuinfo >/tmp/cpu.txt
```

By default, only the content coming to `stdout` is redirected. That happens because the numerical value of the file descriptor should be specified just before the greater than symbol and, when not specified, Bash redirects the standard output. Therefore, using `>` is equivalent to use `1>` (the value of `stdout`'s file descriptor is 1).

To capture the content of `stderr`, the redirect `2>` should be used instead. Most command-line programs send debug information and error messages to the standard error channel. It is possible, for example, to capture the error message triggered by an attempt to read a non-existent file:

```
$ cat /proc/cpu_info 2>/tmp/error.txt
$ cat /tmp/error.txt
cat: /proc/cpu_info: No such file or directory
```

Both `stdout` and `stderr` are redirected to the same target with `&>` or `>&`. It's important to not place any spaces beside the ampersand, otherwise Bash will take it as the instruction to run the process in background and not to perform the redirect.

The target must be a path to a writable file, like `/tmp/cpu.txt`, or a writable file descriptor. A file descriptor target is represented by an ampersand followed by the file descriptor's numerical

value. For example, `1>&2` redirects stdout to stderr. To do the opposite, stderr to stdout, `2>&1` should be used instead.

Although not very useful, given that there is a shorter way to do the same task, it is possible to redirect stderr to stdout and then redirect it to a file. For example, a redirect to write both stderr and stdout to a file named `log.txt` can be written as `>log.txt 2>&1`. However, the main reason for redirecting stderr to stdout is to allow parsing of debug and error messages. It is possible to redirect the standard output of a program to the standard input of another program, but it is not possible to directly redirect the standard error to the standard input of another program. Thus, program's messages sent to stderr first need to be redirected to stdout in order to be read by another program's stdin.

To just discard the output of a command, its content can be redirected to the special file `/dev/null`. For example, `>log.txt 2>/dev/null` saves the contents of stdout in the file `log.txt` and discards the stderr. The file `/dev/null` is writable by any user but no data can be recovered from it, as it is not stored anywhere.

An error message is presented if the specified target is not writable (if the path points to a directory or a read-only file) and no modification to the target is made. However, an output redirect overwrites an existing writable target without any confirmation. Files are overwritten by output redirects unless Bash option `noclobber` is enabled, which can be done for the current session with the command `set -o noclobber` or `set -C`:

```
$ set -o noclobber
$ cat /proc/cpu_info 2>/tmp/error.txt
-bash: /tmp/error.txt: cannot overwrite existing file
```

To unset the `noclobber` option for the current session, run `set +o noclobber` or `set +C`. To make the `noclobber` option persistent, it must be included in the user's Bash profile or in the system-wide profile.

Even with the `noclobber` option enabled it is possible to append redirected data to existent content. This is accomplished with a redirection written with two greater than symbols `>>`:

```
$ cat /proc/cpu_info 2>>/tmp/error.txt
$ cat /tmp/error.txt
cat: /proc/cpu_info: No such file or directory
cat: /proc/cpu_info: No such file or directory
```

In the previous example, the new error message was appended to the existing one in file

/tmp/error.txt. If the file does not exist yet, it will be created with the new data.

The data source of the standard input of a process can be reassigned as well. The less than symbol < is used to redirect the content of a file to the stdin of a process. In this case, data flows from right to left: the reassigned descriptor is assumed to be 0 at the left of the less than symbol and the data source (a path to a file) must be at the right of the less than symbol. The command `uniq`, like most command line utilities for processing text, accepts data sent to stdin by default:

```
$ uniq -c </tmp/error.txt
  2 cat: /proc/cpu_info: No such file or directory
```

The `-c` option makes `uniq` display how many times a repeated line appears in the text. As the numeric value of the redirected file descriptor was suppressed, the example command is equivalent to `uniq -c 0</tmp/error.txt`. To use a file descriptor other than 0 in an input redirect only makes sense in specific contexts, because it is possible for a program to ask for data at file descriptors 3, 4, etc. Indeed, programs can use any integer above 2 as new file descriptors for data input/output. For example, the following C code reads data from file descriptor 3 and just replicates it to file descriptor 4:

NOTE

The program must handle such file descriptors correctly, otherwise it could attempt an invalid read or write operation and crash.

```
#include <stdio.h>

int main(int argc, char **argv){
    FILE *fd_3, *fd_4;
    // Open file descriptor 3
    fd_3 = fdopen(3, "r");
    // Open file descriptor 4
    fd_4 = fdopen(4, "w");
    // Read from file descriptor 3
    char buf[32];
    while ( fgets(buf, 32, fd_3) != NULL ){
        // Write to file descriptor 4
        fprintf(fd_4, "%s", buf);
    }
    // Close both file descriptors
    fclose(fd_3);
    fclose(fd_4);
}
```


To test it, save the sample code as `fd.c` and compile it with `gcc -o fd fd.c`. This program needs file descriptors 3 and 4 to be available so it can read and write to them. As an example, the previously created file `/tmp/error.txt` can be used as the source for file descriptor 3 and the file descriptor 4 can be redirected to stdout:

```
$ ./fd 3</tmp/error.txt 4>&1
cat: /proc/cpu_info: No such file or directory
cat: /proc/cpu_info: No such file or directory
```

From a programmer's perspective, using file descriptors avoids having to deal with option parsing and filesystem paths. The same file descriptor can even be used as input and output. In this case, the file descriptor is defined in the command line with both less than and greater than symbols, like in `3<>/tmp/error.txt`.

Here Document and Here String

Another way to redirect input involve the *Here document* and *Here string* methods. The Here document redirect allows to type multi-line text that will be used as the redirected content. Two less than symbols `<<` indicate a Here document redirect:

```
$ wc -c <<EOF
> How many characters
> in this Here document?
> EOF
43
```

At the right of the two less than symbols `<<` is the ending term EOF. The insertion mode will finish as soon as a line containing only the ending term is entered. Any other term can be used as the ending term, but it is important to not put blank characters between the less than symbol and the ending term. In the example above, the two lines of text were sent to the stdin of `wc -c` command, which displays the characters count. As with input redirects for files, the stdin (file descriptor 0) is assumed if the redirected file descriptor is suppressed.

The Here string method is much like the Here document method, but for one line only:

```
$ wc -c <<< "How many characters in this Here string?"
41
```

In this example, the string is sent to the stdin of `wc -c`, which counts the number of characters.

Strings containing spaces must be inside quotes, otherwise only the first word will be used as the Here string and the remaining ones will be passed as arguments to the command.

Guided Exercises

1. In addition to text files, the command `cat` can also work with binary data, like sending the contents of a block device to a file. Using redirection, how can `cat` send the contents of device `/dev/sdc` to the file `sd.c.img` in the current directory?

2. What's the name of the standard channel redirected by the command `date 1> now.txt`?

3. After trying to overwrite a file using redirection, a user gets an error informing that option `noclobber` is enabled. How can the option `noclobber` be deactivated for the current session?

4. What will be the result of command `cat <<./dev/stdout`?

Explorational Exercises

1. The command `cat /proc/cpu_info` displays an error message because `/proc/cpu_info` is nonexistent. The command `cat /proc/cpu_info 2>1` redirects the error message to where?

2. Will it still be possible to discard content sent to `/dev/null` if the `noclobber` option is enabled for the current shell session?

3. Using `echo`, how could the contents of the variable `$USER` be redirected to the stdin of command `sha1sum`?

4. The Linux kernel keeps symbolic links in `/proc/PID/fd/` to every file opened by a process, where `PID` is the identification number of the corresponding process. How could the system administrator use that directory to verify the location of log files opened by `nginx`, supposing its PID is `1234`?

5. It's possible to do arithmetic calculations using only shell builtin commands, but floating point calculations require specific programs, like `bc` (*basic calculator*). With `bc` it's even possible to specify the number of decimal places, with parameter `scale`. However, `bc` accepts operations only through its standard input, usually entered in interactive mode. Using a Here string, how can the floating point operation `scale=6; 1/3` be sent to the standard input of `bc`?

Summary

This lesson covers methods to run a program redirecting its standard communication channels. Linux processes use these standard channels as generic *file descriptors* to read and to write data, making it possible to arbitrarily change them to files or devices. The lesson goes through the following steps:

- What file descriptors are and the role they play in Linux.
- The standard communication channels of every process: *stdin*, *stdout* and *stderr*.
- How to correctly execute a command using data redirection, both for input and output.
- How to use *Here Documents* and *Here Strings* in input redirections.

The commands and procedures addressed were:

- Redirection operators: `>`, `<`, `>>`, `<<`, `<<<`.
- Commands `cat`, `set`, `uniq` and `wc`.

Answers to Guided Exercises

1. In addition to text files, the command `cat` can also work with binary data, like sending the contents of a block device to a file. Using redirection, how can `cat` send the contents of device `/dev/sdc` to the file `sd.c.img` in the current directory?

```
$ cat /dev/sdc > sd.c.img
```

2. What's the name of the standard channel redirected by the command `date 1> now.txt`?

Standard output or stdout

3. After trying to overwrite a file using redirection, a user gets an error informing that the option `noclobber` is enabled. How can the option `noclobber` be deactivated for the current session?

`set +C` or `set +o noclobber`

4. What will be the result of command `cat <<./dev/stdout`?

Bash will enter Heredoc input mode, then exit when a period appears in a line by itself. The typed text will be redirected to stdout (printed on screen).

Answers to Explorational Exercises

1. The command `cat /proc/cpu_info` displays an error message because `/proc/cpu_info` is nonexistent. The command `cat /proc/cpu_info 2>1` redirects the error message to where?

To a file named `1` in the current directory.

2. Will it still be possible to discard content sent to `/dev/null` if the `noclobber` option is enabled for the current shell session?

Yes. `/dev/null` is a special file not affected by `noclobber`.

3. Using `echo`, how can the contents of the variable `$USER` be redirected to the stdin of command `sha1sum`?

```
$ echo -n "$USER" | sha1sum
```

4. The Linux kernel keeps symbolic links in `/proc/PID/fd/` to every file opened by a process, where `PID` is the identification number of the corresponding process. How could the system administrator use that directory to verify the location of log files opened by `nginx`, supposing its `PID` is `1234`?

By issuing the command `ls -l /proc/1234/fd`, which will display the targets of every symbolic link in the directory.

5. It's possible to do arithmetic calculations using only shell builtin commands, but floating point calculations require specific programs, like `bc` (*basic calculator*). With `bc` it's even possible to specify the number of decimal places, with parameter `scale`. However, `bc` accepts operations only through its standard input, usually entered in interactive mode. Using a Here string, how could the floating point operation `scale=6; 1/3` be sent to the standard input of `bc`?

```
$ bc <<<"scale=6; 1/3"
```



103.4 Lesson 2

Certificate:	LPIC-1 (101)
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.4 Use streams, pipes and redirects
Lesson:	2 of 2

Introduction

One aspect of the Unix philosophy states that each program should have a specific purpose and should not try to incorporate features outside its scope. But keeping things simple does not mean less elaborated results, because different programs can be chained together to produce a combined output. The vertical bar character `|`, also known as the *pipe* symbol, can be used to create a pipeline connecting the output of a program directly into the input of another program, whereas *command substitution* allows to store the output of a program in a variable or use it directly as an argument to another command.

Pipes

Unlike redirects, with pipes data flows from left to right in the command line and the target is another process, not a filesystem path, file descriptor or Here document. The pipe character `|` tells the shell to start all distinct commands at the same time and to connect the output of the previous command to the input of the following command, left to right. For example, instead of using redirects, the content of the file `/proc/cpuinfo` sent to the standard output by `cat` can be piped to the stdin of `wc` with the following command:


```
$ cat /proc/cpuinfo | wc
208      1184      6096
```

In the absence of a path to a file, `wc` counts the number of lines, words and characters it receives on its stdin, as is the case in the example. Many pipes can be present in a compound command. In the following example, two pipes are used:

```
$ cat /proc/cpuinfo | grep 'model name' | uniq
model name      : Intel(R) Xeon(R) CPU           X5355   @ 2.66GHz
```

The content of file `/proc/cpuinfo` produced by `cat /proc/cpuinfo` was piped to the command `grep 'model name'`, which then selects only the lines containing the term `model name`. The machine running the example has many CPUs, so there are repeated lines with `model name`. The last pipe connects `grep 'model name'` to `uniq`, which is responsible for skipping any line equal to the previous one.

Pipes can be combined with redirects in the same command line. The previous example can be rewritten to a simpler form:

```
$ grep 'model name' </proc/cpuinfo | uniq
model name      : Intel(R) Xeon(R) CPU           X5355   @ 2.66GHz
```

The input redirect for `grep` is not strictly necessary as `grep` accepts a file path as argument, but the example demonstrates how to build such combined commands.

Pipes and redirects are exclusive, that is, one source can be mapped to only one target. Yet, it is possible to redirect an output to a file and still see it on the screen with program `tee`. To do it, the first program sends its output to the stdin of `tee` and a file name is provided to the latter to store the data:

```
$ grep 'model name' </proc/cpuinfo | uniq | tee cpu_model.txt
model name      : Intel(R) Xeon(R) CPU           X5355   @ 2.66GHz
$ cat cpu_model.txt
model name      : Intel(R) Xeon(R) CPU           X5355   @ 2.66GHz
```

The output of the last program in the chain, generated by `uniq`, is displayed and stored in the file `cpu_model.txt`. To not overwrite the content of the provided file but to append data to it, the option `-a` must be provided to `tee`.

Only the standard output of a process is captured by a pipe. Let's say you must go through a long compilation process on the screen and at the same time save both the standard output and the standard error to a file for later inspection. Assuming your current directory does not have a *Makefile*, the following command will output an error:

```
$ make | tee log.txt
make: *** No targets specified and no makefile found. Stop.
```

Although shown on the screen, the error message generated by `make` was not captured by `tee` and the file `log.txt` was created empty. A redirect needs to be done before a pipe can capture the `stderr`:

```
$ make 2>&1 | tee log.txt
make: *** No targets specified and no makefile found. Stop.
$ cat log.txt
make: *** No targets specified and no makefile found. Stop.
```

In this example the `stderr` of `make` was redirected to the `stdout`, so `tee` was able to capture it with a pipe, display it on the screen and save it in the file `log.txt`. In cases like this, it may be useful to save the error messages for later inspection.

Command Substitution

Another method to capture the output of a command is *command substitution*. By placing a command inside backquotes, Bash replaces it with its standard output. The following example shows how to use the `stdout` of a program as an argument to another program:

```
$ mkdir `date +%Y-%m-%d`
$ ls
2019-09-05
```

The output of the program `date`, the current date formatted as *year-month-day*, was used as an argument to create a directory with `mkdir`. An identical result is obtained by using `$()` instead of backquotes:

```
$ rmdir 2019-09-05
$ mkdir $(date +%Y-%m-%d)
$ ls
```

2019-09-05

The same method can be used to store the output of a command as a variable:

```
$ OS=`uname -o`
$ echo $OS
GNU/Linux
```

The command `uname -o` outputs the generic name of the current operating system, which was stored in the session variable `OS`. To assign the output of a command to a variable is very useful in scripts, making it possible to store and evaluate the data in many distinct ways.

Depending on the output generated by the replaced command, the builtin command substitution may not be appropriate. A more sophisticated method to use the output of a program as the argument of another program employs an intermediate called `xargs`. The program `xargs` uses the contents it receives via `stdin` to run a given command with the contents as its argument. The following example shows `xargs` running the program `identify` with arguments provided by program `find`:

```
$ find /usr/share/icons -name 'debian*' | xargs identify -format "%f: %wx%h\n"
debian-swirl.svg: 48x48
debian-swirl.png: 22x22
debian-swirl.png: 32x32
debian-swirl.png: 256x256
debian-swirl.png: 48x48
debian-swirl.png: 16x16
debian-swirl.png: 24x24
debian-swirl.svg: 48x48
```

The program `identify` is part of *ImageMagick*, a set of command-line tools to inspect, convert and edit most image file types. In the example, `xargs` took all paths listed by `find` and put them as arguments to `identify`, which then shows the information for each file formatted as required by the option `-format`. The files found by `find` in the example are images containing the distribution logo in a Debian filesystem. `-format` is a parameter to `identify`, not to `xargs`.

Option `-n 1` requires `xargs` to run the given command with only one argument at a time. In the example's case, instead of passing all paths found by `find` as a list of arguments to `identify`, using `xargs -n 1` would execute command `identify` for each path separately. Using `-n 2` would execute `identify` with two paths as arguments, `-n 3` with three paths as arguments and so on. Similarly, when `xargs` process multiline contents — as is the case with input provided by

`find`—the option `-L` can be used to limit how many lines will be used as arguments per command execution.

NOTE

Using `xargs` with option `-n 1` or `-L 1` to process output generated by `find` may be unnecessary. Command `find` has the option `-exec` to run a given command for each search result item.

If the paths have space characters, it is important to run `find` with the option `-print0`. This option instructs `find` to use a null character between each entry so the list can be correctly parsed by `xargs` (the output was suppressed):

```
$ find . -name '*avi' -print0 -o -name '*mp4' -print0 -o -name '*mkv' -print0 | xargs -0 du  
| sort -n
```

The option `-0` tells `xargs` the null character should be used as the separator. That way the file paths given by `find` are correctly parsed even if they have blank or other special characters in it. The previous example shows how to use the command `du` to find out the disk usage of every file found and then sort the results by size. The output was suppressed for conciseness. Note that for each search criteria it is necessary to place the `-print0` option for `find`.

By default, `xargs` places the arguments of the executed command last. To change that behavior, the option `-I` should be used:

```
$ find . -mindepth 2 -name '*avi' -print0 -o -name '*mp4' -print0 -o -name '*mkv' -print0 |  
xargs -0 -I PATH mv PATH ./
```

In the last example, every file found by `find` is moved to the current directory. As the source path(s) must be informed to `mv` before the target path, a substitution term is given to the option `-I` of `xargs` which is then appropriately placed alongside `mv`. By using the null character as separator, it is not necessary to enclose the substitution term with quotes.

Guided Exercises

1. It's convenient to save the execution date of actions performed by automated scripts. Command `date +%Y-%m-%d` shows the current date in *year-month-day* format. How can the output of such a command be stored in a shell variable called `TODAY` using command substitution?
2. Using command `echo`, how can the contents of variable `TODAY` be sent to the standard input of command `sed s/-/. /g`?
3. How could the output of command `date +%Y-%m-%d` be used as a Here string to command `sed s/-/. /g`?
4. Command `convert image.jpeg -resize 25% small/image.jpeg` creates a smaller version of `image.jpeg` and places the resulting image in a likewise named file inside subdirectory `small`. Using `xargs`, how is it possible to perform the same command for every image listed in file `filelist.txt`?

Explorational Exercises

1. A simple backup routine periodically creates an image of partition `/dev/sda1` with `dd < /dev/sda1 > sda1.img`. To perform future data integrity checks, the routine also generates a SHA1 hash of the file with `sha1sum < sda1.img > sda1.sha1`. By adding pipes and command `tee`, how would these two commands be combined into one?

2. Command `tar` is used to archive many files into a single file, preserving directory structure. Option `-T` allows to specify a file containing the paths to be archived. For example, `find /etc -type f | tar -cJ -f /srv/backup/etc.tar.xz -T -` creates a compressed tar file `etc.tar.xz` from the list provided by command `find` (option `-T -` indicates the standard input as the path list). In order to avoid possible parsing errors due to paths containing spaces, what command options should be present for `find` and `tar`?

3. Instead of opening a new remote shell session, command `ssh` can just execute a command indicated as its argument: `ssh user@storage "remote command"`. Given that `ssh` also allows to redirect the standard output of a local program to the standard input of the remote program, how would the `cat` command pipe a local file named `etc.tar.gz` to `/srv/backup/etc.tar.gz` at `user@storage` through `ssh`?

Summary

This lesson covers traditional inter-process communication techniques employed by Linux. *Command pipelining* creates a one way communication channel between two process and *command substitution* allows to store the output of a process in a shell variable. The lesson goes through the following steps:

- How *pipes* can be used to stream the output of a process to the input of another process.
- The purpose of commands `tee` and `xargs`.
- How to capture the output of a process with *command substitution*, storing it in a variable or using it directly as a parameter to another command.

The commands and procedures addressed were:

- Command pipelining with `|`.
- Command substitution with backticks and `$ ( )`.
- Commands `tee`, `xargs` and `find`.

Answers to Guided Exercises

1. It's convenient to save the execution date of actions performed by automated scripts. Command `date +%Y-%m-%d` shows the current date in *year-month-day* format. How can the output of such a command can be stored in a shell variable called `TODAY` using command substitution?

```
$ TODAY=`date +%Y-%m-%d`
```

OR

```
$ TODAY=$(date +%Y-%m-%d)
```

2. Using command `echo`, how can the contents of variable `TODAY` be sent to the standard input of command `sed s/-/. /g`?

```
$ echo $TODAY | sed s/-/. /g
```

3. How could the output of command `date +%Y-%m-%d` be used as a Here string to command `sed s/-/. /g`?

```
$ sed s/-/. /g <<< `date +%Y-%m-%d`
```

OR

```
$ sed s/-/. /g <<< $(date +%Y-%m-%d)
```

4. Command `convert image.jpeg -resize 25% small/image.jpeg` creates a smaller version of `image.jpeg` and places the resulting image in a likewise named file inside subdirectory `small`. Using `xargs`, how is it possible to perform the same command for every image listed in file `filelist.txt`?

```
$ xargs -I IMG convert IMG -resize 25% small/IMG < filelist.txt
```

OR

```
$ cat filelist.txt | xargs -I IMG convert IMG -resize 25% small/IMG
```


Answers to Explorational Exercises

1. A simple backup routine periodically creates an image of partition `/dev/sda1` with `dd < /dev/sda1 > sda1.img`. To perform future data integrity checks, the routine also generates a SHA1 hash of the file with `sha1sum < sda1.img > sda1.sha1`. By adding pipes and command `tee`, how would these two commands be combined into one?

```
# dd < /dev/sda1 | tee sda1.img | sha1sum > sda1.sha1
```

2. Command `tar` is used to archive many files into a single file, preserving directory structure. Option `-T` allows to specify a file containing the paths to be archived. For example, `find /etc -type f | tar -cJ -f /srv/backup/etc.tar.xz -T -` creates a compressed tar file `etc.tar.xz` from the list provided by command `find` (option `-T -` indicates the standard input as the path list). In order to avoid possible parsing errors due to paths containing spaces, what command options should be present for `find` and `tar`?

Options `-print0` and `--null`:

```
$ find /etc -type f -print0 | tar -cJ -f /srv/backup/etc.tar.xz --null -T -
```

3. Instead of opening a new remote shell session, command `ssh` can just execute a command indicated as its argument: `ssh user@storage "remote command"`. Given that `ssh` also allows to redirect the standard output of a local program to the standard input of the remote program, how would the `cat` pipe a local file named `etc.tar.gz` to `/srv/backup/etc.tar.gz` at `user@storage` through `ssh`?

```
$ cat etc.tar.gz | ssh user@storage "cat > /srv/backup/etc.tar.gz"
```

OR

```
$ ssh user@storage "cat > /srv/backup/etc.tar.gz" < etc.tar.gz
```



103.5 Create, monitor and kill processes

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 103.5

Weight

4

Key knowledge areas

- Run jobs in the foreground and background.
- Signal a program to continue running after logout.
- Monitor active processes.
- Select and sort processes for display.
- Send signals to processes.

Partial list of the used files, terms and utilities

- `&`
- `bg`
- `fg`
- `jobs`
- `kill`
- `nohup`
- `ps`
- `top`
- `free`

- `uptime`
- `pgrep`
- `pkill`
- `killall`
- `watch`
- `screen`
- `tmux`



103.5 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.5 Create, monitor and kill processes
Lesson:	1 of 2

Introduction

Every time we invoke a command, one or more processes are started. A well-trained system administrator not only needs to create processes, but also be able to keep track of them and send them different types of signals if and when required. In this lesson we will look at job control and how to monitor processes.

Job Control

Jobs are processes that have been started interactively through a terminal, sent to the background and have not yet finished execution. You can find out about the active jobs (and their status) in your Linux system by running `jobs`:

```
$ jobs
```

The `jobs` command above did not produce any output, which means there are no active jobs at the moment. Let us create our first job by running a command that takes some time to finish executing (the `sleep` command with a parameter of `60`) and — while running — press `Ctrl + Z`:

```
$ sleep 60
^Z
[1]+  Stopped                  sleep 60
```

The execution of the command has been stopped (or — rather — suspended) and the command prompt is available again. You can look for jobs a second time and will find the *suspended* one now:

```
$ jobs
[1]+  Stopped                  sleep 60
```

Let us explain the output:

[1]

This number is the job ID and can be used — preceded by a percentage symbol (%) — to change the job status by the `fg`, `bg` and `kill` utilities (as you will be shown later).

+

The plus sign indicates the current, default job (that is, the last one being suspended or sent to the background). The previous job is flagged with a minus sign (-). Any other prior jobs are not flagged.

Stopped

Description of the job status.

sleep 60

The command or job itself.

With the `-l` option, jobs will additionally display the process ID (PID) right before the status:

```
$ jobs -l
[1]+  1114 Stopped              sleep 60
```

The remaining possible options of jobs are:

-n

Lists only processes that have changed status since the last notification. Possible status include, Running, Stopped, Terminated or Done.

-p

Lists process IDs.

-r

Lists only running jobs.

-s

Lists only stopped (or suspended) jobs.

NOTE Remember, a job has both a *job ID* and a *process ID* (PID).

Job Specification

The `jobs` command as well as other utilities such as `fg`, `bg` and `kill` (that you will see in the next section) need a job specification (or `jobspec`) to act upon a particular job. As we have just seen, this can be — and normally is — the job ID preceded by `%`. However, other job specifications are also possible. Let us have a look at them:

%n

Job whose id number is `n`:

```
$ jobs %1
[1]+  Stopped                  sleep 60
```

%str

Job whose command line starts with `str`:

```
$ jobs %s1
[1]+  Stopped                  sleep 60
```

%?str

Job whose command line contains `str`:

```
$ jobs %?le
[1]+  Stopped                  sleep 60
```

%+ or %%

Current job (the one that was last started in the background or suspended from the

foreground):

```
$ jobs %+
[1]+  Stopped                  sleep 60
```

%-

Previous job (the one that was %+ before the default, current one):

```
$ jobs %-
[1]+  Stopped                  sleep 60
```

In our case, since there is only one job, it is both current and previous.

Job Status: Suspension, Foreground and Background

Once a job is in the background or has been suspended, we can do any of three things to it:

1. Take it to the foreground with fg:

```
$ fg %1
sleep 60
```

fg moves the specified job to the foreground and makes it the current job. Now we can wait until it finishes, stop it again with `Ctrl + Z` or terminate it with `Ctrl + C`.

2. Take it to the background with bg:

```
$ bg %1
[1]+ sleep 60 &
```

Once in the background, the job can be brought back into the foreground with fg or killed (see below). Note the ampersand (&) meaning the job has been sent to the background. As a matter of fact, you can also use the ampersand to start a process directly in the background:

```
$ sleep 100 &
[2] 970
```

Together with the job ID of the new job ([2]), we now get its process ID (970) too. Now both

jobs are running in the background:

```
$ jobs
[1]-  Running                  sleep 60 &
[2]+  Running                  sleep 100 &
```

A bit later the first job finishes execution:

```
$ jobs
[1]-  Done                     sleep 60
[2]+  Running                  sleep 100 &
```

3. Terminate it through a `SIGTERM` signal with `kill`:

```
$ kill %2
```

To make sure the job has been terminated, run `jobs` again:

```
$ jobs
[2]+  Terminated              sleep 100
```

NOTE

If no job is specified, `fg` and `bg` will act upon the current, default one. `kill`, however, always needs a job specification.

Detached Jobs: `nohup`

The jobs we have seen in the previous sections were all attached to the session of the user who invoked them. That means that if the session is terminated, the jobs are gone. However, it is possible to detach jobs from sessions and have them run even after the session is closed. This is achieved with the `nohup` (“no hangup”) command. The syntax is as follows:

```
nohup COMMAND &
```

Remember, the `&` sends the process into the background and frees up the terminal you are working at.

Let us detach the background job `ping localhost` from the current session:


```
$ nohup ping localhost &
[1] 1251
$ nohup: ignoring input and appending output to 'nohup.out'
^C
```

The output shows us the job ID ([1]) and the PID (1251), followed by a message telling us about the file `nohup.out`. This is the default file where `stdout` and `stderr` will be saved. Now we can press `Ctrl + C` to free up the command prompt, close the session, start another one as `root` and use `tail -f` to check if the command is running and output is being written to the default file:

```
$ exit
logout
$ tail -f /home/carol/nohup.out
64 bytes from localhost (::1): icmp_seq=3 ttl=64 time=0.070 ms
64 bytes from localhost (::1): icmp_seq=4 ttl=64 time=0.068 ms
64 bytes from localhost (::1): icmp_seq=5 ttl=64 time=0.070 ms
^C
```

TIP

Instead of using the default `nohup.out` you could have specified the output file of your choice with `nohup ping localhost > /path/to/your/file &`.

If we want to kill the process, we should specify its PID:

```
# kill 1251
```

Process Monitoring

A process or task is an instance of a running program. Thus, you create new processes every time you type in commands into the terminal.

The `watch` command executes a program periodically (2 seconds by default) and allows us to *watch* the program's output change over time. For instance, we can monitor how the load average changes as more processes are run by typing `watch uptime`:

```
Every 2.0s: uptime          debian: Tue Aug 20 23:31:27 2019

23:31:27 up 21 min,  1 user,  load average: 0.00, 0.00, 0.00
```

The command runs until interrupted so we would have to stop it with `Ctrl + C`. We get two lines as

output: the first one corresponds to `watch` and tells us how often the command will be run (Every 2.0s: uptime), what command/program to watch (uptime) as well as the hostname and date (debian: Tue Aug 20 23:31:27 2019). The second line of output is uptime's and includes the time (23:31:27), how much time the system has been up (up 21 min), the number of active users (1 user) and the average system load or number of processes in execution or in waiting state for the last 1, 5 and 15 minutes (load average: 0.00, 0.00, 0.00).

Similarly, you can check on memory use as new processes are created with `watch free`:

```
Every 2.0s: free          debian: Tue Aug 20 23:43:37 2019

23:43:37 up 24 min,  1 user,  load average: 0.00, 0.00, 0.00
      total        used        free      shared  buff/cache   available
Mem:    16274868    493984    14729396        35064     1051488    15462040
Swap:    16777212         0     16777212
```

To change the update interval for `watch` use the `-n` or `--interval` options plus the number of seconds as in:

```
$ watch -n 5 free
```

Now the `free` command will run every 5 seconds.

For more information about the options for `uptime`, `free` and `watch`, please refer to their manual pages.

NOTE

The information provided by `uptime` and `free` is also integrated in the more comprehensive tools `top` and `ps` (see below).

Sending Signals to Processes: `kill`

Every single process has a unique process identifier or PID. One way of finding out the PID of a process is by using the `pgrep` command followed by the process' name:

```
$ pgrep sleep
1201
```

NOTE

A process' identifier can also be discovered through the `pidof` command (e.g. `pidof sleep`).

Similar to `pgrep`, `pkill` command kills a process based on its name:

```
$ pkill sleep
[1]+  Terminated          sleep 60
```

To kill multiple instances of the same process, the `killall` command can be used:

```
$ sleep 60 &
[1] 1246
$ sleep 70 &
[2] 1247
$ killall sleep
[1]-  Terminated          sleep 60
[2]+  Terminated          sleep 70
```

Both `pkill` and `killall` work much in the same way as `kill` in that they send a terminating signal to the specified process(es). If no signal is provided, the default of `SIGTERM` is sent. However, `kill` only takes either a job or a process ID as an argument.

Signals can be specified either by:

- Name:

```
$ kill -SIGHUP 1247
```

- Number:

```
$ kill -1 1247
```

- Option:

```
$ kill -s SIGHUP 1247
```

To have `kill` work in a similar way to `pkill` or `killall` (and save ourselves the commands to find out PIDs first) we can use command substitution:

```
$ kill -1 $(pgrep sleep)
```

As you should already know, an alternative syntax is `kill -1 `pgrep sleep``.

TIP

For an exhaustive list of all `kill` signals and their codes, type `kill -l` into the terminal. Use `-KILL` (`-9` or `-s KILL`) to kill rebel processes when any other signal fails.

top and ps

When it comes to process monitoring, two invaluable tools are `top` and `ps`. Whilst the former produces output dynamically, the latter does it statically. In any case, both are excellent utilities to have a comprehensive view of all processes in the system.

Interacting with top

To invoke `top`, simply type `top`:

```
$ top
```

```
top - 11:10:29 up 2:21, 1 user, load average: 0,11, 0,20, 0,14
Tasks: 73 total, 1 running, 72 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0,0 us, 0,3 sy, 0,0 ni, 99,7 id, 0,0 wa, 0,0 hi, 0,0 si, 0,0 st
KiB Mem : 1020332 total, 909492 free, 38796 used, 72044 buff/cache
KiB Swap: 1046524 total, 1046524 free, 0 used. 873264 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
  436 carol    20   0   42696   3624   3060 R   0,7   0,4   0:00.30 top
     4 root      20   0      0      0      0 S   0,3   0,0   0:00.12 kworker/0:0
  399 root      20   0   95204   6748   5780 S   0,3   0,7   0:00.22 sshd
     1 root      20   0   56872   6596   5208 S   0,0   0,6   0:01.29 systemd
     2 root      20   0      0      0      0 S   0,0   0,0   0:00.00 kthreadd
     3 root      20   0      0      0      0 S   0,0   0,0   0:00.02 ksoftirqd/0
     5 root       0 -20      0      0      0 S   0,0   0,0   0:00.00 kworker/0:0H
     6 root      20   0      0      0      0 S   0,0   0,0   0:00.00 kworker/u2:0
     7 root      20   0      0      0      0 S   0,0   0,0   0:00.08 rcu_sched
     8 root      20   0      0      0      0 S   0,0   0,0   0:00.00 rcu_bh
     9 root      rt    0      0      0      0 S   0,0   0,0   0:00.00 migration/0
    10 root       0 -20      0      0      0 S   0,0   0,0   0:00.00 lru-add-drain
  (...)
```

`top` allows the user some interaction. By default, the output is sorted by the percentage of CPU time used by each process in descending order. This behavior can be modified by pressing the following keys from within `top`:

M

Sort by *memory* usage.

N

Sort by process ID *number*.

T

Sort by running *time*.

P

Sort by *percentage* of CPU usage.

TIP | To switch between descending/ascending order just press R.

Other interesting keys to interact with `top` are:

? or h

Help.

k

Kill a process. `top` will ask for the PID of the process to be killed as well as for the signal to be sent (by default `SIGTERM` or 15).

r

Change the priority of a process (`renice`). `top` will ask you for the `nice` value. Possible values range from -20 through 19, but only the superuser (`root`) can set it to a value which is negative or lower than the current one.

u

List processes from a particular user (by default processes from all users are shown).

c

Show programs' absolute paths and differentiate between userspace processes and kernelspace processes (in square brackets).

V

Forest/hierarchy view of processes.

t and m

Change the look of CPU and memory readings respectively in a four-stage cycle: the first two presses show progress bars, the third hides the bar and the fourth brings it back.

W

Save configuration settings to `~/ .toprc`.

TIP

A fancier and more user-friendly version of `top` is `htop`. Another — perhaps more exhaustive — alternative is `atop`. If not already installed in your system, use your package manager to install them and give them a try.

An Explanation of the output of `top`

`top` output is divided into two areas: the *summary area* and the *task area*.

The Summary Area in `top`

The *summary area* is made up of the the five top rows and gives us the following information:

- `top - 11:10:29 up 2:21, 1 user, load average: 0,11, 0,20, 0,14`
 - current time (in 24-hour format): `11:20:29`
 - uptime (how much time the system has been up and running): `up 2:21`
 - number of users logged in and load average of the CPU for the last 1, 5 and 15 minutes, respectively: `load average: 0,11, 0,20, 0,14`
- `Tasks: 73 total, 1 running, 72 sleeping, 0 stopped, 0 zombie` (information about the processes)
 - total number of processes in active mode: `73 total`
 - running (those being executed): `1 running`
 - sleeping (those waiting to resume execution): `72 sleeping`
 - stopped (by a job control signal): `0 stopped`
 - zombie (those which have completed execution but are still waiting for their parent process to remove them from the process table): `0 zombie`
- `%Cpu(s): 0,0 us, 0,3 sy, 0,0 ni, 99,7 id, 0,0 wa, 0,0 hi, 0,0 si, 0,0 st` (percentage of CPU time spent on)
 - user processes: `0,0 us`
 - system/kernel processes: `0,4 sy`
 - processes set to a *nice* value — the nicer the value, the lower the priority: `0,0 ni`
 - nothing — idle CPU time: `99,7 id`
 - processes waiting for I/O operations: `0,0 wa`

- processes serving hardware interrupts—peripherals sending the processor signals that require attention: `0,0 hi`
- processes serving software interrupts: `0,0 si`
- processes serving other virtual machine's tasks in a virtual environment, hence steal time: `0,0 st`
- KiB Mem : 1020332 total, 909492 free, 38796 used, 72044 buff/cache (memory information in kilobytes)
 - the total amount of memory: `1020332 total`
 - unused memory: `909492 free`
 - memory in use: `38796 used`
 - the memory which is buffered and cached to avoid excessive disk access: `72044 buff/cache`

Notice how the `total` is the sum of the other three values—`free`, `used` and `buff/cache`—(roughly 1 GB in our case).
- KiB Swap: 1046524 total, 1046524 free, 0 used. 873264 avail Mem (swap information in kilobytes)
 - the total amount of swap space: `1046524 total`
 - unused swap space: `1046524 free`
 - swap space in use: `0 used`
 - the amount of swap memory that can be allocated to processes without causing more swapping: `873264 avail Mem`

The Task Area in `top`: Fields and Columns

Below the *summary area* there comes the *task area*, which includes a series of *fields* and *columns* reporting information about the running processes:

PID

Process identifier.

USER

User who issued the command that generated the process.

PR

Priority of process to the kernel.

NI

Nice value of process. Lower values have a higher priority than higher ones.

VIRT

Total amount of memory used by process (including Swap).

RES

RAM memory used by process.

SHR

Shared memory of the process with other processes.

S

Status of process. Values include: S (interruptible sleep — waiting for an event to finish), R (runnable — either executing or in the queue to be executed) or Z (zombie — terminated child processes whose data structures have not yet been removed from the process table).

%CPU

Percentage of CPU used by process.

%MEM

Percentage of RAM used by process, that is, the RES value expressed as a percentage.

TIME+

Total time of activity of process.

COMMAND

Name of command/program which generated the process.

Viewing processes statically: ps

As said above, `ps` shows a snapshot of processes. To see all processes with a terminal (tty), type `ps a`:

```
$ ps a
PID TTY      STAT   TIME COMMAND
386  tty1      Ss+    0:00 /sbin/agetty --noclear tty1 linux
424  tty7      Ss1+   0:00 /usr/lib/xorg/Xorg :0 -seat seat0 (...)
655  pts/0     Ss      0:00 -bash
1186 pts/0     R+      0:00 ps a
```


(...)

An Explanation of `ps` Option Syntax and Output

Concerning options, `ps` can accept three different styles: BSD, UNIX and GNU. Let us see how each of these styles would work when reporting information about a particular process ID:

BSD

Options do not follow any leading dash:

```
$ ps p 811
PID TTY      STAT   TIME COMMAND
811 pts/0    S       0:00  -su
```

UNIX

Options do follow a leading dash:

```
$ ps -p 811
PID TTY      TIME CMD
811 pts/0    00:00:00 bash
```

GNU

Options are followed by double leading dashes:

```
$ ps --pid 811
PID TTY      TIME CMD
811 pts/0    00:00:00 bash
```

In all three cases, `ps` reports information about the process whose PID is 811 — in this case, `bash`.

Similarly, you can use `ps` to search for the processes started by a particular user:

- `ps U carol` (BSD)
- `ps -u carol` (UNIX)
- `ps --user carol` (GNU)

Let us check on the processes started by `carol`:

```
$ ps U carol
PID TTY      STAT   TIME COMMAND
811 pts/0    S       0:00 -su
898 pts/0    R+      0:00 ps U carol
```

She started two processes: `bash (-su)` and `ps (ps U carol)`. The `STAT` column tells us the state of the process (see below).

We can get the best out of `ps` by combining some of its options. A very useful command (producing an output similar to that of `top`) is `ps aux` (BSD style). In this case, processes from all shells (not only the current one) are shown. The meaning of the switches are the following:

a

Show processes that are attached to a `tty` or terminal.

u

Display user-oriented format.

x

Show processes that are not attached to a `tty` or terminal.

```
$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1 204504 6780 ?        Ss   14:04   0:00 /sbin/init
root         2  0.0  0.0      0      0 ?        S    14:04   0:00 [kthreadd]
root         3  0.0  0.0      0      0 ?        S    14:04   0:00 [ksoftirqd/0]
root         5  0.0  0.0      0      0 ?        S<   14:04   0:00 [kworker/0:0H]
root         7  0.0  0.0      0      0 ?        S    14:04   0:00 [rcu_sched]
root         8  0.0  0.0      0      0 ?        S    14:04   0:00 [rcu_bh]
root         9  0.0  0.0      0      0 ?        S    14:04   0:00 [migration/0]
(...)
```

Let us explain the columns:

USER

Owner of process.

PID

Process identifier.

%CPU

Percentage of CPU used.

%MEM

Percentage of physical memory used.

VSZ

Virtual memory of process in KiB.

RSS

Non-swapped physical memory used by process in KiB.

TT

Terminal (tty) controlling the process.

STAT

Code representing the state of process. Apart from S, R and Z (that we saw when describing the output of top), other possible values include: D (uninterruptible sleep—usually waiting for I/O), T (stopped—normally by a control signal). Some extra modifier include: < (high-priority—not nice to other processes), N (low-priority—nice to other processes), or + (in the foreground process group).

STARTED

Time at which the process started.

TIME

Accumulated CPU time.

COMMAND

Command that started the process.

Guided Exercises

1. `oneko` is a nice funny program that displays a cat chasing your mouse cursor. If not already installed in your desktop system, install it using your distribution's package manager. We will use it to study job control.

- Start the program. How do you do that?

- Move the mouse cursor to see how the cat chases it. Now suspend the process. How do you do that? What is the output?

- Check how many jobs you currently have. What do you type? What is the output?

- Now send it to the background specifying its job ID. What is the output? How can you tell the job is running in the background?

- Finally, terminate the job specifying its job ID. What do you type?

2. Discover the PIDs of all the processes spawned by the *Apache HTTPD web server* (`apache2`) with two different commands:

3. Terminate all `apache2` processes without using their PIDs and with two different commands:

4. Suppose you have to terminate all instances of `apache2` and you do not have time to find out what their PIDs are. How would you accomplish that using `kill` with the default `SIGTERM` signal in a one-liner:

5. Start `top` and interact with it by performing the following:

- Show a forest view of processes:

- Show full paths of processes differentiating between userspace and kernelspace:

6. Type the `ps` command to display all processes started by the *Apache HTTPD web server* user (`www-data`):

- Using BSD syntax:

- Using UNIX syntax:

- Using GNU syntax:

Explorational Exercises

1. The `SIGHUP` signal can be used as a way to restart certain daemons. With the *Apache HTTPD web server*—for example—sending `SIGHUP` to the parent process (the one started by `init`) kills off its children. The parent, however, re-reads its configuration files, re-opens log files and spawns a new set of children. Do the following tasks:

- Start the web server:

- Make sure you know the PID of the parent process:

- Make the Apache HTTPD web server restart by sending it the `SIGHUP` signal to the parent process:

- Check that the parent was not killed and that new children have been spawned:

2. Although initially static, the output of `ps` can be *turned* dynamic by combining `ps` and `watch`. We will monitor the *Apache HTTPD web server* for new connections. Before doing the tasks described below it is recommended that you read the description of the `MaxConnectionsPerChild` directive in [Apache MPM Common Directives](#).

- Add the `MaxConnectionsPerChild` directive with a value of `1` in the configuration file of `apache2`—in *Debian* and derivatives that is found in `/etc/apache2/apache2.conf`; in the *CentOS* family, in `/etc/httpd/conf/httpd.conf`. Do not forget to restart `apache2` for the changes to take effect.

- Type in a command that uses `watch`, `ps` and `grep` for `apache2` connections.

- Now open a web browser or use a command line browser such as `lynx` to establish a connection to the web server through its IP address. What do you observe in the output of `watch`?

3. As you have learned, by default `top` sorts the tasks by percentage of CPU usage in descending

order (the higher values on top). This behaviour can be modified with the interactive keys **M** (memory usage), **N** (process unique identifier), **T** (running time) and **P** (percentage of CPU time). However, you can also sort the task list to your liking by launching **top** with the **-o** switch (for more information, check **top**'s **man** page). Now, perform the following tasks:

- Launch **top** so that the tasks are sorted by memory usage:

- Verify that you typed the right command by highlighting the memory column:

4. **ps** also has an **o** switch to specify the columns you want shown. Investigate this option and do the following tasks:

- Launch **ps** so that only information about *user*, *percentage of memory used*, *percentage of CPU time used* and *full command* is shown:

- Now, launch **ps** so that the only information displayed is that of the user and the name of the programs they are using:

Summary

In this lesson you have learned about *jobs* and *job control*. Important facts and concepts to bear in mind are:

- Jobs are processes that are sent to the background.
- Apart from a *process ID*, jobs are also assigned a *job ID* when created.
- To control jobs, a job specification (`jobspec`) is required.
- Jobs can be brought to the foreground, sent to the background, suspended and terminated (or *killed*).
- A job can be detached from the terminal and session in which it was created.

Likewise, we have also discussed the concept of *processes* and *process monitoring*. The most relevant ideas are:

- Processes are running programs.
- Processes can be monitored.
- Different utilities allow us to find out the *process ID* of processes as well as to send them signals to terminate them.
- Signals can be specified by name (e.g. `-SIGTERM`), number (e.g. `-15`) or option (e.g. `-s SIGTERM`).
- `top` and `ps` are very powerful when it comes to monitoring processes. The output of the former is dynamic and keeps updating constantly; on the other hand, `ps` reveals output statically.

Commands used in this lesson:

jobs

Display active jobs and their status.

sleep

Delay for a specific amount of time.

fg

Bring job to the foreground.

bg

Move job to the background.

kill

Terminate job.

nohup

Detach job from session/terminal.

exit

Exit current shell.

tail

Display most recent lines in a file.

watch

Run a command repeatedly (2 seconds cycle by default).

uptime

Display how long the system has been running, the number of current users and system load average.

free

Display memory usage.

pgrep

Look up process ID based on name.

pidof

Look up process ID based on name.

pkill

Send signal to process by name.

killall

Kill process(es) by name.

top

Display Linux processes.

ps

Report a snapshot of the current processes.

Answers to Guided Exercises

1. `oneko` is a nice funny program that displays a cat chasing your mouse cursor. If not already installed in your desktop system, install it using your distribution's package manager. We will use it to study job control.

- Start the program. How do you do that?

By typing `oneko` into the terminal.

- Move the mouse cursor to see how the cat chases it. Now suspend the process. How do you do that? What is the output?

By pressing the key combo `Ctrl + z`:

```
[1]+  Stopped                  oneko
```

- Check how many jobs you currently have. What do you type? What is the output?

```
$ jobs
[1]+  Stopped                  oneko
```

- Now send it to the background specifying its job ID. What is the output? How can you tell the job is running in the background?

```
$ bg %1
[1]+ oneko &
```

The cat is moving again.

- Finally, terminate the job specifying its job ID. What do you type?

```
$ kill %1
```

2. Discover the PIDs of all the processes spawned by the *Apache HTTPD web server* (`apache2`) with two different commands:

```
$ pgrep apache2
```

OR

```
$ pidof apache2
```

3. Terminate all `apache2` processes without using their PIDs and with two different commands:

```
$ pkill apache2
```

OR

```
$ killall apache2
```

4. Suppose you have to terminate all instances of `apache2` and you do not have time to find out what their PIDs are. How would you accomplish that using `kill` with the default `SIGTERM` signal in a one-liner:

```
$ kill $(pgrep apache2)
$ kill `pgrep apache2`
```

OR

```
$ kill $(pidof apache2)
$ kill `pidof apache2`
```

NOTE

Since `SIGTERM` (15) is the default signal, it is not necessary to pass any options to `kill`.

5. Start `top` and interact with it by performing the following:

- Show a forest view of processes:

Press `V`.

- Show full paths of processes differentiating between userspace and kernelspace:

Press `c`.

6. Type the `ps` command to display all processes started by the *Apache HTTPD web server* user (`www-data`):

- Using BSD syntax:

```
$ ps U www-data
```

- Using UNIX syntax:

```
$ ps -u www-data
```

- Using GNU syntax:

```
$ ps --user www-data
```

Answers to Explorational Exercises

1. The `SIGHUP` signal can be used as a way to restart certain daemons. With the *Apache HTTPD web server*—for example—sending `SIGHUP` to the parent process (the one started by `init`) kills off its children. The parent, however, re-reads its configuration files, re-opens log files and spawns a new set of children. Do the following tasks:

- Start the web server:

```
$ sudo systemctl start apache2
```

- Make sure you know the PID of the parent process:

```
$ ps aux | grep apache2
```

The parent process is that started by the `root` user. In our case the one with PID 1653.

- Make the Apache HTTPD web server restart by sending it the `SIGHUP` signal to the parent process:

```
$ kill -SIGHUP 1653
```

- Check that the parent was not killed and that new children have been spawned:

```
$ ps aux | grep apache2
```

Now you should see the parent `apache2` process together with two new children.

2. Although initially static, the output of `ps` can be *turned* dynamic by combining `ps` and `watch`. We will monitor the *Apache HTTPD web server* for new connections. Before doing the tasks described below it is recommended that you read the description of the `MaxConnectionsPerChild` directive in [Apache MPM Common Directives](#).

- Add the `MaxConnectionsPerChild` directive with a value of `1` in the config file of `apache2`—in *Debian* and derivatives that is found in `/etc/apache2/apache2.conf`; in the *CentOS* family, in `/etc/httpd/conf/httpd.conf`. Do not forget to restart `apache2` for the changes to take effect.

The line to include in the config file is `MaxConnectionsPerChild 1`. One way to restart the web server is through `sudo systemctl restart apache2`.

- Type in a command that uses `watch`, `ps` and `grep` for `apache2` connections.

```
$ watch 'ps aux | grep apache2'
```

or

```
$ watch "ps aux | grep apache2"
```

- Now open a web browser or use a command line browser such as `lynx` to establish a connection to the web server through its IP address. What do you observe in the output of `watch`?

One of the child processes owned by `www-data` disappears.

3. As you have learned, by default `top` sorts the tasks by percentage of CPU usage in descending order (the higher values on top). This behaviour can be modified with the interactive keys `M` (memory usage), `N` (process unique identifier), `T` (running time) and `P` (percentage of CPU time). However, you can also sort the task list to your liking by launching `top` with the `-o` switch (for more information, check `top`'s man page). Now, do the following tasks:

- Launch `top` so that the tasks are sorted by memory usage:

```
$ top -o %MEM
```

- Verify that you typed the right command by highlighting the memory column:

Press `x`.

4. `ps` has also an `o` switch to specify the columns you want shown. Investigate about this and do the following tasks:

- Launch `ps` so that only information about *user*, *percentage of memory used*, *percentage of CPU time used* and *full command* is shown:

```
$ ps o user,%mem,%cpu,cmd
```

- Now, launch `ps` so that the only information displayed is that of the user and the name of the programs they are using:

```
$ ps o user,comm
```



103.5 Lesson 2

Certificate:	LPIC-1
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.5 Create, monitor and kill processes
Lesson:	2 of 2

Introduction

The tools and utilities seen in the previous lesson are very useful for process monitoring at large. However, a system administrator may need to go one step further. In this lesson we will discuss the concept of terminal multiplexers and learn about *GNU Screen* and *tmux* as—despite today’s modern and great terminal emulators—multiplexers still preserve some cool, powerful features for a productive system administrator.

Features of Terminal Multiplexers

In electronics, a multiplexer (or *mux*) is a device that allows for multiple inputs to be connected to a single output. Thus, a terminal multiplexer gives us the ability to switch between different inputs as required. Although not exactly the same, *screen* and *tmux* share a series of common features:

- Any successful invocation will result in at least a session which—in turn—will include at least a window. Windows contain programs.
- Windows can be split into regions or panes—which can aid in productivity when working

with various programs simultaneously.

- Ease of control: to run most commands, they use a key combination — the so-called *command prefix* or *command key* — followed by another character.
- Sessions can be detached from their current terminal (that is, programs are sent to the background and continue to run). This guarantees complete execution of programs no matter if we accidentally close a terminal, experience an occasional terminal freeze or even remote connection loss.
- Socket connection.
- Copy mode.
- They are highly customizable.

GNU Screen

In the earlier days of Unix (1970s - 80s) computers basically consisted of terminals connected to a central computer. That was all, no multiple windows or tabs. And that was the reason behind the creation of GNU Screen in 1987: emulate multiple independent *VT100 screens* on a single physical terminal.

Windows

GNU Screen is invoked just by typing `screen` into the terminal. You will first see a welcome message:

```
GNU Screen version 4.05.00 (GNU) 10-Dec-16

Copyright (c) 2010 Juergen Weigert, Sadrul Habib Chowdhury
Copyright (c) 2008, 2009 Juergen Weigert, Michael Schroeder, Micah Cowan, Sadrul Habib
Chowdhury
Copyright (c) 1993-2002, 2003, 2005, 2006, 2007 Juergen Weigert, Michael Schroeder
Copyright (c) 1987 Oliver Laumann
(...)
```

Press Space or Enter to close the message and you will be shown a command prompt:

```
$
```

It may seem nothing has happened but the fact is that `screen` has already created and managing its first session and window. Screen's command prefix is `Ctrl + a`. To see all windows at the bottom

of the terminal display, type `Ctrl + a -w`:

```
0*$ bash
```

There it is, our one and only window so far! Note that the count starts at 0, though. To create another window, type `Ctrl + a -c`. You will see a new prompt. Let us start `ps` in that new window:

```
$ ps
PID TTY          TIME CMD
 974 pts/2        00:00:00 bash
 981 pts/2        00:00:00 ps
```

and type `Ctrl + a -w` again:

```
0-$ bash  1*$ bash
```

There we have our two windows (note the asterisk indicating the one that is being shown at the moment). However, as they were started with Bash, they are both given the same name. Since we invoked `ps` in our current window, let us rename it with that same name. For that, you have to type `Ctrl + a -A` and write the new window name (`ps`) when prompted:

```
Set window's title to: ps
```

Now, let us create another window but provide it a name from the start: `yetanotherwindow`. This is done by invoking `screen` with the `-t` switch:

```
$ screen -t yetanotherwindow
```

You can move between windows in different ways:

- By using `Ctrl + a -n` (go to *next* window) and `Ctrl + a -p` (go to *previous* window).
- By using `Ctrl + a -number` (go to window number *number*).
- By using `Ctrl + a -"` to see a list of all windows. You can move up and down with the arrow keys and select the one you want with Enter:

```
Num Name
```

```
Flags
```

```

0 bash                                $
1 ps                                  $
2 yetanotherwindow

```

While working with windows it is important to remember the following:

- Windows run their programs completely independent of each other.
- Programs will continue to run even if their window is not visible (also when the screen session is detached as we will see shortly).

To remove a window, simply terminate the program running in it (once the last window is removed, screen will itself terminate). Alternatively, use `Ctrl + a -k` while in the window you want to remove; you will be asked for confirmation:

```

Really kill this window [y/n]

Window 0 (bash) killed.

```

Regions

screen can divide a terminal screen up into multiple regions in which to accommodate windows. These divisions can be either horizontal (`Ctrl + a -s`) or vertical (`Ctrl + a -l`).

The only thing the new region will show is just `--` at the bottom meaning it is empty:

```

1 ps                                --

```

To move to the new region, type `Ctrl + a -Tab`. Now you can add a window by any of the methods that we have already seen, for example: `Ctrl + a -2`. Now the `--` should have turned into 2 yetanotherwindow:

```

$ ps                                $
  PID TTY          TIME CMD
 1020 pts/2    00:00:00 bash
 1033 pts/2    00:00:00 ps
$ screen -t yetanotherwindow

```

1 ps

2 yetanotherwindow

Important aspects to keep in mind when working with regions are:

- You move between regions by typing `Ctrl` + `a-Tab`.
- You can terminate all regions except the current one with `Ctrl` + `a-Q`.
- You can terminate the current region with `Ctrl` + `a-X`.
- Terminating a region does not terminate its associated window.

Sessions

So far we have played with a few windows and regions, but all belonging to the same and only session. It is time to start playing with sessions. To see a list of all sessions, type `screen -list` or `screen -ls`:

```
$ screen -list
There is a screen on:
      1037.pts-0.debian      (08/24/19 13:53:35)      (Attached)
1 Socket in /run/screen/S-carol.
```

That is our only session so far:

PID

1037

Name

pts-0.debian (indicating the terminal—in our case a *pseudo terminal slave*—and the hostname).

Status

Attached

Let us create a new session giving it a more descriptive name:

```
$ screen -S "second session"
```

The terminal display will be cleared and you will be given a new prompt. You can check for sessions once more:

```
$ screen -ls
There are screens on:
    1090.second session      (08/24/19 14:38:35)    (Attached)
    1037.pts-0.debian        (08/24/19 13:53:36)    (Attached)
2 Sockets in /run/screen/S-carol.
```

To kill a session, quit out of all its windows or just type the command `screen -S SESSION-PID -X quit` (you can also provide the session name instead). Let us get rid of our first session:

```
$ screen -S 1037 -X quit
```

You will be sent back to your terminal prompt outside of `screen`. But remember, our second session is still alive:

```
$ screen -ls
There is a screen on:
    1090.second session (08/24/19 14:38:35) (Detached)
1 Socket in /run/screen/S-carol.
```

However, since we killed its parent session, it is given a new label: `Detached`.

Session Detachment

For a number of reasons you may want to detach a screen session from its terminal:

- To let your computer at work do its business and connect remotely later from home.
- To share a session with other users.

You detach a session with the key combination `Ctrl + a -d`. You will be taken back into your terminal:

```
[detached from 1090.second session]
$
```

To attach again to the session, you use the command `screen -r SESSION-PID`. Alternatively, you can use the `SESSION-NAME` as we saw above. If there is only one detached session, neither is compulsory:

```
$ screen -r
```

This command suffices to reattach to our second session:

```
$ screen -ls
There is a screen on:
      1090.second session      (08/24/19 14:38:35)      (Attached)
1 Socket in /run/screen/S-carol.
```

Important options for session reattaching:

-d -r

Reattach a session and — if necessary — detach it first.

-d -R

Same as **-d -r** but `screen` will even create the session first if it does not exist.

-d -RR

Same as **-d -R**. However, use the first session if more than one is available.

-D -r

Reattach a session. If necessary, detach and logout remotely first.

-D -R

If a session is running, then reattach (detaching and logging out remotely first if necessary). If it was not running create it and notify the user.

-D -RR

Same as **-D -R** — only stronger.

-d -m

Start `screen` in *detached mode*. This creates a new session but does not attach to it. This is useful for system startup scripts.

-D -m

Same as **-d -m**, but does not fork a new process. The command exits if the session terminates.

Read the manual pages for `screen` to find out about other options.

Copy & Paste: Scrollback Mode

GNU Screen features a copy or *scrollback mode*. Once entered, you can move the cursor in the current window and through the contents of its history using the arrow keys. You can mark text

and copy it across windows. The steps to follow are:

1. Enter copy/scrollback mode: `Ctrl + a-l`.
2. Move to the beginning of the piece of text you want to copy using the arrow keys.
3. Mark the beginning of the piece of text you want to copy: Space.
4. Move to the end of the piece of text you want to copy using the arrow keys.
5. Mark the end of the piece of text you want to copy: Space.
6. Go to the window of your choice and paste the piece of text: `Ctrl + a-l`.

Customization of screen

The system-wide configuration file for screen is `/etc/screenrc`. Alternatively, a user-level `~/.screenrc` can be used. The file includes four main configuration sections:

SCREEN SETTINGS

You can define general settings by specifying the *directive* followed by a space and the *value* as in: `defscrollback 1024`.

SCREEN KEYBINDINGS

This section is quite interesting as it allows you to redefine keybindings that perhaps interfere with your everyday use of the terminal. Use the keyword `bind` followed by a Space, the character to use after the command prefix, another Space and the command as in: `bind l kill` (this setting will change the default way of killing a window to `Ctrl + a-l`).

To display all of screen's bindings, type `Ctrl + a-?` or consult the manual page.

TIP

Of course, you can also change the command prefix itself. For example to go from `Ctrl + a` to `Ctrl + b`, just add this line: `escape ^Bb`.

TERMINAL SETTINGS

This section includes settings related to terminal window sizes and buffers—amongst others. To enable non-blocking mode to better cope with unstable ssh connections, for instance, the following configuration is used: `defnonblock 5`.

STARTUP SCREENS

You can include commands to have various programs running on screen startup; for example: `screen -t top top` (screen will open a window named `top` with `top` inside).

tmux

`tmux` was released in 2007. Although very similar to `screen`, it includes a few notable differences:

- Client-server model: the server supplies a number of sessions, each of which may have a number of windows linked to it that can, in turn, be shared by various clients.
- Interactive selection of sessions, windows and clients via menus.
- The same window can be linked to a number of sessions.
- Availability of both *vim* and *Emacs* key layouts.
- UTF-8 and 256-colour terminal support.

Windows

`tmux` can be invoked by simply typing `tmux` at the command prompt. You will be shown a shell prompt and a status bar at the bottom of the window:

```
[0] 0: bash*
```

```
"debian" 18:53 27-Aug-19
```

Other than the `hostname`, the time and the date, the status bar provides the following information:

Session name

```
[0]
```

Window number

```
0:
```

Window name

`bash*`. By default this is the name of the program running inside the window and — unlike `screen` — `tmux` will automatically update it to reflect the current running program. Note the asterisk indicating that this is the current, visible window.

You can assign session and window names when invoking `tmux`:

```
$ tmux new -s "LPI" -n "Window zero"
```

The status bar will change accordingly:


```
[LPI] 0:Window zero*
19
```

```
"debian" 19:01 27-Aug-
```

tmux's command prefix is `Ctrl + b`. To create a new window, just type `Ctrl + b -c`; you will be taken to a new prompt and the status bar will reflect the new window:

```
[LPI] 0:Window zero- 1:bash*
19
```

```
"debian" 19:02 27-Aug-
```

As Bash is the underlying shell, the new window is given that name by default. Start `top` and see how the name changes to `top`:

```
[LPI] 0:Window zero- 1:top*
```

```
"debian" 19:03 27-Aug-19
```

In any case, you can rename a window with `Ctrl + b -`. When prompted, supply the new name and hit Enter:

```
(rename-window) Window one
```

You can have all windows displayed for selection with `Ctrl + b -w` (use the arrow keys to move up and down and `enter` to select):

```
(0) 0: Window zero- "debian"
(1) 1: Window one* "debian"
```

In a similar fashion to `screen`, we can jump from one window to another with:

```
Ctrl + b -n
```

go to *next* window.

```
Ctrl + b -p
```

go to *previous* window.

```
Ctrl + b -number
```

go to window number *number*.

To get rid of a window, use `Ctrl + b -&`. You will be asked for confirmation:

```
kill-window Window one? (y/n)
```

Other interesting window commands include:

Ctrl + **b** - **f**

find window by name.

Ctrl + **b** - **.**

change window index number.

To read the whole list of commands, consult the manual page.

Panes

The window-splitting facility of `screen` is also present in `tmux`. The resulting divisions are not called *regions* but *panes*, though. The most important difference between regions and panes is that the latter are complete pseudo-terminals linked to a window. This means that killing a pane will also kill its pseudo-terminal and any associated programs running within.

To split a window horizontally, we use **Ctrl** + **b** - **"**:

```
Tasks:  93 total,   1 running,  92 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,   0.0 sy,   0.0 ni,100.0 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
KiB Mem : 4050960 total, 3730920 free,  114880 used,  205160 buff/cache
KiB Swap: 4192252 total, 4192252 free,    0 used.  3716004 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1340	carol	20	0	44876	3400	2800	R	0.3	0.1	0:00.24	top
1	root	20	0	139088	6988	5264	S	0.0	0.2	0:00.50	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.04	ksoftirqd/0
4	root	20	0	0	0	0	S	0.0	0.0	0:01.62	kworker/0:0
5	root	0	-20	0	0	0	S	0.0	0.0	0:00.00	kworker/0:0H
7	root	20	0	0	0	0	S	0.0	0.0	0:00.06	rcu_sched
8	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_bh
9	root	rt	0	0	0	0	S	0.0	0.0	0:00.00	migration/0
10	root	0	-20	0	0	0	S	0.0	0.0	0:00.00	lru-add-drain
11	root	rt	0	0	0	0	S	0.0	0.0	0:00.01	watchdog/0
12	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0

```
$
```

```
$
```

```
[LPI] 0:Window zero- 1:Window one*
Aug-19
```

```
"debian" 19:05 27-
```

To split it vertically, use `Ctrl + b-%`:

```

1 root      20    0  139088   6988   5264 S   0.0   0.2   0:00.50 systemd
2 root      20    0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
3 root      20    0      0      0      0 S   0.0   0.0   0:00.04 ksoftirqd/0
4 root      20    0      0      0      0 S   0.0   0.0   0:01.62 kworker/0:0
5 root       0 -20      0      0      0 S   0.0   0.0   0:00.00 kworker/0:0H
7 root      20    0      0      0      0 S   0.0   0.0   0:00.06 rcu_sched
8 root      20    0      0      0      0 S   0.0   0.0   0:00.00 rcu_bh
9 root      rt    0      0      0      0 S   0.0   0.0   0:00.00 migration/0
10 root     0 -20      0      0      0 S   0.0   0.0   0:00.00 lru-add-drai
n
11 root     rt    0      0      0      0 S   0.0   0.0   0:00.01 watchdog/0
12 root     20    0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
$
$
```

```
[LPI] 0:Window zero- 1:Window one*  
Aug-19
```

```
"debian" 19:05 27-
```

To destroy the current pane (along with its pseudo-terminal running within it, along with any associated programs), use `Ctrl + b - x`. You will be asked for confirmation on the status bar:

```
kill-pane 1? (y/n)
```

Important pane commands:

```
Ctrl + b - ↑, ↓, ←, →
```

move between panes.

```
Ctrl + b - ;
```

move to the last active pane.

```
Ctrl + b - Ctrl + arrow key
```

resize pane by one line.

```
Ctrl + b - Alt + arrow key
```

resize pane by five lines.

```
Ctrl + b - {
```

swap panes (current to previous).

```
Ctrl + b - }
```

swap panes (current to next).

`Ctrl + b - z`

zoom in/out panel.

`Ctrl + b - t`

tmux displays a fancy clock inside the pane (kill it by pressing q).

`Ctrl + b - !`

turn pane into window.

To read the whole list of commands, consult the manual page.

Sessions

To list sessions in tmux you can use `Ctrl + b - s`:

```
(0) + LPI: 2 windows (attached)
```

Alternatively, you can use the `tmux ls` command:

```
$ tmux ls
LPI: 2 windows (created Tue Aug 27 19:01:49 2019) [158x39] (attached)
```

There is only one session (LPI) which includes two windows. Let us create a new session from within our current session. This can be achieved by using `Ctrl + b`, type in `:new` at the prompt, then press Enter. You will be sent to the new session as can be observed on the status bar:

```
[2] 0:bash* "debian" 19:15 27-Aug-19
```

By default tmux named the session 2. To rename it, use `Ctrl + b - s`. When prompted, supply the new name and hit Enter:

```
(rename-session) Second Session
```

You can switch sessions with `Ctrl + b - s` (use the arrow keys and `enter`):

```
(0) + LPI: 2 windows
(1) + Second Session: 1 windows (attached)
```

To kill a session, you can use the command `tmux kill-session -t SESSION-NAME`. If you type the command from within the current, attached session, you will be taken out of `tmux` and back to your initial terminal session:

```
$ tmux kill-session -t "Second Session"
[exited]
$
```

Sessions Detachment

By killing `Second Session` we were thrown outside `tmux`. However, we still have an active session. Ask `tmux` for a listing of sessions and you will surely find it there:

```
$ tmux ls
LPI: 2 windows (created Tue Aug 27 19:01:49 2019) [158x39]
```

However this session is detached from its terminal. We can attach it with `tmux attach -t SESSION-NAME` (attach can be replaced by `at` or — simply — `a`). When there is only a single session, the name specification is optional:

```
$ tmux a
```

You are now back in your session; to detach from it, press `Ctrl + b -d`:

```
[detached (from session LPI)]
$
```

TIP

The same session can be attached to more than one terminal. If you want to attach a session making sure it is first detached from any other terminals, use the `-d` switch: `tmux attach -d -t SESSION-NAME`.

Important commands for session attaching/detaching:

`Ctrl + b -D`

select what client to detach.

`Ctrl + b -r`

refresh the client's terminal.

To read the whole list of commands, consult the manual page.

Copy & Paste: Scrollback Mode

`tmux` also features copy mode in basically the same fashion as `screen` (remember to use `tmux`'s command prefix and not `screen`'s!). The only difference command-wise is that you use `Ctrl` + Space to mark the beginning of the selection and `Alt` + `w` to copy the selected text.

Customization of `tmux`

The configuration files for `tmux` are typically located at `/etc/tmux.conf` and `~/.tmux.conf`. When started, `tmux` sources these files if they exist. There is also the possibility of starting `tmux` with the `-f` switch to supply an alternate configuration file. You can find a `tmux` configuration file example located at `/usr/share/doc/tmux/example_tmux.conf`. The level of customization that you can achieve is really high. Some of the things you can do include:

- Change the prefix key

```
# Change the prefix key to C-a
set -g prefix C-a
unbind C-b
bind C-a send-prefix
```

- Set extra key bindings for windows higher than 9

```
# Some extra key bindings to select higher numbered windows
bind F1 selectw -t:10
bind F2 selectw -t:11
bind F3 selectw -t:12
```

For a comprehensive list of all bindings, type `Ctrl` + `b-?` (press `q` to quit) or consult the manual page.

Guided Exercises

1. Indicate if the following statements/features correspond to GNU Screen, tmux or both:

Feature/Statement	GNU Screen	tmux
Default command prefix is <code>Ctrl + a</code>		
Client-Server Model		
Panes are pseudo-terminals		
Killing a region does not kill its associated window(s)		
Sessions include windows		
Sessions can be detached		

2. Install GNU Screen on your computer (package name: `screen`) and complete the following tasks:

- Start the program. What command do you use?

- Start `top`:

- Using screen's key prefix, open a new window; then, open `/etc/screenrc` using `vi`:

- List the windows at the bottom of the screen:

- Change the name of the current window to `vi`:

- Change the name of the remaining window to `top`. To do that, first display a list of all windows so that you can move up and down and select the right one:

- Check that the names have changed by having the window names displayed at the bottom of

the screen again:

- Now, detach the session and have `screen` create a new one named `ssh`:

- Detach also from `ssh` and have `screen` display the list of sessions:

- Now, attach to the first session using its PID:

- You should be back at the window displaying `top`. Split the window horizontally and move to the new empty region:

- Have `screen` list all windows and select `vi` to be displayed in the new empty region:

- Now, split the current region vertically, move into the newly created empty region and associate it with a brand new window:

- Terminate all regions except the current one (remember, although you kill the regions, the windows are still alive). Then, quit out of all the windows of the current session until the session itself is terminated:

- Finally, have `screen` list its sessions one more time, kill the remaining `ssh` session by PID and check that there are actually no sessions left:

3. Install `tmux` on your computer (package name: `tmux`) and complete the following tasks:

- Start the program. What command do you use?

- Start `top` (note how — in a couple of seconds — the name of the window changes to `top` in the status bar):

- Using `tmux`'s key prefix, open a new window; then, create `~/ .tmux.conf` using `nano`:

- Split the window vertically and reduce the size of the newly created pane a few times:

- Now change the name of the current window to `text editing`; then, have `tmux` display a list with all its sessions:

- Move to the window running `top` and back to the current one using the same key combination:

- Detach from the current session and create a new one whose name is `ssh` and its window name is `ssh window`:

- Detach also from the `ssh` session and have `tmux` display the list of sessions again:

NOTE

From this point on the exercise requires that you use a *remote* machine for `ssh` connections to your local host (a virtual machine is perfectly valid and can prove really practical). Make sure you have `openssh-server` installed and running on your local machine and that at least `openssh-client` is installed on the remote machine.

- Now, start a remote machine and connect via `ssh` with your local host. Once the connection has been established, check for `tmux` sessions:

- On the remote host, attach to the `ssh` session by name:

- Back at your local machine, attach to the `ssh` session by name making sure the connection to the remote host is terminated first:

- Have all sessions displayed for selection and go to your first session (`[0]`). Once there, kill session `ssh` by name:

- Finally, detach from the current session and kill it by name:

Explorational Exercises

1. Both `screen` and `tmux` can enter command line mode through *command prefix* + `:` (we already saw a brief example with `tmux`). Do some research and the following tasks in command line mode:

- Make `screen` enter copy mode:

- Make `tmux` rename the current window:

- Make `screen` close all windows and terminate session:

- Make `tmux` split a pane into two:

- Make `tmux` kill the current window:

2. When you enter copy mode in `screen` not only can you use the arrow keys and `PgUP` or `PgDown` to navigate the current window and the scrollback buffer. There is also the possibility of using a vi-like full screen editor. Using this editor, perform the following tasks:

- Echo `supercalifragilisticexpialidocious` in your `screen` terminal:

- Now, copy the five consecutive characters (left-to-right) in the line right above your cursor:

- Finally, paste the selection (`stice`) back at your command prompt:

3. Suppose you want to share a `tmux` session (`our_session`) with another user. You have created the socket (`/tmp/our_socket`) with the right permissions so that both you and the other user can read and write. What other two conditions should be met in order for the second user to be able to successfully attach the session through `tmux -S /tmp/our_socket a -t`

our_session?

Summary

In this lesson you have learned about *terminal multiplexers* in general and GNU Screen and tmux in particular. Important concepts to remember include:

- Command prefix: `screen` uses `Ctrl + a + character`; `tmux`, `Ctrl + b + character`.
- Structure of sessions, windows and window divisions (regions or panes).
- Copy mode.
- Session detachment: one of the most powerful features of multiplexers.

Commands used in this lesson:

screen

Start a screen session.

tmux

Start a `tmux` session.

Answers to Guided Exercises

1. Indicate if the following statements/features correspond to GNU Screen, tmux or both:

Feature/Statement	GNU Screen	tmux
Default command prefix is <code>Ctrl + a</code>	x	
Client-Server Model		x
Panes are pseudo-terminals		x
Killing a region does not kill its associated window(s)	x	
Sessions include windows	x	x
Sessions can be detached	x	x

2. Install GNU Screen on your computer (package name: `screen`) and complete the following tasks:

- Start the program. What command do you use?

```
screen
```

- Start `top`:

```
top
```

- Using `screen`'s key prefix, open a new window; then, open `/etc/screenrc` using `vi`:

```
Ctrl + a - c
```

```
sudo vi /etc/screenrc
```

- List the windows at the bottom of the screen:

```
Ctrl + a - w
```

- Change the name of the current window to `vi`:

```
Ctrl + a - A. Then we have to type vi and press enter.
```

- Change the name of the remaining window to `top`. To do that, first display a list of all windows so that you can move up and down and select the right one:

First we type `Ctrl + a - "`. Then we use the arrow keys to mark the one that says `0 bash` and press `enter`. Finally, we type `Ctrl + a - A`, type in `top` and press `enter`.

- Check that the names have changed by having the window names displayed at the bottom of the screen again:

```
Ctrl + a - w
```

- Now, detach the session and have `screen` create a new one named `ssh`:

```
Ctrl + a - d screen -S "ssh" and press enter.
```

- Detach also from `ssh` and have `screen` display the list of sessions:

```
Ctrl + a - d screen -list or screen -ls.
```

- Now, attach to the first session using using its PID:

```
screen -r PID-OF-SESSION
```

- You should be back at the window displaying `top`. Split the window horizontally and move to the new empty region:

```
Ctrl + a - S
```

```
Ctrl + a - Tab
```

- Have `screen` list all windows and select `vi` to be displayed in the new empty region:

We use `Ctrl + a - "` to have all windows displayed for selection, mark `vi` and press `enter`.

- Now, split the current region vertically, move into the newly created empty region and associate it with a brand new window:

```
Ctrl + a - |
```

```
Ctrl + a - Tab
```

```
Ctrl + a - c
```

- Terminate all regions except the current one (remember, although you kill the regions, the windows are still alive). Then, quit out of all the windows of the current session until the session itself is terminated:

```
Ctrl + a - q. exit (to exit Bash). Shift + :, then we type quit and press enter (to exit vi). After that,
```


we type `exit` (to exit the underlying Bash shell) `q` (to terminate `top`); then we type `exit` (to exit the underlying Bash shell).

- Finally, have `screen` list its sessions one more time, kill the remaining `ssh` session by PID and check that there are actually no sessions left:

```
screen -list or screen -ls
```

```
screen -S PID-OF-SESSION -X quit
```

```
screen -list or screen -ls
```

3. Install `tmux` on your computer (package name: `tmux`) and complete the following tasks:

- Start the program. What command do you use?

```
tmux
```

- Start `top` (note how — in a couple of seconds — the name of the window changes to `top` in the status bar):

```
top
```

- Using `tmux`'s key prefix, open a new window; then, create `~/ .tmux.conf` using `nano`:

```
Ctrl + b - c nano ~/ .tmux.conf
```

- Split the window vertically and reduce the size of the newly created pane a few times:

```
Ctrl + b - "
```

```
Ctrl + b - Ctrl + i
```

- Now change the name of the current window to `text editing`; then, have `tmux` display a list with all its sessions:

```
Ctrl + b - ,. Then we supply the new name and press enter. Ctrl + b - s or tmux ls.
```

- Move to the window running `top` and back to the current one using the same key combination:

```
Ctrl + b - n OR Ctrl + b - p
```

- Detach from the current session and create a new one whose name is `ssh` and its window name is `ssh window`:

```
Ctrl + b -d tmux new -s "ssh" -n "ssh window"
```

- Detach also from the `ssh` session and have `tmux` display the list of sessions again:

```
Ctrl + b -d tmux ls
```

NOTE

From this point on the exercise requires that you use a *remote* machine for `ssh` connections to your local host (a virtual machine is perfectly valid and can prove really practical). Make sure you have `openssh-server` installed and running on your local machine and that at least `openssh-client` is installed on the remote machine.

- Now, start a remote machine and connect via `ssh` with your local host. Once the connection has been established, check for `tmux` sessions:

On remote host: `ssh local-username@local-ipaddress`. Once connected to the local machine: `tmux ls`.

- On the remote host, attach to the `ssh` session by name:

```
tmux a -t ssh (a can be replaced by at or attach).
```

- Back at your local machine, attach to the `ssh` session by name making sure the connection to the remote host is terminated first:

```
tmux a -d -t ssh (a can be replaced by at or attach).
```

- Have all sessions displayed for selection and go to your first session (`[0]`). Once there, kill session `ssh` by name:

We type `Ctrl + b -s`, use the arrow keys to mark session `0` and press `enter` `tmux kill-session -t ssh`.

- Finally, detach from the current session and kill it by name:

```
Ctrl + b -d tmux kill-session -t 0.
```

Answers to Explorational Exercises

- Both `screen` and `tmux` can enter command line mode through *command prefix* + `:` (we already saw a brief example with `tmux`). Do some research and the following tasks in command line mode:

- Make `screen` enter copy mode:

`Ctrl` + `a` - `:` — then, we type `copy`.

- Make `tmux` rename the current window:

`Ctrl` + `b` - `:` — then, we type `rename-window`.

- Make `screen` close all windows and terminate session:

`Ctrl` + `a` - `:` — then, we type `quit`.

- Make `tmux` split a pane into two:

`Ctrl` + `b` - `:` — then, we type `split-window`.

- Make `tmux` kill the current window:

`Ctrl` + `b` - `:` — then, we type `kill-window`.

- When you enter copy mode in `screen` not only can you use the arrow keys and `PgUp` or `PgDown` to navigate the current window and the scrollback buffer. There is also the possibility of using a vi-like full screen editor. Using this editor, perform the following tasks:

- Echo `supercalifragilisticexpialidocious` in your `screen` terminal:

```
echo supercalifragilisticexpialidocious
```

- Now, copy the five consecutive characters (left-to-right) in the line right above your cursor:

We enter copy mode: `Ctrl` + `a` - `[` or `Ctrl` + `a` - `:` and then type `copy`. Then, we move to the line above using `k` and press `space` to mark beginning of selection. Finally, we move forward four characters using `l` and press `space` again to mark the end of the selection.

- Finally, paste the selection (`stice`) back at your command prompt:

`Ctrl` + `a` - `]`

- Suppose you want to share a `tmux` session (`our_session`) with another user. You have created

the socket (/tmp/our_socket) with the right permissions so that both you and the other user can read and write. What other two conditions should be met in order for the second user to be able to successfully attach the session through `tmux -S /tmp/our_socket a -t our_session`?

Both users must have a group in common, e.g. `multiplexer`. Then, we must have the socket changed into that group as well: `chgrp multiplexer /tmp/our_socket`.



**Linux
Professional
Institute**

103.6 Modify process execution priorities

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 103.6

Weight

2

Key knowledge areas

- Know the default priority of a job that is created.
- Run a program with higher or lower priority than the default.
- Change the priority of a running process.

Partial list of the used files, terms and utilities

- `nice`
- `ps`
- `renice`
- `top`



103.6 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.6 Modify process execution priorities
Lesson:	1 of 1

Introduction

Operating systems able to run more than one process at the same time are called multi-tasking or multi-processing systems. Whilst true simultaneity only happens when more than one processing unit is available, even single processor systems can mimic simultaneity by switching between processes very quickly. This technique is also employed in systems with many equivalent CPUs, or *symmetric multi-processor (SMP)* systems, given that the number of potential concurrent processes greatly exceeds the number of available processor units.

In fact, only one process at a time can control the CPU. However, most process activities are *system calls*, that is, the running process transfers CPU control to an operating system's process so it performs the requested operation. System calls are in charge of all inter-device communication, like memory allocations, reading and writing on filesystems, printing text on the screen, user interaction, network transfers, etc. Transferring CPU control during a system call allows the operating system to decide whether to return CPU control to the previous process or to hand it to another process. As modern CPUs can execute instructions much faster than most external hardware can communicate with each other, a new controlling process can do a lot of CPU work while previously requested hardware responses are still unavailable. To ensure maximum CPU harnessing, multi-processing operating systems keep a dynamic queue of active processes waiting

for a CPU time slot.

Although they allow to significantly improve CPU time utilization, relying solely on system calls to switch between processes is not enough to achieve satisfactory multi-tasking performance. A process that makes no system calls could control the CPU indefinitely. This is why modern operating systems are also *preemptive*, that is, a running process can be put back in the queue so a more important process can control the CPU, even if the running process has not made a system call.

The Linux Scheduler

Linux, as a preemptive multi-processing operating system, implements a scheduler that organizes the process queue. More precisely, the scheduler also decides which queued *thread* will be executed—a process can branch out many independent threads—but process and thread are interchangeable terms in this context. Every process has two predicates that intervene on its scheduling: the *scheduling policy* and the *scheduling priority*.

There are two main types of scheduling policies: *real-time policies* and *normal policies*. Processes under a real-time policy are scheduled by their priority values directly. If a more important process becomes ready to run, a less important running process is preempted and the higher priority process takes control of the CPU. A lower priority process will gain CPU control only if higher priority processes are idle or waiting for hardware response.

Any real-time process has higher priority than a normal process. As a general purpose operating system, Linux runs just a few real-time processes. Most processes, including system and user programs, run under normal scheduling policies. Normal processes usually have the same priority value, but normal policies can define execution priority rules using another process predicate: the *nice value*. To avoid confusion with the dynamic priorities derived from nice values, scheduling priorities are usually called *static scheduling priorities*.

The Linux scheduler can be configured in many different ways and even more intricate ways of establishing priorities exist, but these general concepts always apply. When inspecting and tuning process scheduling, it is important to keep in mind that only processes under normal scheduling policy will be affected.

Reading Priorities

Linux reserves static priorities ranging from 0 to 99 for real-time processes and normal processes are assigned to static priorities ranging from 100 to 139, meaning that there are 39 different priority levels for normal processes. Lower values mean higher priority. The static priority of an active process can be found in the `sched` file, located in its respective directory inside the `/proc`

filesystem:

```
$ grep ^prio /proc/1/sched
prio                :      120
```

As shown in the example, the line beginning with `prio` gives the priority value of the process (the PID 1 process is the *init* or the *systemd* process, the first process the kernel starts during system initialization). The standard priority for normal processes is 120, so that it can be decreased to 100 or increased to 139. The priorities of all running process can be verified with the command `ps -Al` or `ps -el`:

```
$ ps -el
 F S  UID  PID  PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
 4 S   0    1    0  0  80   0 -  9292 -    ?         00:00:00 systemd
 4 S   0   19    1  0  80   0 -  8817 -    ?         00:00:00 systemd-journal
 4 S  104   61    1  0  80   0 - 64097 -    ?         00:00:00 rsyslogd
 4 S   0   63    1  0  80   0 -  7244 -    ?         00:00:00 cron
 1 S   0  126    1  0  80   0 -  4031 -    ?         00:00:00 dhclient
 4 S   0  154    1  0  80   0 -  3937 - pts/0      00:00:00 agetty
 4 S   0  155    1  0  80   0 -  3937 - pts/1      00:00:00 agetty
 4 S   0  156    1  0  80   0 -  3937 - pts/2      00:00:00 agetty
 4 S   0  157    1  0  80   0 -  3937 - pts/3      00:00:00 agetty
 4 S   0  158    1  0  80   0 -  3937 - console   00:00:00 agetty
 4 S   0  160    1  0  80   0 - 16377 -    ?         00:00:00 sshd
 4 S   0  280    0  0  80   0 -  5301 -    ?         00:00:00 bash
 0 R   0  392  280  0  80   0 -  7221 -    ?         00:00:00 ps
```

The `PRI` column indicates the static priority assigned by the kernel. Note, however, that the priority value displayed by `ps` differs from that obtained in the previous example. Due to historical reasons, priorities displayed by `ps` range from -40 to 99 by default, so the actual priority is obtained by adding 40 to it (in particular, $80 + 40 = 120$).

It is also possible to continuously monitor processes currently being managed by the Linux kernel with program `top`. As with `ps`, `top` also displays the priority value differently. To make it easier to identify real-time processes, `top` displays their priority using the formula `PR = -1-real_time_priority` (`real_time_priority` ranges from 1 to 99), thus making all real-time priorities negative, with a negative number or *rt* identifying them. Therefore, normal priorities displayed by `top` range from 0 to 39.

NOTE

To get more details from the `ps` command, additional options can be used. Compare the output from this command to the one from our previous example:


```
$ ps -e -o user,uid,comm,TTY,pid,ppid,pri,pmem,pcpu --sort=-pcpu | head
```

Process Niceness

Every normal process begins with a default nice value of 0 (priority 120). The *nice* name comes from the idea that “nicer” processes allow other processes to run before them in a particular execution queue. Nice numbers range from -20 (less nice, high priority) to 19 (more nice, low priority). Linux also allows the ability to assign different nice values to threads within the same process. The NI column in `ps` output indicates the *nice* number.

Only the root user can decrease the niceness of a process below zero. It’s possible to start a process with a non-standard priority with the command `nice`. By default, `nice` changes the niceness to 10, but it can be specified with option `-n`:

```
$ nice -n 15 tar czf home_backup.tar.gz /home
```

In this example, the command `tar` is executed with a niceness of 15. The command `renice` can be used to change the priority of a running process. The option `-p` indicates the PID number of the target process. For example:

```
# renice -10 -p 2164
2164 (process ID) old priority 0, new priority -10
```

The options `-g` and `-u` are used to modify all the processes of a specific group or user, respectively. With `renice +5 -g users`, the niceness of processes owned by users of the group *users* will be raised in five.

Besides `renice`, the priority of processes can be modified with other programs, like the program `top`. On the `top` main screen, the niceness of a process can be modified by pressing `r` and then the PID number of the process:

```
top - 11:55:21 up 23:38, 1 user, load average: 0,10, 0,04, 0,05
Tasks: 20 total, 1 running, 19 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0,5 us, 0,3 sy, 0,0 ni, 99,0 id, 0,0 wa, 0,2 hi, 0,0 si, 0,0 st
KiB Mem : 4035808 total, 774700 free, 1612600 used, 1648508 buff/cache
KiB Swap: 7999828 total, 7738780 free, 261048 used. 2006688 avail Mem

PID to renice [default pid = 1]
  PID USER      PR  NI   VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   74232   7904   6416 S   0,000 0,196   0:00.12 systemd
```

```
15 root      20    0   67436   6144   5568 S 0,000 0,152   0:00.03 systemd-journal
21 root      20    0   61552   5628   5000 S 0,000 0,139   0:00.01 systemd-logind
22 message+  20    0   43540   4072   3620 S 0,000 0,101   0:00.03 dbus-daemon
23 root      20    0   45652   6204   4992 S 0,000 0,154   0:00.06 wickedd-dhcp4
24 root      20    0   45648   6276   5068 S 0,000 0,156   0:00.06 wickedd-auto4
25 root      20    0   45648   6272   5060 S 0,000 0,155   0:00.06 wickedd-dhcp6
```

The message `PID to renice [default pid = 1]` appears with the first listed process selected by default. To change the priority of another process, type its PID and press Enter. Then, the message `Renice PID 1 to value` will appear (with the requested PID number) and a new nice value can be assigned.

Guided Exercises

1. In a preemptive multi-tasking system, what happens when a lower priority process is occupying the processor and a higher priority process is queued to be executed?

2. Consider the following `top` screen:

```
top - 08:43:14 up 23 days, 12:29, 5 users, load average: 0,13, 0,18, 0,21
Tasks: 240 total, 2 running, 238 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1,4 us, 0,4 sy, 0,0 ni, 98,1 id, 0,0 wa, 0,0 hi, 0,0 si, 0,0 st
MiB Mem : 7726,4 total, 590,9 free, 1600,8 used, 5534,7 buff/cache
MiB Swap: 30517,0 total, 30462,5 free, 54,5 used. 5769,4 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	171420	10668	7612	S	0,0	0,1	9:59.15	systemd
2	root	20	0	0	0	0	S	0,0	0,0	0:02.76	kthreadd
3	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	rcu_par_gp
8	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	mm_percpu_wq
9	root	20	0	0	0	0	S	0,0	0,0	0:49.06	ksoftirqd/0
10	root	20	0	0	0	0	I	0,0	0,0	18:24.20	rcu_sched
11	root	20	0	0	0	0	I	0,0	0,0	0:00.00	rcu_bh
12	root	rt	0	0	0	0	S	0,0	0,0	0:08.17	migration/0
14	root	20	0	0	0	0	S	0,0	0,0	0:00.00	cpuhp/0
15	root	20	0	0	0	0	S	0,0	0,0	0:00.00	cpuhp/1
16	root	rt	0	0	0	0	S	0,0	0,0	0:11.79	migration/1
17	root	20	0	0	0	0	S	0,0	0,0	0:26.01	ksoftirqd/1

What PIDs have real-time priorities?

3. Consider the following `ps -el` listing:

F	S	UID	PID	PPID	C	PRI	NI	ADDR	SZ	WCHAN	TTY	TIME	CMD
4	S	0	1	0	0	80	0	-	42855	-	?	00:09:59	systemd
1	S	0	2	0	0	80	0	-	0	-	?	00:00:02	kthreadd
1	I	0	3	2	0	60	-20	-	0	-	?	00:00:00	rcu_gp
1	S	0	9	2	0	80	0	-	0	-	?	00:00:49	ksoftirqd/0
1	I	0	10	2	0	80	0	-	0	-	?	00:18:26	rcu_sched
1	I	0	11	2	0	80	0	-	0	-	?	00:00:00	rcu_bh
1	S	0	12	2	0	-40	-	-	0	-	?	00:00:08	migration/0

```
1 S      0    14    2 0 80  0 -    0 -    ?    00:00:00 cpuhp/0
5 S      0    15    2 0 80  0 -    0 -    ?    00:00:00 cpuhp/1
```

Which PID has higher priority?

4. After trying to renice a process with `renice`, the following error happens:

```
$ renice -10 21704
renice: failed to set priority for 21704 (process ID): Permission denied
```

What is the probable cause for the error?

Explorational Exercises

1. Changing process priorities is usually required when a process is occupying too much CPU time. Using `ps` with standard options for printing all system processes in long format, what `--sort` flag will sort processes by CPU utilization, increasing order?

2. Command `schedtool` can set all CPU scheduling parameters Linux is capable of or display information for given processes. How can it be used to display the scheduling parameters of process 1750? Also, how can `schedtool` be used to change process 1750 to real-time with priority -90 (as displayed by `top`)?

Summary

This lesson covers how Linux shares CPU time among its managed processes. To ensure best performance, more critical processes must overtake less critical processes. The lesson goes through the following steps:

- Basic concepts about multi-processing systems.
- What is a process scheduler and how Linux implements it.
- What are Linux priorities, nice numbers and their purpose.
- How to read and interpret process priorities in Linux.
- How to change the priority of a process, before and during its execution.

Answers to Guided Exercises

1. In a preemptive multi-tasking system, what happens when a lower priority process is occupying the processor and a higher priority process is queued to be executed?

The lower priority process pauses and the higher priority process is executed instead.

2. Consider the following `top` screen:

```
top - 08:43:14 up 23 days, 12:29, 5 users, load average: 0,13, 0,18, 0,21
Tasks: 240 total, 2 running, 238 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1,4 us, 0,4 sy, 0,0 ni, 98,1 id, 0,0 wa, 0,0 hi, 0,0 si, 0,0 st
MiB Mem : 7726,4 total, 590,9 free, 1600,8 used, 5534,7 buff/cache
MiB Swap: 30517,0 total, 30462,5 free, 54,5 used. 5769,4 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	171420	10668	7612	S	0,0	0,1	9:59.15	systemd
2	root	20	0	0	0	0	S	0,0	0,0	0:02.76	kthreadd
3	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	rcu_par_gp
8	root	0	-20	0	0	0	I	0,0	0,0	0:00.00	mm_percpu_wq
9	root	20	0	0	0	0	S	0,0	0,0	0:49.06	ksoftirqd/0
10	root	20	0	0	0	0	I	0,0	0,0	18:24.20	rcu_sched
11	root	20	0	0	0	0	I	0,0	0,0	0:00.00	rcu_bh
12	root	rt	0	0	0	0	S	0,0	0,0	0:08.17	migration/0
14	root	20	0	0	0	0	S	0,0	0,0	0:00.00	cpuhp/0
15	root	20	0	0	0	0	S	0,0	0,0	0:00.00	cpuhp/1
16	root	rt	0	0	0	0	S	0,0	0,0	0:11.79	migration/1
17	root	20	0	0	0	0	S	0,0	0,0	0:26.01	ksoftirqd/1

What PIDs have real-time priorities?

PIDs 12 and 16.

3. Consider the following `ps -el` listing:

F	S	UID	PID	PPID	C	PRI	NI	ADDR	SZ	WCHAN	TTY	TIME	CMD
4	S	0	1	0	0	80	0	- 42855	-		?	00:09:59	systemd
1	S	0	2	0	0	80	0	-	0	-	?	00:00:02	kthreadd
1	I	0	3	2	0	60	-20	-	0	-	?	00:00:00	rcu_gp
1	S	0	9	2	0	80	0	-	0	-	?	00:00:49	ksoftirqd/0
1	I	0	10	2	0	80	0	-	0	-	?	00:18:26	rcu_sched

1	I	0	11	2	0	80	0	-	0	-	?	00:00:00	rcu_bh
1	S	0	12	2	0	-40	-	-	0	-	?	00:00:08	migration/0
1	S	0	14	2	0	80	0	-	0	-	?	00:00:00	cpuhp/0
5	S	0	15	2	0	80	0	-	0	-	?	00:00:00	cpuhp/1

Which PID has higher priority?

PID 12.

4. After trying to renice a process with `renice`, the following error happens:

```
$ renice -10 21704
renice: failed to set priority for 21704 (process ID): Permission denied
```

What is the probable cause for the error?

Only user root can decrease nice numbers below zero.

Answers to Explorational Exercises

1. Changing process priorities is usually required when a process is occupying too much CPU time. Using `ps` with standard options for printing all system processes in long format, what `--sort` flag will sort processes by CPU utilization, increasing order?

```
$ ps -el --sort=pcpu
```

2. Command `schedtool` can set all CPU scheduling parameters Linux is capable of or display information for given processes. How can it be used to display the scheduling parameters of process 1750? Also, how can `schedtool` be used to change process 1750 to real-time with priority -90 (as displayed by `top`)?

```
$ schedtool 1750
```

```
$ schedtool -R -p 89 1750
```



103.7 Search text files using regular expressions

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 103.7

Weight

3

Key knowledge areas

- Create simple regular expressions containing several notational elements.
- Understand the differences between basic and extended regular expressions.
- Understand the concepts of special characters, character classes, quantifiers and anchors.
- Use regular expression tools to perform searches through a filesystem or file content.
- Use regular expressions to delete, change and substitute text.

Partial list of the used files, terms and utilities

- `grep`
- `egrep`
- `fgrep`
- `sed`
- `regex(7)`



103.7 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.7 Search text files using regular expressions
Lesson:	1 of 2

Introduction

String-searching algorithms are widely used by several data-processing tasks, so much that Unix-like operating systems have their own ubiquitous implementation: *Regular expressions*, often abbreviated to *REs*. Regular expressions consist of character sequences that make up a generic pattern used to locate and sometimes modify a corresponding sequence in a larger string of characters. Regular expressions greatly expand the ability to:

- Write parsing rules to requests in HTTP servers, *nginx* in particular.
- Write scripts that convert text-based datasets to another format.
- Search for occurrences of interest in journal entries or documents.
- Filter markup documents, keeping semantic content.

The simplest regular expression contains at least one *atom*. An atom, so named because it's the basic element of a regular expression, is just a character that may or may not have special meaning. Most ordinary characters are unambiguous, they retain their literal meaning, while others have special meaning:

. (dot)

Atom matches with any character.

^ (caret)

Atom matches with the beginning of a line.

\$ (dollar sign)

Atom matches with the end of a line.

For example, the regular expression `bc`, composed by the literal atoms `b` and `c`, can be found in the string `abcd`, but can not be found in the string `a1cd`. On the other hand, the regular expression `.c` can be found in both strings `abcd` and `a1cd`, as the dot `.` matches with any character.

The caret and dollar sign atoms are used when only matches at the beginning or at the end of the string are of interest. For that reason they are also called *anchors*. For example, `cd` can be found in `abcd`, but `^cd` can not. Similarly, `ab` can be found in `abcd`, but `ab$` can not. The caret `^` is a literal character except when at the beginning and `$` is a literal character except when at the end of the regular expression.

Bracket Expression

There is another type of atom named *bracket expression*. Although not a single character, brackets `[]` (including their content) are considered a single atom. A bracket expression usually is just a list of literal characters enclosed by `[]`, making the atom match any single character from the list. For example, the expression `[1b]` can be found in both strings `abcd` and `a1cd`. To specify characters the atom should not correspond to, the list must begin with `^`, as in `[^1b]`. It is also possible to specify ranges of characters in bracket expressions. For example, `[0-9]` matches digits 0 to 9 and `[a-z]` matches any lowercase letter. Ranges must be used with caution, as they might not be consistent across distinct locales.

Bracket expression lists also accept classes instead of just single characters and ranges. Traditional character classes are:

[:alnum:]

Represents an alphanumeric character.

[:alpha:]

Represents an alphabetic character.

[:ascii:]

Represents a character that fits into the ASCII character set.

[:blank:]

Represents a blank character, that is, a space or a tab.

[:cntrl:]

Represents a control character.

[:digit:]

Represents a digit (0 through 9).

[:graph:]

Represents any printable character except space.

[:lower:]

Represents a lowercase character.

[:print:]

Represents any printable character including space.

[:punct:]

Represents any printable character which is not a space or an alphanumeric character.

[:space:]

Represents white-space characters: space, form-feed (`\f`), newline (`\n`), carriage return (`\r`), horizontal tab (`\t`), and vertical tab (`\v`).

[:upper:]

Represents an uppercase letter.

[:xdigit:]

Represents hexadecimal digits (0 through F).

Character classes can be combined with single characters and ranges, but may not be used as an endpoint of a range. Also, character classes may be used only in bracket expressions, not as an independent atom outside the brackets.

Quantifiers

The reach of an atom, either a single character atom or a bracket atom, can be adjusted using an *atom quantifier*. Atom quantifiers define atom *sequences*, that is, matches occur when a contiguous repetition for the atom is found in the string. The substring corresponding to the match is called a

piece. Notwithstanding, quantifiers and other features of regular expressions are treated differently depending on which standard is being used.

As defined by POSIX, there are two forms of regular expressions: “basic” regular expressions and “extended” regular expressions. Most text related programs in any conventional Linux distribution support both forms, so it is important to know their differences in order to avoid compatibility issues and to pick the most suitable implementation for the intended task.

The `*` quantifier has the same function in both basic and extended REs (atom occurs zero or more times) and it's a literal character if it appears at the beginning of the regular expression or if it's preceded by a backslash `\`. The plus sign quantifier `+` will select pieces containing one or more atom matches in sequence. With the question mark quantifier `?`, a match will occur if the corresponding atom appears once or if it doesn't appear at all. If preceded by a backslash `\`, their special meaning is not considered. Basic regular expressions also support `^` and `$` quantifiers, but they need to be preceded by a backslash. Unlike extended regular expressions, `+` and `?` by themselves are literal characters in basic regular expressions.

Bounds

A *bound* is an atom quantifier that, as the name implies, allows a user to specify precise quantity boundaries for an atom. In extended regular expressions, a bound may appear in three forms:

`{i}`

The atom must appear exactly `i` times (`i` an integer number). For example, `[[:blank:]]{2}` matches with exactly two blank characters.

`{i,}`

The atom must appear at least `i` times (`i` an integer number). For example, `[[:blank:]]{2,}` matches with any sequence of two or more blank characters.

`{i,j}`

The atom must appear at least `i` times and at most `j` times (`i` and `j` integer numbers, `j` greater than `i`). For example, `xyz{2,4}` matches the `xy` string followed by two to four of the `z` character.

In any case, if a substring matches with a regular expression and a longer substring starting at the same point also matches, the longer substring will be considered.

Basic regular expressions also support bounds, but the delimiters must be preceded by `\`: `\{` and `\}`. By themselves, `{` and `}` are interpreted as literal characters. A `\{` followed by a character other

than a digit is a literal character, not the beginning of a bound.

Branches and Back References

Basic regular expressions also differ from extended regular expressions in another important aspect: an extended regular expression can be divided into *branches*, each one an independent regular expression. Branches are separated by `|` and the combined regular expression will match anything that corresponds to any of the branches. For example, `he|him` will match if either substring `he` or `him` are found in the string being examined. Basic regular expressions interpret `|` as a literal character. However, most programs supporting basic regular expressions will allow branches with `\|`.

An extended regular expression enclosed in `()` can be used in a *back reference*. For example, `([[:digit:]]+)\1` will match any regular expression that repeats itself at least once, because the `\1` in the expression is the back reference to the piece matched by the first parenthesized subexpression. If more than one parenthesized subexpression exist in the regular expression, they can be referenced with `\2`, `\3` and so on.

For basic REs, subexpressions must be enclosed by `\(` and `\)`, with `(` and `)` by themselves ordinary characters. The back reference indicator is used like in extended regular expressions.

Searching with Regular Expressions

The immediate benefit offered by regular expressions is to improve searches on filesystems and in text documents. The `-regex` option of command `find` allows to test every path in a directory hierarchy against a regular expression. For example,

```
$ find $HOME -regex '.*\/\..*' -size +100M
```

searches for files greater than 100 megabytes (100 units of 1048576 bytes), but only in paths inside the user's home directory that do contain a match with `.*\/\..*`, that is, a `/.` surrounded by any other number of characters. In other words, only hidden files or files inside hidden directories will be listed, regardless of the position of `/.` in the corresponding path. For case insensitive regular expressions, the `-iregex` option should be used instead:

```
$ find /usr/share/fonts -regextype posix-extended -iregex
'.*(dejavu|liberation).*sans.*(italic|oblique).*'
/usr/share/fonts/dejavu/DejaVuSansCondensed-BoldOblique.ttf
/usr/share/fonts/dejavu/DejaVuSansCondensed-Oblique.ttf
/usr/share/fonts/dejavu/DejaVuSans-BoldOblique.ttf
```

```
/usr/share/fonts/dejavu/DejaVuSans-Oblique.ttf
/usr/share/fonts/dejavu/DejaVuSansMono-BoldOblique.ttf
/usr/share/fonts/dejavu/DejaVuSansMono-Oblique.ttf
/usr/share/fonts/liberation/LiberationSans-BoldItalic.ttf
/usr/share/fonts/liberation/LiberationSans-Italic.ttf
```

In this example, the regular expression contains branches (written in *extended* style) to list only specific font files under the `/usr/share/fonts` directory hierarchy. Extended regular expressions are not supported by default, but `find` allows for them to be enabled with `-regextype posix-extended` or `-regextype egrep`. The default RE standard for `find` is *findutils-default*, which is virtually a basic regular expression clone.

It is often necessary to pass the output of a program to command `less` when it doesn't fit on the screen. Command `less` splits its input in pages, one screenful at a time, allowing the user to easily navigate the text up and down. In addition, `less` also allows a user to perform regular expression based searches. This feature is notably important because `less` is the default paginator used for many everyday tasks, like inspecting journal entries or consulting manual pages. When reading a manual page, for instance, pressing the `/` key will open a search prompt. This is a typical scenario in which regular expressions are useful, as command options are listed just after a page margin in the general manual page layout. However, the same option might appear many times through the text, making literal searches unfeasible. Regardless of that, typing `^[[:blank:]]*-o—` or more simply: `^ *-o—` in the search prompt will jump immediately to option the `-o` section (if it exists) after pressing Enter, thus allowing one to consult an option description more rapidly.

Guided Exercises

1. What extended regular expression would match any email address, like `info@example.org`?
2. What extended regular expression would only match any IPv4 address in the standard dotted-quad format, like `192.168.15.1`?
3. How can the `grep` command be used to list the contents of file `/etc/services`, discarding all comments (lines starting with `#`)?
4. The file `domains.txt` contains a list of domain names, one per line. How would the `egrep` command be used to list only `.org` or `.com` domains?

Explorational Exercises

1. From the current directory, how would the `find` command use an extended regular expression to search for all files not containing a standard file suffix (file names not ending in `.txt` or `.c`, for example)?

2. Command `less` is the default paginator for displaying long text files in the shell environment. By typing `/`, a regular expression can be entered in the search prompt to jump to the first corresponding match. In order to stay in the current document position and only highlight the corresponding matches, what key combination should be entered at the search prompt?

3. In `less`, how would it be possible to filter the output so only lines which match a regular expression get displayed?

Summary

This lesson covers the general Linux support for regular expressions, a widely used standard whose pattern matching capabilities are supported by most text related programs. The lesson goes through the following steps:

- What a regular expression is.
- The main components of a regular expression.
- The differences between regular and extended regular expressions.
- How to perform simple text and file searches using regular expressions.

Answers to Guided Exercises

1. What extended regular expression would match any email address, like `info@example.org`?

```
egrep "\S+@\S+\.\S+"
```

2. What extended regular expression would only match any IPv4 address in the standard dotted-quad format, like `192.168.15.1`?

```
egrep "[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}"
```

3. How can the `grep` command be used to list the contents of file `/etc/services`, discarding all comments (lines starting with `#`)?

```
grep -v ^# /etc/services
```

4. The file `domains.txt` contains a list of domain names, one per line. How would the `egrep` command be used to list only `.org` or `.com` domains?

```
egrep ".org$|.com$" domains.txt
```

Answers to Explorational Exercises

1. From the current directory, how would the `find` command use an extended regular expression to search for all files not containing a standard file suffix (file names not ending in `.txt` or `.c`, for example)?

```
find . -type f -regextype egrep -not -regex '.*\.[[:alnum:]]{1,}$'
```

2. Command `less` is the default paginator for displaying long text files in the shell environment. By typing `/`, a regular expression can be entered in the search prompt to jump to the first corresponding match. In order to stay in the current document position and only highlight the corresponding matches, what key combination should be entered at the search prompt?

Pressing `Ctrl` + `k` before entering the search expression.

3. In `less`, how would it be possible to filter the output so only lines which match a regular expression get displayed?

By pressing `&` and entering the search expression.



103.7 Lesson 2

Certificate:	LPIC-1
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.7 Search text files using regular expressions
Lesson:	2 of 2

Introduction

Streaming data through a chain of piped commands allows for the application of compound filters based on regular expressions. Regular expressions are an important technique used not only in system administration, but also in *data mining* and related areas. Two commands are specially suited to manipulate files and text data using regular expressions: `grep` and `sed`. `grep` is a pattern finder and `sed` is a stream editor. They are useful by themselves, but it is when working together with other processes that they stand out.

The Pattern Finder: `grep`

One of the most common uses of `grep` is to facilitate the inspection of long files, using the regular expression as a filter applied to each line. It can be used to show only the lines starting with a certain term. For example, `grep` can be used to investigate a configuration file for kernel modules, listing only option lines:

```
$ grep '^options' /etc/modprobe.d/alsa-base.conf
options snd-pcsp index=-2
options snd-usb-audio index=-2
```

```
options bt87x index=-2
options cx88_alsa index=-2
options snd-atiixp-modem index=-2
options snd-intel8x0m index=-2
options snd-via82xx-modem index=-2
```

The pipe `|` character can be employed to redirect the output of a command directly to `grep`'s input. The following example uses a bracket expression to select lines from `fdisk -l` output, starting with `Disk /dev/sda` or `Disk /dev/sdb`:

```
# fdisk -l | grep '^Disk /dev/sd[ab] '
Disk /dev/sda: 320.1 GB, 320072933376 bytes, 625142448 sectors
Disk /dev/sdb: 7998 MB, 7998537728 bytes, 15622144 sectors
```

The mere selection of lines with matches may not be appropriate for a particular task, requiring adjustments to `grep`'s behavior through its options. For example, option `-c` or `--count` tells `grep` to show how many lines had matches:

```
# fdisk -l | grep '^Disk /dev/sd[ab] ' -c
2
```

The option can be placed before or after the regular expression. Other important `grep` options are:

-c or --count

Instead of displaying the search results, only display the total count for how many times a match occurs in any given file.

-i or --ignore-case

Turn the search case-insensitive.

-f FILE or --file=FILE

Indicate a file containing the regular expression to use.

-n or --line-number

Show the number of the line.

-v or --invert-match

Select every line, except those containing matches.

-H or --with-filename

Print also the name of the file containing the line.

-z or --null-data

Rather than have `grep` treat input and output data streams as separate lines (using the *newline* by default) instead take the input or output as a sequence of lines. When combining output from the `find` command using its `-print0` option with the `grep` command, the `-z` or `--null-data` option should be used to process the stream in the same manner.

Although activated by default when multiple file paths are given as input, the option `-H` is not activated for single files. That may be critical in special situations, like when `grep` is called directly by `find`, for instance:

```
$ find /usr/share/doc -type f -exec grep -i '3d modeling' "{}" \; | cut -c -100
artistic aspects of 3D modeling. Thus this might be the application you are
This major approach of 3D modeling has not been supported
oce is a C++ 3D modeling library. It can be used to develop CAD/CAM softwares, for instance
[FreeCad
```

In this example, `find` lists every file under `/usr/share/doc` then passes each one to `grep`, which in turn performs a case-insensitive search for `3d modeling` inside the file. The pipe to `cut` is there just to limit output length to 100 columns. Note, however, that there is no way of knowing from which file the lines came from. This issue is solved by adding `-H` to `grep`:

```
$ find /usr/share/doc -type f -exec grep -i -H '3d modeling' "{}" \; | cut -c -100
/usr/share/doc/openscad/README.md:artistic aspects of 3D modeling. Thus this might be the
applicatio
/usr/share/doc/openscg/doc/publications.html:This major approach of 3D modeling has not been
support
```

Now it is possible to identify the files where each match was found. To make the listing even more informative, leading and trailing lines can be added to lines with matches:

```
$ find /usr/share/doc -type f -exec grep -i -H -1 '3d modeling' "{}" \; | cut -c -100
/usr/share/doc/openscad/README.md-application Blender), OpenSCAD focuses on the CAD aspects
rather t
/usr/share/doc/openscad/README.md:artistic aspects of 3D modeling. Thus this might be the
applicatio
/usr/share/doc/openscad/README.md-looking for when you are planning to create 3D models of
machine p
```



```

/usr/share/doc/opencsg/doc/publications.html-3D graphics library for Constructive Solid
Geometry (CS
/usr/share/doc/opencsg/doc/publications.html:This major approach of 3D modeling has not been
support
/usr/share/doc/opencsg/doc/publications.html-by real-time computer graphics until recently.

```

The option `-1` instructs `grep` to include one line before and one line after when it finds a line with a match. These extra lines are called *context lines* and are identified in the output by a minus sign after the file name. The same result can be obtained with `-C 1` or `--context=1` and other context line quantities may be indicated.

There are two complementary programs to `grep`: `egrep` and `fgrep`. The program `egrep` is equivalent to the command `grep -E`, which incorporates extra features other than the basic regular expressions. For example, with `egrep` it is possible to use extended regular expression features, like branching:

```

$ find /usr/share/doc -type f -exec egrep -i -H -1 '3d (modeling|printing)' "{}" \; | cut -c
-100
/usr/share/doc/opencad/README.md-application Blender), OpenSCAD focuses on the CAD aspects
rather t
/usr/share/doc/opencad/README.md:artistic aspects of 3D modeling. Thus this might be the
applicatio
/usr/share/doc/opencad/README.md-looking for when you are planning to create 3D models of
machine p
/usr/share/doc/opencad/RELEASE_NOTES.md-* Support for using 3D-Mouse / Joystick / Gamepad
input dev
/usr/share/doc/opencad/RELEASE_NOTES.md:* 3D Printing support: Purchase from a print
service partne
/usr/share/doc/opencad/RELEASE_NOTES.md-* New export file formats: SVG, 3MF, AMF
/usr/share/doc/opencsg/doc/publications.html-3D graphics library for Constructive Solid
Geometry (CS
/usr/share/doc/opencsg/doc/publications.html:This major approach of 3D modeling has not been
support
/usr/share/doc/opencsg/doc/publications.html-by real-time computer graphics until recently.

```

In this example either `3D modeling` or `3D printing` will match the expression, case-insensitive. To display only the parts of a text stream that match the expression used by `egrep`, use the `-o` option.

The program `fgrep` is equivalent to `grep -F`, that is, it does not parse regular expressions. It is useful in simple searches where the goal is to match a literal expression. Therefore, special characters like the dollar sign and the dot will be taken literally and not by their meanings in a

regular expression.

The Stream Editor: sed

The purpose of the `sed` program is to modify text-based data in a non-interactive way. It means that all the editing is made by predefined instructions, not by arbitrarily typing directly into a text displayed on the screen. In modern terms, `sed` can be understood as a template parser: given a text as input, it places custom content at predefined positions or when it finds a match for a regular expression.

`Sed`, as the name implies, is well suited for text streamed through pipelines. Its basic syntax is `sed -f SCRIPT` when editing instructions are stored in the file `SCRIPT` or `sed -e COMMANDS` to execute `COMMANDS` directly from the command line. If neither `-f` or `-e` are present, `sed` uses the first non-option parameter as the script file. It is also possible to use a file as the input just by giving its path as an argument to `sed`.

`sed` instructions are composed of a single character, possibly preceded by an address or followed by one or more options, and are applied to each line at a time. Addresses can be a single line number, a regular expression, or a range of lines. For example, the first line of a text stream can be deleted with `1d`, where `1` specifies the line where the delete command `d` will be applied. To clarify `sed`'s usage, take the output of the command `factor `seq 12``, which returns the prime factors for numbers 1 to 12:

```
$ factor `seq 12`
1:
2: 2
3: 3
4: 2 2
5: 5
6: 2 3
7: 7
8: 2 2 2
9: 3 3
10: 2 5
11: 11
12: 2 2 3
```

Deleting the first line with `sed` is accomplished by `1d`:

```
$ factor `seq 12` | sed 1d
2: 2
```

```
3: 3
4: 2 2
5: 5
6: 2 3
7: 7
8: 2 2 2
9: 3 3
10: 2 5
11: 11
12: 2 2 3
```

A range of lines can be specified with a separating comma:

```
$ factor `seq 12` | sed 1,7d
8: 2 2 2
9: 3 3
10: 2 5
11: 11
12: 2 2 3
```

More than one instruction can be used in the same execution, separated by semicolons. In this case, however, it is important to enclose them with parenthesis so the semicolon is not interpreted by the shell:

```
$ factor `seq 12` | sed "1,7d;11d"
8: 2 2 2
9: 3 3
10: 2 5
12: 2 2 3
```

In this example, two deletion instructions were executed, first on lines ranging from 1 to 7 and then on line 11. An address can also be a regular expression, so only lines with a match will be affected by the instruction:

```
$ factor `seq 12` | sed "1d;/:.*2.*/d"
3: 3
5: 5
7: 7
9: 3 3
11: 11
```

The regular expression `:.*2.*` matches with any occurrence of the number 2 anywhere after a colon, causing the deletion of lines corresponding to numbers with 2 as a factor. With `sed`, anything placed between slashes (/) is considered a regular expression and by default all basic RE is supported. For example, `sed -e "/^#/d" /etc/services` shows the contents of the file `/etc/services` without the lines beginning with `#` (comment lines).

The delete instruction `d` is only one of the many editing instructions provided by `sed`. Instead of deleting a line, `sed` can replace it with a given text:

```
$ factor `seq 12` | sed "1d;/:.*2.*/c REMOVED"
REMOVED
3: 3
REMOVED
5: 5
REMOVED
7: 7
REMOVED
9: 3 3
REMOVED
11: 11
REMOVED
```

The instruction `c REMOVED` simply replaces a line with the text `REMOVED`. In the example's case, every line with a substring matching the regular expression `:.*2.*` is affected by instruction `c REMOVED`. Instruction `a TEXT` copies text indicated by `TEXT` to a new line after the line with a match. The instruction `r FILE` does the same, but copies the contents of the file indicated by `FILE`. Instruction `w` does the opposite of `r`, that is, the line will be appended to the indicated file.

By far the most used `sed` instruction is `s/FIND/REPLACE/`, which is used to replace a match to the regular expression `FIND` with text indicated by `REPLACE`. For example, the instruction `s/hda/sda/` replaces a substring matching the literal RE `hda` with `sda`. Only the first match found in the line will be replaced, unless the flag `g` is placed after the instruction, as in `s/hda/sda/g`.

A more realistic case study will help to illustrate `sed`'s features. Suppose a medical clinic wants to send text messages to its customers, reminding them of their scheduled appointments for the next day. A typical implementation scenario relies on a professional instant message service, which provides an API to access the system responsible for delivering the messages. These messages usually originate from the same system that runs the application controlling customer's appointments, triggered by a specific time of the day or some other event. In this hypothetical situation, the application could generate a file called `appointments.csv` containing tabulated data with all the appointments for the next day, then used by `sed` to render the text messages

from a template file called `template.txt`. CSV files are a standard way of export data from database queries, so sample appointments could be given as follows:

```
$ cat appointments.csv
"NAME","TIME","PHONE"
"Carol","11am","55557777"
"Dave","2pm","33334444"
```

The first line holds the labels for each column, which will be used to match the tags inside the sample template file:

```
$ cat template.txt
Hey <NAME>, don't forget your appointment tomorrow at <TIME>.
```

The less than `<` and greater than `>` signs were put around labels just to help identify them as tags. The following Bash script parses all enqueued appointments using `template.txt` as the message template:

```
#!/bin/bash

TEMPLATE=`cat template.txt`
TAGS=(`sed -ne '1s/^"//;1s/","/\n/g;1s/"$//p' appointments.csv`)
mapfile -t -s 1 ROWS < appointments.csv
for (( r = 0; r < ${#ROWS[*]}; r++ ))
do
    MSG=$TEMPLATE
    VALS=(`sed -e 's/^"//;s/","/\n/g;s/"$//' <<<${ROWS[$r]}`)
    for (( c = 0; c < ${#TAGS[*]}; c++ ))
    do
        MSG=`sed -e "s/<${TAGS[$c]}>/${VALS[$c]}/g" <<<"$MSG"`
    done
    echo curl --data message="\$MSG" --data phone="\${VALS[2]}" https://mysmsprovider/api
done
```

An actual production script would also handle authentication, error checking and logging, but the example has basic functionality to start with. The first instructions executed by `sed` are applied only to the first line—the address `1` in `1s/^"//;1s/","/\n/g;1s/"$//p`—to remove the leading and trailing quotes—`1s/^"//` and `1s/"$//`—and to replace field separators with a newline character: `1s/","/\n/g`. Only the first line is needed for loading column names, so non-matching lines will be suppressed by option `-n`, requiring flag `p` to be placed after the last `sed`

command to print the matching line. The tags are then stored in the `TAGS` variable as a Bash array. Another Bash array variable is created by the command `mapfile` to store the lines containing the appointments in the array variable `ROWS`.

A `for` loop is employed to process each appointment line found in `ROWS`. Then, quotes and separators in the appointment—the appointment is in variable `${ROWS[$r]}` used as a *here string*—are replaced by `sed`, similarly to the commands used to load the tags. The separated values for the appointment are then stored in the array variable `VALS`, where array subscripts 0, 1 and 2 correspond to values for `NAME`, `TIME` and `PHONE`.

Finally, a nested `for` loop walks through the `TAGS` array and replaces each tag found in the template with its corresponding value in `VALS`. The `MSG` variable holds a copy of the rendered template, updated by the substitution command `s/<${TAGS[$c]}>/${VALS[$c]}/g` on every loop pass through `TAGS`.

This results in a rendered message like: "Hey Carol, don't forget your appointment tomorrow at 11am." The rendered message can then be sent as a parameter through a HTTP request with `curl`, as a mail message or any other similar method.

Combining grep and sed

Commands `grep` and `sed` can be used together when more complex text mining procedures are required. As a system administrator, you may want to inspect all the login attempts to a server, for example. The file `/var/log/wtmp` records all logins and logouts, whilst the file `/var/log/btmp` records the failed login attempts. They are written in a binary format, which can be read by the commands `last` and `lastb`, respectively.

The output of `lastb` shows not only the username used in the bad login attempt, but its IP address as well:

```
# lastb -d -a -n 10 --time-format notime
user      ssh:notty      (00:00)      81.161.63.251
nrostagn  ssh:notty      (00:00)      vmd60532.contaboserver.net
pi        ssh:notty      (00:00)      132.red-88-20-39.staticip.rima-tde.net
pi        ssh:notty      (00:00)      132.red-88-20-39.staticip.rima-tde.net
pi        ssh:notty      (00:00)      46.6.11.56
pi        ssh:notty      (00:00)      46.6.11.56
nps       ssh:notty      (00:00)      vmd60532.contaboserver.net
narmadan  ssh:notty      (00:00)      vmd60532.contaboserver.net
nominati  ssh:notty      (00:00)      vmd60532.contaboserver.net
nominati  ssh:notty      (00:00)      vmd60532.contaboserver.net
```

Option `-d` translates the IP number to the corresponding hostname. The hostname may provide clues about the ISP or hosting service used to perform these bad login attempts. Option `-a` puts the hostname in the last column, which facilitates the filtering yet to be applied. Option `--time-format notime` suppresses the time when the login attempt occurred. Command `lastb` can take some time to finish if there were too many bad login attempts, so the output was limited to ten entries with the option `-n 10`.

Not all remote IPs have a hostname associated to it, so reverse DNS does not apply to them and they can be dismissed. Although you could write a regular expression to match the expected format for a hostname at the end of the line, it is probably simpler to write a regular expression to match with either a letter from the alphabet or with a single digit at the end of the line. The following example shows how the command `grep` takes the listing at its standard input and removes the lines without hostnames:

```
# lastb -d -a --time-format notime | grep -v '[0-9]$\ ' | head -n 10
nvidia  ssh:notty      (00:00)  vmd60532.contaboserver.net
n_tonson ssh:notty      (00:00)  vmd60532.contaboserver.net
nrostagn ssh:notty      (00:00)  vmd60532.contaboserver.net
pi      ssh:notty      (00:00)  132.red-88-20-39.staticip.rima-tde.net
pi      ssh:notty      (00:00)  132.red-88-20-39.staticip.rima-tde.net
nps     ssh:notty      (00:00)  vmd60532.contaboserver.net
narmadan ssh:notty      (00:00)  vmd60532.contaboserver.net
nominati ssh:notty      (00:00)  vmd60532.contaboserver.net
nominati ssh:notty      (00:00)  vmd60532.contaboserver.net
nominati ssh:notty      (00:00)  vmd60532.contaboserver.net
```

Command `grep` option `-v` shows only the lines that don't match with the given regular expression. A regular expression matching any line ending with a number (i.e. `[0-9]$\`) will capture only the entries without a hostname. Therefore, `grep -v '[0-9]$\ '` will show only the lines ending with a hostname.

The output can be filtered even further, by keeping only the domain name and removing the other parts from each line. Command `sed` can do it with a substitution command to replace the whole line with a back-reference to the domain name in it:

```
# lastb -d -a --time-format notime | grep -v '[0-9]$\ ' | sed -e 's/.* \(.*)$/\1/' | head -n 10
vmd60532.contaboserver.net
vmd60532.contaboserver.net
vmd60532.contaboserver.net
132.red-88-20-39.staticip.rima-tde.net
```

```
132.red-88-20-39.staticip.rima-tde.net
vmd60532.contaboserver.net
vmd60532.contaboserver.net
vmd60532.contaboserver.net
vmd60532.contaboserver.net
vmd60532.contaboserver.net
```

The escaped parenthesis in `. * \(. * \)$` tells `sed` to remember that part of the line, that is, the part between the last space character and the end of the line. In the example, this part is referenced with `\1` and used to replace the entire line.

It's clear that most remote hosts try to login more than once, thus the same domain name repeats itself. To suppress the repeated entries, first they need to be sorted (with command `sort`) then passed to the command `uniq`:

```
# lastb -d -a --time-format notime | grep -v '[0-9]$\ ' | sed -e 's/. * \(. * \)$/\1/' | sort |
uniq | head -n 10
116-25-254-113-on-nets.com
132.red-88-20-39.staticip.rima-tde.net
145-40-33-205.power-speed.at
tor.laquadrature.net
tor.momx.site
ua-83-226-233-154.bbcust.telenor.se
vmd38161.contaboserver.net
vmd60532.contaboserver.net
vmi488063.contaboserver.net
vmi515749.contaboserver.net
```

This shows how different commands can be combined to produce the desired outcome. The hostname list can then be used to write blocking firewall rules or to take other measures to enforce the security of the server.

Guided Exercises

1. Command `last` shows a listing of last logged in users, including their origin IPs. How would the `egrep` command be used to filter `last` output, showing only occurrences of an IPv4 address, discarding any additional information in the corresponding line?
2. What option should be given to `grep` in order to correctly filter the output generated by command `find` executed with option `-print0`?
3. Command `uptime -s` shows the last date when the system was powered on, as in `2019-08-05 20:13:22`. What will be the result of command `uptime -s | sed -e 's/(.*) (.*)/\1/'`?
4. What option should be given to `grep` so it counts matching lines instead of displaying them?

Explorational Exercises

1. The basic structure of an HTML file starts with elements `html`, `head` and `body`, for example:

```
<html>
<head>
  <title>News Site</title>
</head>
<body>
  <h1>Headline</h1>
  <p>Information of interest.</p>
</body>
</html>
```

Describe how addresses could be used in `sed` to display only the `body` element and its contents.

2. What `sed` expression will remove all tags from an HTML document, keeping only the rendered text?

3. Files with extension `.ovpn` are very popular to configure VPN clients as they contain not only the settings, but also the contents of keys and certificates for the client. These keys and certificates are originally in separate files, so they need to be copied into the `.ovpn` file. Given the following excerpt of a `.ovpn` template:

```
client
dev tun
remote 192.168.1.155 1194
<ca>
ca.crt
</ca>
<cert>
client.crt
</cert>
<key>
client.key
</key>
<tls-auth>
ta.key
</tls-auth>
```

Assuming files `ca.crt`, `client.crt`, `client.key` and `ta.key` are in the current directory, how would the template configuration be modified by `sed` to replace each filename by its content?

Summary

This lesson covers the two most important Linux commands related to regular expressions: `grep` and `sed`. Scripts and compound commands rely on `grep` and `sed` to perform a wide range of text filtering and parsing tasks. The lesson goes through the following steps:

- How to use `grep` and its variations such as `egrep` and `fgrep`.
- How to use `sed` and its internal instructions to manipulate text.
- Examples of regular expression applications using `grep` and `sed`.

Answers to Guided Exercises

1. Command `last` shows a listing of last logged in users, including their origin IPs. How would the `egrep` command be used to filter `last` output, showing only occurrences of an IPv4 address, discarding any additional information in the corresponding line?

```
last -i | egrep -o '[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}\.[0-9]{1,3}'
```

2. What option should be given to `grep` in order to correctly filter the output generated by command `find` executed with option `-print0`?

The option `-z` or `--null-data`, as in `find . -print0 | grep -z expression`.

3. Command `uptime -s` shows the last date when the system was powered on, as in `2019-08-05 20:13:22`. What will be the result of command `uptime -s | sed -e 's/(.*) (.*)/\1/'`?

An error will occur. By default, parenthesis should be escaped to use backreferences in `sed`.

4. What option should be given to `grep` so it counts matching lines instead of displaying them?

Option `-c`.

Answers to Explorational Exercises

1. The basic structure of an HTML file starts with elements `html`, `head` and `body`, for example:

```
<html>
<head>
  <title>News Site</title>
</head>
<body>
  <h1>Headline</h1>
  <p>Information of interest.</p>
</body>
</html>
```

Describe how addresses could be used in `sed` to display only the `body` element and its contents.

To only show `body`, the addresses should be `/<body>/,/<\/body>/`, as in `sed -n -e '/<body>/,/<\/body>/p'`. Option `-n` is given to `sed` so it doesn't print lines by default, hence the command `p` at the end of `sed` expression to print matching lines.

2. What `sed` expression will remove all tags from an HTML document, keeping only the rendered text?

The `sed` expression `s/<[^>]*>/ /g` will replace any content enclosed in `<>` by an empty string.

3. Files with extension `.ovpn` are very popular to configure VPN clients as they contain not only the settings, but also the contents of keys and certificates for the client. These keys and certificates are originally in separate files, so they need to be copied into the `.ovpn` file. Given the following excerpt of a `.ovpn` template:

```
client
dev tun
remote 192.168.1.155 1194
<ca>
ca.crt
</ca>
<cert>
client.crt
</cert>
<key>
client.key
</key>
```

```
<tls-auth>
ta.key
</tls-auth>
```

Assuming files `ca.crt`, `client.crt`, `client.key` and `ta.key` are in the current directory, how would the template configuration be modified by `sed` to replace each filename by its content?

The command

```
sed -r -e 's/([^\.]*)\.(crt|key)$/cat \1.\2/e' < client.template > client.ovpn
```

replaces any line terminating in `.crt` or `.key` with the content of a file whose name equals the line. Option `-r` tells `sed` to use extended regular expressions, whilst `e` at the end of the expression tells `sed` to replace matches with the output of command `cat \1.\2`. The backreferences `\1` and `\2` correspond to the filename and extension found in the match.



103.8 Basic file editing

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 103.8

Weight

3

Key knowledge areas

- Navigate a document using vi.
- Understand and use vi modes.
- Insert, edit, delete, copy and find text in vi.
- Awareness of Emacs, nano and vim.
- Configure the standard editor.

Partial list of the used files, terms and utilities

- vi
- /, ?
- h, j, k, l
- i, o, a
- d, p, y, dd, yy
- ZZ, :w!, :q!
- EDITOR



Linux
Professional
Institute

103.8 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	103 GNU and Unix Commands
Objective:	103.8 Basic file editing
Lesson:	1 of 1

Introduction

In most Linux distributions, `vi` — abbreviation for “visual” — is pre-installed and it is the standard editor in the shell environment. Vi is an interactive text editor, it shows the file content on the screen as it is being edited. As such, it allows the user to move through and to make modifications anywhere in the document. However, unlike visual editors from the graphical desktop, the `vi` editor is a shell application with keyboard shortcuts to every editing task.

An alternative to `vi`, called `vim` (*vi improved*), is sometimes used as a modern replacement for `vi`. Among other improvements, `vim` offers support for syntax highlighting, multilevel undo/redo and multi-document editing. Although more resourceful, `vim` is fully backwards compatible with `vi`, making both indistinguishable for most tasks.

The standard way to start `vi` is to give it a path to a file as a parameter. To jump directly to a specific line, its number should be informed with a plus sign, as in `vi +9 /etc/fstab` to open `/etc/fstab/` and place the cursor at the 9th line. Without a number, the plus sign by itself places the cursor at the last line.

`vi`'s interface is very simple: all space available in the terminal window is occupied to present a

file, normally informed as a command argument, to the user. The only visual clues are a footer line showing the current position of the cursor and a tilde ~ indicating where the file ends. There are different execution modes for vi where program behavior changes. The most common are: *insert mode* and *normal mode*.

Insert Mode

The insert mode is straightforward: text appears on the screen as it is typed on the keyboard. It is the type of interaction most users expect from a text editor, but it is not how vi first presents a document. To enter the insert mode, the user must execute an insertion command in the normal mode. The `Esc` key finishes the insert mode and returns to normal mode, the default vi mode.

NOTE

If you are interested to know more on the other execution modes, open vi and type:

```
:help vim-modes-intro
```

Normal Mode

Normal mode—also known as command mode—is how vi starts by default. In this mode, keyboard keys are associated with commands for navigation and text manipulation tasks. Most commands in this mode are unique keys. Some of the keys and their functions on normal mode are:

0, \$

Go to the beginning and end of the line.

1G, G

Go to the beginning and end of the document.

(,)

Go to the beginning and end of the sentence.

{, }

Go to the beginning and end of the paragraph.

w, W

Jump word and jump word including punctuation.

h, j, k, l

Left, down, up, right.

e or E

Go to the end of current word.

/, ?

Search forward and backwards.

i, I

Enter the insert mode before the current cursor position and at the beginning of the current line.

a, A

Enter the insert mode after the current cursor position and at the end of the current line.

o, O

Add a new line and enter the insert mode in the next line or in the previous line.

s, S

Erase the character under the cursor or the entire line and enter the insert mode.

c

Change the character(s) under the cursor.

r

Replace the character under the cursor.

x

Delete the selected characters or the character under the cursor.

v, V

Start a new selection with the current character or the entire line.

y, yy

Copy (yanks) the character(s) or the entire line.

p, P

Paste copied content, after or before the current position.

u

Undo the last action.

Ctrl-R

Redo the last action.

ZZ

Close and save.

ZQ

Close and do not save.

If preceded by a number, the command will be executed the same number of times. For example, press `3yy` to copy the current line plus the following two, press `d5w` to delete the current word and the following 4 words, and so on.

Most editing tasks are combinations of multiple commands. For example, the key sequence `vey` is used to copy a selection starting at the current position until the end of the current word. Command repetition can also be used in combinations, so `v3ey` would copy a selection starting at the current position until the end of the third word from there.

`vi` can organize copied text in registers, allowing to keep distinct contents at the same time. A register is specified by a character preceded by `"` and once created it's kept until the end of the current session. The key sequence `"ly` creates a register containing the current selection, which will be accessible through the key `l`. Then, the register `l` may be pasted with `"lp`.

There is also a way to set custom marks at arbitrary positions along the text, making it easier to quickly jump between them. Marks are created by pressing the key `m` and then a key to address the current position. With that done, the cursor will come back to the marked position when `'` followed by the chosen key are pressed.

Any key sequence can be recorded as a macro for future execution. A macro can be recorded, for example, to surround a selected text in double-quotes. First, a string of text is selected and the key `q` is pressed, followed by a register key to associate the macro with, like `d`. The line `recording @d` will appear in the footer line, indicating that the recording is on. It is assumed that some text is already selected, so the first command is `x` to remove (and automatically copy) the selected text. The key `i` is pressed to insert two double quotes at the current position, then `Esc` returns to normal mode. The last command is `P`, to re-insert the deleted selection just before the last double-quote. Pressing `q` again will end the recording. Now, a macro consisting of key sequence `x, i, "", Esc` and `P` will execute every time keys `@d` are pressed in normal mode, where `d` is the register key associated with the macro.

However, the macro will be available only during the current session. To make macros persistent, they should be stored in the configuration file. As most modern distributions use *vim* as the *vi* compatible editor, the user's configuration file is `~/.vimrc`. Inside `~/.vimrc`, the line `let @d = 'xi'"^'[P'` will set the register `d` to the key sequence inside single-quotes. The same register previously assigned to a macro can be used to paste its key sequence.

Colon Commands

The normal mode also supports another set of *vi* commands: the *colon commands*. Colon commands, as the name implies, are executed after pressing the colon key `:` in normal mode. Colon commands allow the user to perform searches, to save, to quit, to run shell commands, to change *vi* settings, etc. To go back to the normal mode, the command `:visual` must be executed or the Enter key pressed without any command. Some of the most common colon commands are indicated here (the initial is not part of the command):

:s/REGEX/TEXT/g

Replaces all the occurrences of regular expression `REGEX` with `TEXT` in the current line. It accepts the same syntax of command `sed`, including addresses.

:!

Run a following shell command.

:quit or :q

Exit the program.

:quit! or :q!

Exit the program without saving.

:wq

Save and exit.

:exit or :x or :e

Save and exit, if needed.

:visual

Go back to navigation mode.

The standard *vi* program is capable of doing most text editing tasks, but any other non-graphical editor can be used to edit text files in the shell environment.

TIP | Novice users may have difficulty memorizing *vi*'s command keys all at once.

Distributions adopting `vim` also have the command `vimtutor`, which uses `vim` itself to open a step-by-step guide to the main activities. The file is an editable copy that can be used to practice the commands and progressively get used to them.

Alternative Editors

Users unfamiliar with `vi` may have difficulties adapting to it, since its operation is not intuitive. A simpler alternative is GNU `nano`, a small text editor that offers all basic text editing features like undo/redo, syntax coloring, interactive search-and-replace, auto-indentation, line numbers, word completion, file locking, backup files, and internationalization support. Unlike `vi`, all key presses are just inserted in the document being edited. Commands in `nano` are given by using the `Ctrl` key or the Meta key (depending on the system, Meta is `Alt` or `⌘`).

Ctrl-6 or Meta-A

Start a new selection. It's also possible to create a selection by pressing Shift and moving the cursor.

Meta-6

Copy the current selection.

Ctrl-K

Cut the current selection.

Ctrl-U

Paste copied content.

Meta-U

Undo.

Meta-E

Redo.

Ctrl-

Replace the text at the selection.

Ctrl-T

Start a spell-checking session for the document or current selection.

Emacs is another very popular text editor for the shell environment. Whilst text is inserted just by typing it, like in `nano`, navigation through the document is assisted by keyboard commands, like in `vi`. Emacs includes many features that makes it more than just a text editor. It is also an IDE

(*integrated development environment*) capable of compiling, running, and testing programs. Emacs can be configured as an email, news or RSS client, making it an authentic productivity suite.

The shell itself will run a default text editor, usually `vi`, every time it is necessary. This is the case, for example, when `crontab -e` is executed to edit *cronjobs*. Bash uses the session variables `VISUAL` or `EDITOR` to find out the default text editor for the shell environment. For example, the command `export EDITOR=nano` defines `nano` as the default text editor in the current shell session. To make this change persistent across sessions, the command should be included in `~/.bash_profile`.

Guided Exercises

1. `vi` is most used as an editor for configuration files and source code, where indentation helps to identify sections of text. A selection can be indented to the left by pressing `<` and to the right by pressing `>`. What keys should be pressed in normal mode to indent the current selection three steps to the left?

2. An entire line can be selected by pressing `V` in `vi` normal mode. However, the terminating newline character is also included. What keys should be pressed in normal mode to select from the starting character until, but not including, the newline character?

3. How should `vi` be executed in the command line to open `~/ .bash_profile` and jump straight to the last line?

4. What keys should be pressed in `vi` normal mode to delete characters from the current cursor position until the next period character?

Explorational Exercises

1. `vim` allows to select blocks of text with arbitrary width, not only sections with entire lines. By pressing `Ctrl + v` in normal mode, a selection is made by moving the cursor up, down, left and right. Using this method, how would a block starting at the first character in the current line, containing the next eight columns and five lines of text, be deleted?

2. A `vi` session was interrupted by an unexpected power failure. When reopening the file, `vi` prompts the user if they want to recover the swap file (an automatic copy made by `vi`). What should the user do to discard the swap file?

3. In a `vim` session, a line was previously copied to the register `l`. What key combination would record a macro in register `a` to paste the line in register `l` immediately before the current line?

Summary

This lesson covers the standard text editor for the Linux shell environment: the `vi` editor. Whilst intimidating for the unfamiliar user, `vi` has features that make it a good choice for technical and non-technical text editing. The lesson goes through the following steps:

- `vi` basic usage and useful features.
- What is `vim` — the improved `vi` — and other alternative editors.
- How to define the default text editor for the shell environment.

The commands and procedures addressed were:

- Editor `vi` and its improved version `vim`.
- Basic text editing in `vi`.
- Alternative editors `emacs` and `nano`.

Answers to Guided Exercises

1. `vi` is most used as an editor for configuration files and source code, where indentation helps to identify sections of text. A selection can be indented to the left by pressing `<` and to the right by pressing `>`. What keys should be pressed in normal mode to indent the current selection three steps to the left?

The keys `3<`, meaning three steps to the left.

2. An entire line can be selected by pressing `V` in `vi` normal mode. However, the terminating newline character is also included. What keys should be pressed in normal mode to select from the starting character until, but not including, the newline character?

The keys `0v$h`, meaning `0` (“jump to start of a line”), `v` (“start character selection”), `$` (“go to end of line”) and `h` (“go back one position”).

3. How should `vi` be executed in the command line to open `~/ .bash_profile` and jump straight to the last line?

The command `vi + ~/ .bash_profile` will open the file and place the cursor at its last line.

4. What keys should be pressed in `vi` normal mode to delete characters from the current cursor position until the next period character?

The keys `dt.`, meaning `d` (“start deletion”), `t` (“jump to the following character”) and `.` (period character).

Answers to Explorational Exercises

1. `vim` allows to select blocks of text with arbitrary width, not only sections with entire lines. By pressing `Ctrl-V` in normal mode, a selection is made by moving the cursor up, down, left and right. Using this method, how would a block starting at the first character in the current line, containing the next eight columns and five lines of text, be deleted?

The combination `0`, `Ctrl-V` and `8l5jd` will select and delete the corresponding block.

2. A `vi` session was interrupted by an unexpected power failure. When reopening the file, `vi` prompts the user if they want to recover the swap file (an automatic copy made by `vi`). What should the user do to discard the swap file?

Press `d` when prompted by `vi`.

3. In a `vim` session, a line was previously copied to the register `l`. What key combination would record a macro in register `a` to paste the line in register `l` immediately before the current line?

The combination `qa"lPq`, meaning `q` (“start macro recording”), `a` (“assign register `a` to macro”), `"l` (“select text in register `l`”), `P` (“paste before the current line”) and `q` (“end macro recording”).



**Linux
Professional
Institute**

Topic 104: Devices, Linux Filesystems, Filesystem Hierarchy Standard



104.1 Create partitions and filesystems

Reference to LPI objectives

LPIC-1 v5, Exam 101, Objective 104.1

Weight

2

Key knowledge areas

- Manage MBR and GPT partition tables
- Use various `mkfs` commands to create various filesystems such as:
 - `ext2/ext3/ext4`
 - `XFS`
 - `VFAT`
 - `exFAT`
- Basic feature knowledge of `Btrfs`, including multi-device filesystems, compression and subvolumes.

Partial list of the used files, terms and utilities

- `fdisk`
- `gdisk`
- `parted`
- `mkfs`
- `mkswap`



104.1 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	104 Devices, Linux Filesystems, Filesystem Hierarchy Standard
Objective:	104.1 Create partitions and filesystems
Lesson:	1 of 1

Introduction

On any operating system, a disk needs to be partitioned before it can be used. A partition is a logical subset of the physical disk, and information about partitions are stored in a partition table. This table includes information about the first and last sectors of the partition and its type, and further details on each partition.

Usually each partition is seen by an operating system as a separate “disk”, even if they all reside in the same physical media. On Windows systems they are assigned letters like C: (historically the main disk), D: and so on. On Linux each partition is assigned to a directory under /dev, like /dev/sda1 or /dev/sda2.

In this lesson, you will learn how to create, delete, restore and resize partitions using the three most common utilities (fdisk, gdisk and parted), how to create a filesystem on them and how to create and set up a *swap partition* or *swap file* to be used as virtual memory.

NOTE

For historical reasons, through this lesson we will refer to storage media as “disks”, even though modern storage systems, like SSDs and Flash Storage, do not contain

any “disks” at all.

Understanding MBR and GPT

There are two main ways of storing partition information on hard disks. The first one is MBR (*Master Boot Record*), and the second one is GPT (*GUID Partition Table*).

MBR

This is a remnant from the early days of MS-DOS (more specifically, PC-DOS 2.0 from 1983) and for decades was the standard partitioning scheme on PCs. The partition table is stored on the first sector of a disk, called the *Boot Sector*, along with a boot loader, which on Linux systems is usually the *GRUB* bootloader. But MBR has a series of limitations that hinder its use on modern systems, like the inability to address disks of more than 2 TB in size, and the limit of only 4 primary partitions per disk.

GUID

A partitioning system that addresses many of the limitations of MBR. There is no practical limit on disk size, and the maximum number of partitions are limited only by the operating system itself. It is more commonly found on more modern machines that use UEFI instead of the old PC BIOS.

During system administration tasks it is highly possible that you will find both schemes in use, so it is important to know how to use the tools associated with each one to create, delete or modify partitions.

Managing MBR Partitions with FDISK

The standard utility for managing MBR partitions on Linux is `fdisk`. This is an interactive, menu-driven utility. To use it, type `fdisk` followed by the device name corresponding to the disk you want to edit. For example, the command

```
# fdisk /dev/sda
```

would edit the partition table of the first SATA-connected device (`sda`) on the system. Keep in mind that you need to specify the device corresponding to the physical disk, not one of its partitions (like `/dev/sda1`).

NOTE

All disk-related operations in this lesson need to be done as the user `root` (the system administrator), or with root privileges using `sudo`.

When invoked, `fdisk` will show a greeting, then a warning and it will wait for your commands.

```
# fdisk /dev/sda
Welcome to fdisk (util-linux 2.33.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Command (m for help):
```

The warning is important. You can create, edit or delete partitions at will, but *nothing* will be written to disk unless you use the write (`w`) command. So you can “practice” without risk of losing data, as long as you stay clear of the `w` key. To exit `fdisk` without saving changes, use the `q` command.

NOTE

That being said, you should never practice on an important disk, as there are always risks. Use a spare external disk, or a USB flash drive instead.

Printing the Current Partition Table

The command `p` is used to print the current partition table. The output is something like this:

```
Command (m for help): p
Disk /dev/sda: 111.8 GiB, 120034123776 bytes, 234441648 sectors
Disk model: CT120BX500SSD1
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x97f8fef5

Device      Boot      Start        End    Sectors   Size Id Type
/dev/sda1                4096 226048942 226044847 107.8G 83 Linux
/dev/sda2          226048944 234437550   8388607     4G 82 Linux swap / Solaris
```

Here is the meaning of each column:

Device

The device assigned to the partition.

Boot

Shows whether the partition is “bootable” or not.

Start

The sector where the partition starts.

End

The sector where the partition ends.

Sectors

The total number of sectors in the partition. Multiply it by the sector size to get the partition size in bytes.

Size

The size of the partition in “human readable” format. In the example above, values are in gigabytes.

Id

The numerical value representing the partition type.

Type

The description for the partition type.

Primary vs Extended Partitions

On an MBR disk, you can have 2 main types of partitions, *primary* and *extended*. Like we said before, you can have only 4 primary partitions on the disk, and if you want to make the disk “bootable”, the first partition must be a primary one.

One way to work around this limitation is to create an extended partition that acts as a container for *logical* partitions. You could have, for example, a primary partition, an extended partition occupying the remainder of the disk space and five logical partitions inside it.

For an operating system like Linux, primary and extended partitions are treated exactly in the same way, so there are no “advantages” of using one over the other.

Creating a Partition

To create a partition, use the `n` command. By default, partitions will be created at the start of unallocated space on the disk. You will be asked for the partition type (primary or extended), first sector and last sector.

For the first sector, you can usually accept the default value suggested by `fdisk`, unless you need a partition to start at a specific sector. Instead of specifying the last sector, you can specify a size

followed by the letters K, M, G, T or P (Kilo, Mega, Giga, Tera or Peta). So, if you want to create a 1 GB partition, you could specify +1G as the Last sector, and `fdisk` will size the partition accordingly. See this example for the creation of a primary partition:

```
Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-3903577, default 2048): 2048
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-3903577, default 3903577): +1G
```

Checking for Unallocated Space

If you do not know how much free space there is on the disk, you can use the `F` command to show the unallocated space, like so:

```
Command (m for help): F
Unpartitioned space /dev/sdd: 881 MiB, 923841536 bytes, 1804378 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes

   Start      End Sectors  Size
2099200 3903577 1804378  881M
```

Deleting Partitions

To delete a partition, use the `d` command. `fdisk` will ask you for the number of the partition you want to delete, *unless there is only one partition on the disk*. In this case, this partition will be *selected and deleted immediately*.

Be aware that if you delete an extended partition, all the logical partitions inside it will also be deleted.

Mind the Gap!

Keep in mind that when creating a new partition with `fdisk`, the maximum size will be limited to the maximum amount of *contiguous* unallocated space on the disk. Say, for example, that you have the following partition map:

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdd1		2048	1050623	1048576	512M	83	Linux
/dev/sdd2		1050624	2099199	1048576	512M	83	Linux
/dev/sdd3		2099200	3147775	1048576	512M	83	Linux

Then you delete partition 2 and check for free space:

```
Command (m for help): F
Unpartitioned space /dev/sdd: 881 MiB, 923841536 bytes, 1804378 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes

   Start      End Sectors  Size
1050624 2099199 1048576 512M
3147776 3903577 755802 369M
```

Adding up the size of the unallocated space, in theory we have 881 MB available. But see what happens when we try to create a 700 MB partition:

```
Command (m for help): n
Partition type
  p   primary (2 primary, 0 extended, 2 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (2,4, default 2): 2
First sector (1050624-3903577, default 1050624):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (1050624-2099199, default 2099199): +700M
Value out of range.
```

That happens because the largest contiguous unallocated space on the disk is the 512 MB block that belonged to partition 2. Your new partition cannot “reach over” partition 3 to use some of the unallocated space after it.

Changing the Partition Type

Occasionally, you may need to change the partition type, especially when dealing with disks that will be used on other operating systems and platforms. This is done with the command `t`, followed by the number of the partition you wish to change.

The partition type must be specified by its corresponding hexadecimal code, and you can see a list

of all the valid codes by using the command `l`.

Do not confuse the partition type with the filesystem used on it. Although at first there was a relation between them, today you cannot assume this to be true. A Linux partition, for example, can contain any Linux-native filesystem, like *ext4* or *ReiserFS*.

TIP | Linux partitions are type `83` (Linux). Swap partitions are type `82` (Linux Swap).

Managing GUID Partitions with GDISK

The utility `gdisk` is the equivalent of `fdisk` when dealing with GPT partitioned disks. In fact, the interface is modeled after `fdisk`, with an interactive prompt and the same (or very similar) commands.

Printing the Current Partition Table

The command `p` is used to print the current partition table. The output is something like this:

```
Command (? for help): p
Disk /dev/sdb: 3903578 sectors, 1.9 GiB
Model: DataTraveler 2.0
Sector size (logical/physical): 512/512 bytes
Disk identifier (GUID): AB41B5AA-A217-4D1E-8200-E062C54285BE
Partition table holds up to 128 entries
Main partition table begins at sector 2 and ends at sector 33
First usable sector is 34, last usable sector is 3903544
Partitions will be aligned on 2048-sector boundaries
Total free space is 1282071 sectors (626.0 MiB)
```

Number	Start (sector)	End (sector)	Size	Code	Name
1	2048	2099199	1024.0 MiB	8300	Linux filesystem
2	2623488	3147775	256.0 MiB	8300	Linux filesystem

Right from the start, we notice a few different things:

- Each disk has a unique Disk Identifier (GUID). This is a 128 bit hexadecimal number, assigned randomly when the partition table is created. Since there are 3.4×10^{38} possible values to this number, the chances that 2 random disks have the same GUID are pretty slim. The GUID can be used to identify which filesystems to mount at boot time (and where), eliminating the need to use the device path to do so (like `/dev/sdb`).
- See the phrase `Partition table holds up to 128 entries`? That's right, you can have up to 128 partitions on a GPT disk. Because of this, there is no need for *primary* and *extended*

partitions.

- The free space is listed on the last line, so there is no need for an equivalent of the `F` command from `fdisk`.

Creating a Partition

The command to create a partition is `n`, just as in `fdisk`. The main difference is that besides the partition number and the first and last sector (or size), you can also specify the partition type during the creation. GPT partitions support many more types than MBR. You can check a list of all the supported types by using the `l` command.

Deleting a Partition

To delete a partition, type `d` and the partition number. Unlike `fdisk`, the first partition will not be automatically selected if it is the only one on the disk.

On GPT disks, partitions can be easily reordered, or “sorted”, to avoid gaps in the numbering sequence. To do this, simply use the `s` command. For example, imagine a disk with the following partition table:

Number	Start (sector)	End (sector)	Size	Code	Name
1	2048	2099199	1024.0 MiB	8300	Linux filesystem
2	2099200	2361343	128.0 MiB	8300	Linux filesystem
3	2361344	2623487	128.0 MiB	8300	Linux filesystem

If you delete the second partition, the table would become:

Number	Start (sector)	End (sector)	Size	Code	Name
1	2048	2099199	1024.0 MiB	8300	Linux filesystem
3	2361344	2623487	128.0 MiB	8300	Linux filesystem

If you use the `s` command, it would become:

Number	Start (sector)	End (sector)	Size	Code	Name
1	2048	2099199	1024.0 MiB	8300	Linux filesystem
2	2361344	2623487	128.0 MiB	8300	Linux filesystem

Notice that the third partition became the second one.

Recovery Options

GPT disks store backup copies of the GPT header and partition table, making it easy to recover disks in case this data has been damaged. `gdisk` provides features to aid in those recovery tasks, accessed with the `r` command.

You can rebuild a corrupt main GPT header or partition table with `b` and `c`, respectively, or use the main header and table to rebuild a backup with `d` and `e`. You can also convert a MBR to a GPT with `f`, and do the opposite with `g`, among other operations. Type `?` in the recovery menu to get a list of all the available recovery commands and descriptions about what they do.

Creating File Systems

Partitioning the disk is only the first step towards being able to use a disk. After that, you will need to format the partition with a filesystem before using it to store data.

A filesystem controls how the data is stored and accessed on the disk. Linux supports many filesystems, some native, like the `ext` (Extended Filesystem) family, while others come from other operating systems like `FAT` from MS-DOS, `NTFS` from Windows NT, `HFS` and `HFS+` from Mac OS, etc.

The standard tool used to create a filesystem on Linux is `mkfs`, which comes in many “flavors” according to the filesystem it needs to work with.

Creating an ext2/ext3/ext4 Filesystem

The *Extended Filesystem* (`ext`) was the first filesystem for Linux, and through the years was replaced by new versions called `ext2`, `ext3` and `ext4`, which is currently the default filesystem for many Linux distributions.

The utilities `mkfs.ext2`, `mkfs.ext3` and `mkfs.ext4` are used to create `ext2`, `ext3` and `ext4` filesystems. In fact, all of these “utilities” exist only as symbolic links to another utility called `mke2fs`. `mke2fs` alters its defaults according to the name it is called by. As such, they all have the same behavior and command line parameters.

The most simple form of usage is:

```
# mkfs.ext2 TARGET
```

Where `TARGET` is the name of the partition where the filesystem should be created. For example, to create an `ext3` filesystem on `/dev/sdb1` the command would be:

```
# mkfs.ext3 /dev/sdb1
```

Instead of using the command corresponding to the filesystem you wish to create, you can pass the `-t` parameter to `mke2fs` followed by the filesystem name. For example, the following commands are equivalent, and will create an ext4 filesystem on `/dev/sdb1`.

```
# mkfs.ext4 /dev/sdb1
# mke2fs -t ext4 /dev/sdb1
```

Command Line Parameters

`mke2fs` supports a wide range of command line parameters and options. Here are some of the most significant ones. All of them also apply to `mkfs.ext2`, `mkfs.ext3` and `mkfs.ext4`:

-b SIZE

Sets the size of the data blocks in the device to `SIZE`, which can be 1024, 2048 or 4096 bytes per block.

-c

Checks the target device for bad blocks before creating the filesystem. You can run a thorough, but much slower check by passing this parameter twice, as in `mkfs.ext4 -c -c TARGET`.

-d DIRECTORY

Copies the contents of the specified directory to the root of the new filesystem. Useful if you need to “pre-populate” the disk with a predefined set of files.

-F

Danger, Will Robinson! This option will *force* `mke2fs` to create a filesystem, even if the other options passed to it or the target are dangerous or make no sense at all. If specified twice (as in `-F -F`) it can even be used to create a filesystem on a device which is mounted or in use, which is a very, very bad thing to do.

-L VOLUME_LABEL

Will set the volume label to the one specified in `VOLUME_LABEL`. This label must be at most 16 characters long.

-n

This is a truly useful option that simulates the creation of the filesystem, and displays what would be done if executed without the `n` option. Think of it as a “trial” mode. Good to check

things out before actually committing any changes to disk.

-q

Quiet mode. `mke2fs` will run normally, but will not produce any output to the terminal. Useful when running `mke2fs` from a script.

-U ID

This will set the UUID (*Universally Unique Identifier*) of a partition to the value specified as ID. UUIDs are 128 bit numbers in hexadecimal notation that serve to uniquely identify a partition to the operating system. This number is specified as a 32-digit string in the format 8-4-4-4-12, meaning 8 digits, hyphen, 4 digits, hyphen, 4 digits, hyphen, 4 digits, hyphen, 12 digits, like `D249E380-7719-45A1-813C-35186883987E`. Instead of an ID you can also specify parameters like `clear` to clear the filesystem UUID, `random`, to use a randomly generated UUID, or `time` to create a time-based UUID.

-v

Verbose mode, prints much more information during operation than usual. Useful for debugging purposes.

Creating an XFS Filesystem

XFS is a high-performance filesystem originally developed by Silicon Graphics in 1993 for its IRIX operating system. Due to its performance and reliability features, it is commonly used for servers and other environments that require high (or guaranteed) filesystem bandwidth.

Tools for managing XFS filesystems are part of the `xfsprogs` package. This package may need to be installed manually, as it is not included by default in some Linux distributions. Others, like Red Hat Enterprise Linux 7, use XFS as the default filesystem.

XFS filesystems are divided into at least 2 parts, a *log section* where a log of all filesystem operations (commonly called a *Journal*) are maintained, and the *data section*. The log section may be located inside the data section (the default behavior), or even on a separate disk altogether, for better performance and reliability.

The most basic command to create an XFS filesystem is `mkfs.xfs TARGET`, where `TARGET` is the partition you want the filesystem to be created in. For example: `mkfs.xfs /dev/sda1`.

As in `mke2fs`, `mkfs.xfs` supports a number of command line options. Here are some of the most common ones.

-b size=VALUE

Sets the block size on the filesystem, in bytes, to the one specified in `VALUE`. The default value is 4096 bytes (4 KiB), the minimum is 512, and the maximum is 65536 (64 KiB).

-m crc=VALUE

Parameters starting with `-m` are metadata options. This one enables (if `VALUE` is 1) or disables (if `VALUE` is 0) the use of CRC32c checks to verify the integrity of all metadata on the disk. This enables better error detection and recovery from crashes related to hardware issues, so it is enabled by default. The performance impact of this check should be minimal, so normally there is no reason to disable it.

-m uuid=VALUE

Sets the partition UUID to the one specified as `VALUE`. Remember that UUIDs are 32-character (128 bits) numbers in hexadecimal base, specified in groups of 8, 4, 4, 4 and 12 digits separated by dashes, like `1E83E3A3-3AE9-4AAC-BF7E-29DFFECD36C0`.

-f

Force the creation of a filesystem on the target device even if a filesystem is detected on it.

-l logdev=DEVICE

This will put the log section of the filesystem on the specified device, instead of inside the data section.

-l size=VALUE

This will set the size of the log section to the one specified in `VALUE`. The size can be specified in bytes, and suffixes like `m` or `g` can be used. `-l size=10m`, for example, will limit the log section to 10 Megabytes.

-q

Quiet mode. In this mode, `mkfs.xfs` will not print the parameters of the file system being created.

-L LABEL

Sets the filesystem label, which can be at most 12 characters long.

-N

Similar to the `-n` parameter of `mke2fs`, will make `mkfs.xfs` print all the parameters for the creation of the file system, without actually creating it.

Creating a FAT or VFAT Filesystem

The FAT filesystem originated from MS-DOS, and through the years has received many revisions culminating on the FAT32 format released in 1996 with Windows 95 OSR2.

VFAT is an extension of the FAT16 format with support for long (up to 255 characters) file names. Both filesystems are handled by the same utility, `mkfs.fat`. `mkfs.vfat` is an alias to it.

The FAT filesystem has important drawbacks which restrict its use on large disks. FAT16, for example, supports volumes of at most 4 GB and a maximum file size of 2 GB. FAT32 ups the volume size to up to 2 PB, and the maximum file size to 4 GB. Because of this, FAT filesystems are today more commonly used on small flash drives or memory cards (up to 2 GB in size), or legacy devices and OSs that do not support more advanced filesystems.

The most basic command for the creation of a FAT filesystem is `mkfs.fat TARGET`, where `TARGET` is the partition you want the filesystem to be created in. For example: `mkfs.fat /dev/sdc1`.

Like other utilities, `mkfs.fat` supports a number of command line options. Below are the most important ones. A full list and description of every option can be read in the manual for the utility, with the command `man mkfs.fat`.

-c

Checks the target device for bad blocks before creating the filesystem.

-C FILENAME BLOCK_COUNT

Will create the file specified in `FILENAME` and then create a FAT filesystem inside it, effectively creating an empty “disk image”, that can be later written to a device using a utility such as `dd` or mounted as a loopback device. When using this option, the number of blocks in the filesystem (`BLOCK_COUNT`) must be specified after the device name.

-F SIZE

Selects the size of the FAT (*File Allocation Table*), between 12, 16 or 32, i.e., between FAT12, FAT16 or FAT32. If not specified, `mkfs.fat` will select the appropriate option based on the filesystem size.

-n NAME

Sets the volume label, or name, for the filesystem. This can be up to 11 characters long, and the default is no name.

-v

Verbose mode. Prints much more information than usual, useful for debugging.

NOTE

`mkfs.fat` *cannot* create a “bootable” filesystem. According to the manual page, “this isn’t as easy as you might think” and will not be implemented.

Creating an exFAT Filesystem

exFAT is a filesystem created by Microsoft in 2006 that addresses one of the most important limitations of FAT32: file and disk size. On exFAT, the maximum file size is 16 exabytes (from 4 GB on FAT32), and the maximum disk size is 128 petabytes.

As it is well supported by all three major operating systems (Windows, Linux and mac OS), it is a good choice where interoperability is needed, like on large capacity flash drives, memory cards and external disks. In fact, it is the default filesystem, as defined by the *SD Association*, for SDXC memory cards larger than 32 GB.

The default utility for creating exFAT filesystems is `mkfs.exfat`, which is a link to `mkexfatfs`. The most basic command is `mkfs.exfat TARGET`, where `TARGET` is the partition you want the filesystem to be created in. For example: `mkfs.exfat /dev/sdb2`.

Contrary to the other utilities discussed in this lesson, `mkfs.exfat` has very few command line options. They are:

-i VOL_ID

Sets the Volume ID to the value specified in `VOL_ID`. This is a 32-Bit hexadecimal number. If not defined, an ID based on the current time is set.

-n NAME

Sets the volume label, or name. This can have up to 15 characters, and the default is no name.

-p SECTOR

Specifies the first sector of the first partition on the disk. This is an optional value, and the default is zero.

-s SECTORS

Defines the number of physical sectors per cluster of allocation. This must be a power of two, like 1, 2, 4, 8, and so on.

Getting to Know the Btrfs Filesystem

Btrfs (officially the *B-Tree Filesystem*, pronounced as “Butter FS”, “Better FS” or even “Butterfuss”, your pick) is a filesystem that has been in development since 2007 specifically for Linux by the Oracle Corporation and other companies, including Fujitsu, Red Hat, Intel and SUSE, among

others.

There are many features that make Btrfs attractive on modern systems where massive amounts of storage are common. Among these are multiple device support (including striping, mirroring and striping+mirroring, as in a RAID setup), transparent compression, SSD optimizations, incremental backups, snapshots, online defragmentation, offline checks, support for subvolumes (with quotas), deduplication and much more.

As it is a *copy-on-write* filesystem it is very resilient to crashes. And on top of that, Btrfs is simple to use, and well supported by many Linux distributions. And some of them, like SUSE, use it as the default filesystem.

NOTE

On a traditional filesystem, when you want to overwrite part of a file the new data is put directly over the old data that it is replacing. On a *copy-on-write* filesystem the new data is written to free space on disk, then the file's original metadata is updated to refer to the new data and only then the old data is freed up, as it is no longer needed. This reduces the chances of data loss in case of a crash, as the old data is only discarded after the filesystem is absolutely sure that it is no longer needed and the new data is in place.

Creating a Btrfs Filesystem

The utility `mkfs.btrfs` is used to create a Btrfs filesystem. Using the command without any options creates a Btrfs filesystem on a given device, like so:

```
# mkfs.btrfs /dev/sdb1
```

TIP

If you do not have the `mkfs.btrfs` utility on your system, look for `btrfs-progs` in your distribution's package manager.

You can use the `-L` to set a label (or name) for your filesystem. Btrfs labels can have up to 256 characters, except for newlines:

```
# mkfs.btrfs /dev/sdb1 -L "New Disk"
```

TIP

Enclose the label in quotes (like above) if it contains spaces.

Note this peculiar thing about Btrfs: you can pass *multiple* devices to the `mkfs.btrfs` command. Passing more than one device will span the filesystem over all the devices which is similar to a RAID or LVM setup. To specify how metadata will be distributed in the disk array, use the `-m`

parameter. Valid parameters are `raid0`, `raid1`, `raid5`, `raid6`, `raid10`, `single` and `dup`.

For example, to create a filesystem spanning `/dev/sdb1` and `/dev/sdc1`, concatenating the two partitions into one big partition, use:

```
# mkfs.btrfs -d single -m single /dev/sdb /dev/sdc
```

WARNING

Filesystems spanning multiple partitions such as above might look advantageous at first but are not a good idea from a data safety standpoint, as a failure on a single disk of the array means certain data loss. The risk gets bigger the more disks you use, as you also have more possible points of failure.

Managing Subvolumes

Subvolumes are like filesystems inside filesystems. Think of them as a directory which can be mounted as (and treated like) a separate filesystem. Subvolumes make organization and system administration easier, as each one of them can have separate quotas or snapshot rules.

NOTE

Subvolumes are not partitions. A partition allocates a fixed space on a drive. This can lead to problems further down the line, like one partition running out of space when another one has plenty of space left. Not so with subvolumes, as they “share” the free space from their root filesystem, and grow as needed.

Suppose you have a Btrfs filesystem mounted on `/mnt/disk`, and you wish to create a subvolume inside it to store your backups. Let’s call it BKP:

```
# btrfs subvolume create /mnt/disk/BKP
```

Next we list the contents of the `/mnt/disk` filesystem. You will see that we have a new directory, named after our subvolume.

```
$ ls -lh /mnt/disk/
total 0
drwxr-xr-x 1 root  root    0 jul 13 17:35 BKP
drwxrwxr-x 1 carol carol 988 jul 13 17:30 Images
```

NOTE

Yes, subvolumes can *also* be accessed like any other directory.

We can check that the subvolume is active, with the command:

```
# btrfs subvolume show /mnt/disk/BKP/
Name:          BKP
UUID:          e90a1afe-69fa-da4f-9764-3384f66fa32e
Parent UUID:    -
Received UUID:  -
Creation time:  2019-07-13 17:35:40 -0300
Subvolume ID:   260
Generation:     23
Gen at creation: 22
Parent ID:      5
Top level ID:   5
Flags:          -
Snapshot(s):
```

You can mount the subvolume on `/mnt/BKP` by passing the `-t btrfs -o subvol=NAME` parameter to the `mount` command:

```
# mount -t btrfs -o subvol=BKP /dev/sdb1 /mnt/bkp
```

NOTE The `-t` parameter specifies the filesystem type to be mounted.

Working with Snapshots

Snapshots are just like subvolumes, but pre-populated with the contents from the volume from which the snapshot was taken.

When created, a snapshot and the original volume have exactly the same content. But from that point in time, they will diverge. Changes made to the original volume (like files added, renamed or deleted) will not be reflected on the snapshot, and vice-versa.

Keep in mind that a snapshot *does not* duplicate the files, and initially takes almost no disk space. It simply duplicates the filesystem tree, while pointing to the original data.

The command to create a snapshot is the same used to create a subvolume, just add the `snapshot` parameter after `btrfs subvolume`. The command below will create a snapshot of the Btrfs filesystem mounted in `/mnt/disk` in `/mnt/disk/snap`:

```
# btrfs subvolume snapshot /mnt/disk /mnt/disk/snap
```

Now, imagine you have the following contents in `/mnt/disk`:

```
$ ls -lh
total 2,8M
-rw-rw-r-- 1 carol carol 109K jul 10 16:22 Galaxy_Note_10.png
-rw-rw-r-- 1 carol carol 484K jul  5 15:01 geminoid2.jpg
-rw-rw-r-- 1 carol carol 429K jul  5 14:52 geminoid.jpg
-rw-rw-r-- 1 carol carol 467K jul  2 11:48 LG-G8S-ThinQ-Mirror-White.jpg
-rw-rw-r-- 1 carol carol 654K jul  2 11:39 LG-G8S-ThinQ-Range.jpg
-rw-rw-r-- 1 carol carol  94K jul  2 15:43 Memoji_Comparativo.jpg
-rw-rw-r-- 1 carol carol 112K jul 10 16:20 Note10Plus.jpg
drwx----- 1 carol carol  366 jul 13 17:56 snap
-rw-rw-r-- 1 carol carol 118K jul 11 16:36 Twitter_Down_20190711.jpg
-rw-rw-r-- 1 carol carol 324K jul  2 15:22 Xiaomi_Memoji.png
```

Notice the snap directory, containing the snapshot. Now let us remove some files, and check the directory contents:

```
$ rm LG-G8S-ThinQ-*
$ ls -lh
total 1,7M
-rw-rw-r-- 1 carol carol 109K jul 10 16:22 Galaxy_Note_10.png
-rw-rw-r-- 1 carol carol 484K jul  5 15:01 geminoid2.jpg
-rw-rw-r-- 1 carol carol 429K jul  5 14:52 geminoid.jpg
-rw-rw-r-- 1 carol carol  94K jul  2 15:43 Memoji_Comparativo.jpg
-rw-rw-r-- 1 carol carol 112K jul 10 16:20 Note10Plus.jpg
drwx----- 1 carol carol  366 jul 13 17:56 snap
-rw-rw-r-- 1 carol carol 118K jul 11 16:36 Twitter_Down_20190711.jpg
-rw-rw-r-- 1 carol carol 324K jul  2 15:22 Xiaomi_Memoji.png
```

However, if you check inside the snap directory, the files you deleted are still there and can be restored if needed.

```
$ ls -lh snap/
total 2,8M
-rw-rw-r-- 1 carol carol 109K jul 10 16:22 Galaxy_Note_10.png
-rw-rw-r-- 1 carol carol 484K jul  5 15:01 geminoid2.jpg
-rw-rw-r-- 1 carol carol 429K jul  5 14:52 geminoid.jpg
-rw-rw-r-- 1 carol carol 467K jul  2 11:48 LG-G8S-ThinQ-Mirror-White.jpg
-rw-rw-r-- 1 carol carol 654K jul  2 11:39 LG-G8S-ThinQ-Range.jpg
-rw-rw-r-- 1 carol carol  94K jul  2 15:43 Memoji_Comparativo.jpg
-rw-rw-r-- 1 carol carol 112K jul 10 16:20 Note10Plus.jpg
-rw-rw-r-- 1 carol carol 118K jul 11 16:36 Twitter_Down_20190711.jpg
```



```
-rw-rw-r-- 1 carol carol 324K jul  2 15:22 Xiaomi_Mimoji.png
```

It is also possible to create read-only snapshots. They work exactly like writable snapshots, with the difference that the contents of the snapshot cannot be changed, they are “frozen” in time. Just add the `-r` parameter when creating the snapshot:

```
# btrfs subvolume snapshot -r /mnt/disk /mnt/disk/snap
```

A Few Words on Compression

Btrfs supports transparent file compression, with three different algorithms available to the user. This is done automatically on a per-file basis, as long as the filesystem is mounted with the `-o compress` option. The algorithms are smart enough to detect incompressible files and will not try to compress them, saving system resources. So on a single directory you may have compressed and uncompressed files together. The default compression algorithm is ZLIB, but LZO (faster, worse compression ratio) or ZSTD (faster than ZLIB, comparable compression) are available, with multiple compression levels (see the corresponding objective on mount options).

Managing Partitions with GNU Parted

GNU Parted is a very powerful partition editor (hence the name) that can be used to create, delete, move, resize, rescue and copy partitions. It can work with both GPT and MBR disks, and cover almost all of your disk management needs.

There are many graphical front-ends that make working with `parted` much easier, like *GParted* for GNOME-based desktop environments and the *KDE Partition Manager* for KDE Desktops. However, you should learn how to use `parted` on the command line, since in a server setting you can never count on a graphical desktop environment being available.

WARNING

Unlike `fdisk` and `gdisk`, `parted` makes changes to the disk *immediately* after the command is issued, without waiting for another command to write the changes to disk. When practicing, it is wise to do so on an empty or spare disk or flash drive, so there is no risk of data loss should you make a mistake.

The simplest way to start using `parted` is by typing `parted DEVICE`, where `DEVICE` is the device you want to manage (`parted /dev/sdb`). The program starts an interactive command line interface like `fdisk` and `gdisk` with a (`parted`) prompt for you to enter commands.

```
# parted /dev/sdb
GNU Parted 3.2
```

```
Using /dev/sdb
Welcome to GNU Parted! Type 'help' to view a list of commands.

(parted)
```

WARNING

Be careful! If you do not specify a device, `parted` will automatically select the primary disk (usually `/dev/sda`) to work with.

Selecting Disks

To switch to a different disk than the one specified on the command line, you can use the `select` command, followed by the device name:

```
(parted) select /dev/sdb
Using /dev/sdb
```

Getting Information

The `print` command can be used to get more information about a specific partition or even all of the block devices (disks) connected to your system.

To get information about the currently selected partition, just type `print`:

```
(parted) print
Model: ATA CT120BX500SSD1 (scsi)
Disk /dev/sda: 120GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:

Number  Start   End     Size    Type     File system  Flags
  1      2097kB  116GB   116GB   primary  ext4
  2      116GB   120GB   4295MB  primary  linux-swap(v1)
```

You can get a list of all block devices connected to your system using `print devices`:

```
(parted) print devices
/dev/sdb (1999MB)
/dev/sda (120GB)
/dev/sdc (320GB)
```

```
/dev/mapper/cryptswap (4294MB)
```

To get information about all connected devices at once, you can use `print all`. If you wish to know how much free space there is in each one of them, you can use `print free`:

```
(parted) print free
Model: ATA CT120BX500SSD1 (scsi)
Disk /dev/sda: 120GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:
```

Number	Start	End	Size	Type	File system	Flags
	32.3kB	2097kB	2065kB		Free Space	
1	2097kB	116GB	116GB	primary	ext4	
	116GB	116GB	512B		Free Space	
2	116GB	120GB	4295MB	primary	linux-swap(v1)	
	120GB	120GB	2098kB		Free Space	

Creating a Partition Table on an Empty Disk

To create a partition table on an empty disk, use the `mklabel` command, followed by the partition table type that you want to use.

There are many supported partition table types, but the main types you should know of are `msdos` which is used here to refer to an MBR partition table, and `gpt` to refer to a GPT partition table. To create an MBR partition table, type:

```
(parted) mklabel msdos
```

And to create a GPT partition table, the command is:

```
(parted) mklabel gpt
```

Creating a Partition

To create a partition the command `mkpart` is used, using the syntax `mkpart PARTTYPE FSTYPE START END`, where:

PARTTYPE

Is the partition type, which can be `primary`, `logical` or `extended` in case an MBR partition table is used.

FSTYPE

Specifies which filesystem will be used on this partition. Note that `parted` will *not* create the filesystem. It just sets a flag on the partition which tells the OS what kind of data to expect from it.

START

Specifies the exact point on the device where the partition begins. You can use different units to specify this point. `2s` can be used to refer to the second sector of the disk, while `1m` refers to the beginning of the first megabyte of the disk. Other common units are `B` (bytes) and `%` (percentage of the disk).

END

Specifies the end of the partition. Note that this is *not* the size of the partition, this is *the point on the disk where it ends*. For example, if you specify `100m` the partition will end 100 MB after the start of the disk. You can use the same units as in the `START` parameter.

So, the command:

```
(parted) mkpart primary ext4 1m 100m
```

Creates a primary partition of type `ext4`, starting at the first megabyte of the disk, and ending after the 100th megabyte.

Removing a Partition

To remove a partition, use the command `rm` followed by the partition number, which you can display using the `print` command. So, `rm 2` would remove the second partition on the currently selected disk.

Recovering Partitions

`parted` can recover a deleted partition. Consider you have the following partition structure:

Number	Start	End	Size	File system	Name	Flags
1	1049kB	99.6MB	98.6MB	ext4	primary	
2	99.6MB	200MB	100MB	ext4	primary	

Number	Start	End	Size	File system	Name	Flags
3	200MB	300MB	99.6MB	ext4	primary	

By accident, you removed partition 2 using `rm 2`. To recover it, you can use the `rescue` command, with the syntax `rescue START END`, where `START` is the approximate location where the partition started, and `END` the approximate location where it ended.

`parted` will scan the disk in search of partitions, and offer to restore any that are found. In the example above the partition 2 started at 99,6 MB and ended at 200 MB. So you can use the following command to recover the partition:

```
(parted) rescue 90m 210m
Information: A ext4 primary partition was found at 99.6MB -> 200MB.
Do you want to add it to the partition table?

Yes/No/Cancel? y
```

This will recover the partition and its contents. Note that `rescue` can only recover partitions that have a filesystem installed on them. Empty partitions are not detected.

Resizing ext2/3/4 Partitions

`parted` can be used to resize partitions to make them bigger or smaller. However, there are some caveats:

- During resizing the partition must be unused and unmounted.
- You need enough free space *after* the partition to grow it to the size you want.

The command is `resizepart`, followed by the partition number and where it should end. For example, if you have the following partition table:

Number	Start	End	Size	File system	Name	Flags
1	1049kB	99.6MB	98.6MB	ext4	primary	
2	99.6MB	200MB	100MB	ext4		
3	200MB	300MB	99.6MB	ext4	primary	

Trying to grow partition 1 using `resizepart` would trigger an error message, because with the new size partition 1 would overlap with partition 2. However partition 3 can be resized as there is free space after it, which can be verified with the command `print free`:

```
(parted) print free
```

```
Model: Kingston DataTraveler 2.0 (scsi)
Disk /dev/sdb: 1999MB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:
```

Number	Start	End	Size	File system	Name	Flags
	17.4kB	1049kB	1031kB	Free Space		
1	1049kB	99.6MB	98.6MB	ext4	primary	
2	99.6MB	200MB	100MB	ext4		
3	200MB	300MB	99.6MB	ext4	primary	
	300MB	1999MB	1699MB	Free Space		

So you can use the following command to resize partition 3 to 350 MB:

```
(parted) resizepart 3 350m
```

```
(parted) print
```

```
Model: Kingston DataTraveler 2.0 (scsi)
Disk /dev/sdb: 1999MB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:
```

Number	Start	End	Size	File system	Name	Flags
1	1049kB	99.6MB	98.6MB	ext4	primary	
2	99.6MB	200MB	100MB	ext4		
3	200MB	350MB	150MB	ext4	primary	

Remember that the new end point is specified counting from the start of the disk. So, because partition 3 ended at 300 MB, now it needs to end at 350 MB.

But resizing the partition is only one part of the task. You also need to resize the filesystem that resides in it. For ext2/3/4 filesystems this is done with the `resize2fs` command. In the case of the example above, partition 3 still shows the “old” size when mounted:

```
$ df -h /dev/sdb3
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sdb3	88M	1.6M	80M	2%	/media/carol/part3

To adjust the size the command `resize2fs DEVICE SIZE` can be used, where `DEVICE`

corresponds to the partition you want to resize, and **SIZE** is the new size. If you omit the size parameter, it will use all of the available space of the partition. Before resizing, it is advised to unmount the partition.

In the example above:

```
$ sudo resize2fs /dev/sdb3
resize2fs 1.44.6 (5-Mar-2019)
Resizing the filesystem on /dev/sdb3 to 146212 (1k) blocks.
The filesystem on /dev/sdb3 is now 146212 (1k) blocks long.

$ df -h /dev/sdb3
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdb3       135M   1.6M  123M   2% /media/carol/part3
```

To *shrink* a partition, the process needs to be done in the reverse order. *First* you resize the filesystem to the new, smaller size, then you resize the partition itself using `parted`.

WARNING

Pay attention when shrinking partitions. If you get the order of things wrong, you will lose data!

In our example:

```
# resize2fs /dev/sdb3 88m
resize2fs 1.44.6 (5-Mar-2019)
Resizing the filesystem on /dev/sdb3 to 90112 (1k) blocks.
The filesystem on /dev/sdb3 is now 90112 (1k) blocks long.

# parted /dev/sdb3
(parted) resizepart 3 300m
Warning: Shrinking a partition can cause data loss, are you sure
you want to continue?

Yes/No? y

(parted) print
Model: Kingston DataTraveler 2.0 (scsi)
Disk /dev/sdb: 1999MB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:
```

Number	Start	End	Size	File system	Name	Flags
1	1049kB	99.6MB	98.6MB	ext4	primary	
2	99.6MB	200MB	100MB	ext4		
3	200MB	300MB	99.7MB	ext4	primary	

TIP

Instead of specifying a new size, you can use the `-M` parameter of `resize2fs` to adjust the size of the filesystem so it is just big enough for the files on it.

Creating Swap Partitions

On Linux, the system can swap memory pages from RAM to disk as needed, storing them on a separate space usually implemented as a separate partition on a disk, called the *swap partition* or simply *swap*. This partition needs to be of a specific type, and set-up with a proper utility (`mkswap`) before it can be used.

To create the swap partition using `fdisk` or `gdisk`, just proceed as if you were creating a regular partition, as explained before. The only difference is that you will need to change the partition type to *Linux swap*.

- On `fdisk` use the `t` command. Select the partition you want to use and change its type to `82`. Write changes to disk and quit with `w`.
- On `gdisk` the command to change the partition type is also `t`, but the code is `8200`. Write changes to disk and quit with `w`.

If you are using `parted`, the partition should be identified as a swap partition during creation, just use `linux-swap` as the filesystem type. For example, the command to create a 500 MB swap partition, starting at 300 MB on disk is:

```
(parted) mkpart primary linux-swap 301m 800m
```

Once the partition is created and properly identified, just use `mkswap` followed by the device representing the partition you want to use, like:

```
# mkswap /dev/sda2
```

To enable swap on this partition, use `swapon` followed by the device name:

```
# swapon /dev/sda2
```


Similarly, `swapoff`, followed by the device name, will disable swap on that device.

Linux also supports the use of swap *files* instead of partitions. Just create an empty file of the size you want using `dd` and then use `mkswap` and `swapon` with this file as the target.

The following commands will create a 1 GB file called `myswap` in the current directory, filled with zeroes, and then set-up and enable it as a swap file.

Create the swap file:

```
$ dd if=/dev/zero of=myswap bs=1M count=1024
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 7.49254 s, 143 MB/s
```

`if=` is the input file, the source of the data that will be written to the file. In this case it is the `/dev/zero` device, which provides as many NULL characters as requested. `of=` is the output file, the file that will be created. `bs=` is the size of the data blocks, here specified in Megabytes, and `count=` is the amount of blocks to be written to the output. 1024 blocks of 1 MB each equals 1 GB.

```
# mkswap myswap
Setting up swapspace version 1, size = 1024 MiB (1073737728 bytes)
no label, UUID=49c53bc4-c4b1-4a8b-a613-8f42cb275b2b

# swapon myswap
```

Using the commands above, this swap file will be used only during the current system session. If the machine is rebooted, the file will still be available, but will not be automatically loaded. You can automate that by adding the new swap file to `/etc/fstab`, which we will discuss in a later lesson.

TIP

Both `mkswap` and `swapon` will complain if your swap file has insecure permissions. The recommended file permission flag is `0600`. Owner and group should be `root`.

Guided Exercises

1. Which partitioning scheme should be used to partition a 3 TB hard disk into three 1 GB partitions? Why?

2. On `gdisk`, how can we find out how much space is available on the disk?

3. What would be the command to create an ext3 filesystem, checking for bad blocks before, with the label `MyDisk` and a random UUID, on the device `/dev/sdc1`?

4. Using `parted`, what is the command to create a 300 MB ext4 partition, starting at 500 MB on the disk?

5. Imagine you have 2 partitions, one on `/dev/sda1` and the other on `/dev/sda2`, both 20 GB in size. How can you use them on a single Btrfs filesystem, in such a way that the contents of one partition will be automatically mirrored on the other, like on a RAID1 setup? How big will the filesystem be?

Explorational Exercises

1. Consider a 2 GB disk with an MBR partition table and the following layout:

```
Disk /dev/sdb: 1.9 GiB, 1998631936 bytes, 3903578 sectors
Disk model: DataTraveler 2.0
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x31a83a48
```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdb1		2048	1050623	1048576	512M	83	Linux
/dev/sdb3		2099200	3147775	1048576	512M	83	Linux

Can you create a 600 MB partition on it? Why?

2. On a disk at `/dev/sdc`, we have a first partition of 1 GB, containing about 256 MB of files. Using `parted`, how can you shrink it so it has just enough space for the files?

3. Imagine you have a disk at `/dev/sdb`, and you want to create a 1 GB swap partition at the start of it. So, using `parted`, you create the partition with `mkpart primary linux-swap 0 1024M`. Then, you enable swap on this partition with `swapon /dev/sdb1`, but get the following error message:

```
swapon: /dev/sdb1: read swap header failed
```

What went wrong?

4. During the course of this lesson, you were trying out some commands in `parted` but, by mistake, deleted the 3rd partition on your hard disk. You know that it came after a 250 MB UEFI partition and a 4 GB swap partition, and was 10 GB in size. Which command can you use to recover it?

5. Imagine you have a 4 GB unused partition on `/dev/sda3`. Using `fdisk`, what would be the sequence of operations to turn it into an active swap partition?

Summary

In this lesson, you learned:

- How to create an MBR partition table on a disk with `fdisk`, and how to use it to create and delete partitions.
- How to create an MBR partition table on a disk with `gdisk`, and how to use it to create and delete partitions.
- How to create ext2, ext3, ext4, XFS, VFAT and exFAT partitions.
- How to use `parted` to create, delete and recover partitions on both MBR and GPT disks.
- How to use `resize` ext2, ext3, ext4 and Btrfs partitions.
- How to create, set-up and activate swap partitions and swap files.

The following commands were discussed in this lesson:

- `fdisk`
- `gdisk`
- `mkfs.ext2`, `mkfs.ext3`, `mkfs.ext4`, `mkfs.xfs`, `mkfs.vfat` and `mkfs.exfat`
- `parted`
- `btrfs`
- `mkswap`
- `swapon` and `swapoff`

Answers to Guided Exercises

1. Which partitioning scheme should be used to partition a 3 TB hard disk into three 1 GB partitions? Why?

GPT, as MBR supports at most 2 TB hard disks.

2. On `gdisk`, how can we find out how much space is available on the disk?

Use `p` (print). The total free space will be shown as the last line of information before the partition table itself.

3. What would be the command to create an ext3 filesystem, checking for bad blocks before, with the label `MyDisk` and a random UUID, on the device `/dev/sdc1`?

The command would be `mkfs.ext3 -c -L MyDisk -U random /dev/sdc1`. Alternatively, `mke2fs -t ext3` can also be used instead of `mkfs.ext3`

4. Using `parted`, what is the command to create a 300 MB ext4 partition, starting at 500 MB on the disk?

Use `mkpart primary ext4 500m 800m`. Remember that you will have to create the filesystem using `mkfs.ext4`, as `parted` does not do this.

5. Imagine you have 2 partitions, one on `/dev/sda1` and the other on `/dev/sdb1`, both 20 GB in size. How can you use them on a single Btrfs filesystem, in such a way that the contents of one partition will be automatically mirrored on the other, like on a RAID1 setup? How big will the filesystem be?

Use `mkfs.btrfs /dev/sda1 /dev/sdb1 -m raid1`. The resulting filesystem will be 20 GB in size, as one partition acts simply as a mirror of the other.

Answers to Explorational Exercises

1. Consider a 2 GB disk with an MBR partition table and the following layout:

```
Disk /dev/sdb: 1.9 GiB, 1998631936 bytes, 3903578 sectors
Disk model: DataTraveler 2.0
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x31a83a48
```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/sdb1		2048	1050623	1048576	512M	83	Linux
/dev/sdb3		2099200	3147775	1048576	512M	83	Linux

Can you create a 600 MB partition on it? Why?

You cannot, because there is not enough contiguous space. The first clue that something is “off” is the list of devices: you have `/dev/sdb1` and `/dev/sdb3`, but no `/dev/sdb2`. So, something is missing.

Then, you need to look where a partition ends and the other one begins. Partition one ends at sector `1050623`, and partition 2 starts at `2099200`. That’s a “gap” of `1048577` sectors. At `512` bytes per sector, that’s `536.871.424` bytes. If you divide it by `1024` you get `524.288` Kilobytes. Divide by `1024` again and you get... `512` MB. This is the size of the “gap”.

If the disk is 2 GB, then we have at most another `512` MB after partition 3. Even if we have in total around 1 GB unallocated, the biggest contiguous block is `512` MB. So, there is no space for a `600` MB partition.

2. On a disk at `/dev/sdc`, we have a first partition of 1 GB, containing an `ext4` filesystem with `241` MB of files. Using `parted`, how can you shrink it so it has just enough space for the files?

This is a multi-part operation. First you have to shrink the filesystem using `resize2fs`. Instead of specifying the new size directly, you can use the `-M` parameter so it is just “big enough”. So: `resize2fs -M /dev/sdc1`.

Then, you resize the partition itself with `parted` using `resizepart`. Since it is the first partition, we can assume that it starts at zero and ends at `241` MB. So the command is `resizepart 1 241M`.

3. Imagine you have a disk at `/dev/sdb`, and you want to create a 1 GB partition at the start of it. So, using `parted`, you create the partition with `mkpart primary linux-swap 0 1024M`. Then, you enable swap on this partition with `swapon /dev/sdb1`, but get the following error message:

```
swapon: /dev/sdb1: read swap header failed
```

What went wrong?

You created a partition of the correct type (`linux-swap`), but remember that `mkpart` *does not create a filesystem*. You forgot to set-up the partition as a swap space first with `mkswap` before using it.

4. During the course of this lesson, you were trying out some commands in `parted` but, by mistake, deleted the 3rd partition on your hard disk. You know that it came after a 250 MB EFI partition and a 4 GB swap partition, and was 10 GB in size. Which `parted` command can you use to recover it?

Don't panic, you have all the information you need to recover the partition, just use `rescue` and do the math. You had 250 MB + 4.096 MB (4×1024) before it, so the start point should be around 4346 MB. Plus 10.240 MB (10×1024) in size, it should end at 14.586 MB. So, `rescue 4346m 14586m` should do the trick. You might need to give some "slack" to `rescue`, starting a little early and ending a little late, depending on the geometry of your disk.

5. Imagine you have a 4 GB unused partition on `/dev/sda3`. Using `fdisk`, what would be the sequence of operations to turn it into an active swap partition?

First, change the partition type to "Linux Swap" (82), write your changes to disk and quit. Then, use `mkswap` to set up the partition as a swap area. Then, use `swapon` to enable it.



Linux
Professional
Institute

104.2 Maintain the integrity of filesystems

Reference to LPI objectives

LPIC-1 version 5.0, Exam 101, Objective 104.2

Weight

2

Key knowledge areas

- Verify the integrity of filesystems.
- Monitor free space and inodes.
- Repair simple filesystem problems.

Partial list of the used files, terms and utilities

- `du`
- `df`
- `fsck`
- `e2fsck`
- `mke2fs`
- `tune2fs`
- `xfs_repair`
- `xfs_fsr`
- `xfs_db`



104.2 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	104 Devices, Linux Filesystems, Filesystem Hierarchy Standard
Objective:	104.2 Maintain the integrity of filesystems
Lesson:	1 of 1

Introduction

Modern Linux filesystems are journaled. This means that every operation is registered in an internal log (the *journal*) before it is executed. If the operation is interrupted due to a system error (like a kernel panic, power failure, etc.) it can be reconstructed by checking the journal, avoiding filesystem corruption and loss of data.

This greatly reduces the need for manual filesystem checks, but they may still be needed. Knowing the tools used for this (and the corresponding parameters) may represent the difference between dinner at home with your family or an all-nighter in the server room at work.

In this lesson, we will discuss the tools available to monitor filesystem usage, optimize its operation and how to check and repair damage.

Checking Disk Usage

There are two commands that can be used to check how much space is being used and how much is left on a filesystem. The first one is `du`, which stands for “disk usage”.

`du` is recursive in nature. In its most basic form, the command will simply show how many 1 Kilobyte blocks are being used by the current directory and all its subdirectories:

```
$ du
4816 .
```

This is not very helpful, so we can request more “human readable” output by adding the `-h` parameter:

```
$ du -h
4.8M .
```

By default, `du` only shows the usage count for directories (considering all files and subdirectories inside it). To show an individual count for all files in the directory, use the `-a` parameter:

```
$ du -ah
432K ./geminoid.jpg
508K ./Linear_B_Hero.jpg
468K ./LG-G8S-ThinQ-Mirror-White.jpg
656K ./LG-G8S-ThinQ-Range.jpg
60K ./Stranger3_Titulo.png
108K ./Baidu_Banho.jpg
324K ./Xiaomi_Mimoji.png
284K ./Mi_CC_9e.jpg
96K ./Mimoji_Comparativo.jpg
32K ./Xiaomi FCC.jpg
484K ./geminoid2.jpg
108K ./Mimoji_Abre.jpg
88K ./Mi8_Hero.jpg
832K ./Tablet_Linear_B.jpg
332K ./Mimoji_Comparativo.png
4.8M .
```

The default behaviour is to show the usage of every subdirectory, then the total usage of the current directory, *including* subdirectories:

```
$ du -h
4.8M ./Temp
6.0M .
```

In the example above, we can see that the subdirectory `Temp` occupies 4.8 MB and the current directory, *including* `Temp`, occupies 6.0 MB. But how much space do the *files* in the current directory occupy, excluding the subdirectories? For that we have the `-s` parameter:

```
$ du -Sh
4.8M    ./Temp
1.3M    .
```

TIP

Keep in mind that command line parameters are case-sensitive: `-s` is different from `-S`.

If you want to keep this distinction between the space used by the files in the current directory and the space used by subdirectories, but *also* want a grand total at the end, you can add the `-c` parameter:

```
$ du -Shc
4.8M    ./Temp
1.3M    .
6.0M    total
```

You can control how “deep” the output of `du` should go with the `-d N` parameter, where `N` describes the levels. For example, if you use the `-d 1` parameter, it will show the current directory and its subdirectories, but not the subdirectories of those.

See the difference below. Without `-d`:

```
$ du -h
216K    ./somedir/anotherdir
224K    ./somedir
232K    .
```

And limiting the depth to one level with `-d 1`:

```
$ du -h -d1
224K    ./somedir
232K    .
```

Please note that even if `anotherdir` is not being shown, its size is still being taken into account.

You may wish to exclude some types of files from the count, which you can do with

`--exclude="PATTERN"`, where `PATTERN` is the pattern against which you wish to match. Consider this directory:

```
$ du -ah
124K    ./ASM68K.EXE
2.0M    ./Contra.bin
36K     ./fixheadr.exe
4.0K    ./README.txt
2.1M    ./Contra_NEW.bin
4.0K    ./Built.bat
8.0K    ./Contra_Main.asm
4.2M    .
```

Now, we will use `--exclude` to filter out every file with the `.bin` extension:

```
$ du -ah --exclude="*.bin"
124K    ./ASM68K.EXE
36K     ./fixheadr.exe
4.0K    ./README.txt
4.0K    ./Built.bat
8.0K    ./Contra_Main.asm
180K    .
```

Note that the total no longer reflects the size of the excluded files.

Checking for Free Space

`du` works at the files level. There is another command that can show you disk usage, and how much space is available, at the filesystem level. This command is `df`.

The command `df` will provide a list of all of the available (already mounted) filesystems on your system, including their total size, how much space has been used, how much space is available, the usage percentage and where it is mounted:

```
$ df
Filesystem      1K-blocks      Used Available Use% Mounted on
udev            2943068         0    2943068  0% /dev
tmpfs           595892        2496     593396  1% /run
/dev/sda1       110722904    25600600    79454800  25% /
tmpfs           2979440     951208     2028232  32% /dev/shm
tmpfs           5120         0         5120   0% /run/lock
```

```

tmpfs          2979440      0   2979440    0% /sys/fs/cgroup
tmpfs          595888     24   595864    1% /run/user/119
tmpfs          595888    116   595772    1% /run/user/1000
/dev/sdb1      89111     1550   80824    2% /media/carol/part1
/dev/sdb3      83187     1550   75330    3% /media/carol/part3
/dev/sdb2      90827     1921   82045    3% /media/carol/part2
/dev/sdc1     312570036 233740356 78829680 75% /media/carol/Samsung Externo

```

However, showing the size in 1 KB blocks is not very user-friendly. Like on `du`, you can add the `-h` parameters to get a more “human readable” output:

```

$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            2.9G     0  2.9G   0% /dev
tmpfs           582M   2.5M  580M   1% /run
/dev/sda1       106G   25G   76G  25% /
tmpfs           2.9G  930M   2.0G  32% /dev/shm
tmpfs           5.0M     0   5.0M   0% /run/lock
tmpfs           2.9G     0   2.9G   0% /sys/fs/cgroup
tmpfs           582M    24K  582M   1% /run/user/119
tmpfs           582M   116K  582M   1% /run/user/1000
/dev/sdb1        88M   1.6M   79M   2% /media/carol/part1
/dev/sdb3        82M   1.6M   74M   3% /media/carol/part3
/dev/sdb2        89M   1.9M   81M   3% /media/carol/part2
/dev/sdc1       299G  223G   76G  75% /media/carol/Samsung Externo

```

You can also use the `-i` parameter to show used/available inodes instead of blocks:

```

$ df -i
Filesystem      Inodes   IUsed   IFree IUse% Mounted on
udev            737142    547  736595    1% /dev
tmpfs           745218    908  744310    1% /run
/dev/sda6       6766592 307153 6459439    5% /
tmpfs           745218    215  745003    1% /dev/shm
tmpfs           745218     4  745214    1% /run/lock
tmpfs           745218    18  745200    1% /sys/fs/cgroup
/dev/sda1        62464    355   62109    1% /boot
tmpfs           745218    43  745175    1% /run/user/1000

```

One useful parameter is `-T`, which will also print the type of each filesystem:

```
$ df -hT
```

Filesystem	Type	Size	Used	Avail	Use%	Mounted on
udev	devtmpfs	2.9G	0	2.9G	0%	/dev
tmpfs	tmpfs	582M	2.5M	580M	1%	/run
/dev/sda1	ext4	106G	25G	76G	25%	/
tmpfs	tmpfs	2.9G	930M	2.0G	32%	/dev/shm
tmpfs	tmpfs	5.0M	0	5.0M	0%	/run/lock
tmpfs	tmpfs	2.9G	0	2.9G	0%	/sys/fs/cgroup
tmpfs	tmpfs	582M	24K	582M	1%	/run/user/119
tmpfs	tmpfs	582M	116K	582M	1%	/run/user/1000
/dev/sdb1	ext4	88M	1.6M	79M	2%	/media/carol/part1
/dev/sdb3	ext4	82M	1.6M	74M	3%	/media/carol/part3
/dev/sdb2	ext4	89M	1.9M	81M	3%	/media/carol/part2
/dev/sdc1	fuseblk	299G	223G	76G	75%	/media/carol/Samsung Externo

Knowing the type of the filesystem you can filter the output. You can show only filesystems of a given type with `-t TYPE`, or exclude filesystems of a given type with `-x TYPE`, like in the examples below.

Excluding tmpfs filesystems:

```
$ df -hx tmpfs
```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	2.9G	0	2.9G	0%	/dev
/dev/sda1	106G	25G	76G	25%	/
/dev/sdb1	88M	1.6M	79M	2%	/media/carol/part1
/dev/sdb3	82M	1.6M	74M	3%	/media/carol/part3
/dev/sdb2	89M	1.9M	81M	3%	/media/carol/part2
/dev/sdc1	299G	223G	76G	75%	/media/carol/Samsung Externo

Showing only ext4 filesystems:

```
$ df -ht ext4
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sda1	106G	25G	76G	25%	/
/dev/sdb1	88M	1.6M	79M	2%	/media/carol/part1
/dev/sdb3	82M	1.6M	74M	3%	/media/carol/part3
/dev/sdb2	89M	1.9M	81M	3%	/media/carol/part2

You can also customize the output of `df`, selecting what should be displayed and in which order, using the `--output=` parameter followed by a comma separated list of fields you wish to display.

Some of the available fields are:

source

The device corresponding to the filesystem.

fstype

The filesystem type.

size

The total size of the filesystem.

used

How much space is being used.

avail

How much space is available.

pcent

The usage percentage.

target

Where the filesystem is mounted (mount point).

If you want an output showing the target, source, type and usage, you can use:

```
$ df -h --output=target,source,fstype,pcent
Mounted on      Filesystem      Type      Use%
/dev            udev            devtmpfs   0%
/run            tmpfs           tmpfs      1%
/               /dev/sda1       ext4       25%
/dev/shm        tmpfs           tmpfs      32%
/run/lock       tmpfs           tmpfs      0%
/sys/fs/cgroup  tmpfs           tmpfs      0%
/run/user/119   tmpfs           tmpfs      1%
/run/user/1000  tmpfs           tmpfs      1%
/media/carol/part1  /dev/sdb1       ext4       2%
/media/carol/part3  /dev/sdb3       ext4       3%
/media/carol/part2  /dev/sdb2       ext4       3%
/media/carol/Samsung Externo /dev/sdc1       fuseblk    75%
```

df can also be used to check inode information, by passing the following fields to **--output=**:

itotal

The total number of inodes in the filesystem.

iused

The number of used inodes in the filesystem.

iavail

The number of available inodes in the filesystem.

ipcent

The percentage of used inodes in the filesystem.

For example:

```
$ df --output=source,fstype,itotal,iused,ipcent
Filesystem      Type      Inodes   IUsed IUse%
udev            devtmpfs  735764    593   1%
tmpfs           tmpfs     744858   1048   1%
/dev/sda1       ext4      7069696 318651   5%
tmpfs           tmpfs     744858    222   1%
tmpfs           tmpfs     744858     3    1%
tmpfs           tmpfs     744858    18   1%
tmpfs           tmpfs     744858    22   1%
tmpfs           tmpfs     744858    40   1%
```

Maintaining ext2, ext3 and ext4 Filesystems

To check a filesystem for errors (and hopefully fix them), Linux provides the `fsck` utility (think of “filesystem check” and you will never forget the name). In its most basic form, you can invoke it with `fsck` followed by the filesystem’s location you want to check:

```
# fsck /dev/sdb1
fsck from util-linux 2.33.1
e2fsck 1.44.6 (5-Mar-2019)
DT_2GB: clean, 20/121920 files, 369880/487680 blocks
```

WARNING

NEVER run `fsck` (or related utilities) on a mounted filesystem. If this is done anyway, data may be lost.

`fsck` itself will not check the filesystem, it will merely call the appropriate utility for the

filesystem type to do so. In the example above, since a filesystem type was not specified, `fsck` assumed an `ext2/3/4` filesystem by default, and called `e2fsck`.

To specify a filesystem, use the `-t` option, followed by the filesystem name, like in `fsck -t vfat /dev/sdc`. Alternatively, you may call a filesystem-specific utility directly, like `fsck.msdos` for FAT filesystems.

TIP | Type `fsck` followed by `Tab` twice to see a list of all related commands on your system.

`fsck` can take some command-line arguments. These are some of the most common:

-A

This will check all filesystems listed in `/etc/fstab`.

-C

Displays a progress bar when checking a filesystem. Currently only works on `ext2/3/4` filesystems.

-N

This will print what would be done and exit, without actually checking the filesystem.

-R

When used in conjunction with `-A`, this will skip checking the root filesystem.

-V

Verbose mode, prints more information than usual during operation. This is useful for debugging.

The specific utility for `ext2`, `ext3` and `ext4` filesystems is `e2fsck`, also called `fsck.ext2`, `fsck.ext3` and `fsck.ext4` (those three are merely links to `e2fsck`). By default, it runs in interactive mode: when a filesystem error is found, it stops and asks the user what to do. The user must type `y` to fix the problem, `n` to leave it unfixed or `a` to fix the current problem and all subsequent ones.

Of course sitting in front of a terminal waiting for `e2fsck` to ask what to do is not a productive use of your time, especially if you are dealing with a big filesystem. So, there are options that cause `e2fsck` to run in non-interactive mode:

-p

This will attempt to automatically fix any errors found. If an error that requires intervention from the system administrator is found, `e2fsck` will provide a description of the problem and exit.

-y

This will answer y (yes) to all questions.

-n

The opposite of -y. Besides answering n (no) to all questions, this will cause the filesystem to be mounted read-only, so it cannot be modified.

-f

Forces `e2fsck` to check a filesystem even if it is marked as “clean”, i.e. has been correctly unmounted.

Fine Tuning an ext Filesystem

ext2, ext3 and ext4 filesystems have a number of parameters that can be adjusted, or “tuned”, by the system administrator to better suit the system needs. The utility used to display or modify these parameters is called `tune2fs`.

To see the current parameters for any given filesystem, use the `-l` parameter followed by the device representing the partition. The example below shows the output of this command on the first partition of the first disk (`/dev/sda1`) of a machine:

```
# tune2fs -l /dev/sda1
tune2fs 1.44.6 (5-Mar-2019)
Filesystem volume name: <none>
Last mounted on: /
Filesystem UUID: 6e2c12e3-472d-4bac-a257-c49ac07f3761
Filesystem magic number: 0xEF53
Filesystem revision #: 1 (dynamic)
Filesystem features: has_journal ext_attr resize_inode dir_index filetype
needs_recovery extent 64bit flex_bg sparse_super large_file huge_file dir_nlink extra_isize
metadata_csum
Filesystem flags: signed_directory_hash
Default mount options: user_xattr acl
Filesystem state: clean
Errors behavior: Continue
Filesystem OS type: Linux
Inode count: 7069696
Block count: 28255605
Reserved block count: 1412780
Free blocks: 23007462
Free inodes: 6801648
First block: 0
```

```
Block size:                4096
Fragment size:             4096
Group descriptor size:     64
Reserved GDT blocks:      1024
Blocks per group:         32768
Fragments per group:      32768
Inodes per group:         8192
Inode blocks per group:    512
Flex block group size:     16
Filesystem created:        Mon Jun 17 13:49:59 2019
Last mount time:           Fri Jun 28 21:14:38 2019
Last write time:           Mon Jun 17 13:53:39 2019
Mount count:               8
Maximum mount count:       -1
Last checked:              Mon Jun 17 13:49:59 2019
Check interval:            0 (<none>)
Lifetime writes:           20 GB
Reserved blocks uid:       0 (user root)
Reserved blocks gid:       0 (group root)
First inode:               11
Inode size:                256
Required extra isize:      32
Desired extra isize:       32
Journal inode:             8
First orphan inode:        5117383
Default directory hash:    half_md4
Directory Hash Seed:       fa95a22a-a119-4667-a73e-78f77af6172f
Journal backup:            inode blocks
Checksum type:             crc32c
Checksum:                  0xe084fe23
```

ext filesystems have *mount counts*. The count is increased by 1 each time the filesystem is mounted, and when a threshold value (the *maximum mount count*) is reached the system will be automatically checked with `e2fsck` on the next boot.

The maximum mount count can be set with the `-c N` parameter, where N is the number of times the filesystem can be mounted without being checked. The `-C N` parameter sets the number of times the system has been mounted to the value of N. Note that command line parameters are case-sensitive, so `-c` is different from `-C`.

It is also possible to define a time interval between checks, with the `-i` parameter, followed by a number and the letters d for days, w for weeks and m for months. For example, `-i 10d` would check the filesystem at the next reboot every 10 days. Use zero as the value to disable this feature.

`-L` can be used to set a label for the filesystem. This label can have up to 16 characters. The `-U` parameter sets the UUID for the filesystem, which is a 128 bit hexadecimal number. In the example above, the UUID is `6e2c12e3-472d-4bac-a257-c49ac07f3761`. Both the label and UUID can be used instead of the device name (like `/dev/sda1`) to mount the filesystem.

The option `-e BEHAVIOUR` defines the kernel behaviour when a filesystem error is found. There are three possible behaviours:

continue

Will continue execution normally.

remount-ro

Will remount the filesystem as read-only.

panic

Will cause a kernel panic.

The default behaviour is to `continue`. `remount-ro` might be useful in data-sensitive applications, as it will immediately stop writes to the disk, avoiding more potential errors.

`ext3` filesystems are basically `ext2` filesystems with a journal. Using `tune2fs` you can add a journal to an `ext2` filesystem, thus converting it to `ext3`. The procedure is simple, just pass the `-j` parameter to `tune2fs`, followed by the device containing the filesystem:

```
# tune2fs -j /dev/sda1
```

Afterwards, when mounting the converted filesystem, do not forget to set the type to `ext3` so the journal can be used.

When dealing with journaled filesystems, the `-J` parameter allows you to use extra parameters to set some journal options, like `-J size=` to set the journal size (in megabytes), `-J location=` to specify where the journal should be stored (either a specific block, or a specific position on the disk with suffixes like `M` or `G`) and even put the journal on an external device with `-J device=`.

You can specify multiple parameters at once by separating them with a comma. For example: `-J size=10,location=100M,device=/dev/sdb1` will create a 10 MB Journal at the 100 MB position on the device `/dev/sdb1`.

WARNING

`tune2fs` has a “brute force” option, `-f`, which will force it to complete an operation even if errors are found. Needless to say, this should be only used with extreme caution.

Maintaining XFS Filesystems

For XFS filesystems, the equivalent of `fsck` is `xfs_repair`. If you suspect that something is wrong with the filesystem, the first thing to do is to scan it for damage.

This can be done by passing the `-n` parameter to `xfs_repair`, followed by the device containing the filesystem. The `-n` parameter means “no modify”: the filesystem will be checked, errors will be reported but no repairs will be made:

```
# xfs_repair -n /dev/sdb1
Phase 1 - find and verify superblock...
Phase 2 - using internal log
    - zero log...
    - scan filesystem freespace and inode maps...
    - found root inode chunk
Phase 3 - for each AG...
    - scan (but do not clear) agi unlinked lists...
    - process known inodes and perform inode discovery...
    - agno = 0
    - agno = 1
    - agno = 2
    - agno = 3
    - process newly discovered inodes...
Phase 4 - check for duplicate blocks...
    - setting up duplicate extent list...
    - check for inodes claiming duplicate blocks...
    - agno = 1
    - agno = 3
    - agno = 0
    - agno = 2
No modify flag set, skipping phase 5
Phase 6 - check inode connectivity...
    - traversing filesystem ...
    - traversal finished ...
    - moving disconnected inodes to lost+found ...
Phase 7 - verify link counts...
No modify flag set, skipping filesystem flush and exiting.
```

If errors are found, you can proceed to do the repairs without the `-n` parameter, like so: `xfs_repair /dev/sdb1`.

`xfs_repair` accepts a number of command line options. Among them:

-l LOGDEV and -r RTDEV

These are needed if the filesystem has external log and realtime sections. In this case, replace LOGDEV and RTDEV with the corresponding devices.

-m N

Is used to limit the memory usage of `xfs_repair` to N megabytes, something which can be useful on server settings. According to the man page, by default `xfs_repair` will scale its memory usage as needed, up to 75% of the system's physical RAM.

-d

The “dangerous” mode will enable the repair of filesystems that are mounted read-only.

-v

You may have guessed it: verbose mode. Each time this parameter is used, the “verbosity” is increased (for example, `-v -v` will print more information than just `-v`).

Note that `xfs_repair` is unable to repair filesystems with a “dirty” log. You can “zero out” a corrupt log with the `-L` parameter, but keep in mind that this is a *last resort* as it may result in filesystem corruption and data loss.

To debug an XFS filesystem, the utility `xfs_db` can be used, like in `xfs_db /dev/sdb1`. This is mostly used to inspect various elements and parameters of the filesystem.

This utility has an interactive prompt, like `parted`, with many internal commands. A help system is also available: type `help` to see a list of all commands, and `help` followed by the command name to see more information about the command.

Another useful utility is `xfs_fsr`, which can be used to reorganize (“defragment”) an XFS filesystem. When executed without any extra arguments it will run for two hours and try to defragment all mounted, writable XFS filesystems listed on the `/etc/mtab/` file. You may need to install this utility using the package manager for your Linux distribution, as it may not be part of a default install. For more information consult the corresponding man page.

Guided Exercises

1. Using `du`, how can we check how much space is being used by just the files on the current directory?

2. Using `df`, list information for every ext4 filesystem, with the outputs including the following fields, in order: device, mount point, total number of inodes, number of available inodes, percentage of free space.

3. What is the command to run `e2fsck` on `/dev/sdc1` in non-interactive mode, while trying to automatically fix most errors?

4. Suppose `/dev/sdb1` is an ext2 filesystem. How can you convert it to ext3, and at the same time reset its mount count and change its label to UserData?

5. How can you check for errors on an XFS filesystem, *without* repairing any damage found?

Explorational Exercises

1. Consider you have an ext4 filesystem on `/dev/sda1` with the following parameters, obtained with `tune2fs`:

```
Mount count:          8
Maximum mount count:  -1
```

What will happen at the next boot if the command `tune2fs -c 9 /dev/sda1` is issued?

2. Consider the following output of `du -h`:

```
$ du -h
216K    ./somedir/anotherdir
224K    ./somedir
232K    .
```

How much space is occupied by just the files on the current directory? How could we rewrite the command to show this information more clearly?

3. What would happen to the ext2 filesystem `/dev/sdb1` if the command below is issued?

```
# tune2fs -j /dev/sdb1 -J device=/dev/sdc1 -i 30d
```

4. How can we check for errors on a XFS filesystem on `/dev/sda1` that has a log section on `/dev/sdc1`, without actually making any repairs?

5. What is the difference between the `-T` and `-t` parameters for `df`?

Summary

In this lesson, you learned:

- How to check for used and free space on a filesystem.
- How to tailor the output of `df` to suit your needs.
- How to check the integrity of and repair a filesystem with `fsck` and `e2fsck`.
- How to fine tune an ext filesystem with `tune2fs`.
- How to check and repair XFS filesystems with `xfs_repair`.

The following commands were discussed in this lesson:

du

View the amount of disk space in use on a filesystem.

df

View the amount of disk space that is available (free) on a filesystem.

fsck

The filesystem check repair utility.

e2fsck

The filesystem check repair utility specific to extended (ext2/3/4) filesystems.

tune2fs

Modifies filesystem parameters on an extended (ext2/3/4) filesystem.

xfs_repair

The equivalent of `fsck` for XFS filesystems.

xfs_db

This utility is used to view various parameters of an XFS filesystem.

Answers to Guided Exercises

1. Using `du`, how can we check how much space is being used by just the files on the current directory?

First, use the `-S` parameter to separate the output of the current directory from its subdirectories. Then, use `-d 0` to limit the output depth to zero, meaning “no subdirectories”. Do not forget `-h` to get an output in an “human-readable” format:

```
$ du -S -h -d 0
```

or

```
$ du -Shd 0
```

2. Using `df`, list information for every ext4 filesystem, with the outputs including the following fields, in order: device, mount point, total number of inodes, number of available inodes, percentage of free space.

You can filter filesystems with the `-t` option followed by the filesystem name. To get the output needed, use `--output=source,target,itotal,iavail,pcent`. So, the answer is:

```
$ df -t ext4 --output=source,target,itotal,iavail,pcent
```

3. What is the command to run `e2fsck` on `/dev/sdc1` in non-interactive mode, while trying to automatically fix most errors?

The parameter to automatically try to fix most errors is `-p`. So the answer is:

```
# e2fsck -p /dev/sdc1
```

4. Suppose `/dev/sdb1` is an ext2 filesystem. How can you convert it to ext3 and at the same time reset its mount count and change its label to `UserData`?

Remember that converting an ext2 filesystem to ext3 is just a matter of adding a journal, which can be done with the `-j` parameter. To reset the mount count, use `-C 0`. To change the label use `-L UserData`. The correct answer is:

```
# tune2fs -j -C 0 -L UserData /dev/sdb1
```

5. How can you check for errors on an XFS filesystem, *without* repairing any damage found?

Use the `-n` parameter, like in `xfs -n`, followed by the corresponding device.

Answers to Explorational Exercises

1. Consider you have an ext4 filesystem on `/dev/sda1` with the following parameters, obtained with `tune2fs`:

```
Mount count:          8
Maximum mount count:  -1
```

What will happen at the next boot if the command `tune2fs -c 9 /dev/sda1` is issued?

The command will set the maximum mount count for the filesystem to 9. Since the mount count is currently 8, the next system boot will cause a filesystem check.

2. Consider the following output of `du -h`:

```
$ du -h
216K    ./somedir/anotherdir
224K    ./somedir
232K    .
```

How much space is occupied by just the files on the current directory? How could we rewrite the command to show this information more clearly?

Of the total of 232 K used, 224 K are used by the subdirectory `somedir` and its subdirectories. So, excluding those, we have 8K being occupied by the files on the current directory. This information can be shown more clearly by using the `-S` parameter, which will separate the directories in the count.

3. What would happen to the ext2 filesystem `/dev/sdb1` if the command below is issued?

```
# tune2fs -j /dev/sdb1 -J device=/dev/sdc1 -i 30d
```

A journal will be added to `/dev/sdb1`, converting it to ext3. The journal will be stored on the device `/dev/sdc1` and the filesystem will be checked every 30 days.

4. How can we check for errors on a XFS filesystem on `/dev/sda1` that has a log section on `/dev/sdc1`, without actually making any repairs?

Use `xfs_repair`, followed by `-l /dev/sdc1` to indicate the device containing the log section, and `-n` to avoid making any changes.

```
# xfs_repair -l /dev/sdc1 -n
```

5. What is the difference between the `-T` and `-t` parameters for `df`?

The parameter `-T` will include the type of each filesystem in the output of `df`. `-t` is a filter, and will show only filesystems of the given type on the output, excluding all others.



Linux
Professional
Institute

104.3 Control mounting and unmounting of filesystems

Reference to LPI objectives

LPIC-1 version 5.0, Exam 101, Objective 104.3

Weight

3

Key knowledge areas

- Manually mount and unmount filesystems.
- Configure filesystem mounting on bootup.
- Configure user mountable removable filesystems.
- Use of labels and UUIDs for identifying and mounting file systems.
- Awareness of systemd mount units.

Partial list of the used files, terms and utilities

- `/etc/fstab`
- `/media/`
- `mount`
- `umount`
- `blkid`
- `lsblk`



104.3 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	104 Devices, Linux Filesystems, Filesystem Hierarchy Standard
Objective:	104.3 Control mounting and unmounting of filesystems
Lesson:	1 of 1

Introduction

Up until now, you learned how to partition disks and how to create and maintain filesystems on them. However before a filesystem can be accessed on Linux, it needs to be *mounted*.

This means attaching the filesystem to a specific point in your system's directory tree, called a *mount point*. Filesystems can be mounted manually or automatically and there are many ways to do this. We will learn about some of them in this lesson.

Mounting and Unmounting Filesystems

The command to manually mount a filesystem is called `mount` and its syntax is:

```
mount -t TYPE DEVICE MOUNTPOINT
```

Where:

TYPE

The type of the filesystem being mounted (e.g. ext4, btrfs, exfat, etc.).

DEVICE

The name of the partition containing the filesystem (e.g. /dev/sdb1)

MOUNTPOINT

Where the filesystem will be mounted. The mounted-on directory need not be empty, although it must exist. Any files in it, however, will be inaccessible by name while the filesystem is mounted.

For example, to mount a USB flash drive containing an exFAT filesystem located on /dev/sdb1 to a directory called flash under your home directory, you could use:

```
# mount -t exfat /dev/sdb1 ~/flash/
```

TIP

Many Linux systems use the Bash shell, and on those the tilde ~ on the path to the mountpoint is a shorthand for the current user's home directory. If the current user is named john, for example, it will be replaced by /home/john.

After mounting, the contents of the filesystem will be accessible under the ~/flash directory:

```
$ ls -lh ~/flash/
total 469M
-rwxrwxrwx 1 root root 454M jul 19 09:49 lineage-16.0-20190711-MOD-quark.zip
-rwxrwxrwx 1 root root 16M jul 19 09:44 twrp-3.2.3-mod_4-quark.img
```

Listing Mounted Filesystems

If you type just mount, you will get a list of all the filesystems currently mounted on your system. This list can be quite large, because in addition to the disks attached to your system, it also contains a number of run-time filesystems in memory that serve various purposes. To filter the output, you can use the -t parameter to list only filesystems of the corresponding type, like below:

```
# mount -t ext4
/dev/sda1 on / type ext4 (rw,noatime,errors=remount-ro)
```

You can specify multiple filesystems at once by separating them with a comma:

```
# mount -t ext4,fuseblk  
/dev/sda1 on / type ext4 (rw,noatime,errors=remount-ro)  
/dev/sdb1 on /home/carol/flash type fuseblk  
(rw,nosuid,nodev,relatime,user_id=0,group_id=0,default_permissions,allow_other,blksize=4096)  
[DT_8GB]
```

The output in the examples above can be described in the format:

```
SOURCE on TARGET type TYPE OPTIONS
```

Where **SOURCE** is the partition which contains the filesystem, **TARGET** is the directory where it is mounted, **TYPE** is the filesystem type and **OPTIONS** are the options passed to the **mount** command at mount time.

Additional Command Line Parameters

There are many command line parameters that can be used with **mount**. Some of the most used ones are:

-a

This will mount all filesystems listed in the file **/etc/fstab** (more on that in the next section).

-o or --options

This will pass a list of comma-separated *mount options* to the **mount** command, which can change how the filesystem will be mounted. These will also be discussed alongside **/etc/fstab**.

-r or -ro

This will mount the filesystem as read-only.

-w or -rw

This will mount the filesystem as writable.

To unmount a filesystem, use the **umount** command, followed by the device name or the mount point. Considering the example above, the commands below are interchangeable:

```
# umount /dev/sdb1  
# umount ~/flash
```

Some of the command line parameters to `umount` are:

-a

This will unmount all filesystems listed in `/etc/fstab`.

-f

This will force the unmounting of a filesystem. This may be useful if you mounted a remote filesystem that has become unreachable.

-r

If the filesystem cannot be unmounted, this will try to make it read-only.

Dealing with Open Files

When unmounting a filesystem, you may encounter an error message stating that the target is busy. This will happen if any files on the filesystem are open. However, it may not be immediately obvious where an open file is located, or what is accessing the filesystem.

In such cases you may use the `lsof` command, followed by the name of the device containing the filesystem, to see a list of processes accessing it and which files are open. For example:

```
# umount /dev/sdb1
umount: /media/carol/External_Drive: target is busy.

# lsof /dev/sdb1
COMMAND  PID   USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
evince   3135 carol   16r   REG   8,17 21881768 5195 /media/carol/External_Drive/Documents/E-
Books/MagPi40.pdf
```

`COMMAND` is the name of the executable that opened the file, and `PID` is the process number. `NAME` is the name of the file that is open. In the example above, the file `MagPi40.pdf` is opened by the program `evince` (a PDF viewer). If we close the program, then we will be able to unmount the filesystem.

Before the `lsof` output appears GNOME users may see a warning message in the terminal window.

NOTE

```
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1000/gvfs
Output information may be incomplete.
```

`lsuf` tries to process all mounted filesystems. This warning message is raised because `lsuf` has encountered a GNOME Virtual file system (GVFS). This is a special case of a filesystem in user space (FUSE). It acts as a bridge between GNOME, its APIs and the kernel. No one—not even root—can access one of these file systems, apart from the owner who mounted it (in this case, GNOME). You can ignore this warning.

Where to Mount?

You can mount a filesystem anywhere you want. However, there are some good practices that should be followed to make system administration easier.

Traditionally, `/mnt` was the directory under which all external devices would be mounted and a number of pre-configured “anchor points” for common devices, like CD-ROM drives (`/mnt/cdrom`) and floppy disks (`/mnt/floppy`) existed under it.

This has been superseded by `/media`, which is now the default mount point for any user-removable media (e.g. external disks, USB flash drives, memory card readers, etc.) connected to the system.

On most modern Linux distributions and desktop environments, removable devices are automatically mounted under `/media/USER/LABEL` when connected to the system, where `USER` is the username and `LABEL` is the device label. For example, a USB flash drive with the label `FlashDrive` connected by the user `john` would be mounted under `/media/john/FlashDrive/`. The way this is handled is different depending on the desktop environment.

That being said, whenever you need to *manually* mount a filesystem, it is good practice to mount it under `/mnt`.

Mounting Filesystems on Bootup

The file `/etc/fstab` contains descriptions about the filesystems that can be mounted. This is a text file, where each line describes a filesystem to be mounted, with six fields per line in the following order:

```
FILESYSTEM MOUNTPOINT TYPE OPTIONS DUMP PASS
```

Where:

FILESYSTEM

The device containing the filesystem to be mounted. Instead of the device, you can specify the UUID or label of the partition, something which we will discuss later on.

MOUNTPOINT

Where the filesystem will be mounted.

TYPE

The filesystem type.

OPTIONS

Mount options that will be passed to `mount`.

DUMP

Indicates whether any ext2, ext3 or ext4 filesystems should be considered for backup by the `dump` command. Usually it is zero, meaning they should be ignored.

PASS

When non-zero, defines the order in which the filesystems will be checked on bootup. Usually it is zero.

For example, the first partition on the first disk of a machine could be described as:

```
/dev/sda1 / ext4 noatime,errors
```

The mount options on `OPTIONS` are a comma-separated list of parameters, which can be generic or filesystem specific. Among the generic ones we have:

atime and noatime

By default, every time a file is read the access time information is updated. Disabling this (with `noatime`) can speed up disk I/O. Do not confuse this with the modification time, which is updated every time a file is written to.

auto and noauto

Whether the filesystem can (or can not) be mounted automatically with `mount -a`.

defaults

This will pass the options `rw`, `suid`, `dev`, `exec`, `auto`, `nouser` and `async` to `mount`.

dev and nodev

Whether character or block devices in the mounted filesystem should be interpreted.

exec and noexec

Allow or deny permission to execute binaries on the filesystem.

user and nouser

Allows (or not) an ordinary user to mount the filesystem.

group

Allows a user to mount the filesystem if the user belongs to the same group which owns the device containing it.

owner

Allows a user to mount a filesystem if the user owns the device containing it.

suid and nosuid

Allow, or not, SETUID and SETGID bits to take effect.

ro and rw

Mount a filesystem as read-only or writable.

remount

This will attempt to remount an already mounted filesystem. This is not used on `/etc/fstab`, but as a parameter to `mount -o`. For example, to remount the already mounted partition `/dev/sdb1` as read-only, you could use the command `mount -o remount,ro /dev/sdb1`. When remounting, you do not need to specify the filesystem type, only the device name or the mount point.

sync and async

Whether to do all I/O operations to the filesystem synchronously or asynchronously. `async` is usually the default. The manual page for `mount` warns that using `sync` on media with a limited number of write cycles (like flash drives or memory cards) may shorten the life span of the device.

Using UUIDs and Labels

Specifying the name of the device containing the filesystem to mount may introduce some problems. Sometimes the same device name may be assigned to another device depending on when, or where, it was connected to your system. For example, a USB flash drive on `/dev/sdb1` may be assigned to `/dev/sdc1` if plugged on another port, or after another flash drive.

One way to avoid this is to specify the label or UUID (*Universally Unique Identifier*) of the volume. Both are specified when the filesystem is created and will not change, unless the filesystem is destroyed or manually assigned a new label or UUID.

The command `lsblk` can be used to query information about a filesystem and find out the label

and UUID associated to it. To do this, use the `-f` parameter, followed by the device name:

```
$ lsblk -f /dev/sda1
NAME FSTYPE LABEL UUID                                FSAVAIL FSUSE% MOUNTPOINT
sda1 ext4          6e2c12e3-472d-4bac-a257-c49ac07f3761    64,9G    33% /
```

Here is the meaning of each column:

NAME

Name of the device containing the filesystem.

FSTYPE

Filesystem type.

LABEL

Filesystem label.

UUID

Universally Unique Identifier (UUID) assigned to the filesystem.

FSAVAIL

How much space is available in the filesystem.

FSUSE%

Usage percentage of the filesystem.

MOUNTPOINT

Where the filesystem is mounted.

In `/etc/fstab` a device can be specified by its UUID with the `UUID=` option, followed by the UUID, or with `LABEL=`, followed by the label. So, instead of:

```
/dev/sda1 / ext4 noatime,errors
```

You would use:

```
UUID=6e2c12e3-472d-4bac-a257-c49ac07f3761 / ext4 noatime,errors
```

Or, if you have a disk labeled `homedisk`:

```
LABEL=homedisk /home ext4 defaults
```

The same syntax can be used with the `mount` command. Instead of the device name, pass the UUID or label. For example, to mount an external NTFS disk with the UUID `56C11DCC5D2E1334` on `/mnt/external`, the command would be:

```
$ mount -t ntfs UUID=56C11DCC5D2E1334 /mnt/external
```

Mounting Disks with Systemd

Systemd is the init process, the first process to run on many Linux distributions. It is responsible for spawning other processes, starting services and bootstrapping the system. Among many other tasks, *systemd* can also be used to manage the mounting (and automounting) of filesystems.

To use this feature of *systemd*, you need to create a configuration file called a *mount unit*. Each volume to be mounted gets its own mount unit and they need to be placed in `/etc/systemd/system/`.

Mount units are simple text files with the `.mount` extension. The basic format is shown below:

```
[Unit]
Description=

[Mount]
What=
Where=
Type=
Options=

[Install]
WantedBy=
```

Description=

Short description of the mount unit, something like `Mounts the backup disk`.

What=

What should be mounted. The volume must be specified as `/dev/disk/by-uuid/VOL_UUID` where `VOL_UUID` is the UUID of the volume.

Where=

Should be the full path to where the volume should be mounted.

Type=

The filesystem type.

Options=

Mount options that you may wish to pass, these are the same used with the `mount` command or in `/etc/fstab`.

WantedBy=

Used for dependency management. In this case, we will use `multi-user.target`, which means that whenever the system boots into a multi-user environment (a normal boot) the unit will be mounted.

Our previous example of the external disk could be written as:

```
[Unit]
Description=External data disk

[Mount]
What=/dev/disk/by-uuid/56C11DCC5D2E1334
Where=/mnt/external
Type=ntfs
Options=defaults

[Install]
WantedBy=multi-user.target
```

But we are not done yet. To work correctly, the mount unit *must* have the same name as the mount point. In this case, the mount point is `/mnt/external`, so the file should be named `mnt-external.mount`.

After that, you need to restart the `systemd` daemon with the `systemctl` command, and start the unit:

```
# systemctl daemon-reload
# systemctl start mnt-external.mount
```

Now the contents of the external disk should be available on `/mnt/external`. You can check the

status of the mounting with the command `systemctl status mnt-external.mount`, like below:

```
# systemctl status mnt-external.mount
• mnt-external.mount - External data disk
  Loaded: loaded (/etc/systemd/system/mnt-external.mount; disabled; vendor preset: enabled)
  Active: active (mounted) since Mon 2019-08-19 22:27:02 -03; 14s ago
    Where: /mnt/external
    What: /dev/sdb1
   Tasks: 0 (limit: 4915)
  Memory: 128.0K
   CGroup: /system.slice/mnt-external.mount

ago 19 22:27:02 pop-os systemd[1]: Mounting External data disk...
ago 19 22:27:02 pop-os systemd[1]: Mounted External data disk.
```

The `systemctl start mnt-external.mount` command will only enable the unit for the current session. If you want to enable it on every boot, replace `start` with `enable`:

```
# systemctl enable mnt-external.mount
```

Automounting a Mount Unit

Mount units can be automounted whenever the mount point is accessed. To do this, you need an `.automount` file, alongside the `.mount` file describing the unit. The basic format is:

```
[Unit]
Description=

[Automount]
Where=

[Install]
WantedBy=multi-user.target
```

Like before, `Description=` is a short description of the file, and `Where=` is the mountpoint. For example, an `.automount` file for our previous example would be:

```
[Unit]
Description=Automount for the external data disk
```

```
[Automount]
Where=/mnt/external

[Install]
WantedBy=multi-user.target
```

Save the file with the same name as the mount point (in this case, `mnt-external.automount`), reload `systemd` and start the unit:

```
# systemctl daemon-reload
# systemctl start mnt-external.automount
```

Now whenever the `/mnt/external` directory is accessed, the disk will be mounted. Like before, to enable the automount on every boot you would use:

```
# systemctl enable mnt-external.automount
```

Guided Exercises

1. Using `mount`, how can you mount an `ext4` filesystem on `/dev/sdc1` to `/mnt/external` as read-only, using the `noatime` and `async` options?

2. When unmounting a filesystem at `/dev/sdd2`, you get the `target is busy` error message. How can you find out which files on the filesystem are open, and what processes opened them?

3. Consider the following entry in `/etc/fstab`: `/dev/sdb1 /data ext4 noatime,noauto,async`. Will this filesystem be mounted if the command `mount -a` is issued? Why?

4. How can you find out the UUID of a filesystem under `/dev/sdb1`?

5. How can you use `mount` to remount as read-only an exFAT filesystem with the UUID `6e2c12e3-472d-4bac-a257-c49ac07f3761`, mounted at `/mnt/data`?

6. How can you get a list of all `ext3` and `ntfs` filesystems currently mounted on a system?

Explorational Exercises

1. Consider the following entry in `/etc/fstab`: `/dev/sdc1 /backup ext4 noatime,nouser,async`. Can a user mount this filesystem with the command `mount /backup`? Why?

2. Consider a remote filesystem mounted at `/mnt/server`, which has become unreachable due to a loss of network connectivity. How could you force it to be unmounted, or mounted as read-only if this is not possible?

3. Write an `/etc/fstab` entry that would mount a `btrfs` volume with the label `Backup` on `/mnt/backup`, with default options and without allowing the execution of binaries from it.

4. Consider the following `systemd` mount unit:

```
[Unit]
Description=External data disk

[Mount]
What=/dev/disk/by-uuid/56C11DCC5D2E1334
Where=/mnt/external
Type=ntfs
Options=defaults

[Install]
WantedBy=multi-user.target
```

- What would be an equivalent `/etc/fstab` entry for this filesystem?

5. What should be the file name for the unit above, so it can be used by `systemd`? Where should it be placed?

Summary

In this lesson, you learned how to mount and unmount filesystems, manually or automatically. Some of the commands and concepts explained were:

- `mount` (mounts a device to a location)
- `umount` (unmounts a device)
- `lsof` (lists the processes accessing a filesystem)
- `/mnt` and `/media` directories
- `/etc/fstab`
- `lsblk` (lists the type and UUID of a filesystem)
- How to mount a filesystem using its UUID or label.
- How to mount a filesystem using systemd mount units.
- How to automount a filesystem using systemd mount units.

Answers to Guided Exercises

1. Using `mount`, how can you mount an `ext4` filesystem on `/dev/sdc1` to `/mnt/external` as read-only, using the `noatime` and `async` options?

```
# mount -t ext4 -o noatime,async,ro /dev/sdc1 /mnt/external
```

2. When unmounting a filesystem at `/dev/sdd2`, you get the `target is busy` error message. How can you find out which files on the filesystem are open, and what processes opened them?

Use `lsdf` followed by the device name:

```
$ lsdf /dev/sdd2
```

3. Consider the following entry in `/etc/fstab`: `/dev/sdb1 /data ext4 noatime,noauto,async`. Will this filesystem be mounted if the command `mount -a` is issued? Why?

It will not be mounted. The key is the `noauto` parameter, which means this entry be ignored by `mount -a`.

4. How can you find out the UUID of a filesystem under `/dev/sdb1`?

Use `lsblk -f`, followed by the filesystem name:

```
$ lsblk -f /dev/sdb1
```

5. How can you use `mount` to remount as read-only an exFAT filesystem with the UUID `6e2c12e3-472d-4bac-a257-c49ac07f3761`, mounted at `/mnt/data`?

Since the filesystem is mounted, you do not need to worry about the filesystem type or the ID, just use the `remount` option with the `ro` (read-only) parameter and the mount point:

```
# mount -o remount,ro /mnt/data
```

6. How can you get a list of all `ext3` and `ntfs` filesystems currently mounted on a system?

Use `mount -t`, followed by a comma-separated list of filesystems:

```
# mount -t ext3,ntfs
```


Answers to Explorational Exercises

1. Consider the following entry in `/etc/fstab`: `/dev/sdc1 /backup ext4 noatime,nouser,async`. Can a user mount this filesystem with the command `mount /backup`? Why?

No, the parameter `nouser` will not allow ordinary users to mount this filesystem.

2. Consider a remote filesystem mounted at `/mnt/server`, which has become unreachable due to a loss of network connectivity. How could you force it to be unmounted, or mounted as read-only if this is not possible?

Pass the `-f` and `-r` parameters to `umount`. The command would be `umount -f -r /mnt/server`. Remember that you can group parameters, so `umount -fr /mnt/server` would also work.

3. Write an `/etc/fstab` entry that would mount an `btrfs` volume with the label `Backup` on `/mnt/backup`, with default options and without allowing the execution of binaries from it.

The line should be `LABEL=Backup /mnt/backup btrfs defaults,noexec`

4. Consider the following `systemd` mount unit:

```
[Unit]
Description=External data disk

[Mount]
What=/dev/disk/by-uuid/56C11DCC5D2E1334
Where=/mnt/external
Type=ntfs
Options=defaults

[Install]
WantedBy=multi-user.target
```

- What would be an equivalent `/etc/fstab` entry for this filesystem?

The entry would be: `UUID=56C11DCC5D2E1334 /mnt/external ntfs defaults`

5. What should be the file name for the unit above, so it can be used by `systemd`? Where should it be placed?

The filename must be the same as the mount point, so `mnt-external.mount`, placed in `/etc/systemd/system`.



104.5 Manage file permissions and ownership

Reference to LPI objectives

LPIC-1 version 5.0, Exam 101, Objective 104.5

Weight

3

Key knowledge areas

- Manage access permissions on regular and special files as well as directories.
- Use access modes such as suid, sgid and the sticky bit to maintain security.
- Know how to change the file creation mask.
- Use the group field to grant file access to group members.

Partial list of the used files, terms and utilities

- `chmod`
- `umask`
- `chown`
- `chgrp`



104.5 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	104 Devices, Linux Filesystems, Filesystem Hierarchy Standard
Objective:	104.5 Manage file permissions and ownership
Lesson:	1 of 1

Introduction

Being a multi-user system, Linux needs some way to track who owns each file and whether or not a user is allowed to perform actions on a file. This is to ensure the privacy of users who might want to keep the contents of their files confidential, as well as to ensure collaboration by making certain files accessible to multiple users.

This is done through a three-level permissions system. Every file on disk is owned by a user and a user group and has three sets of permissions: one for its owner, one for the group who owns the file and one for everyone else. In this lesson, you will learn how to query the permissions for a file, the meaning of these permissions and how to manipulate them.

Querying Information about Files and Directories

The command `ls` is used to get a listing of the contents of any directory. In this basic form, all you get are the filenames:

```
$ ls
```

```
Another_Directory picture.jpg text.txt
```

But there is much more information available for each file, including its type, size, ownership and more. To see this information you must ask `ls` for a “long form” listing, using the `-l` parameter:

```
$ ls -l
total 536
drwxrwxr-x 2 carol carol 4096 Dec 10 15:57 Another_Directory
-rw----- 1 carol carol 539663 Dec 10 10:43 picture.jpg
-rw-rw-r-- 1 carol carol 1881 Dec 10 15:57 text.txt
```

Each column on the output above has a meaning. Let us have a look at the columns relevant for this lesson.

- The *first* column on the listing shows the file type and permissions. For example, on `drwxrwxr-x`:
 - The first character, `d`, indicates the file type.
 - The next three characters, `rw``x`, indicate the permissions for the owner of the file, also referred to as *user* or `u`.
 - The next three characters, `rw``x`, indicate the permissions of the *group* owning the file, also referred to as `g`.
 - The last three characters, `r``-``x`, indicate the permissions for anyone else, also known as *others* or `o`.

TIP

It is also common to hear the *others* permission set referred to as *world* permissions, as in “Everyone else in the world has these permissions”.

- The *third* and *fourth* columns show ownership information: respectively the user and group that own the file.
- The *seventh* and last column shows the file name.

The *second* column indicates the number of hard links pointing to that file. The *fifth* column shows the filesize. The *sixth* column shows the date and time the file was last modified. But these columns are not relevant for the current topic.

What about Directories?

If you try to query information about a directory using `ls -l`, it will show you a listing of the directory’s contents instead:

```
$ ls -l Another_Directory/
total 0
-rw-r--r-- 1 carol carol 0 Dec 10 17:59 another_file.txt
```

To avoid this and query information about the directory itself, add the `-d` parameter to `ls`:

```
$ ls -l -d Another_Directory/
drwxrwxr-x 2 carol carol 4096 Dec 10 17:59 Another_Directory/
```

Viewing Hidden Files

The directory listing we have retrieved using `ls -l` before is incomplete:

```
$ ls -l
total 544
drwxrwxr-x 2 carol carol 4096 Dec 10 17:59 Another_Directory
-rw----- 1 carol carol 539663 Dec 10 10:43 picture.jpg
-rw-rw-r-- 1 carol carol 1881 Dec 10 15:57 text.txt
```

There are three other files in that directory, but they are hidden. On Linux, files whose name starts with a period (.) are automatically hidden. To see them we need to add the `-a` parameter to `ls`:

```
$ ls -l -a
total 544
drwxrwxr-x 3 carol carol 4096 Dec 10 16:01 .
drwxrwxr-x 4 carol carol 4096 Dec 10 15:56 ..
drwxrwxr-x 2 carol carol 4096 Dec 10 17:59 Another_Directory
-rw----- 1 carol carol 539663 Dec 10 10:43 picture.jpg
-rw-rw-r-- 1 carol carol 1881 Dec 10 15:57 text.txt
-rw-r--r-- 1 carol carol 0 Dec 10 16:01 .thisIsHidden
```

The file `.thisIsHidden` is simply hidden because its name starts with `.`

The directories `.` and `..` however are special. `.` is a pointer to the current directory. And `..` is a pointer to the parent directory, the one which contains the current one. In Linux, every directory contains at least these two directories.

TIP

You can combine multiple parameters for `ls` (and many other Linux commands). `ls`

`-l -a` can, for example, be written as `ls -la`.

Understanding Filetypes

We have already mentioned that the first letter in each output of `ls -l` describes the type of the file. The three most common file types are:

- (normal file)

A file can contain data of any kind and help to manage this data. Files can be modified, moved, copied and deleted.

d (directory)

A directory contains other files or directories and helps to organize the file system. Technically, directories are a special kind of file.

l (symbolic link)

This “file” is a pointer to another file or directory elsewhere in the filesystem.

In addition to these, there are three other file types that you should at least know about, but are out of scope for this lesson:

b (block device)

This file stands for a virtual or physical device, usually disks or other kinds of storage devices, such as the first hard disk which might be represented by `/dev/sda`.

c (character device)

This file stands for a virtual or physical device. Terminals (like the main terminal on `/dev/ttyS0`) and serial ports are common examples of character devices.

s (socket)

Sockets serve as “conduits” passing information between two programs.

WARNING

Do not alter any of the permissions on block devices, character devices or sockets, unless you know what you are doing. This may prevent your system from working!

Understanding Permissions

In the output of `ls -l` the file permissions are shown right after the filetype, as three groups of three characters each, in the order `r`, `w` and `x`. Here is what they mean. Keep in mind that a dash `-` represents the lack of a permission.

Permissions on Files

r

Stands for *read* and has an octal value of 4 (do not worry, we will discuss octals shortly). This means permission to open a file and read its contents.

w

Stands for *write* and has an octal value of 2. This means permission to edit or delete a file.

x

Stands for *execute* and has an octal value of 1. This means that the file can be run as an executable or script.

So, for example, a file with permissions `rw-` can be read and written to, but cannot be executed.

Permissions on Directories

r

Stands for *read* and has an octal value of 4. This means permission to read the directory's contents, like filenames. But it *does not* imply permission to read the files themselves.

w

Stands for *write* and has an octal value of 2. This means permission to create or delete files in a directory.

Keep in mind that you cannot make these changes with *write* permissions alone, but also need *execute* permission (x) to change to the directory.

x

Stands for *execute* and has an octal value of 1. This means permission to enter a directory, but not to list its files (for that `r` is needed).

The last bit about directories may sound a bit confusing. Let us imagine, for example, that you have a directory named `Another_Directory`, with the following permissions:

```
$ ls -ld Another_Directory/
d--x--x--x 2 carol carol 4,0K Dec 20 18:46 Another_Directory
```

Also imagine that inside this directory you have a shell script called `hello.sh`:


```
-rwxr-xr-x 1 carol carol 33 Dec 20 18:46 hello.sh
```

If you are the user `carol` and try to list the contents of `Another_Directory`, you will get an error message, as your user lacks read permission for that directory:

```
$ ls -l Another_Directory/  
ls: cannot open directory 'Another_Directory/': Permission denied
```

However, the user `carol` *does have* execute permissions, which means that she can enter the directory. Therefore, the user `carol` can access files inside the directory, as long as she has the correct permissions *for the respective file*. Let us assume the user has full permissions (`rwx`) for the script `hello.sh`. Then she *can* run the script, even though she *cannot* read the contents of the directory containing it if she knows the complete filename:

```
$ sh Another_Directory/hello.sh  
Hello LPI World!
```

As we said before, permissions are specified in sequence: first for the owner of the file, then for the owning group, and then for other users. Whenever someone tries to perform an action on the file, the permissions are checked in the same fashion.

First the system checks if the current user owns the file, and if this is true it applies the first set of permissions only. Otherwise, it checks if the current user belongs to the group owning the file. In that case, it applies the second set of permissions only. In any other case, the system will apply the third set of permissions.

This means that if the current user is the owner of the file, only the owner permissions are effective, even if the group or other permissions are more permissive than the owner's permissions.

Modifying File Permissions

The command `chmod` is used to modify the permissions for a file, and takes at least two parameters: the first one describes which permissions to change, and the second one points to the file or directory where the change will be made. Keep in mind that only the owner of the file, or the system administrator (`root`) can change the permissions on a file.

The permissions to change can be described in two different ways, or “modes”.

The first one, called *symbolic mode* offers fine grained control, allowing you to add or revoke a single permission without modifying others on the set. The other mode, called *octal mode*, is easier to remember and quicker to use if you wish to set all permission values at once.

Both modes will lead to the same end result. So, for example, the commands:

```
$ chmod ug+rw-x,o-rwx text.txt
```

and

```
$ chmod 660 text.txt
```

will produce exactly the same output, a file with the permissions set:

```
-rw-rw---- 1 carol carol 765 Dec 20 21:25 text.txt
```

Now, let us see how each mode works.

Symbolic Mode

When describing which permissions to change in *symbolic mode* the first character(s) indicate(s) whose permissions you will alter: the ones for the user (u), for the group (g), for others (o) and/or for everyone (a).

Then you need to tell the command what to do: you can grant a permission (+), revoke a permission (-), or set it to a specific value (=).

Lastly, you specify which permission you wish to act on: read (r), write (w), or execute (x).

For example, imagine we have a file called `text.txt` with the following permission set:

```
$ ls -l text.txt
-rw-r--r-- 1 carol carol 765 Dec 20 21:25 text.txt
```

If you wish to grant write permissions to members of the group owning the file, you would use the `g+w` parameter. It is easier if you think about it this way: “For the group (g), grant (+) write permissions (w)”. So, the command would be:

```
$ chmod g+w text.txt
```

Let us check the result with `ls`:

```
$ ls -l text.txt
-rw-rw-r-- 1 carol carol 765 Dec 20 21:25 text.txt
```

Do you wish to remove read permissions for the owner of the same file? Think about it as: “For the user (u), revoke (-) read permissions (r)”. So the parameter is `u-r`, like so:

```
$ chmod u-r text.txt
$ ls -l text.txt
--w-rw-r-- 1 carol carol 765 Dec 20 21:25 text.txt
```

What if we want to set the permissions exactly as `rw-` for everyone? Then think of it as: “For all (a), set exactly (=) read (r), write (w), and no execute (-)”. So:

```
$ chmod a=rw- text.txt
$ ls -l text.txt
-rw-rw-rw- 1 carol carol 765 Dec 20 21:25 text.txt
```

Of course, it is possible to modify multiple permissions at the same time. In this case, separate them with a comma (,):

```
$ chmod u+rw,g-x text.txt
$ ls -lh text.txt
-rwxrw-rw- 1 carol carol 765 Dec 20 21:25 text.txt
```

The example above can be read as: “For the user (u), grant (+) read, write and execute (rwx) permissions, for the group (g), revoke (-) execute permissions (x)”.

When run on a directory, `chmod` modifies only the directory’s permissions. `chmod` also has a recursive mode, which is useful for when you want to change the permissions for “all files inside a directory and its subdirectories”. To use this, add the parameter `-R` after the command name, before the permissions to change:

```
$ chmod -R u+rwx Another_Directory/
```

This command can be read as: “Recursively (-R), for the user (u), grant (+) read, write and execute (rwx) permissions”.

WARNING

Be careful and think twice before using the -R switch, as it is easy to change permissions on files and directories which you do not want to change, especially on directories with a large number of files and subdirectories.

Octal Mode

In *octal mode*, the permissions are specified in a different way: as a three-digit value on octal notation, a base-8 numeric system.

Each permission has a corresponding value, and they are specified in the following order: first comes read (r), which is 4, then write (w), which is 2 and last is execute (x), represented by 1. If there is no permission, use the value zero (0). So, a permission of rwx would be 7 (4+2+1) and r-x would be 5 (4+0+1).

The first of the three digits on the permission set represents the permissions for the user (u), the second for the group (g) and the third for others (o). If we wanted to set the permissions for a file to rw-rw- ---, the octal value would be 660:

```
$ chmod 660 text.txt
$ ls -l text.txt
-rw-rw---- 1 carol carol 765 Dec 20 21:25 text.txt
```

Besides this, the syntax in *octal mode* is the same as in *symbolic mode*, the first parameter represents the permissions you wish to change, and the second parameter points to the file or directory where the change will be made.

TIP | If a permission value is *odd*, the file surely is executable!

Which syntax should you use? The *octal mode* is recommended if you want to change the permissions to a specific value, for example 640 (rw- r-- ---).

The *symbolic mode* is more useful if you want to flip just a specific value, regardless of the current permissions for the file. For example, you can add execute permissions for the user using just `chmod u+x script.sh` without regard to, or even touching, the current permissions for the group and others.

Modifying File Ownership

The command `chown` is used to modify the ownership of a file or directory. The syntax is quite simple:

```
chown USERNAME:GROUPNAME FILENAME
```

For example, let us check a file called `text.txt`:

```
$ ls -l text.txt
-rw-rw---- 1 carol carol 1881 Dec 10 15:57 text.txt
```

The user who owns the file is `carol`, and the group is also `carol`. Now, we will change the group owning the file to some other group, like `students`:

```
$ chown carol:students text.txt
$ ls -l text.txt
-rw-rw---- 1 carol students 1881 Dec 10 15:57 text.txt
```

Keep in mind that the user who owns a file does not need to belong to the group who owns a file. In the example above, the user `carol` does not need to be a member of the `students` group.

The user or group permission set can be omitted if you do not wish to change them. So, to change just the group owning a file you would use `chown :students text.txt`. To change just the user, the command would be `chown carol: text.txt` or just `chown carol text.txt`. Alternatively, you could use the command `chgrp students text.txt`.

Unless you are the system administrator (root), you cannot change ownership of a file to another user or group you do not belong to. If you try to do this, you will get the error message `Operation not permitted`.

Querying Groups

Before changing the ownership of a file, it might be useful to know which groups exist on the system, which users are members of a group and to which groups a user belongs.

To see which groups exist on your system, type `getent group`. The output will be similar to this one (the output has been abbreviated):

```
$ getent group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,rigues
tty:x:5:rigues
disk:x:6:
lp:x:7:
mail:x:8:
news:x:9:
uucp:x:10:rigues
```

If you want to know to which groups a user belongs, add the username as a parameter to `groups`:

```
$ groups carol
carol : carol students cdrom sudo dip plugdev lpadmin sambashare
```

To do the reverse (see which users belong to a group) use `groupmems`. The parameter `-g` specifies the group, and `-l` will list all of its members:

```
# groupmems -g cdrom -l
carol
```

TIP

`groupmems` can only be run as root, the system administrator. If you are not currently logged in as root, add `sudo` before the command.

Default Permissions

Let us try an experiment. Open a terminal window and create an empty file with the following command:

```
$ touch testfile
```

Now, let us take a look at the permissions for this file. They *may* be different on your system, but let us assume they look like the following:

```
$ ls -lh testfile
```

```
-rw-r--r-- 1 carol carol 0 jul 13 21:55 testfile
```

The permissions are `rw-r--r--`: *read* and *write* for the user, and *read* for the group and others, or `644` in octal mode. Now, try creating a directory:

```
$ mkdir testdir
$ ls -lhd testdir
drwxr-xr-x 2 carol carol 4,0K jul 13 22:01 testdir
```

Now the permissions are `rw-r-xr-x`: *read*, *write* and *execute* for the user, *read* and *execute* for the group and others, or `755` in octal mode.

No matter where you are in the filesystem every file or directory you create will get those same permissions. Did you ever wonder where they come from?

They come from the *user mask* or `umask`, which sets the default permissions for every file created. You can check the current values with the `umask` command:

```
$ umask
0022
```

But that does not look like `rw-r--r--`, or even `644`. Maybe we should try with the `-S` parameter, to get an output in symbolic mode:

```
$ umask -S
u=rwx,g=rx,o=rx
```

Those are the same permissions our test directory got in one of the examples above. But, why is it that when we created a file the permissions were different?

Well, it makes no sense to set global execute permissions for everyone on any file by default, right? Directories need execute permissions (otherwise you cannot enter them), but files do not, so they do not get them. Hence the `rw-r--r--`.

Besides displaying the default permissions, `umask` can also be used to change them for your current shell session. For example, if we use the command:

```
$ umask u=rwx,g=rwx,o=
```

Every new directory will inherit the permissions `rw-rwx---`, and every file `rw-rw----` (as they do not get execute permissions). If you repeat the examples above to create a `testfile` and `testdir` and check the permissions, you should get:

```
$ ls -lhd test*
drwxrwx--- 2 carol carol 4,0K jul 13 22:25 testdir
-rw-rw---- 1 carol carol    0 jul 13 22:25 testfile
```

And if you check the `umask` without the `-S` (symbolic mode) parameter, you get:

```
$ umask
0007
```

The result does not look familiar because the values used are different. Here is a table with every value and its respective meaning:

Value	Permission for Files	Permission for Directories
0	<code>rw-</code>	<code>rwX</code>
1	<code>rw-</code>	<code>rw-</code>
2	<code>r--</code>	<code>r-x</code>
3	<code>r--</code>	<code>r--</code>
4	<code>-w-</code>	<code>-wX</code>
5	<code>-w-</code>	<code>-w-</code>
6	<code>---</code>	<code>--X</code>
7	<code>---</code>	<code>---</code>

As you can see, `007` corresponds to `rw-rwx---`, exactly as we requested. The leading zero can be ignored.

Special Permissions

Besides the read, write and execute permissions for user, group and others, each file can have three other *special permissions* which can alter the way a directory works or how a program runs. They can be specified either in symbolic or octal mode, and are as follows:

Sticky Bit

The sticky bit, also called the *restricted deletion flag*, has the octal value 1 and in symbolic mode is represented by a `t` within the other's permissions. This applies only to directories, and has no effect on normal files. On Linux it prevents users from removing or renaming a file in a directory unless they own that file or directory.

Directories with the sticky bit set show a `t` replacing the `x` on the permissions for *others* on the output of `ls -l`:

```
$ ls -ld Sample_Directory/
drwxr-xr-t 2 carol carol 4096 Dec 20 18:46 Sample_Directory/
```

In octal mode, the special permissions are specified using a 4-digit notation, with the first digit representing the special permission to act upon. For example, to set the sticky bit (value 1) for the directory `Another_Directory` in octal mode, with permissions 755, the command would be:

```
$ chmod 1755 Another_Directory
$ ls -ld Another_Directory
drwxr-xr-t 2 carol carol 4,0K Dec 20 18:46 Another_Directory
```

Set GID

Set GID, also known as SGID or Set Group ID bit, has the octal value 2 and in symbolic mode is represented by an `s` on the *group* permissions. This can be applied to executable files or directories. On files, it will make the process run with the privileges of the group who owns the file. When applied to directories, it will make every file or directory created under it inherit the group from the parent directory.

Files and directories with SGID bit show an `s` replacing the `x` on the permissions for the *group* on the output of `ls -l`:

```
$ ls -l test.sh
-rwxr-sr-x 1 carol root 33 Dec 11 10:36 test.sh
```

To add SGID permissions to a file in symbolic mode, the command would be:

```
$ chmod g+s test.sh
$ ls -l test.sh
```

```
-rwxr-sr-x 1 carol root    33 Dec 11 10:36 test.sh
```

The following example will help you better understand the effects of SGID on a directory. Suppose we have a directory called `Sample_Directory`, owned by the user `carol` and the group `users`, with the following permission structure:

```
$ ls -ldh Sample_Directory/
drwxr-xr-x 2 carol users 4,0K Jan 18 17:06 Sample_Directory/
```

Now, let us change to this directory and, using the command `touch`, create an empty file inside of it. The result would be:

```
$ cd Sample_Directory/
$ touch newfile
$ ls -lh newfile
-rw-r--r-- 1 carol carol 0 Jan 18 17:11 newfile
```

As we can see, the file is owned by the user `carol` and group `carol`. But, if the directory had the SGID permission set, the result would be different. First, let us add the SGID bit to the `Sample_Directory` and check the results:

```
$ sudo chmod g+s Sample_Directory/
$ ls -ldh Sample_Directory/
drwxr-sr-x 2 carol users 4,0K Jan 18 17:17 Sample_Directory/
```

The `s` on the group permissions indicates that the SGID bit is set. Now, we will change to this directory and, again, create an empty file with the `touch` command:

```
$ cd Sample_Directory/
$ touch emptyfile
$ ls -lh emptyfile
-rw-r--r-- 1 carol users 0 Jan 18 17:20 emptyfile
```

The group who owns the file is `users`. This is because the SGID bit made the file inherit the group owner of its parent directory, which is `users`.

Set UID

SUID, also known as Set User ID, has the octal value 4 and is represented by an `s` on the *user* permissions in symbolic mode. It only applies to files and has no effect on directories. Its behavior is similar to the SGID bit, but the process will run with the privileges of the *user* who owns the file. Files with the SUID bit show a `s` replacing the `x` on the permissions for the user on the output of `ls -l`:

```
$ ls -ld test.sh
-rwsr-xr-x 1 carol carol 33 Dec 11 10:36 test.sh
```

You can combine multiple special permissions on one parameter. So, to set SGID (value 2) and SUID (value 4) in octal mode for the script `test.sh` with permissions 755, you would type:

```
$ chmod 6755 test.sh
```

And the result would be:

```
$ ls -lh test.sh
-rwsr-sr-x 1 carol carol 66 Jan 18 17:29 test.sh
```

TIP

If your terminal supports color, and these days most of them do, you can quickly see if these special permissions are set by glancing at the output of `ls -l`. For the sticky bit, the directory name might be shown in a black font with blue background. The same applies for files with the SGID (yellow background) and SUID (red background) bits. Colors may be different depending on which Linux distribution and terminal settings you use.

Guided Exercises

1. Create a directory named `emptydir` using the command `mkdir emptydir`. Now, using `ls`, list the permissions for the directory `emptydir`.

2. Create an empty file named `emptyfile` with the command `touch emptyfile`. Now, using `chmod` in symbolic mode, add execute permissions for the owner of the file `emptyfile`, and remove write and execute permissions for everyone else. Do this using only one `chmod` command.

3. What would the default permissions for a file be if the `umask` value is set to `027`?

4. Let us assume a file named `test.sh` is a shell script with the following permissions and ownership:

```
-rwxr-sr-x 1 carol root    33 Dec 11 10:36 test.sh
```

- What are the permissions for the owner of the file?
-
- Using the octal notation, what would the syntax of `chmod` be to “unset” the special permission granted to this file?

5. Consider this file:

```
$ ls -l /dev/sdb1
brw-rw---- 1 root disk 8, 17 Dec 21 18:51 /dev/sdb1
```

Which type of file is `sdb1`? Who can write to it?

6. Consider the following 4 files:

```
drwxr-xr-t 2 carol carol 4,0K Dec 20 18:46 Another_Directory
```

```
----r--r-- 1 carol carol    0 Dec 11 10:55 foo.bar
-rw-rw-r-- 1 carol carol 1,2G Dec 20 18:22 HugeFile.zip
drwxr-sr-x 2 carol users   4,0K Jan 18 17:26 Sample_Directory
```

Write down the corresponding permissions for each file and directory using octal mode using the 4-digit notation.

Another_Directory	
foo.bar	
HugeFile.zip	
Sample_Directory	

Explorational Exercises

1. Try this on a terminal: create an empty file called `emptyfile` with the command `touch emptyfile`. Now “zero out” the permissions for the file with `chmod 000 emptyfile`. What will happen if you change the permissions for `emptyfile` by passing only *one* value for `chmod` in octal mode, like `chmod 4 emptyfile`? And if you use two, as in `chmod 44 emptyfile`? What can we learn about the way `chmod` reads the numerical value?

2. Consider the permissions for the temporary directory on a Linux system, `/tmp`:

```
$ ls -l /tmp
drwxrwxrwt 19 root root 16K Dec 21 18:58 tmp
```

User, group and others have full permissions. But can a regular user delete *any* files inside this directory? Why is this the case?

3. A file called `test.sh` has the following permissions: `-rwsr-xr-x`, meaning the SUID bit is set. Now, run the following commands:

```
$ chmod u-x test.sh
$ ls -l test.sh
-rwsr-xr-x 1 carol carol 33 Dec 11 10:36 test.sh
```

What did we do? What does the uppercase `S` mean?

4. How would you create a directory named `Box` where all the files are automatically owned by the group `users`, and can only be deleted by the user who created them?

Summary

In this lesson you have learned how to use `ls` to get (and decode) information about file permissions, how to control or change who can create, delete or modify a file with `chmod`, both in *octal* and *symbolic* modes, how to change the ownership of files with `chown` and `chgrp` and how to query and change the default permissions mask for files and directories with `umask`

The following commands were discussed in this lesson:

`ls`

List files, optionally including details such as permissions.

`chmod`

Change the permissions of a file or directory.

`chown`

Change the owning user and/or group of a file or directory.

`chgrp`

Change the owning group of a file or directory.

`umask`

Query, or set, the default permissions mask for files and directories

Answers to Guided Exercises

1. Create a directory named `emptydir` using the command `mkdir emptydir`. Now, using `ls`, list the permissions for the directory `emptydir`.

Add the `-d` parameter to `ls` to see the file attributes of a directory, instead of listing its contents. So, the answer is:

```
ls -l -d emptydir
```

Bonus points if you merged the two parameters in one, as in `ls -ld emptydir`.

2. Create an empty file named `emptyfile` with the command `touch emptyfile`. Now, using `chmod` on symbolic mode, add execute permissions for the owner of the file `emptyfile`, and remove write and execute permissions for everyone else. Do this using only one `chmod` command.

Think about it this way:

- “For the user who owns the file (u) add (+) execute (x) permissions”, so `u+x`.
- “For the group (g) and other users (o), remove (-) write (w) and execute (x) permissions”, so `go-wx`.

To combine these two sets of permissions, we add a comma between them. So the final result is:

```
chmod u+x,go-wx emptyfile
```

3. What would the default permissions be for a file if the `umask` value is set to `027`?

The permissions would be `rw-r-----`

4. Let us assume a file named `test.sh` is a shell script with the following permissions and ownership:

```
-rwxr-sr-x 1 carol root    33 Dec 11 10:36 test.sh
```

- What are the permissions for the owner of the file?

The permissions for the owner (2nd to 4th characters in the output of `ls -l`) are `rwX`, so the

answer is: “to read, to write to and to execute the file”.

- Using the octal notation, which should be the syntax of `chmod` to “unset” the special permission granted to this file?

We can “unset” the special permissions by passing a 4th digit, `0`, to `chmod`. The current permissions are `755`, so the command should be `chmod 0755`.

5. Consider this file:

```
$ ls -l /dev/sdb1
brw-rw---- 1 root disk 8, 17 Dec 21 18:51 /dev/sdb1
```

Which type of file is `sdb1`? Who can write to it?

The first character on the output of `ls -l` shows the type of file. `b` is a *block device*, usually a disk (internal or external), connected to the machine. The owner (`root`) and any users of the group `disk` can write to it.

6. Consider the following 4 files:

```
drwxr-xr-t 2 carol carol 4,0K Dec 20 18:46 Another_Directory
----r--r-- 1 carol carol    0 Dec 11 10:55 foo.bar
-rw-rw-r-- 1 carol carol 1,2G Dec 20 18:22 HugeFile.zip
drwxr-sr-x 2 carol users 4,0K Jan 18 17:26 Sample_Directory
```

Write down the corresponding permissions for each file and directory using octal mode using the 4-digit notation.

The corresponding permissions, in octal mode, are as follows:

Another_Directory	1755. 1 for the sticky bit, 755 for the regular permissions (<code>rw</code> for the user, <code>r-x</code> for group and others).
foo.bar	0044. No special permissions (so the first digit is <code>0</code>), no permissions for the user (<code>---</code>) and just read (<code>r--</code>) for group and others.
HugeFile.zip	0664. No special permissions, so the first digit is <code>0</code> . 6 (<code>rw-</code>) for the user and group, 4 (<code>r--</code>) for the others.

Sample_Directory	2755. 2 for the SGID bit, 7 (rwx) for the user, 5 (r-x) for the group and others.
------------------	---

Answers to Explorational Exercises

1. Try this on a terminal: create an empty file called `emptyfile` with the command `touch emptyfile`. Now “zero out” the permissions for the file with `chmod 000 emptyfile`. What will happen if you change the permissions for `emptyfile` by passing only *one* value for `chmod` in octal mode, like `chmod 4 emptyfile`? And if you use two, as in `chmod 44 emptyfile`? What can we learn about the way `chmod` reads the numerical value?

Remember that we “zeroed out” the permissions for `emptyfile`. So, its initial state would be:

```
----- 1 carol carol  0 Dec 11 10:55 emptyfile
```

Now, let us try the first command, `chmod 4 emptyfile`:

```
$ chmod 4 emptyfile
$ ls -l emptyfile
-----r-- 1 carol carol 0 Dec 11 10:55 emptyfile
```

See? The permissions for *others* were changed. And what if we try two digits, like in `chmod 44 emptyfile`?

```
$ chmod 44 emptyfile
$ ls -l emptyfile
----r--r-- 1 carol carol 0 Dec 11 10:55 emptyfile
```

Now, the permissions for *group* and *others* were affected. From this, we can conclude that in octal mode `chmod` reads the value “backwards”, from the least significant digit (*others*) to the most significant one (*user*). If you pass one digit, you modify the permissions for *others*. With two digits you modify *group* and *others*, and with three you modify *user*, *group* and *others* and with four digits you modify *user*, *group*, *others* and the special permissions.

2. Consider the permissions for the temporary directory on a Linux system, `/tmp`:

```
$ ls -l /tmp
drwxrwxrwt 19 root root 16K Dec 21 18:58 tmp
```

User, group and others have full permissions. But can a regular user delete *any* files inside this directory? Why is this the case?

`/tmp` is what we call a *world writeable* directory, meaning that any user can write to it. But we don't want one user messing with files created by others, so the *sticky bit* is set (as indicated by the `t` on the permissions for *others*). This means that a user can delete files on `/tmp`, but only those created by himself.

3. A file called `test.sh` has the following permissions: `-rwsr-xr-x`, meaning the SUID bit is set. Now, run the following commands:

```
$ chmod u-x test.sh
$ ls -l test.sh
-rwsr-xr-x 1 carol carol 33 Dec 11 10:36 test.sh
```

What did we do? What does the uppercase `S` mean?

We removed execute permissions for the user who owns the file. The `s` (or `t`) takes the place of the `x` on the output of `ls -l`, so the system needs a way to show if the user has execute permissions or not. It does this by changing the case of the special character.

A lowercase `s` on the first group of permissions means that the user who owns the file has execute permissions and that the SUID bit is set. An uppercase `S` means that the user who owns the file lacks (-) execute permissions and that the SUID bit is set.

The same can be said for SGID, a lowercase `s` on the second group of permissions means that the group who owns the file has execute permissions and that the SGID bit is set. An uppercase `S` means that the group who owns the file lacks (-) execute permissions and that the SGID bit is set.

This is also true for the sticky bit, represented by the `t` on the third group of permissions. Lowercase `t` means sticky bit set and that others have execute permissions. Uppercase `T` means sticky bit set and that others do not have execute permissions.

4. How would you create a directory named `Box` where all the files are automatically owned by the group `users`, and can only be deleted by the user who created them?

This is a multi-step process. The first step is to create the directory:

```
$ mkdir Box
```

We want every file created inside this directory to be automatically assigned to the group `users`. We can do this by setting this group as the owner of the directory, and then by setting the SGID bit on it. We also need to make sure that any member of the group can write to that

directory.

Since we do not care about what the other permissions are, and want to “flip” only the special bits, it makes sense to use the symbolic mode:

```
$ chown :users Box/  
$ chmod g+wx Box/
```

Note that if your current user does not belong to the group `users`, you will have to use the command `sudo` before the commands above to do the change as root.

Now for the last part, making sure that only the user who created a file is allowed to delete it. This is done by setting the sticky bit (represented by a `t`) on the directory. Remember that it is set on the permissions for others (`o`).

```
$ chmod o+t Box/
```

The permissions on the directory `Box` should be as follows:

```
drwxrwsr-t 2 carol users 4,0K Jan 18 19:09 Box
```

Of course, you can specify SGID and the sticky bit using only one `chmod` command:

```
$ chmod g+wx,o+t Box/
```

Bonus points if you thought of that.



104.6 Create and change hard and symbolic links

Reference to LPI objectives

LPIC-1 version 5.0, Exam 101, Objective 104.6

Weight

2

Key knowledge areas

- Create links.
- Identify hard and/or soft links.
- Copying versus linking files.
- Use links to support system administration tasks.

Partial list of the used files, terms and utilities

- `ln`
- `ls`



104.6 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	104 Devices, Linux Filesystems, Filesystem Hierarchy Standard
Objective:	104.6 Create and change hard and symbolic links
Lesson:	1 of 1

Introduction

On Linux some files get a special treatment either because of the place they are stored in, such as temporary files, or the way they interact with the filesystem, like links. In this lesson you will learn what links are and how to manage them.

Understanding Links

As already mentioned, on Linux everything is treated as a file. But there is a special kind of file, called a *link*, and there are two types of links on a Linux system:

Symbolic links

Also called *soft links*, they point to the path of another file. If you delete the file the link points to (called *target*) the link will still exist, but it “stops working”, as it now points to “nothing”.

Hard links

Think of a hard link as a second name for the original file. They are *not* duplicates, but instead

are an additional entry in the filesystem pointing to the same place (inode) on the disk.

TIP

An *inode* is a data structure that stores attributes for an object (like a file or directory) on a filesystem. Among those attributes are permissions, ownership and on which blocks of the disk the data for the object is stored. Think of it as an entry on an index, hence the name, which comes from “index node”.

Working with Hard Links

Creating Hard Links

The command to create a hard link on Linux is `ln`. The basic syntax is:

```
$ ln TARGET LINK_NAME
```

The `TARGET` must exist already (this is the file the link will point to), and if the target is not on the current directory, or if you want to create the link elsewhere, you *must* specify the full path to it. For example, the command:

```
$ ln target.txt /home/carol/Documents/hardlink
```

will create a file named `hardlink` on the directory `/home/carol/Documents/`, linked to the file `target.txt` on the current directory.

If you leave out the last parameter (`LINK_NAME`), a link with the same name as the target will be created in the current directory.

Managing Hard Links

Hard links are entries in the filesystem which have different names but point to the same data on disk. All such names are equal and can be used to refer to a file. If you change the contents of one of the names, the contents of all other names pointing to that file change since all these names point to the very same data. If you delete one of the names, the other names will still work.

This happens because when you “delete” a file the data is not actually erased from the disk. The system simply deletes the entry on the filesystem table pointing to the inode corresponding to the data on the disk. But if you have a second entry pointing to the same inode, you can still get to the data. Think of it as two roads converging on the same point. Even if you block or redirect one of the roads, you can still reach the destination using the other.

You can check this by using the `-i` parameter of `ls`. Consider the following contents of a directory:

```
$ ls -li
total 224
3806696 -r--r--r-- 2 carol carol 111702 Jun  7 10:13 hardlink
3806696 -r--r--r-- 2 carol carol 111702 Jun  7 10:13 target.txt
```

The number before the permissions is the inode number. See that both the file `hardlink` and the file `target.txt` have the same number (3806696)? This is because one is a hard link of the other.

But which one is the original and which one is the link? You cannot really tell, as for all practical purposes they are the same.

Note that every hard link pointing to a file increases the *link count* of the file. This is the number right after the permissions on the output of `ls -l`. By default, every file has a link count of 1 (directories have a count of 2), and every hard link to it increases the count by one. So, that is the reason for the link count of 2 on the files in the listing above.

In contrast to symbolic links, you can only create hard links to files, and both the link and target must reside in the same filesystem.

Moving and Removing Hard Links

Since hard links are treated as regular files, they can be deleted with `rm` and renamed or moved around the filesystem with `mv`. And since a hard link points to the same inode of the target, it can be moved around freely, without fear of “breaking” the link.

Symbolic Links

Creating Symbolic Links

The command used to create a symbolic link is also `ln`, but with the `-s` parameter added. Like so:

```
$ ln -s target.txt /home/carol/Documents/softlink
```

This will create a file named `softlink` in the directory `/home/carol/Documents/`, pointing to the file `target.txt` on the current directory.

As with hard links, you can omit the link name to create a link with the same name as the target in the current directory.

Managing Symbolic Links

Symbolic links point to another path in the filesystem. You can create soft links to files *and* directories, even on different partitions. It is pretty easy to spot a symbolic link with the output of the `ls` command:

```
$ ls -lh
total 112K
-rw-r--r-- 1 carol carol 110K Jun  7 10:13 target.txt
lrwxrwxrwx 1 carol carol  12 Jun  7 10:14 softlink -> target.txt
```

In the example above, the first character on the permissions for the file `softlink` is `l`, indicating a symbolic link. Furthermore, just after the filename you see the name of the target the link points to, the file `target.txt`.

Note that on file and directory listings, soft links themselves always show the permissions `rwX` for the user, the group and others, but in practice the access permissions for them are the same as those for the target.

Moving and Removing Symbolic Links

Like hard links, symbolic links can be removed using `rm` and moved around or renamed using `mv`. However, special care should be taken when creating them, to avoid “breaking” the link if it is moved from its original location.

When creating symbolic links you should be aware that unless a path is fully specified the location of the target is interpreted as *relative* to the location of the link. This may create problems if the link, or the file it points to, is moved.

This is easier to understand with an example. Say you have a file named `original.txt` in the current directory, and you want to create a symbolic link to it called `softlink`. You could use:

```
$ ln -s original.txt softlink
```

And apparently all would be well. Let us check with `ls`:

```
$ ls -lh
total 112K
-r--r--r-- 1 carol carol 110K Jun  7 10:13 original.txt
lrwxrwxrwx 1 carol carol  12 Jun  7 19:23 softlink -> original.txt
```

See how the link is constructed: `softlink` points to `(→) original.txt`. However, let's see what happens if you move the link to the previous directory and try to display its contents using the command `less`:

```
$ mv softlink ../
$ less ../softlink
../softlink: No such file or directory
```

Since the path to `original.txt` was not specified, the system assumes that it is on the same directory as the link. When this is no longer true, the link stops working.

The way to prevent this is to always specify the full path to the target when creating the link:

```
$ ln -s /home/carol/Documents/original.txt softlink
```

This way, no matter where you move the link to it will still work, because it points to the absolute location of the target. Check with `ls`:

```
$ ls -lh
total 112K
lrwxrwxrwx 1 carol carol 40 Jun  7 19:34 softlink -> /home/carol/Documents/original.txt
```

Guided Exercises

1. What is the parameter for `chmod` in *symbolic* mode to enable the sticky bit on a directory?

2. Imagine there is a file named `document.txt` in the directory `/home/carol/Documents`. What is the command to create a symbolic link to it named `text.txt` on the current directory?

3. Explain the difference between a hard link to a file and a copy of this file.

Explorational Exercises

1. Imagine that inside a directory you create a file called `recipes.txt`. Inside this directory, you will also create a hard link to this file, called `receitas.txt`, and a symbolic (or *soft*) link to this called `rezepte.txt`.

```
$ touch recipes.txt
$ ln recipes.txt receitas.txt
$ ln -s recipes.txt rezepte.txt
```

The contents of the directory should be like so:

```
$ ls -lhi
total 160K
5388833 -rw-r--r-- 4 carol carol 77K jun 17 17:25 receitas.txt
5388833 -rw-r--r-- 4 carol carol  0K jun 17 17:25 recipes.txt
5388837 lrwxrwxrwx 1 carol carol  12 jun 17 17:25 rezepte.txt -> receitas.txt
```

Remember that, as a hard link, `receitas.txt` points to the same inode that `recipes.txt` is assigned. What would happen to the soft link `rezepte.txt` if the file `receitas.txt` is deleted? Why?

2. Imagine you have a flash drive plugged into your system, and mounted on `/media/youruser/FlashA`. You want to create a link called `schematics.pdf` in your home directory, pointing to the file `esquema.pdf` on the root of the flash drive. So, you type the command:

```
$ ln /media/youruser/FlashA/esquema.pdf ~/schematics.pdf
```

What would happen? Why?

3. Consider the following output of `ls -lah`:

```
$ ls -lah
total 3,1M
drwxr-xr-x 2 carol carol 4,0K jun 17 17:27 .
drwxr-xr-x 5 carol carol 4,0K jun 17 17:29 ..
-rw-rw-r-- 1 carol carol 2,8M jun 17 15:45 compressed.zip
-rw-r--r-- 4 carol carol  77K jun 17 17:25 document.txt
-rw-rw-r-- 1 carol carol 216K jun 17 17:25 image.png
```

```
-rw-r--r-- 4 carol carol 77K jun 17 17:25 text.txt
```

- How many links point to the file `document.txt`?

- Are they soft or hard links?

- Which parameter should you pass to `ls` to see which inode each file occupies?

4. Imagine you have in your `~/Documents` directory a file named `clients.txt` containing some client names, and a directory named `somedir`. Inside this there is a *different* file *also* named `clients.txt` with different names. To replicate this structure, use the following commands.

```
$ cd ~/Documents
$ echo "John, Michael, Bob" > clients.txt
$ mkdir somedir
$ echo "Bill, Luke, Karl" > somedir/clients.txt
```

You then create a link inside `somedir` named `partners.txt` pointing to this file, with the commands:

```
$ cd somedir/
$ ln -s clients.txt partners.txt
```

So, the directory structure is:

```
Documents
|-- clients.txt
`-- somedir
    |-- clients.txt
    `-- partners.txt -> clients.txt
```

Now, you move `partners.txt` from `somedir` to `~/Documents`, and list its contents.

```
$ cd ~/Documents/
$ mv somedir/partners.txt .
$ less partners.txt
```

Will the link still work? If so, which file will have its contents listed? Why?

5. Consider the following files:

```
-rw-r--r-- 1 carol carol 19 Jun 24 11:12 clients.txt
lrwxrwxrwx 1 carol carol 11 Jun 24 11:13 partners.txt -> clients.txt
```

What are the access permissions for `partners.txt`? Why?

Summary

In this lesson, you learned:

- What links are.
- The difference between *symbolic* and *hard* links.
- How to create links.
- How to move, rename or remove these links.

The following commands were discussed on this lesson:

- `ln`: The “link” command. By itself, this command creates a hard link. With the `-s` switch a *symbolic* or *soft* link can be created. Remember that hard links can only reside on the same partition and filesystem, and symbolic links can traverse partitions and filesystems (even network attached storage).
- The `-i` parameter to `ls`, which allows one to view the inode number for a file.

Answers to Guided Exercises

1. What is the parameter for `chmod` in *symbolic* mode to enable the sticky bit on a directory?

The symbol for the sticky bit in symbolic mode is `t`. Since we want to enable (add) this permission to the directory, the parameter should be `+t`.

2. Imagine there is a file named `document.txt` in the directory `/home/carol/Documents`. What is the command to create a symbolic link to it named `text.txt` on the current directory?

`ln -s` is the command to create a symbolic link. Since you should specify the full path to the file you are linking to, the command is:

```
$ ln -s /home/carol/Documents/document.txt text.txt
```

3. Explain the difference between a hard link to a file and a copy of this file.

A hard link is just another name for a file. Even though it looks like a duplicate of the original file, for all purposes both the link and the original are the same, as they point to the same data on disk. Changes made on the contents of the link will be reflected on the original, and vice-versa. A copy is a completely independent entity, occupying a different place on disk. Changes to the copy will not be reflected on the original, and vice-versa.

Answers to Explorational Exercises

1. Imagine that inside a directory you create a file called `recipes.txt`. Inside this directory, you will also create a hard link to this file, called `receitas.txt`, and a symbolic (or *soft*) link to this called `rezepte.txt`.

```
$ touch recipes.txt
$ ln recipes.txt receitas.txt
$ ln -s receitas.txt rezepte.txt
```

The contents of the directory should be like so:

```
$ ls -lhi
total 160K
5388833 -rw-r--r-- 4 carol carol 77K jun 17 17:25 receitas.txt
5388833 -rw-r--r-- 4 carol carol  0K jun 17 17:25 recipes.txt
5388837 lrwxrwxrwx 1 carol carol  12 jun 17 17:25 rezepte.txt -> receitas.txt
```

Remember that, as a hard link, `receitas.txt` points to the same inode that `recipes.txt`. What would happen to the soft link `rezepte.txt` if the file `receitas.txt` is deleted? Why?

The soft link `rezepte.txt` would stop working. This is because soft links point to names, not inodes, and the name `receitas.txt` no longer exists, even if the data is still on the disk under the name `recipes.txt`.

2. Imagine you have a flash drive plugged into your system, and mounted on `/media/youruser/FlashA`. You want to create a link called `schematics.pdf` in your home directory, pointing to the file `esquema.pdf` on the root of the flash drive. So, you type the command:

```
$ ln /media/youruser/FlashA/esquema.pdf ~/schematics.pdf
```

What would happen? Why?

The command would fail. The error message would be `Invalid cross-device link`, and it makes the reason clear: hard links cannot point to a target in a different partition or device. The only way to create a link like this is to use a *symbolic* or *soft* link, adding the `-s` parameter to `ln`.

3. Consider the following output of `ls -lah`:

```
$ ls -lah
total 3,1M
drwxr-xr-x 2 carol carol 4,0K jun 17 17:27 .
drwxr-xr-x 5 carol carol 4,0K jun 17 17:29 ..
-rw-rw-r-- 1 carol carol 2,8M jun 17 15:45 compressed.zip
-rw-r--r-- 4 carol carol 77K jun 17 17:25 document.txt
-rw-rw-r-- 1 carol carol 216K jun 17 17:25 image.png
-rw-r--r-- 4 carol carol 77K jun 17 17:25 text.txt
```

- How many links point to the file `document.txt`?

Every file starts with a link count of 1. Since the link count for the file is 4, there are three links pointing to that file.

- Are they soft or hard links?

They are hard links, since soft links do not increase the link count of a file.

- Which parameter should you pass to `ls` to see which inode each file occupies?

The parameter is `-li`. The inode will be shown as the first column in the output of `ls`, like below:

```
$ ls -lahi
total 3,1M
5388773 drwxr-xr-x 2 carol carol 4,0K jun 17 17:27 .
5245554 drwxr-xr-x 5 carol carol 4,0K jun 17 17:29 ..
5388840 -rw-rw-r-- 1 carol carol 2,8M jun 17 15:45 compressed.zip
5388833 -rw-r--r-- 4 carol carol 77K jun 17 17:25 document.txt
5388837 -rw-rw-r-- 1 carol carol 216K jun 17 17:25 image.png
5388833 -rw-r--r-- 4 carol carol 77K jun 17 17:25 text.txt
```

4. Imagine you have in your `~/Documents` directory a file named `clients.txt` containing some client names, and a directory named `somedir`. Inside this there is a *different* file *also* named `clients.txt` with different names. To replicate this structure, use the following commands.

```
$ cd ~/Documents
$ echo "John, Michael, Bob" > clients.txt
$ mkdir somedir
$ echo "Bill, Luke, Karl" > somedir/clients.txt
```

You then create a link inside `somedir` named `partners.txt` pointing to this file, with the commands:

```
$ cd somedir/  
$ ln -s clients.txt partners.txt
```

So, the directory structure is:

```
Documents  
|-- clients.txt  
`-- somedir  
    |-- clients.txt  
    `-- partners.txt -> clients.txt
```

Now, you move `partners.txt` from `somedir` to `~/Documents`, and list its contents.

```
$ cd ~/Documents/  
$ mv somedir/partners.txt .  
$ less partners.txt
```

Will the link still work? If so, which file will have its contents listed? Why?

This is a “tricky” one, but the link will work, and the file listed will be the one in `~/Documents`, containing the names John, Michael, Bob.

Remember that since you did not specify the full path to the target `clients.txt` when creating the soft link `partners.txt`, the target location will be interpreted as being relative to the location of the link, which in this case is the current directory.

When the link was moved from `~/Documents/somedir` to `~/Documents`, it should stop working, since the target was no longer in the same directory as the link. However, it just so happens that there is a file named `clients.txt` on `~/Documents`, so the link will point to this file, instead of the original target inside `~/somedir`.

To avoid this, always specify the full path to the target when creating a symbolic link.

5. Consider the following files:

```
-rw-r--r-- 1 carol carol 19 Jun 24 11:12 clients.txt  
lrwxrwxrwx 1 carol carol 11 Jun 24 11:13 partners.txt -> clients.txt
```

What are the access permissions for `partners.txt`? Why?

The access permissions for `partners.txt` are `rw-r--`, as links always inherit the same access permissions as the target.



104.7 Find system files and place files in the correct location

Reference to LPI objectives

LPIC-1 version 5.0, Exam 101, Objective 104.7

Weight

2

Key knowledge areas

- Understand the correct locations of files under the FHS.
- Find files and commands on a Linux system.
- Know the location and purpose of important file and directories as defined in the FHS.

Partial list of the used files, terms and utilities

- `find`
- `locate`
- `updatedb`
- `whereis`
- `which`
- `type`
- `/etc/updatedb.conf`



104.7 Lesson 1

Certificate:	LPIC-1
Version:	5.0
Topic:	104 Devices, Linux Filesystems, Filesystem Hierarchy Standard
Objective:	104.7 Find system files and place files in the correct location
Lesson:	1 of 1

Introduction

Linux distributions come in all shapes and sizes, but one thing that almost all of them share is that they follow the *Filesystem Hierarchy Standard* (FHS), which defines a “standard layout” for the filesystem, making interoperability and system administration much easier. In this lesson, you will learn more about this standard, and how to find files on a Linux system.

The Filesystem Hierarchy Standard

The Filesystem Hierarchy Standard (FHS) is an effort by the Linux Foundation to standardize the directory structure and directory contents on Linux systems. Compliance with the standard is not mandatory, but most distributions follow it.

NOTE

Those interested in the details of filesystem organization can read the FHS 3.0 specification, available in multiple formats at: <http://refspecs.linuxfoundation.org/fhs.shtml>

According to the standard, the basic directory structure is as follows:

/

This is the root directory, the topmost directory in the hierarchy. Every other directory is located inside it. A filesystem is often compared to a “tree”, so this would be the “trunk” to which all branches are connected.

/bin

Essential binaries, available to all users.

/boot

Files needed by the boot process, including the Initial RAM Disk (initrd) and the Linux kernel itself.

/dev

Device files. These can be either physical devices connected to the system (for example, `/dev/sda` would be the first SCSI or SATA disk) or virtual devices provided by the kernel.

/etc

Host-specific configuration files. Programs may create subdirectories under `/etc` to store multiple configuration files if needed.

/home

Each user in the system has a “home” directory to store personal files and preferences, and most of them are located under `/home`. Usually, the home directory is the same as the username, so the user John would have his directory under `/home/john`. The exceptions are the superuser (root), which has a separate directory (`/root`) and some system users.

/lib

Shared libraries needed to boot the operating system and to run the binaries under `/bin` and `/sbin`.

/media

User-mountable removable media, like flash drives, CD and DVD-ROM readers, floppy disks, memory cards and external disks are mounted under here.

/mnt

Mount point for temporarily mounted filesystems.

/opt

Application software packages.

/root

Home directory for the superuser (root).

/run

Run-time variable data.

/sbin

System binaries.

/srv

Data served by the system. For example, the pages served by a web server could be stored under `/srv/www`.

/tmp

Temporary files.

/usr

Read-only user data, including data needed by some secondary utilities and applications.

/proc

Virtual filesystem containing data related to running processes.

/var

Variable data written during system operation, including print queue, log data, mailboxes, temporary files, browser cache, etc.

Keep in mind that some of those directories, like `/etc`, `/usr` and `/var`, contain a whole hierarchy of subdirectories under them.

Temporary Files

Temporary files are files used by programs to store data that are only needed for a short time. This can be the data of running processes, crash logs, scratch files from an autosave, intermediary files used during a file conversion, cache files and such.

Location of Temporary Files

Version 3.0 of the *Filesystem Hierarchy Standard* (FHS) defines standard locations for temporary

files on Linux systems. Each location has a different purpose and behavior, and it is recommended that developers follow the conventions set by the FHS when writing temporary data to disk.

/tmp

According to the FHS, programs should not assume that files written here will be preserved between invocations of a program. The recommendation is that this directory be cleared (all files erased) during system boot-up, although this is not mandatory.

/var/tmp

Another location for temporary files, but this one *should not be cleared* during the system boot-up. Files stored here will usually persist between reboots.

/run

This directory contains run-time variable data used by running processes, such as process identifier files (.pid). Programs that need more than one run-time file may create subdirectories here. This location *must be cleared* during system boot-up. The purpose of this directory was once served by /var/run, and on some systems /var/run may be a symbolic link to /run.

Note that there is nothing which prevents a program from creating temporary files elsewhere on the system, but it is good practice to respect the conventions set by the FHS.

Finding Files

To search for files on a Linux system, you can use the `find` command. This is a very powerful tool, full of parameters that can suit its behaviour and modify output exactly to your needs.

To start, `find` needs two arguments: a starting point and what to look for. For example, to search for all files in the current directory (and subdirectories) whose name ends in `.jpg` you can use:

```
$ find . -name '*.jpg'
./pixel_3a_seethrough_1.jpg
./Mate3.jpg
./Expert.jpg
./Pentaro.jpg
./Mate1.jpg
./Mate2.jpg
./Sala.jpg
./Hotbit.jpg
```

This will match any file whose last four characters of the name are `.jpg`, no matter what comes before it, as `*` is a wildcard for “anything”. However, see what happens if another `*` is added at the end of the pattern:

```
$ find . -name '*.jpg*'
./pixel_3a_seethrough_1.jpg
./Pentaro.jpg.zip
./Mate3.jpg
./Expert.jpg
./Pentaro.jpg
./Mate1.jpg
./Mate2.jpg
./Sala.jpg
./Hotbit.jpg
```

The file `Pentaro.jpg.zip` (highlighted above) was not included in the previous listing, because even if it contains `.jpg` on its name, it did not match the pattern as there were extra characters after it. The new pattern means “anything `.jpg` anything”, so it matches.

TIP Keep in mind that the `-name` parameter is case sensitive. If you wish to do a case-insensitive search, use `-iname`.

The `*.jpg` expression must be placed inside single quotes, to avoid the shell from interpreting the pattern itself. Try without the quotes and see what happens.

By default, `find` will begin at the starting point and descend through any subdirectories (and subdirectories of those subdirectories) found. You can restrict this behaviour with the `-maxdepth N` parameters, where `N` is the maximum number of levels.

To search only the current directory, you would use `-maxdepth 1`. Suppose you have the following directory structure:

```
directory
├─ clients.txt
├─ partners.txt -> clients.txt
└─ somedir
   └─ anotherdir
      └─ clients.txt
```

To search inside `somedir`, you would need to use `-maxdepth 2` (the current directory +1 level down). To search inside `anotherdir`, `-maxdepth 3` would be needed (the current directory +2

levels down). The parameter `-mindepth N` works in the opposite way searching only in directories *at least* `N` levels down.

The `-mount` parameter can be used to avoid `find` going down inside mounted filesystems. You can also restrict the search to specific filesystem types using the `-fstype` parameter. So `find /mnt -fstype exfat -iname "*report*"` would only search inside exFAT filesystems mounted under `/mnt`.

Searching for Attributes

You can use the parameters below to search for files with specific attributes, like ones that are writable by your user, have a specific set of permissions or are of a certain size:

-user USERNAME

Matches files owned by the user `USERNAME`.

-group GROUPNAME

Matches files owned by the group `GROUPNAME`.

-readable

Matches files that are readable by the current user.

-writable

Matches files that are writable by the current user.

-executable

Matches files that are executable by the current user. In the case of directories, this will match any directory that the user can enter (`x` permission).

-perm NNNN

This will match any files that have exactly the `NNNN` permission. For example, `-perm 0664` will match any files which the user and group can read and write to and which others can read (or `rw-rw-r--`).

You can add a `-` before `NNNN` to check for files that have *at least* the permission specified. For example, `-perm -644` would match files that have at least `644` (`rw-r--`) permissions. This includes a file with `664` (`rw-rw-r--`) or even `775` (`rw-rwx`).

-empty

Will match empty files and directories.

-size N

Will match any files of N size, where N by default is a number of 512-byte blocks. You can add suffixes to N for other units: Nc will count the size in bytes, Nk in kibibytes (KiB, multiples of 1024 bytes), NM in mebibytes (MiB, multiples of 1024 * 1024) and NG for gibibytes (GiB, multiples of 1024 * 1024 * 1024).

Again, you can add the + or - prefixes (here meaning *bigger than* and *smaller than*) to search for relative sizes. For example, -size -10M will match any file less than 10 MiB in size.

For example, to search for files under your home directory which contain the case insensitive pattern report in any part of the name, have 0644 permissions, have been accessed 10 days ago and which size is at least 1 Mib, you could use

```
$ find ~ -iname "*report*" -perm 0644 -atime 10 -size +1M
```

Searching by Time

Besides searching for attributes you can also perform searches by time, finding files that were accessed, had their attributes changed or were modified during a specific period of time. The parameters are:

-amin N, -cmin N, -mmin N

This will match files that have been accessed, had attributes changed or were modified (respectively) N minutes ago.

-atime N, -ctime N, -mtime N

This will match files that were accessed, had attributes changed or were modified N*24 hours ago.

For -cmin N and -ctime N, any attribute change will cause a match, including a change in permissions, reading or writing to the file. This makes these parameters especially powerful, since practically any operation involving the file will trigger a match.

The following example would match any file in the current directory which has been modified less than 24 hours ago and is bigger than 100 MiB:

```
$ find . -mtime -1 -size +100M
```

Using `locate` and `updatedb`

`locate` and `updatedb` are commands that can be used to quickly find a file matching a given pattern on a Linux system. But unlike `find`, `locate` will not search the filesystem for the pattern: instead, it looks it up on a database built by running the `updatedb` command. This gives you very quick results, but they may be imprecise depending on when the database was last updated.

The simplest way to use `locate` is to just give it a pattern to search for. For example, to find every JPEG image on your system, you would use `locate jpg`. The list of results can be quite lengthy, but should look like this:

```
$ locate jpg
/home/carol/Downloads/Expert.jpg
/home/carol/Downloads/Hotbit.jpg
/home/carol/Downloads/Mate1.jpg
/home/carol/Downloads/Mate2.jpg
/home/carol/Downloads/Mate3.jpg
/home/carol/Downloads/Pentaro.jpg
/home/carol/Downloads/Sala.jpg
/home/carol/Downloads/pixel_3a_seethrough_1.jpg
/home/carol/Downloads/jpg_specs.doc
```

When asked for the `jpg` pattern, `locate` will show anything that contains this pattern, no matter what comes before or after it. You can see an example of this in the file `jpg_specs.doc` in the listing above: it contains the pattern, but the extension is not `jpg`.

TIP Remember that with `locate` you are matching patterns, not file extensions.

By default the pattern is case sensitive. This means that files containing `.JPG` would not be shown since the pattern is in lowercase. To avoid this, pass the `-i` parameter to `locate`. Repeating our previous example:

```
$ locate -i .jpg
/home/carol/Downloads/Expert.jpg
/home/carol/Downloads/Hotbit.jpg
/home/carol/Downloads/Mate1.jpg
/home/carol/Downloads/Mate1_old.JPG
/home/carol/Downloads/Mate2.jpg
/home/carol/Downloads/Mate3.jpg
/home/carol/Downloads/Pentaro.jpg
/home/carol/Downloads/Sala.jpg
```

```
/home/carol/Downloads/pixel_3a_seethrough_1.jpg
```

Notice that the file `Mate1_old.JPG`, in bold above, was not present in the previous listing.

You can pass multiple patterns to `locate`, just separate them with spaces. The example below would do a case-insensitive search for any files matching the `zip` and `jpg` patterns:

```
$ locate -i zip jpg
/home/carol/Downloads/Expert.jpg
/home/carol/Downloads/Hotbit.jpg
/home/carol/Downloads/Mate1.jpg
/home/carol/Downloads/Mate1_old.JPG
/home/carol/Downloads/Mate2.jpg
/home/carol/Downloads/Mate3.jpg
/home/carol/Downloads/OPENMSXPIHAT.zip
/home/carol/Downloads/Pentaro.jpg
/home/carol/Downloads/Sala.jpg
/home/carol/Downloads/gbs-control-master.zip
/home/carol/Downloads/lineage-16.0-20190711-MOD-quark.zip
/home/carol/Downloads/pixel_3a_seethrough_1.jpg
/home/carol/Downloads/jpg_specs.doc
```

When using multiple patterns, you can request `locate` to show only files that match *all* of them. This is done with the `-A` option. The following example would show any file matching the `.jpg` and the `.zip` patterns:

```
$ locate -A .jpg .zip
/home/carol/Downloads/Pentaro.jpg.zip
```

If you wish to count the number of files that match a given pattern instead of showing their full path you can use the `-c` option. For example, to count the number of `.jpg` files on a system:

```
$ locate -c .jpg
1174
```

One problem with `locate` is that it only shows entries present in the database generated by `updatedb` (located in `/var/lib/mlocate.db`). If the database is outdated, the output could show files that have been deleted since the last time it was updated. One way to avoid this is to add the `-e` parameter, which will make it check to see if the file still exists before showing it on the output.

Of course, this will not solve the problem of files created *after* the last database update not showing up. For this you will have to update the database with the command `updatedb`. How long this will take will depend on the amount of files of your disk.

Controlling the Behavior of `updatedb`

The behavior of `updatedb` can be controlled by the file `/etc/updatedb.conf`. This is a text file where each line controls one variable. Blank lines are ignored and lines that start with the `#` character are treated as comments.

PRUNES=

Any filesystem types indicated after this parameter will not be scanned by `updatedb`. The list of types should be separated by spaces, and the types themselves are case-insensitive, so `NFS` and `nfs` are the same.

PRUNENAMES=

This is a space-separated list of directory names that should not be scanned by `updatedb`.

PRUNEPATHS=

This is a list of path names that should be ignored by `updatedb`. The path names must be separated by spaces and specified in the same way they would be shown by `updatedb` (for example, `/var/spool /media`)

PRUNE_BIND_MOUNTS=

This is a simple `yes` or `no` variable. If set to `yes` bind mounts (directories mounted elsewhere with the `mount --bind` command) will be ignored.

Finding Binaries, Manual Pages and Source Code

`which` is a very useful command that shows the full path to an executable. For example, if you want to locate the executable for `bash`, you could use:

```
$ which bash
/usr/bin/bash
```

If the `-a` option is added the command will show all pathnames that match the executable. Observe the difference:

```
$ which mkfs.ext3
/usr/sbin/mkfs.ext3
```



```
$ which -a mkfs.ext3
/usr/sbin/mkfs.ext3
/sbin/mkfs.ext3
```

TIP

To find which directories are in the `PATH` use the `echo $PATH` command. This will print (echo) the contents of the variable `PATH` (`$PATH`) to your terminal.

`type` is a similar command which will show information about a binary, including where it is located and its type. Just use `type` followed by the command name:

```
$ type locate
locate is /usr/bin/locate
```

The `-a` parameter works in the same way as in `which`, showing all pathnames that match the executable. Like so:

```
$ type -a locate
locate is /usr/bin/locate
locate is /bin/locate
```

And the `-t` parameter will show the file type of the command which can be `alias`, `keyword`, `function`, `builtin` or `file`. For example:

```
$ type -t locate
file

$ type -t ll
alias

$ type -t type
type is a built-in shell command
```

The command `whereis` is more versatile and besides binaries can also be used to show the location of man pages or even source code for a program (if available in your system). Just type `whereis` followed by the binary name:

```
$ whereis locate
locate: /usr/bin/locate /usr/share/man/man1/locate.1.gz
```

The results above include binaries (`/usr/bin/locate`) and compressed manual pages (`/usr/share/man/man1/locate.1.gz`).

You can quickly filter the results using commandline switches like `-b`, which will limit them to only the binaries, `-m`, which will limit them to only man pages, or `-s`, which will limit them to only the source code. Repeating the example above, you would get:

```
$ whereis -b locate
locate: /usr/bin/locate

$ whereis -m locate
locate: /usr/share/man/man1/locate.1.gz
```

Guided Exercises

1. Imagine a program needs to create a single-use temporary file that will never be needed again after the program is closed. What would be the correct directory to create this file in?

2. Which is the temporary directory that *must* be cleared during the boot process?

3. Using `find`, search only the current directory for files that are writable by the user, have been modified in the last 10 days and are bigger than 4 GiB.

4. Using `locate`, find any files containing both the patterns `report` and either `updated`, `update` or `updating` on their names.

5. How can you find where the manpage for `ifconfig` is stored?

6. What variable needs to be added to `/etc/updatedb.conf` to make `updatedb` ignore `ntfs` filesystems?

7. A system administrator wishes to mount an internal disk (`/dev/sdc1`). According to the FHS, under which directory should this disk be mounted?

Explorational Exercises

1. When `locate` is used, the results are pulled from a database generated by `updatedb`. However this database can be outdated, causing `locate` to show files that no longer exist. How can you make `locate` show only existing files on its output?
2. Find any file on the current directory or subdirectories up to 2 levels down, excluding mounted filesystems, that contain the pattern `Status` or `statute` on their names.
3. Limiting the search to `ext4` filesystems, find any files under `/mnt` that have at least execute permissions for the group, are readable for the current user and had any attribute changed in the last 2 hours.
4. Find empty files which were created more than 30 days ago and are at least two levels down from the current directory
5. Consider that the users `carol` and `john` are part of the group `mkt`. Find on `john`'s home directory any files that are also readable by `carol`.

Summary

In this lesson you learned about the basic organization of the filesystem on a Linux machine, according to the FHS, and how to find binaries and files, either by name or by attributes. The following commands were discussed in this lesson:

find

A versatile command used to find files and folders based on a variety of search criteria.

locate

A utility that uses a local database that contains the locations for locally stored files.

updatedb

Updates the local database used by the `locate` command.

which

Displays the full path to an executable.

whereis

Displays the locations of manual pages, binaries and source code on the system.

type

Displays the location of a binary and the type of application that it is (such as a program that is installed, a built-in Bash program and more).

Answers to Guided Exercises

1. Imagine a program needs to create a single-use temporary file that will never be needed again after the program is closed. What would be the correct directory to create this file in?

Since we don't care about the file after the program finishes running, the correct directory is `/tmp`.

2. Which is the temporary directory that *must* be cleared during the boot process?

The directory is `/run` or, on some systems, `/var/run`.

3. Using `find`, search only the current directory for files that are writable by the user, have been modified in the last 10 days and are bigger than 4 GiB.

For this you will need the `-writable`, `-mtime` and `-size` parameters:

```
find . -writable -mtime -10 -size +4G
```

4. Using `locate`, find any files containing both the patterns `report` and either `updated`, `update` or `updating` on their names.

Since `locate` needs to match all patterns, use the `-A` option:

```
locate -A "report" "updat"
```

5. How can you find where the manpage for `ifconfig` is stored?

Use the `-m` parameter for `whereis`:

```
whereis -m ifconfig
```

6. What variable needs to be added to `/etc/updatedb.conf` to make `updatedb` ignore `ntfs` filesystems?

The variable is `PRUNEFs` followed by the filesystem type: `PRUNEFs=ntfs`

7. A system administrator wishes to mount an internal disk (`/dev/sdc1`). According to the FHS, under which directory should this disk be mounted?

In practice, the disk can be mounted anywhere. However, the FHS recommends that temporary

mounts be done under `/mnt`

Answers to Explorational Exercises

1. When `locate` is used, the results are pulled from a database generated by `updatedb`. However, this database can be outdated, causing `locate` to show files that no longer exist. How can you make `locate` show only existing files on its output?

Add the `-e` parameter to `locate`, as in `locate -e PATTERN`.

2. Find any file on the current directory or subdirectories up to 2 levels down, excluding mounted filesystems, that contain the pattern `Status` or `statute` on their names.

Remember that for `-maxdepth` you have to also consider the current directory, so we want three levels (the current plus 2 levels down):

```
find . -maxdepth 3 -mount -iname "*statu*"
```

3. Limiting the search to `ext4` filesystems, find any files under `/mnt` that have at least execute permissions for the group, are readable for the current user and had any attribute changed in the last 2 hours.

Use the `-fstype` parameter of `mount` to limit the search to specific filesystem types. A file readable by the current user would have at least 4 in the first digit of the permissions, and one executable by the group would have at least 1 on the second digit. Since we don't care about the permissions for others, we can use 0 for the third digit. Use `-cmin N` to filter recent attribute changes, remembering that N is specified in minutes. So:

```
find /mnt -fstype ext4 -perm -410 -cmin -120
```

4. Find empty files which were modified more than 30 days ago and are at least two levels down from the current directory

The parameter `-mindepth N` can be used to limit the search to at least N levels down, but remember that you have to include the current directory in the count. Use `-empty` to check for empty files, and `-mtime N` to check for modification time. So:

```
find . -empty -mtime +30 -mindepth 3
```

5. Consider that the users `carol` and `john` are part of the group `mkt`. Find on `john's` home directory any files that are also readable by `carol`.

Considering they are members of the same group, we need at least an `r` (4) on the group permissions, and we do not care about the others. So:

```
find /home/john -perm -040
```

Imprint

© 2026 by Linux Professional Institute (LPI): Learning Materials, “LPIC-1 (101) (Version 5.0)”.

PDF generated: 2026-08-07

This work is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND 4.0). To view a copy of this license, visit

<https://creativecommons.org/licenses/by-nc-nd/4.0/>

While Linux Professional Institute has used good faith efforts to ensure that the information and instructions contained in this work are accurate, Linux Professional Institute disclaims all responsibility for errors or omissions, including without limitation responsibility for damages resulting from the use of or reliance on this work. Use of the information and instructions contained in this work is at your own risk. If any code samples or other technology this work contains or describes is subject to open source licenses or the intellectual property rights of others, it is your responsibility to ensure that your use thereof complies with such licenses and/or rights.

The LPI Learning Materials are an initiative of Linux Professional Institute (<https://lpi.org>). Learning Materials and their translations can be found at <https://learning.lpi.org>.

For questions and comments on this edition as well as on the entire project write an email to: learning@lpi.org.